

OPERATIONAL SOFTWARE
FOR MANUFACTURERS & JOB SHOPS



DBA MANUFACTURING USER GUIDE

- *Cut costs*
- *Manage inventory*
- *Deliver on time*

0602

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1. INTRODUCTION

Welcome!

Welcome to DBA Manufacturing, an operational software package for manufacturers and job shops. DBA provides a powerful set of tools that will help you cut costs, manage inventory, and deliver on time.

This user guide is designed to give you a conceptual grasp of the system. The more up-front time you spend studying the system, the more smoothly and quickly the implementation process will be.

About this user guide

This user guide focuses on giving you a conceptual grasp of the system and does not attempt to cover all the operational details of the DBA system. Once you understand how the system works, learning to use the individual programs that comprise it is easy.

Use on-line help for step-by-step instructions

Step-by-step instructions for all the system's individual programs are found in DBA's extensive on-line help system.

To get help on a topic while in DBA, click *Help* on the master menu and navigate to your topic using the table of contents or index or search functions or click *Help* within any program (or press F1) and the on-line Help will open directly with the Help for that program.

Any help topic can be printed out, if desired.

Accounting

DBA Manufacturing is designed to function independently from accounting, thus allowing you to keep your present accounting system. Accounting is available as an optional add-on to DBA Manufacturing and is covered by a separate user guide.

Because they are related to the sales order-work order-inventory cycle, we recommend that you invoice and track customer payments within DBA. You can, however, shut-off DBA's sales system and continue to invoice and track customer payments within your accounting system.

A dozen journal entries at the end of each period are all that is required to update your accounting system to reflect all the activities of the manufacturing system. The specifics are covered in chapter 30, *Using DBA with Your Accounting System*.

Support services

A full range of support services is available to help insure a successful implementation. Phone support is provided by experienced support staff who can

answer your questions and help guide you through the implementation process. For on-site assistance, independent consultants are available in most areas of the country who can help you with training and consulting. Additionally, some of our independent consultants offer group training classes. For a listing of consultants and their services, see our web site (www.DBAsoftware.com).

How to get technical support

Requests for technical support are made by calling our 800-434-0809 support line and are handled on a call back basis in the order received. Phone support is provided on a prepayment basis or on a per-call basis via credit card. For more information on rates and policies, see the *DBA Support Policies* sheet that came with your shipment.

DBA forms

DBA forms are provided by *Fortune Forms, Inc* (800-884-8804). A universal form is available that offers the advantage of using a single form for all printed forms. Pre-printed forms are available for invoices, statements, purchase orders, and shipping labels. Call *Fortune Forms* directly for any questions related to DBA compatible printed products.

Importing your data

DBA comes with a *Data Exchange* module that helps you import the following master files: inventory, customers, vendors, bills of materials, and routings. If you wish to import more than just these master files or if you want your data imported for you, a listing of independent consultants who can help with data import and custom programming can be found on our web site (www.DBAsoftware.com).

Third party add-on products

Because DBA is a widely used system with open source code, add-on products are available from third parties that expand the usefulness of the system. A listing of these add-on products can be found on our web site (www.DBAsoftware.com).

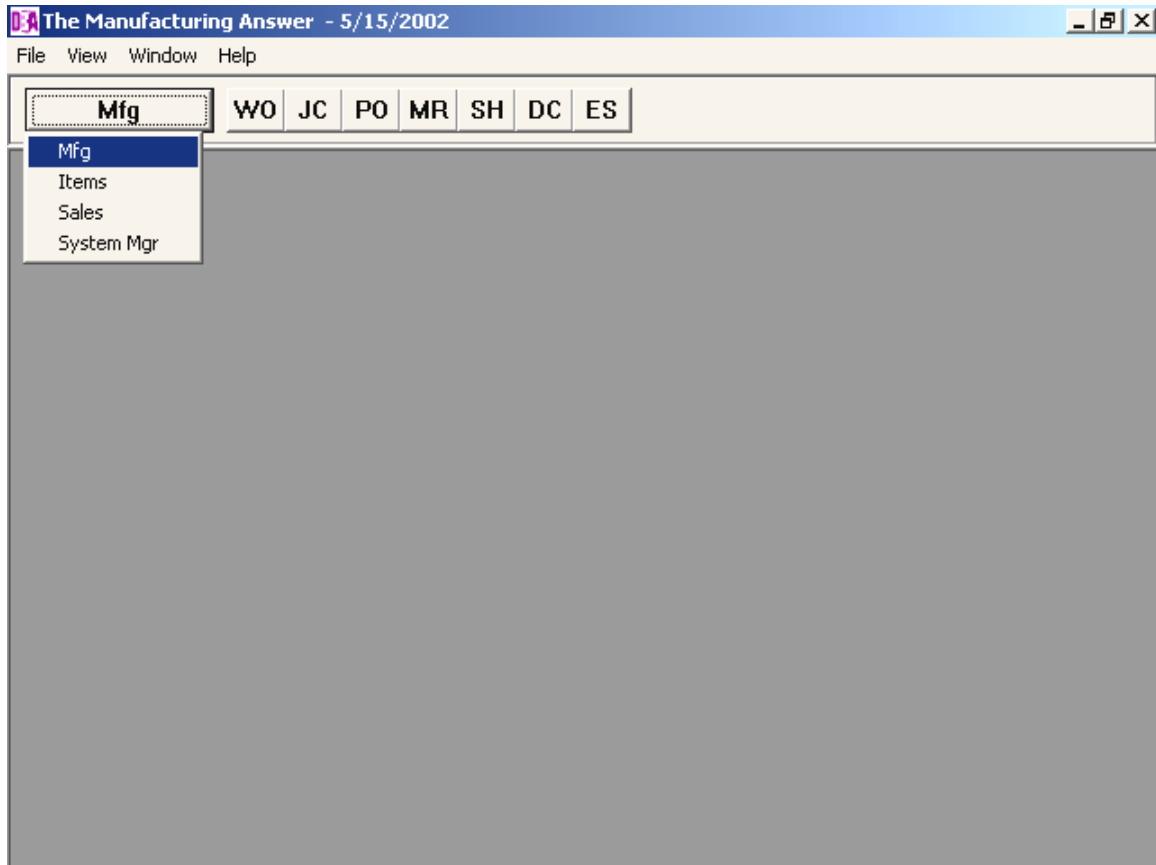
Make sure you have Internet access and e-mail

All customer communications are sent by e-mail and service packs are distributed via our web site. You should therefore make sure you have Internet access and e-mail capability.

Send us your suggestions

We continually strive to make DBA the best product of its kind. A great many of its features have come from customer suggestions and we welcome your ideas. We look forward to working with you and growing together in the years to come.

2. HOW TO USE THE SOFTWARE



Main Menu

Starting the system

To start DBA click Windows *Start*, then *Programs*, then the *DBA Software* program group, then *DBA Manufacturing*. The first screen presented will be the DBA logo and version number. If you have password security activated, you will first be prompted for a *Logon Code* and *Password* followed by a screen showing your user registration information (name and address, licensed number of users, start date, etc.). Press any key and you will be taken to the DBA main menu. If you do not press a key, after several seconds the registration screen will automatically close itself.

WARNING: Do not click the "X" in the upper right corner to close the registration screen. Instead, press any key or wait until it closes itself.

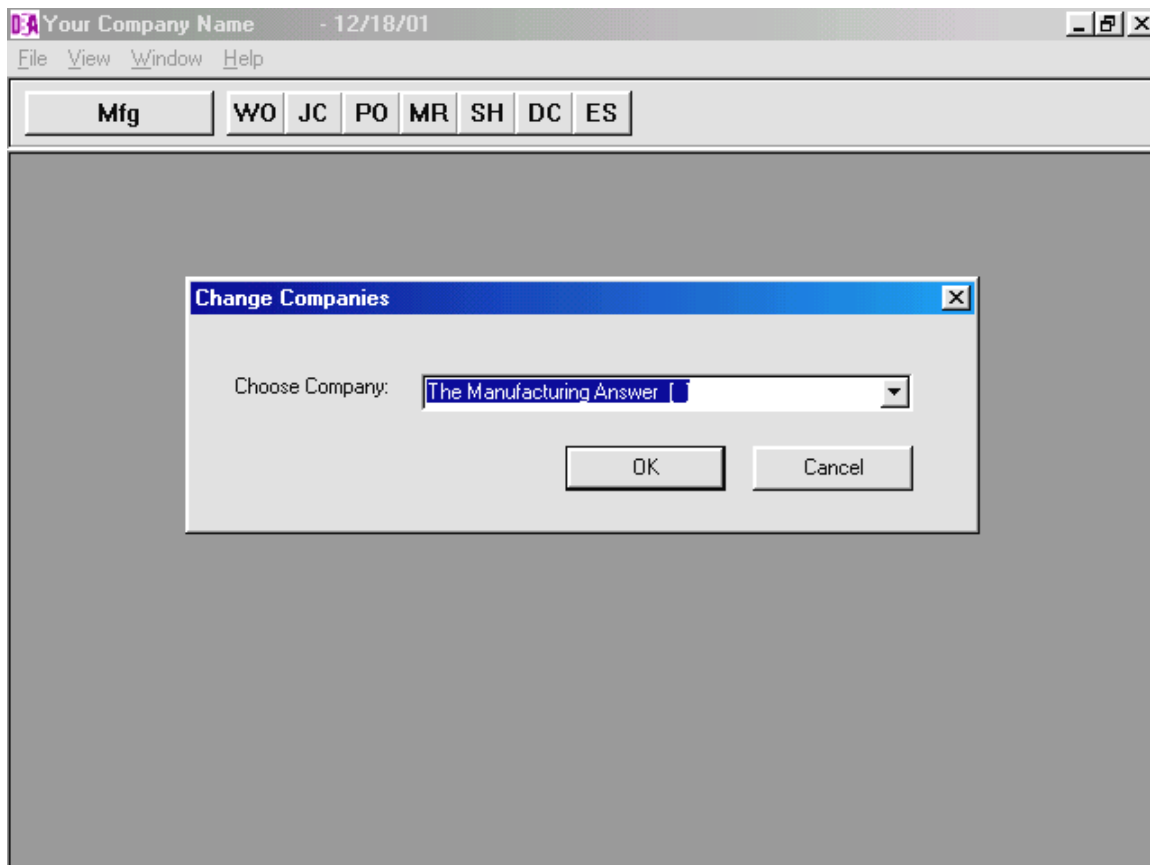
An alternative way to start DBA from Windows is to go to Windows *Run* and type the drive letter and path to your DBA files, followed by \TP5WDBA, then click *OK*. For example, if your DBAMFG subdirectory is on drive F, you would type F:\DBAMFG\TP5WDBA.

Changing companies

DBA supports virtually an unlimited number of companies. To change companies, click on *File* in the upper left corner, then on *Change Company*. You will be presented with a list of companies in a window (the company that you are currently logged into will *not* be included in the list). Click on your choice and then click *OK*. The current company name is always displayed on the banner at the very top of the screen.

The default company that you see upon system startup is normally reserved for your active primary company and is normally not numbered (and therefore is sometimes referred to as the “blank” company). The system comes with a sample company number 99, *The Manufacturing Answer*.

For more information on creating and using multiple companies, see chapter 26, *Using Multiple Companies*.



Changing companies

Exiting the system

You can exit DBA three ways: one, by clicking on the X button on the upper right of the screen; two, by clicking on *File* in the upper left and then on *Exit*; three, by

double clicking on the DBA icon on the very upper left of the screen. You cannot exit DBA until all application programs (menu selections) have been closed first.

Using DBA's master menus

DBA is organized into four master menus: *Mfg*, *Items*, *Sales*, and *System Mgr*.

You can switch from menu to menu by clicking on the menu button on the far left, which will then display a pulldown menu with a listing of the other master menus. Click on your choice or press the alpha character underlined on each choice

You can also click on *File* to get a listing of the master menus on the *File* pulldown menu. Click on your choice or press the alpha character underlined on each choice.

Click on any module button and you will be presented with a pull down menu of that module's programs. Once the pulldown menu is displayed, you can either click on your program choice, press the letter to the left of your program choice, or you can use your arrow keys to highlight your choice and then press <Enter>.

View function

The *View* function, located at the top of the master menus, allows you to change the sized of your DBA program screens to compensate for different sized monitors and screen resolutions.

DBA is pre-set to run in 640 x 480 screen resolution, in which case the DBA program screens occupy the full width of the monitor. If you have a monitor set to a higher resolution the DBA program screens will be smaller in relation to the overall screen size – the higher the resolution the smaller the DBA program screens will appear to be.

You can make the DBA screens larger to compensate for higher resolution by clicking on *View* at the top of any of the master menus. Three screen size settings are available: *640x480 – Small*, *1024x768 – Medium*, and *1280x1024 – Large*. Click on a screen size and your selection will be check marked. You will receive a message indicating that you must exit and re-start DBA for the new screen size to take effect.

NOTE: DBA screens require the Windows *Small Fonts* setting to display properly. If you see field names and boxes not lining up with the entry fields, your screens are most likely set to *Large Fonts*. To check your fonts setting, click *Start*, *Control Panel*, the *Display* icon, then the *Settings* tab. Change the *Font size* setting to *Small Fonts*, then click *OK*.

View also lets you select how you want multiple DBA programs to be displayed – in a stacking or cascading fashion. Click on either *Stack Screens* or *Cascade Screens* – the current selection is marked with a checkmark.

NOTE: The screen resolution and *Stack* and *Cascade* settings are stored in each workstation's TP5WIN.INI file (located in the local WINDOWS or WINNT subdirectory) and thus allows each workstation to have its own settings.

Window function

At the top of the master menus is a function called *Window*. The *Window* function provides a listing of open DBA programs and lets you switch from one program to another. If you click on *Window* you will see all open programs listed, with the one currently displayed on the screen marked with a checkmark. Clicking on another program will checkmark it and bring it to the front of the “stack” and it will now be displayed on the screen. If you click on *Cascade*, the program screens will change from being stacked directly on top of one another to a cascading view.

Using on-line help

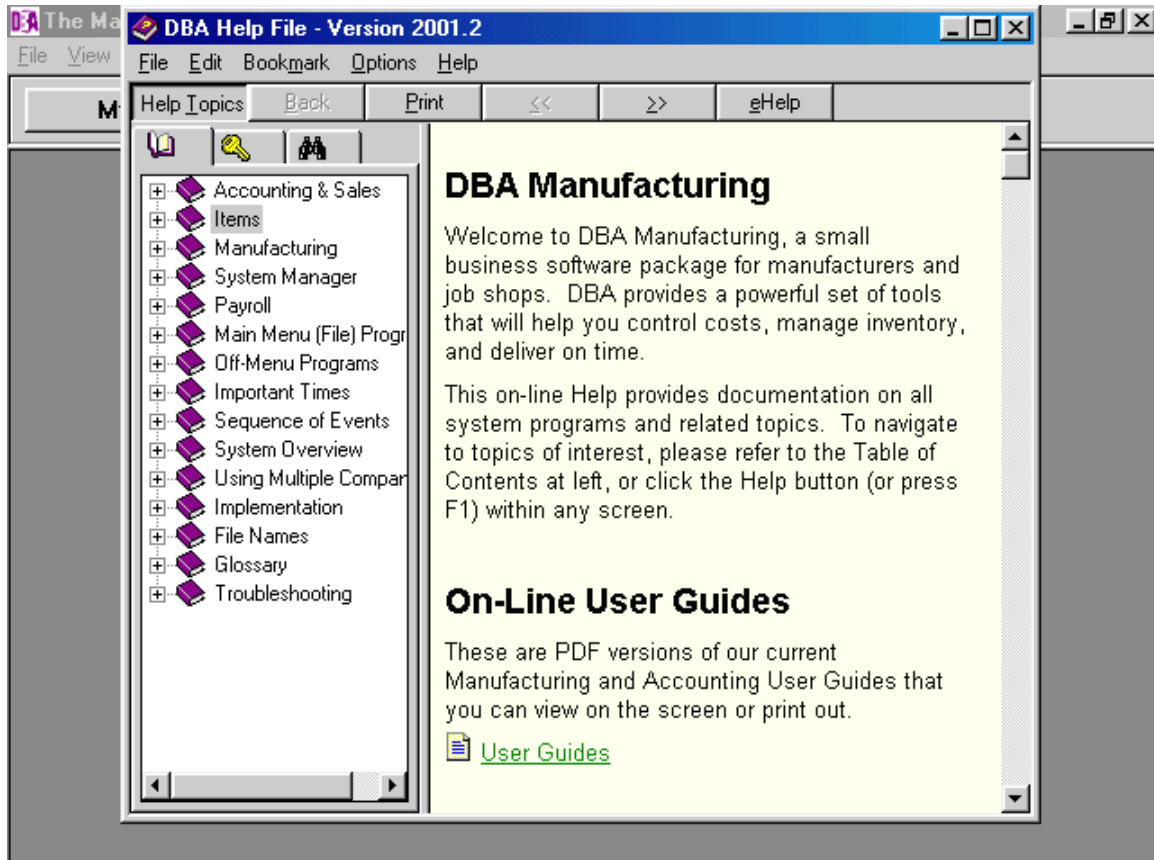
Documentation on each of the system programs (purpose of the program, field explanations, general program operation) is found in DBA's on-line help file. The help file is accessed from any of the master menus by clicking on *Help* at the top of each menu or by clicking *Help* (or pressing F1) from within any program that does not have field specific help.

Once you click on *Help* you will be presented with the *Table of Contents*. Three or four clicks at the most will take you to the documentation on any particular program. You can advance forward from topic-to-topic by clicking on the >> button and backwards from topic-to-topic by clicking on the << button. Clicking on the *Back* button returns you to whatever previous screen you were on.

Within a topic you can use your <PgDn> and <PgUp> keys to move down or up a screen at a time or you can use your mouse on the scrolling bar on the right hand side to move up or down through the topic. Generally the documentation on each program is organized into a single help topic.

If you will be referring to the on-line help frequently it can remain loaded in the background and does not have to be opened each time you wish to refer to it. Any time you want to access help, simply click on the *DBA Help* icon on your Windows taskbar the help screen will become the active window in front of your DBA screen.

A useful feature of on-line help is the *Search* function. If you cannot find a particular topic through the table of contents or submenus, you can click on the *Search* button and then enter the topic you are looking for. With each keystroke the display below shows all topics that begin with the letters you have entered. Once you find the topic you are looking for you can double click on it and you will be taken to that topic.



Help main menu

If you wish to get a hard copy printout of any help topic, click on the *Print* button. The user manual is supplied in a 3-ring binder format so that you can insert any of these printouts along with personal notes, report samples, etc., to create your own customized user guide.

Standard buttons & function keys

Clickable buttons and/or their function key equivalents perform various functions throughout DBA such as calling up lookups, saving records, exiting the program, etc. You can either click on screen buttons (mostly located along the bottom of each screen) or press an equivalent function key. The standard buttons are described below and their equivalent function keys are shown in parentheses.

Help (F1)

Displays whatever help may be available within each program or opens the on-line help for the program if field specific help does not exist within the program.

Lookup (F2) and “Magnifying Glass” Icon

Clicking this button displays a pop-up window that lets you search files for things such as item numbers, customer codes, vendor codes, etc. See *Using Lookups*

later in this chapter for more detail on how to use Lookups. An alternative to clicking on the *Lookup* button on the bottom of the screen is to use the “magnifying glass” icon that is located to the left of any field that contains a lookup. You must first move your cursor to the field itself, at which time the magnifying glass icon turns from gray to green in color, meaning that it is active. Once the icon is active, you can click on it to display the lookup on that field.

Clear (F3)

Clicking this button clears all fields on the screen. This function does not affect the files in memory; in other words the information you just cleared from the screen is not deleted from the data files.

Delete (F4)

Clicking this button deletes the entire record currently on the screen. This option differs from the *Clear* function in that it deletes the record from the data files, rather than from the screen only. USE WITH CAUTION.

First (F5)

Clicking on this button retrieves the first record in the file. It can only be used on a key (searchable) field.

Last (F6)

Clicking on this button retrieves the last record in the file. It can only be used on a key (searchable) field

Previous (F7)

Clicking on this button scrolls you backward through the file one record at a time. A *First* (F5), *Last* (F6), or *Find* (F9) must have been selected before this option can be used.

Next (F8)

Clicking on this button scrolls you forward through the file one record at a time. A *First* (F5), *Last* (F6), or *Find* (F9) must have been selected before this option can be used.

Find (F9)

When your cursor is positioned on a key (searchable) field you can enter one or a few characters and then click on this button to find the closest match in the file. For example, if you type ‘51’ in an account field and then click on the *Find* button, the first account beginning with 51 will be displayed. If an exact match is not found the program will find the next closest match.

Save (F10)

Clicking on this button saves the current information on the screen to the data

files. In most programs once you have completed the screen you will click on this button to save your entries, unless automatically prompted to do so.

Print (F10)

All reports in DBA have an entry screen in which you can enter a variety of from/thru ranges and other selection criteria to control what appears on your report. If you wish to print all entries and avoid pressing <Enter> through all the fields, most reports offer a *Print* button shortcut that bypasses the selection fields and prints all the information.

Exit

Clicking on the *Exit* button allows you to escape any screen without saving any information that has been entered or changed. As an alternative you can also press the <Esc> key to do the same thing.

Esc

Pressing the <Esc> key allows you to escape any screen without saving any information that has been entered or changed. The <Esc> key provides an alternative to clicking on the *Exit* button.

Up Arrow

Moves the cursor to the previous field.

Down Arrow

Moves the cursor to the next field. You can also press <Enter> to move forward from field to field.

Left Arrow

Moves the cursor to the next character position in a field.

Right Arrow

Moves the cursor to the previous character position in a field.

Backspace

Deletes the character to the left of the cursor and shifts all characters to the right of the cursor one position to the left.

Delete

Deletes the character to right of the cursor and shifts all characters to the right of the cursor one position to the left.

Insert

Inserts a space at the current cursor location and shifts all characters to the right of

the cursor one position.

CTRL+R - Restore Field

Restores the original entry in the current field. If you have changed the entry in a field and the cursor has not left that field yet, this option will restore the entry as it was before you changed it. Once the cursor leaves the field you can no longer use this option.

Printing

Before doing any printing make sure your printers are configured correctly for use with DBA. See appendix A, *Setting up Printing*, for details.

All reports and forms printing programs in DBA have an entry screen in which you can enter a variety of from/thru ranges and other selection criteria to control what appears on your report or form. If you wish to print all entries and avoid pressing <Enter> through all the fields, most reports offer a *Print* button shortcut that bypasses the selection fields and prints all the information.

NOTE: When you leave a *From* entry blank it means you want the record selection to begin at the beginning of the file. When you leave a *Thru* entry blank it means you want to select records all the way through to the end of the file. If you leave both ranges blank, it means you want to include all records in the file.

At the present time most DBA reports use a text-based non-proportional font. Forms, labels, a few reports and checks, however, optionally print in graphical versions. The printing screen is slightly different between the two and is described as follows.

Printing reports

Whenever you print a text-based report or form you are presented with a choice of four buttons: *List*, *Printer*, *Disk*, and *Cancel*.

List button

If you click on the *List* button or press <Enter> to accept the *List* default, your report will print to the screen. Various controls are available to help you move around and view your report.

If you click on the left arrow icon you will move one page backward. Clicking on the right arrow icon moves you one page forward. Clicking on the left-arrow-bar icon moves you to the first page of your report; clicking on the right-arrow-bar icon moves you to the last page of your report.

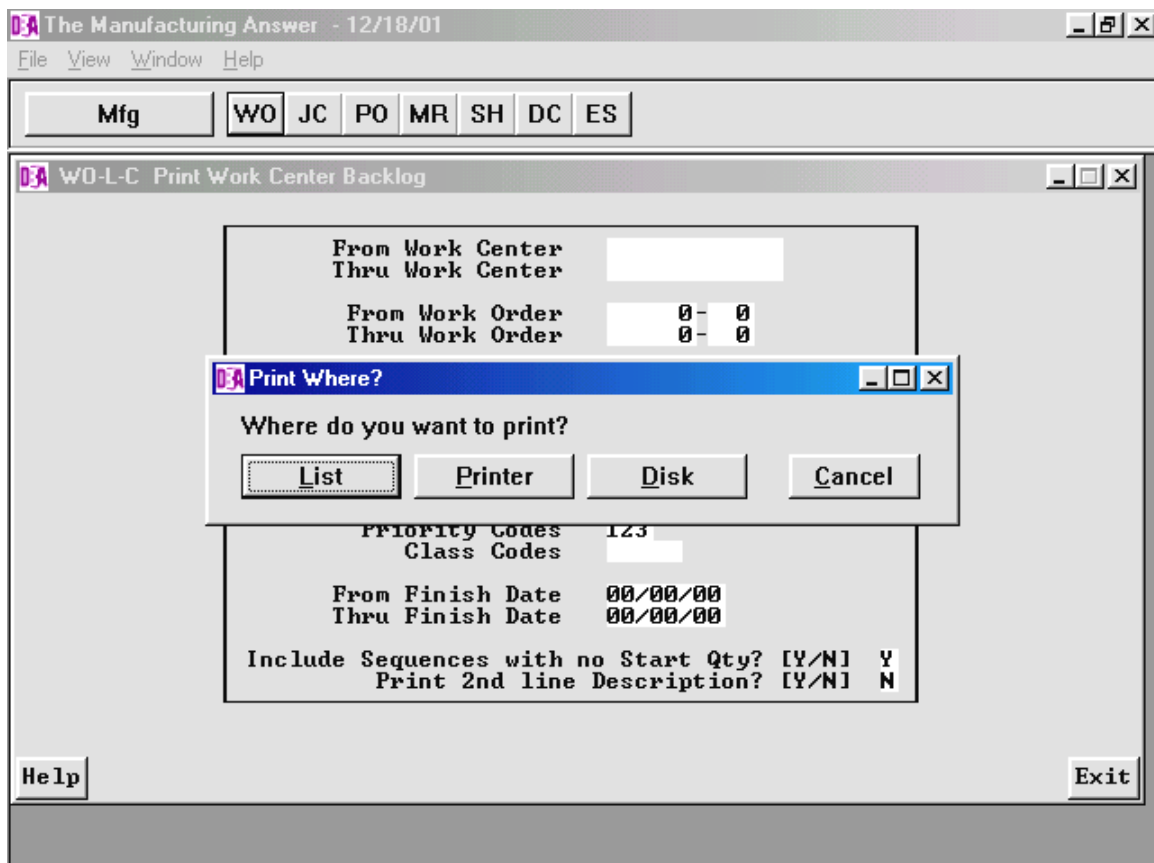
Clicking on the magnifying glass icon with the plus sign inside of it provides a “zoom in” function where each click makes your report progressively larger in size. Clicking on the magnifying glass icon with the minus sign inside of it

provides a “zoom out” function where each click makes your report progressively smaller in size so that more of the report appears in the screen.

Clicking on the printer icon will send your report to the printer and allow selection of one, all, or a range of pages to be printed. Clicking on the *Go to Page#* button allows you to enter a specific page number and will cause the screen to display that particular page.

To see the bottom or right hand portions of wide reports you can use your mouse to manipulate the scroll bars on the bottom and right side of the screen.

To exit the *List* function, either press <Esc> or click on the X icon at the top of the screen.



List, Printer, Disk Buttons

Printer button

When you click on the *Printer* button you are presented with a window that lists your default Windows printer. Click on OK or press <Enter> if this is the printer to which you wish to send the report. If not, click on the *Select* button where you are presented with a standard Windows printer selection window. Choose a specific printer and click OK. You are returned to the first window that now lists the printer you just selected. Click on OK and your report will be sent to the

printer. If the report is a wide report, you will be prompted in a window whether to print in condensed format or not. You can click on the appropriate button or press <Enter> to accept the condensed default.

PAGE: 1

INVENTORY LISTING
Tuesday December 18, 2001 05

Product Code From: - Thru:
Type From: - Thru:
Category From: - Thru:
Class From: - Thru:

Item Number	Description	T	Cat	Class	Stk U/M	Sel U/M	Pur U/M
*	NON-STOCK ITEM	N	W	WAG	EA	EA	EA
1000	DESK, OFFICE	F	F	FURN	EA	EA	EA
1005	DESK FRAME ASSEMBLY	A	F	FURN	EA	EA	EA
1005-1	LEFT BRACE ASSY	B	F	FURN	EA	EA	EA
1005-2	RIGHT BRACE ASSY	B	F	FURN	EA	EA	EA
1005-3	CENTER BRACE ASSY	B	F	FURN	EA	EA	EA
1006	DESK LEG	A	F	FURN	EA	EA	EA
1006-1	CASTER	R	F	HARD	EA	EA	EA
1100	DESKTOP	O	F	FURN	EA	EA	EA
1105	DESKTOP, 4 X 5	A	F	FURN	EA	EA	EA
1110	DESKTOP, 5 X 6	A	F	FURN	EA	EA	EA
1115	DESKTOP, 6 X 8	A	F	FURN	EA	EA	EA
1150	DESKTOP EDGE	O	F	FURN	EA	EA	EA
1155	ROUNDED EDGE	M	F	FURN	EA	EA	EA

List screen for standard reports

Disk button

The *Disk* option allows you to save the report to disk as a text file. When you click on the *Disk* button you are presented with a window that lets you enter a *Print to Disk File Name*. Enter a file name of your choosing. Do not enter a path in front of the file name and do not use the *Search* button to select a path; this confuses the runtime engine. Your file will be saved in your DBAMFG folder. If you want to move it to another folder, do so with Windows Explorer after the file is created.

Printing graphical forms, labels, and checks

If in the various default programs (SD-M, *Sales Orders Defaults*, SD-P, *Accounts Receivable Defaults*, SD-C, *Purchase Orders Defaults*, SD-G, *Estimating Defaults*) you have selected graphical formats for any forms, labels, or checks, you are presented with a choice of three buttons: *List*, *Printer*, and *Cancel*.

R Print Preview [73 %] [1] [Close]

The Manufacturing Answer
999 Success Avenue
Suite 100
Industry, CA 91789

QUOTATION

Quote No: 1
Quote Date: 08/17/94
Page: 1

HART MFG COMPANY
385 South Lemon Ave.
#E-148
Walnut, CA 91789

Attention:
Michael Hart

Cust RFQ No: 999
Expiration Date: 11/17/94
Quoted by:

Job Number:
Class: AA
Status: A

Part Number	Description	UM	Rev
W-100	RED WAGON	EA	A

Quantity

Price

Page 1 of 1

List screen for graphical forms, checks, labels

List button

If you click on the *List* button or press <Enter> to accept the *List* default, your form will print on the screen. Various controls are available to help you move around and view your form.

If you click on the left arrow icon you will move one page backward. Clicking on the right arrow icon moves you one page forward. Clicking on the left-arrow-bar icon moves you to the first page of your report; clicking on the right-arrow-bar icon moves you to the last page of your report. The current page number is displayed in between the left and right arrow icons.

You can change the size of the displayed form by clicking on any of the three icons just to the right of the printer icon to get a small (46%), medium (93%), or full sized (100%) view of the form, or you can enter a percentage number in the field to the right of the icons to more precisely control the size of the display.

If you click on the printer icon you will be presented with a standard Windows print selection screen that allows you to choose the number of copies, page range, or printer.

When you are done viewing or printing the form, click on the *Close* button to return to the menu.

Printer button

When you click on the *Printer* button you will be presented with a standard Windows print selection screen that allows you to choose the number of copies, page range, or printer. Once you complete your entries and click OK, the program will present a message “accessing data” and will then print the form and return you to the menu.

You can print forms directly via the *Printer* button or you can view the form first via the *List* button and then print from *List* mode by clicking on the printer icon.

Using lookups

Lookups are provided throughout the system to help you select items such as item numbers, customer codes, vendor codes, etc. Whenever lookups are available on a field you will see a “magnifying glass” icon to its left, and you will see a *Lookup* button on the bottom of the screen. You can click on either one to bring up a lookup. To use the magnifying glass icon, you must first move your cursor to the field itself, at which time the magnifying glass icon turns from gray to green in color, meaning that it is active. Once the icon is active, you can click on it to display the lookup on that field.

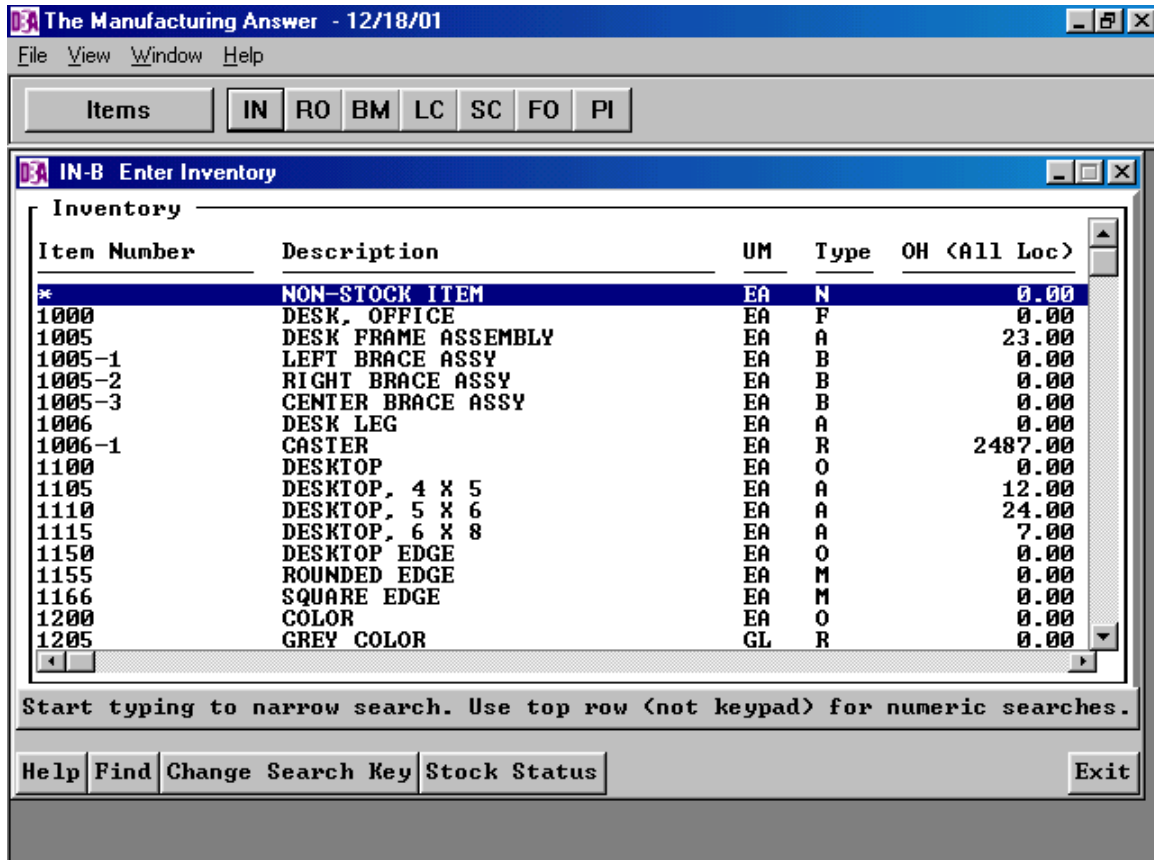
As an alternative to the magnifying glass icon, you can click on the *Lookup* button or press the F2 key.

After selecting the lookup by either method, you will then be presented with a *Search Key* list of various search options. For example, a customer can be searched by customer code, customer name, state, zip code, contact name, telephone, or class. Click on the search key of your choice or highlight it and press <Enter>.

The file you are searching on will be presented in a window and will be sorted in ascending order within the search key selected. At this point you can begin typing characters to narrow down your search. The cursor moves to the top of the list. As you type each character the list resorts itself and progressively gets closer to the item you are looking for. For example, if you are looking for a customer whose name begins with an H, type an H and the list resorts itself on the screen beginning with the first customer record starting with an H. If the next letter you type is an O, the list will resort and will start with the first customer whose name begins with HO. Each character you type progressively narrows down your search.

Be aware that some search keys are case sensitive, such as the *Customer Name*. In the case of the customer name must use your <Shift> key when typing the first character (since the first character of customer names is always capitalized) or else nothing will happen if you type a lower case letter.

NOTE: when entering numeric data the progressive search feature only works on numbers entered via the numeric keys across the top row of the keyboard. It will not work with the number pad keys.
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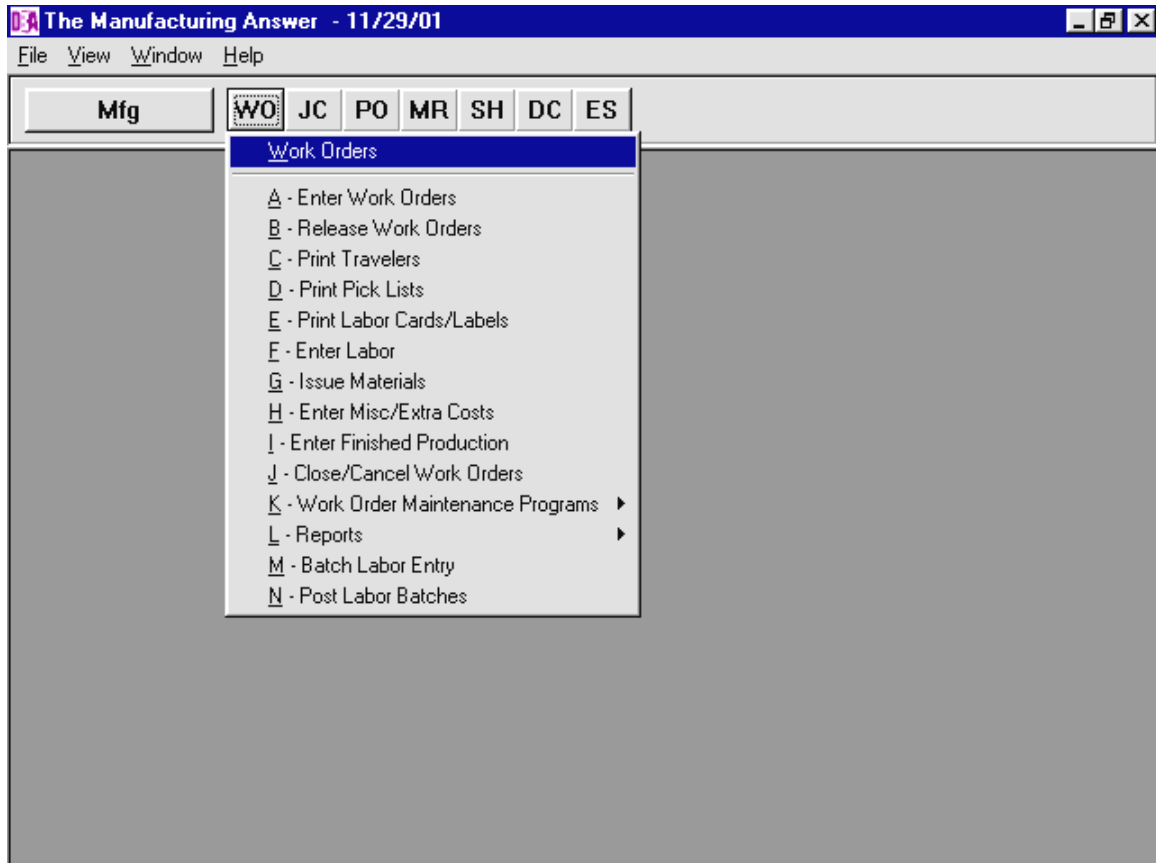
Inventory lookup by Item Number

You can use your mouse on the vertical scroll bar to go to advance to any point in the file (in some lookups you will only be able to go to the beginning or end of the file only). Your <PgDn> and <PgUp> keys can be used to advance forwards or backwards through the file a screen at a time, or your up and down arrow keys can be used to move forwards and backwards a line at a time. Your right and left arrow keys can be used to move right and left within the window to view additional columns of information.

While in a search window you can change the search key option without having to exit the lookup and start over. You can click on the *Change Search Key* button and the *Search Key* list will be presented for another selection.

Once you see the record you want, either double click on it or highlight it and press <Enter> and it will be inserted into the field on the screen.

3. HOW THE MANUFACTURING SYSTEM WORKS



Even though this is a long chapter, we suggest you take the time and study it to before entering data to avoid time-wasting mistakes and to get the most out of your system implementation.

Work orders are the foundation of the manufacturing system

Work orders are the foundation of the manufacturing system. A work order tells the system that you intend to manufacture a specific quantity of some item at a particular time.

When you create the work order, any material or labor needed for that work order is allocated by the system against available inventory and labor resources.

During the course of production materials can be purchased or issued to the work order, outside processing (such as painting, heat treating, etc.) can be purchased for the work order, and setup and direct labor can be reported or issued to the work order.

When items are finished, all these costs are rolled up into a total cost for the items and then those items can be transferred to inventory. Once in inventory, the items can be sold (invoiced) or can be issued to other work orders (such as in the case of

subassemblies).

How are work orders generated?

In DBA you can generate work orders six different ways.

- If you are a make-to-order manufacturer, such as job shop repeating a previous order or a product line manufacturer who builds to order, you can enter items on sales orders and convert them to work orders through the *Convert Sales Orders to Work Orders* program.
- If you have quoted your customer using the *Estimating* module, you can convert your estimate to a work order and a sales order through the *Convert Estimates* program.
- If you are using the *Materials Requirements* module to plan your requirements, the *Generate Material Requirements* program will suggest which items need work orders, which can then be automatically generated through the *Generate Work Orders* program.
- If you want to manually enter work orders, you can do so directly through the *Enter Work Orders* program.
- If you are a custom manufacturer using the *Features & Options* module to specify product features and options, you can convert sales orders to work orders using the *Convert Sales Orders to Work Orders* program.
- Regardless of how the top assembly work order is generated, if you manufacture a product with levels of subassemblies, you can use the *Create Multi-Assembly Work Orders* program to explode through the bill of materials and create the work orders for lower level subassemblies.

How do work orders allocate against available materials and labor?

When a work order gets created, DBA takes a copy of the item's bill of material and makes a work order bill of material. In the same fashion it takes a copy of the item's routing and makes a work order routing.

If for any reason you wish to modify the work order bill of material (such as to change a color or some other component) or work order routing (such as to change an inside operation to one where it is sent to an outside vendor) just for this one work order, you can do so without affecting your standard bill of material and routing. It is this separation of the work order files from the standard files that gives DBA the flexibility to handle custom manufacturing requirements.

Once the work order bill of material exists, the *Allocated* field (meaning that parts are *allocated* for work orders) in the inventory file is updated for each of the bill of material components. If you run the *Reorder Report* you will see these allocations listed and can prepare purchase orders or work orders as required. If you use the *Material Requirements* module for your planning, the *Generate Material Requirements* program will suggest purchase orders or work orders that are needed to insure that material is on-hand to meet these requirements.

Once the work order routing exists, the labor and outside processing sequences (operations) join the backlog of orders against your factory work centers and outside processing work centers. Various reports in the system will list or display these work center backlogs and can compare them against available work center capacity.

How is material issued to or purchased for work orders?

Material can be issued to work orders from inventory or can be purchased for specific work orders.

You can issue material from inventory three different ways: component issues, kit issues, or via backflushing. With component issues you issue one component at a time, usually based on pick lists or material tickets that come in from the shop floor as components are physically pulled from the shelf. With a kit issue you can issue entire sets of the work order bill of material in one transaction. With backflushing you issue complete sets of the work order bill of material after-the-fact when you report finished production. Component issues and kit issues are performed via the *Issue Materials* program. Backflushing can take place when you run the *Enter Finished Production* program or by sequence when labor and quantity completed is reported.

Component issues are generally used when small numbers of components are involved that are picked at different times during the work order process. Kit issues are ideal for large numbers of components that are picked all at once and allow you to remove them from inventory at the time they are pulled.

Backflushing involves the fewest number of steps and is appropriate where overall production time is relatively short and it is ok to deduct components from on-hand inventory after-the-fact.

You may wish to purchase material for specific work orders rather than receive to and issue material from stock. Job shops in particular make this a common practice. When you enter a purchase order you can optionally specify work order numbers on each line item. When the items are received, any that have work order numbers specified on them bypass inventory and are charged directly to the work order and are recorded having been issued.

How is labor reported to work orders?

Labor (setup and direct labor) is collected via time sheets or labor tickets and is reported to work orders via the *Enter Labor* or *Batch Labor Entry* programs, or if you are using the *Data Collection* module, it can be entered directly by production employees on the factory floor through the *Enter Labor/Production* program.

When entering labor you must specify the work order number, employee, sequence (operation), setup time or production time, and the quantity completed and/or scrapped.

The "light manufacturing" method of handling labor

For manufacturing simple products, labor can be set up with an item number(s)

and included as a component(s) in the work order bill of material as an alternative to defining labor within routings. Most companies using this approach do not track labor and instead are using it as rough estimate in order to get labor into the cost of the finished item. In this case the labor component would be included as part of kit issues or backflushes (see previous section) and would be treated the same as material components.

This simple approach to handling labor is sometimes referred to as "light manufacturing."

When using light manufacturing, you can choose to track actual labor, if necessary. In this case you can set up the system in *Work Orders Defaults* so that labor is excluded from kit issues and backflushes. You can then track your labor (on time sheets or time cards) and issue it in units of decimal hours as a single component issue via the *Issue Materials* program.

Only use light manufacturing for ultra-simple products

DBA offers the light manufacturing alternative because many manufacturers coming from general accounting systems are accustomed to using that method.

Using routings to define labor, however, offer several advantages over the light manufacturing method of defining labor as bill of material components.

First, using routings can be kept just as simple as the bill of material method. At a minimum, all that is needed is a single work center, perhaps called "factory." You can enter a single sequence called "labor" and then enter a single hourly amount for the total labor that goes into the product. This simplified routing entry takes no more time than entering labor components in a bill of material.

Using routings, though, enables you to define multiple sequences (operations) and gives you a platform for entering documentation as to how your products are manufactured that for purposes of printing on shop travelers.

If you do any significant amount of outside processing (plating, heat treating, etc.), routings almost have to be used as the best and easiest way to handle the purchasing and costing of these services. Trying to use type T item numbers in your bills of material to account for outside processing is a clumsy process that requires unique item numbers for every process for each item you make, whereas using routings involves no item numbers at all.

Unless you have an ultra-simple product line with little or no outside processing, we do not recommend using the light manufacturing approach.

The light manufacturing method is not appropriate for job shops at all, because job shops need the detailed shop travelers and work center scheduling that can only be accomplished by using routings.

How is outside processing reported to work orders?

The ordering of outside processing is done using a special type of purchase order known as a *service order*. With service orders you do not use item numbers for

outside processing. Instead, you use the item number of the item being processed as the purchase order line item. When entering a service order you are required to specify the work order number and the appropriate outside processing sequence (operation) number from the work order routing. When the processed items are received, they are charged directly to the work order and are recorded as completed.

Entering finished production completes the work order process

When manufactured items are completed, they are ready to be transferred to inventory, from which they can then be sold (invoiced) or issued to another work order (as in the case of a subassembly). Entering finished production is done through the WO-I, *Enter Finished Production*, program.

It is not necessary for the entire work order quantity to be completed in order to enter finished production. In fact, it is common to enter partial completions before the entire work order is completed so that rush orders can be shipped or subassemblies can be issued to other work orders. When entering a partial completion, usually the work order costs are incomplete. The *Enter Finished Production* program will calculate an estimated unit cost based on work order costs reported to date combined with originally estimated costs, or, if you have established standard costs for your items, you can choose to use the standard cost as an alternative to the estimated cost.

On the very last transaction that is made when the work order is finally completed, the program can make a balancing entry through adjustment of the last unit cost so that all costs that were charged to the work order equal the total cost of the finished goods that were transferred to inventory. In other words, this last balancing transaction ensures that “all dollars in equal all dollars out.” If not, as in that case of reporting all finished production at standard cost, closing the work order makes the required balancing transaction to a WIP Variance account.

Tracking work order status and performance

The *Work Order Status* report tracks the status of each work order and the *Work Order Schedule* gives you a master listing of all your open work orders. The *Work Center Backlog* report gives you a listing by work center of all uncompleted sequences (operations) and is often used by shop foremen as a dispatch list to determine which work orders should be worked on next. The *Projected Shipments* report gives you a projection of completed production by month or by customer. The *Print/Post Labor to Payroll* report lists all your direct labor hours and can post those hours into *CheckMark Payroll* so as to avoid duplicate data entry.

The *Job Cost* report compares work order estimated costs and profit with actual costs and profit. The *Profit Projection* report gives you an estimated profit midway through a work order based on costs reported to date. Various transactions listings are available through the *Labor Transactions*, *Overhead Transactions*, *Material Issues*, *Outside Purchases*, and *Finished Production*

reports. The *Labor Efficiency* report analyzes past performance by sequence (operation) and is helpful in setting time standards within your routings. Production performance can be tracked via the Work Order History, Production by Work Center, Production by Machine, and Production by Tool reports. The WIP Summary report is used by accounting to reconcile WIP inventory with the accounting system's general ledger.

Production planning

DBA provides a variety of tools that assist in the complex process of materials planning — meaning the process of assuring that materials are available at the time needed for production. Without material you cannot manufacture anything.

Companies with short term planning requirements where work orders are completed in relatively quick production cycles and material have short lead times tend to use either the reorder report, bill of material availability report, or work order shortage report to determine which items need to be purchased or manufactured. These reports will list any items that are not on hand and will show you any other work orders that may also need the same items.

By far the most powerful materials planning and scheduling tool available to you is the *Generate Material Requirements* program, which takes all requirements, whether from existing customer orders, work orders, or forecasts, and then generates materials requirements down through all levels of the product structures by comparing projected on-hand inventory with material needed, suggesting specific purchase orders and work orders as it goes.

Besides recommending when to place purchase orders and create work orders, the *Generate Material Requirements* program functions as a powerful scheduling tool that advises you when to delay or expedite existing orders in response to changes elsewhere in the system. If you have multiple levels in your bills of materials and common subassemblies and components among different products, scheduling these interdependent items so that you don't end up bringing in material too early (tying up capital) or too late (delaying production) is fantastically complicated and almost impossible to keep track of manually. The *Generate Material Requirements* program keeps all these interdependent relationships synchronized with one another.

The *Generate Material Requirements* program is highly recommended as a planning tool. To use it effectively you must have accurate bills of material and be dedicated to maintaining realistic and accurate due dates.

Scheduling

Working hand-in-hand with the *Generate Material Requirements* program is the *Scheduling* module. The *Scheduling* module offers two types of scheduling: finite and infinite.

The finite scheduling program determines work order finish dates based on your available capacity. Open work orders are first sorted in priority order and then the factory work centers are loaded, one work order at a time in routing sequence

order. The last routing sequence's finish date determines the overall scheduled finish date of each work order.

The infinite scheduling program leaves work order start and finish dates fixed and determines start and finish dates on the routing sequences without regard for plant capacity. Work center loads can be viewed in a graphical display so that you can compare available capacity against your commitments.

If your company has limited short-term ability to change capacity and quotes realistic due dates that reflect your current production capability, you would use finite scheduling. If your company quotes fixed due dates and meets those dates by adapting your capacity as needed (such as adding extra shifts or overtime), then you would use infinite scheduling.

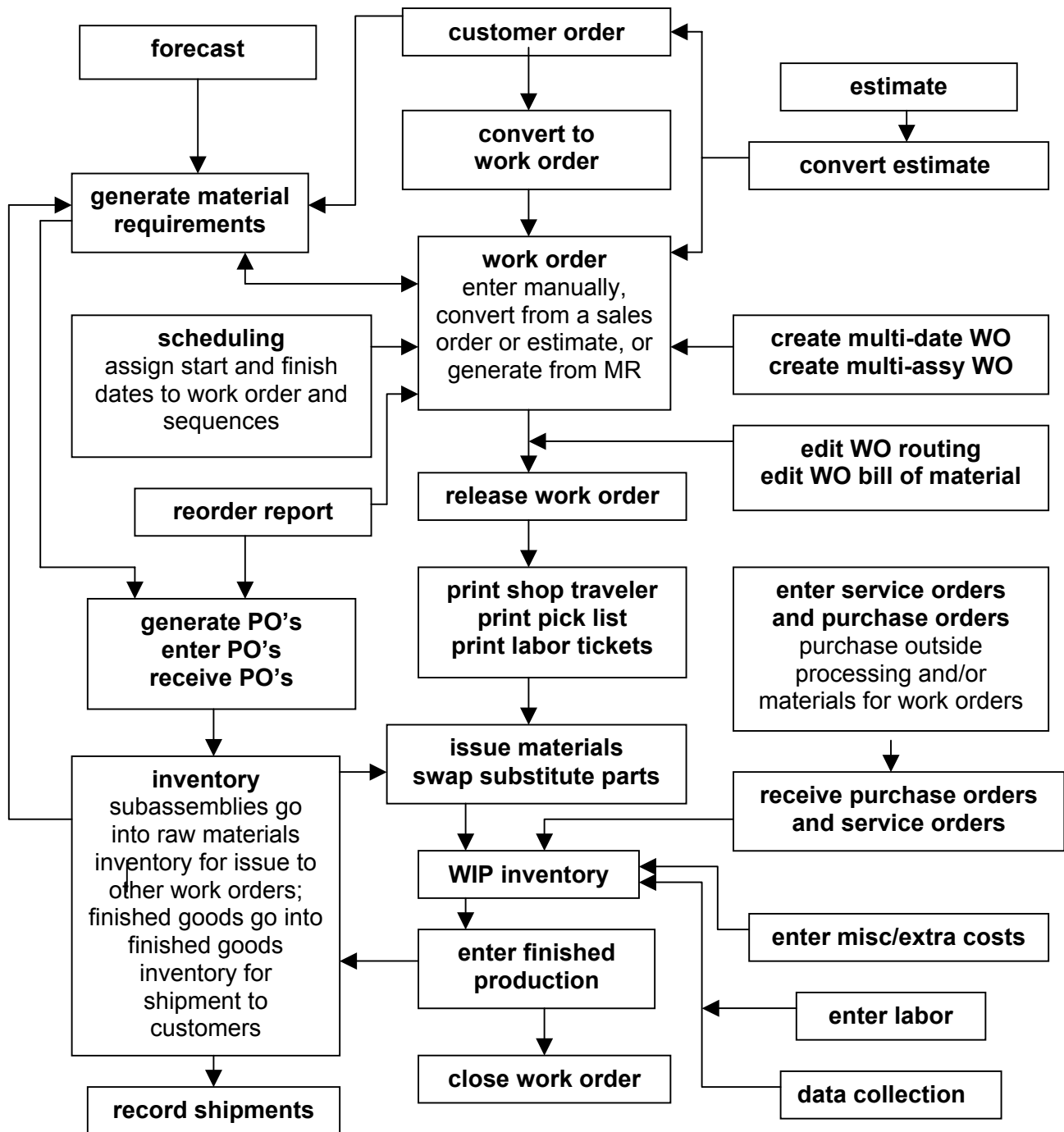
As new work orders get scheduled and existing work orders get re-scheduled, the next time you run *Generate Material Requirements* it will reflect the new start and finish dates and will compare those against the interdependent demand of related subassemblies and materials and will let you know if related work orders and purchase orders need to be re-scheduled also.

Summary

Work orders are the foundation of the manufacturing system. When work orders are created, the materials and labor needed are allocated against available inventory and work center capacity. You can issue material from inventory or purchase material and outside processing services directly for work orders. Labor can be reported or issued to work orders via data entry or direct entry by workers using data collection. As production is completed items are transferred to on-hand inventory where they can be sold to customers or issued to other work orders.

The production planning system lets you use simple tools such as the reorder report or you can use the advanced planning and scheduling capabilities offered by the *Material Requirements* module. Complementing *Material Requirements* is the *Scheduling* module, which helps you schedule your work orders within your available plant capacity.

Flow chart - Manufacturing system



4. USING DBA AS A JOB SHOP

Job shops, by their nature, depend on being efficient to make money because the prices they receive are dictated by competitive bidding whereby the market sets selling prices. Therefore job shops make money by producing at the lowest cost and by offering superior delivery.

DBA offers you a complete manufacturing system designed specifically for job shops with functions such as work orders, routings, estimating, purchasing, and scheduling. This section gives you an overview of how DBA can be used by job shops of all types.

Estimating

DBA comes with an *Estimating* module that can be used for two purposes: actually calculating customer quotes or simply for setting up jobs. Some people who have been doing their own estimating for many years and have developed their own personal techniques do not like to conform to a different system. If this is the case with you, you can continue to quote the way you always have.

The *Estimating* module, though, offers a great way to set up jobs that you actually get confirmed orders for because in one program you can set up customers on-the-fly, item numbers on-the-fly, and you can set up the routing and/or bill of material. Once the estimate is completed, the convert estimates program automatically creates a sales order and a factory work order without any additional entry.

The *Estimating* module is also useful if you ever repeat a job. You can bring it back up in the enter estimates program, make any changes that may have occurred since the last time you ran the job, and then you can convert it once again into a new sales order and new work order.

In summary, you can use the *Estimating* module as a tool to actually prepare and calculate customer quotes, but if you prefer to use your own method of quoting, it still serves as an excellent all-in-one program for setting up your jobs.

Job costing

Accurate job costing is the goal of most job shops for obvious reasons—in a sense it's like getting a report card at the end of each job telling you how you did. Job costs come from four sources: labor, materials, outside processing, and overhead.

An integrated system like DBA makes it easy to collect these costs so that job costing is both prompt and accurate. Let's examine how this is done with each of the four cost categories.

Labor can be collected on time sheets or time cards and can be entered manually by a clerical person at the end of the day or the next day via the enter labor program. All you need is to set the job up with a work order routing. At its simplest the routing could consist of one operation named "make item" and all the labor could be posted to that one operation. More typically the routing will

contain a variety of operations and the time is posted to each of the various operations. All of this capability is built into DBA and can be used just as soon as you set up your work center and employee files.

Many shops are now using DBA's *Data Collection* module in which employees can use the keyboard or scan bar-coded travelers to enter their own labor on a real time basis directly into the computer. Most shops only need one or two computers out in the shop to do this. It not only saves the clerical time entering time cards manually, but for some shops the fact that job status is updated in real time is highly valuable, especially when scheduling is tightly managed and information 24 hours old is not good enough.

With DBA you can choose to charge jobs at employees' actual wage rates or you can charge jobs at work center standard rates. Some companies prefer standard rates so that some jobs don't get charged disproportionately just because a high-paid employee happened to work on it instead of a lower-paid one.

Materials can be entered on purchase orders for specific jobs and when received those costs get charged directly to those work orders. Or, you can issue materials out of inventory.

Outside processing can be ordered via a special type of purchase order coded as a *service order*; just like with materials when the service order parts are received back from the vendor, those costs get charged to the job.

Finally, you can burden your labor with an overhead factor (an hourly amount or a percentage of labor) either using the same factor throughout the factory or you can apply different overhead rates to different work centers.

Routings and shop travelers

Almost all job shops use routings to break jobs down into operations and to produce a shop traveler. With DBA you can create a permanent routing that can be used over and over again for repeat jobs, or you can create a work order routing that is only used on a one-time basis for that one work order.

One nice feature that DBA offers is a routing generator. You can set up a series of operation templates that number and describe all your common operations (shearing, punching, bending, etc.). When creating a new routing you can bring these templates up in a window and mark them one-by-one in the sequence you plan to use them. Once you close the window the program automatically creates a complete routing out of these operation templates that you can now go into and edit if it needs any customizing for that particular job. With creative use of operation templates you can create complex routings quickly.

What about inventory?

Many job shops that purchase material strictly for orders see an inventory module in the system and assume that it means they have to keep an inventory. This is not the case.

The *Inventory* module is where items and supporting functions such as customer

cross-reference numbers, dimensions, etc. are set up and maintained.

If you ever do have some inventory, such as left over material, or if you choose to make a longer production run and ship scheduled shipments out of stock, you can use the *Inventory* module to keep track of any inventory on hand. Inventory tracking is available if you need it, but is not required should you choose to keep zero inventory on-hand.

Materials planning and scheduling

Job shops with long bills of material and multi-level bills of material are provided with a variety of tools within DBA to assist in the complex process of materials planning - meaning the process of assuring that materials are available at the time needed for production. Without material you cannot manufacture anything.

Companies with short term planning requirements where jobs are completed in relatively quick production cycles and materials have short lead times generally use either DBA's reorder report, bill of material availability report, or work order shortage report to determine which materials need to be purchased. These reports will list any materials that are not on hand and will show you any other order work orders that may also need the same material. This allows you to begin combining purchases for better prices rather than just buying specific amounts for each job. In job shop manufacturing any savings you make on material go directly to the bottom line.

By far the most powerful materials planning and scheduling tool available to you is the *Generate Material Requirements* program, which factors in purchase lead times and work order start dates and tells you exactly when purchase orders should be placed (or work orders for subassemblies) in order for material to arrive in time when it is needed.

The *Generate Material Requirements* program is also valuable in that if work orders get delayed or expedited it will advise you to delay or expedite all supporting purchase orders and work orders so that purchasing and production stay coordinated with one another and you don't end up bringing in material too early (tying up capital) or too late (delaying production). The program also shows all requirements for a particular material and gives you the opportunity to combine requirements into single purchase orders for better prices.

Another timesaving feature of the *Material Requirements* module is that it comes with a purchase order generator that can automatically create purchase without entering them manually. When buying many materials from many vendors this can be a tremendous time saver. If, like many electronic assemblers, you need to get vendor quotes on items before purchasing them, the *Material Requirements* module also has an RFQ generator, which generates printed requests for quotes.

Although many job shops are at first more comfortable with the more familiar reorder and shortage reports (which resemble check lists) once they get used to the *Material Requirements* module they find it to be a superior tool in that it is more specific as to when and how much to order, it has the ability to combine

requirements for better prices, and it offers scheduling advice such as when to delay or expedite orders.

Purchasing

Almost all job shops have to purchase material or outside processing services such as plating, painting, heat treating, etc. While many shops are in the habit of picking up the phone and calling in orders, a good purchase order system in the long run actually reduces the work involved by giving you not only a good tracking system, but it also automatically transfers costs to jobs when items are received or invoiced.

Shop floor control and job scheduling

Job shops constantly receive calls from customers asking about the status of their work orders. If you record labor on a timely basis either through manual time card entry or real-time data collection, you also know exactly where each job is at any given time and you can give the customer an accurate answer without having to walk out in the shop to find out what each job's status is. This for many companies is the biggest single time saver of the system.

The other nice thing about the work order system is that you can get lists of all open jobs by work center in priority order to help you determine what should be run next in each given area of the factory. You can keep this simple by using the work center backlog report or you can use the *Scheduling* module to actually generate start and stop dates on routing operations. A variety of reports are available to help you know if schedules can be met and whether you need to change due dates, work overtime, add employees, etc.

Do item numbers have to be used?

The answer is yes and no. All manufacturing systems use item numbers (part numbers) to have something to which to attach costs, routings, bills of material, and track inventory, past history, etc. Yet many job shops balk at having to create item numbers. Common complaints are "I'm going to buy this material one time and may never use it again in my career? Why should I have to take the time and set it up with an item number when I'll never see the thing again?" "Setting up item numbers takes too much time and slows me down getting orders prepared." "I'm a job shop and purchase everything for the job. I never have an actual inventory. Why do I need to bother with item numbers?"

A good rule of thumb is to set up and use item numbers on materials that are commonly used and ordered on more than a one-time basis.

Item numbers should be used for manufactured items

The one item for which you should create an actual item number is the item you are manufacturing. Some companies use the customer's item number; others assign their own number, often referred to as a "job number." It doesn't matter which one you use. Having an item number allows you to attach a permanent

routing to that item so that if you ever manufacture the item again, you already have all the specifications on file.

If you are a job shop where the items you manufacture have complex bills of materials or multiple level bills of material, you definitely want to take time to create item numbers in order to manage the purchasing and tracking requirements involved. You must use item numbers if you want to use tools such as the material requirements program, the reorder report, work order shortage report, and bill of material availability report.

How to get started

Job shops can get up and running much quicker than manufacturers because they have much less data to enter in order to start using the system.

The first thing to do is to set up your work centers and employee file. Once you have work centers you are ready to actually begin entering work orders, tracking labor, and purchasing materials and outside services. Customers, vendors, and item numbers can be added as needed and built up over time, or you can enter them all at once if you wish.

Keep in mind that whenever a new system is being implemented it will always seem like it is more work at first than was involved with the old manual system, especially if you are not used to using a formal purchasing system. Over time, though, as you start using the daily reports that track purchases, costs, labor, schedules, etc., you will soon wonder how you ever operated without them. You will also see ultimately it takes less work using formal purchase orders and work orders than the alternative of keeping everything in your head and operating in a constant expediting mode.

Certainly any company that wants to grow will find that it is much easier to accommodate additional activity and still deliver the same quality and performance that your customers received when you were a smaller company if you get formal procedures in place now rather than later.

5. USING DBA AS A MANUFACTURER

For reference purposes we define a manufacturer as a company that has a product line of its own. A manufacturer may build to inventory or to order or both, and may have standardized product or customized products that are configured to order.

DBA is designed to accommodate all of these methods of manufacturing; in fact, it is common for companies today to both build to order and to inventory. In some aspects you must function both as a job shop when you are building to order and a manufacturer when you are building to inventory.

Let's review each of the major elements of the system and how they work for a manufacturer.

Planning system accommodates manufacturers' needs

DBA's planning system is intimately linked with the production system. For example, you can enter customer orders and then convert them automatically to work orders. You can enter orders with multiple shipments scheduled well into the future. You can use the *Features & Options* module to custom configure products with features and options and automatically convert these special orders into work orders. And customer order requirements are part of the planning system, which can include a variety of reorder and shortage reports as well as a complete materials requirements capability, described later in this section.

Standard and actual costing

DBA offers manufacturers both a standard cost and actual cost system. Standard costs can be established on all materials (which can periodically be globally updated with your last or average cost) and labor and outside processing costs can either be maintained in routings or can be stored against special type item numbers and included in bills of material.

Once you have standard costs established at the component level, the *Print/Rollup Standard Costs* program rolls up these costs through all levels of the bills of material and calculates total costs for all subassemblies and finished goods. Whenever a cost changes on a material that is used in many different products, all you have to do is change its standard cost, run a cost rollup, and all items using that material get re-costed immediately.

Many manufactured products have complex, multi-level bills of material with a numerous subassemblies. A standard cost system makes it easy to establish accurate costs on products to assist you in pricing decisions and help you determine what your margins are.

During the manufacturing process where subassemblies are made to be used on possibly many different products, it is very difficult to track actual costs up through many levels of a product structure. At the time one assembly is being issued to the next level up, the costs may not even be fully reported yet. For that

reason many manufacturers choose to report all finished production at standard cost from one level to the next and then at month end adjust inventory for the variances between actual costs and standard costs. DBA tracks actual costs for materials, labor, and outside processing (plus provides for an overhead burden) on all work orders and provides job cost and month end reports that show all the variances between actual and standard cost.

Work Orders

For shop floor control, scheduling, work order tracking, and accurate labor costs, DBA provides a *Work Orders* module that is part of a comprehensive manufacturing system comprised of routings, work orders, data collection, and scheduling.

Work orders are entered in the *Work Orders* module. Labor is reported either manually or via the *Data Collection* module, materials are issued from inventory or purchased for work orders, and outside processing is tracked and reported via the *Purchase Orders* module.

The light manufacturing alternative for handling labor

For manufacturing simple products, labor can be set up with an item number(s) and included as a component(s) in the work order bill of material as an alternative to defining labor within routings. Most companies using this approach do not track labor and instead are using it as rough estimate in order to get labor into the cost of the finished item. In this case the labor component would be included as part of kit issues or backflushes (see previous section) and would be treated the same as material components.

This simple approach to handling labor is sometimes referred to as "light manufacturing."

Light manufacturing is ideal for companies where the labor content is a small part of products or for companies with small labor forces that jump around from work order to work order such that conventional labor tracking is not feasible.

Material requirements and scheduling

DBA provides a variety of tools that assist in the complex process of materials planning - meaning the process of assuring that materials are available at the time needed for production. Without material you cannot manufacture anything.

Companies with short term planning requirements where work orders are completed in relatively quick production cycles and material have short lead times tend to use either DBA's reorder report, bill of material availability report, or work order shortage report to determine which items need to be purchased or manufactured. These reports will list any items that are not on hand and will show you any other work orders that may also need the same items.

By far the most powerful materials planning and scheduling tool available to you is the *Generate Material Requirements* program, which takes all requirements,

whether from existing sales orders, work orders, or forecasts, and then generates materials requirements down through all levels of the product structures by comparing projected on-hand inventory with material needed, suggesting specific purchase orders and work orders as it goes.

Besides recommending when to place purchase orders and create work orders, the *Generate Material Requirements* program functions as a powerful scheduling tool that advises you when to delay or expedite existing orders in response to changes elsewhere in the system. If you have multiple levels in your bills of materials and common subassemblies and components among different products, scheduling these interdependent items so that you don't end up bringing in material too early (tying up capital) or too late (delaying production) is fantastically complicated and almost impossible to keep track of manually. The *Generate Material Requirements* program keeps all these interdependent relationships synchronized with one another.

The *Material Requirements* module is highly recommended as a planning tool. To use it effectively you must have accurate bills of material and be dedicated to maintaining realistic and accurate due dates and on-hand inventory balances.

The *Material Requirements* module works hand in hand with the *Scheduling* module, which assigns start and finish dates at both the work order and routing sequence levels. You can schedule work orders manually or automatically using the infinite scheduling or finite scheduling programs.

How to get started

The first thing to do is to enter master files for inventory items and bills of material. Routings can be implemented later. Once these two files are created you can begin using work orders and purchase orders and standard product costing and then evolve over time to more advanced functions such as routings, labor tracking, scheduling, and material requirements.

Before rushing into an implementation give careful consideration to your numbering schemes, especially item numbers, because once created they are difficult to change down the road. See *Implementation*, chapter 7, for more details on setting up item numbers and other numbering schemes.

The toughest part in implementing a manufacturing system is getting all the data entered necessary to set up inventory, bills of material, and routings. Once these files are created using the actual programs is not difficult. A good strategy is to focus on getting all your manufacturing master files fully entered, begin using work orders and purchase orders, then evolve over time into using routings, labor tracking, scheduling, and material requirements.

6. HOW TO GET STARTED

The most common question asked by first time users of a comprehensive system such as DBA is “how do I get started?” When you first look at a system with as many modules and features as DBA it can seem overwhelming, but when you break the system down into its component parts it is far simpler than it first appears.

The first thing to be aware of is that DBA is not creating any new areas of activity that will increase your workload, but instead is giving you a system to plug into that formalizes what every company already does. All companies have a planning system whether they know it or not, it’s just that in some companies it is highly informal and relies more on expediting, hot lists, spreadsheets, etc.—but it is still a system.

If you are moving over to DBA from a system that already has full accounting and manufacturing, you will implement far more quickly because you are already comfortable with routings, bills of material, and work orders. If it is your first system, however, some of this will seem foreign to you and may even seem to be extra work without an immediate payoff. This is a natural reaction that will pass over time as the benefits of an integrated system start becoming apparent.

Study the user guide first

We’ve tried with this user guide to help you get a conceptual grasp of the system. We suggest you review the overview chapters 3-5 first. After reviewing the overview chapters you should study the chapters on each module to get a feel for how they function.

Once you have studied the above-mentioned chapters you can then move on to chapter 7, *Implementation*, which provides step-by-step implementation guidelines.

If you purchased DBA accounting, consider implementing manufacturing first

If you are using DBA Manufacturing with your existing accounting system, you can skip this section.

If you plan on using DBA’s accounting add-on, it is not necessary to implement accounting and manufacturing at the same time. It is far easier to implement manufacturing first and tackle accounting later than it is to implement both systems at the same time.

Build your master files

The manufacturing system can be implemented gradually in steps. If you have a product line the most important things to do are to enter your item numbers and bills of material. This might take several weeks, but it is the foundation of the manufacturing system and is the basis by which inventory and costs will be

controlled. You can't do much of anything until these master files are completed.

For many manufacturers just getting standard costs entered and having the ability to have the computer roll up the costs to get total standard costs on products is a big step forward. Many companies really don't know what their product costs are because it is so much work calculating it by hand. The cost rollup program updates costs across all your products whenever the cost of a key component or material changes.

If you are a job shop you can focus on getting work centers and standard operations set up in the operation templates file so that you can early on begin using work orders and shop travelers.

Implement in phases

To get benefit from the manufacturing system you do not have to implement the entire system all at once.

One of the easiest things to get started with is using *Purchase Orders*. Purchase orders can be used whether the work order system is running or not. All you need to start are vendors and item numbers, both of which can be added on-the-fly as you go, if necessary. Suddenly you have a complete purchase order tracking system and you are starting to accumulate purchase history.

The work order system can be kept very simple at first. Some companies simply open and close work orders without any detail in order to start getting open order lists and rough schedules. Bills of material and routings can be added later when you are ready to use them.

When bills of materials are ready, enter starting stock balances, begin using work order bills of material, and suddenly you have a fully functioning inventory system in place.

If you are a job shop you can immediately start creating work order routings and produce your shop travelers on the computer. Job costing can come later when you are ready to report labor to jobs and have procedures in place to do so.

Save advanced functions for later

Job costing (including data collection), scheduling, and material requirements are advanced functions that are not suitable for implementation until the basics are in place. The basics include having accurate bills of materials and routings, timely and accurate labor and materials reporting, and a company-wide devotion to maintaining accurate start and due dates. Only when these basics are in place can your company move forward to advanced functions. So, don't worry about these advanced functions at first and concentrate instead on the basics. This gives you fewer things to think about at first and simplifies the immediate task at hand.

Follow the checklist

When you do actually begin implementation carefully follow the implementation checklist found in chapter 7, *Implementation*. It walks you through the

implementation phases in the proper sequence and insures that everything is set up correctly, including all your default settings. Setting things up correctly in the beginning will save you a lot of headaches later.

Ask our help

When you have questions during the implementation period, don't hesitate to call our Technical Support department for help (800-434-0809). Most of our support staff have personally been through several implementations and can give you valuable advice that can save you a lot of time.

Above all, stay patient and remember that the implementation period is always a strain because for a while everyone is doing extra work in keeping the company running and working on the implementation at the same time. Once you get through it, though, things normalize and you one day will wonder how you ever managed things without all the tools you now have at your disposal.

7. IMPLEMENTATION

This chapter covers all the steps required to implement the DBA manufacturing system.

If you purchased DBA accounting, implement it separately

If you are using DBA Manufacturing with your existing accounting system, you can skip this section.

It is not necessary to implement accounting and manufacturing at the same time. In fact, doing so makes the overall implementation significantly more complex. Most companies are better off focusing on manufacturing first and postponing the accounting implementation until manufacturing is running smoothly.

For accounting implementation guidelines, see the separate *Accounting User Guide* in the back of this binder.

Using DBA's sales system is optional

DBA Manufacturing comes with a complete sales system that includes sales orders, invoicing, and tracking of customer payments through Accounts Receivable. While these activities can also be performed in your accounting system, doing them within DBA is more convenient.

To use DBA's planning system effectively, sales orders are needed to compare orders with existing inventory. Invoicing within DBA automatically relieves finished goods inventory. And, tracking customer payments within DBA eliminates having to enter customers and AR details in your accounting system.

Some companies, however, especially job shops, do not have much inventory and only generate a few invoices a day. In this case, you might want to continue generating invoices and tracking customer payments within your accounting system.

To do this, go to AI-A, *Accounting Configuration*, and deactivate Accounts Receivable processing. Then go to PS-D, *Activate/Deactivate Modules*, and deactivate the *Sales Orders*, *Sales Analysis*, *Sales Commissions*, and *EDI Interface* modules. Do your invoicing in your accounting system. To relieve finished goods inventory in accordance with your invoices, do so using IN-C, *Enter Inventory Adjustments*.

Planning up front will save you time

Implementing a manufacturing system involves many elements. Spending time up front planning the implementation will save you significant time over the long run by helping you avoid time-wasting and costly mistakes.

You can implement manufacturing in stages

Unlike an accounting system where the core modules are implemented all at once,

you can implement the manufacturing system in stages. For example, if you are a manufacturer you might start using work orders with bills of materials, focusing primarily on materials planning on inventory, postponing routings and labor tracking until later. For production planning you may start with the reorder report and later use the *Material Requirements* module. At some point you can add work centers and routings to track labor. Once you are tracking labor you can move on to using *Scheduling* and *Data Collection*.

If you are a job shop (where you can stop and set up each job as it comes up) you would set up your work centers and use routings to track labor from day one. Once things are running smoothly, you can get more out of the system by moving on to *Scheduling*, *Material Requirements*, and *Data Collection*.

Implementing a manufacturing system can be a lot of work, especially for manufacturers with product lines. The programs themselves are not difficult to learn and use – entering the initial data is what takes the time. You are literally documenting what, where, and how all your finished goods and subassemblies are manufactured. Once this documentation process is done, though, tasks that used to take agonizingly long to complete – costing, production planning, scheduling, inventory management – will be done in a fraction of the time it takes now.

The implementation process consists of eight phases, described below. Over the rest of this chapter each phase and its component parts will be described in detail. At the end of the chapter is an *Implementation Checklist* for your reference.

Phase 1 – System orientation

Before entering any data we suggest you browse this user guide and familiarize yourself with the system. Of particular importance are the overview chapters 3-5.

Phase 2 – Develop numbering schemes

DBA uses various codes to identify things such as vendors, item numbers, etc. If you do not already have such codes developed, give careful thought to how you want these codes numbered and organized.

Phase 3 – Enter master files

Master data files must be set up for vendors, items, bills of material, etc.

Phase 4 – Enter module defaults

Each module has default settings that allow you to specify various preferences within the application programs.

Phase 5 – Set up printing and order any forms

In this phase you will get set up for printing forms, labels, and reports, and you will order any printed forms that are needed.

Phase 6 – Train personnel to use the programs

During phase 6 you should see that your personnel are trained to use the various

system programs.

Phase 7 – Enter open orders and on-hand inventory

When you are ready to begin using the system you need to enter your open purchase orders and work orders, as well as beginning inventory balances.

Phase 1 – System orientation

Study this user guide

Before entering any data we suggest you study this user guide and familiarize yourself with the system. Of particular importance is chapter 3, *How the Manufacturing System Works*, which provides a detailed overview of the system as well as flow charts.

Look at sample company 99

A sample company 99, *The Manufacturing Answer*, is supplied with the system. The sample company allows you to print reports and run programs without having to wait for all your own data to be entered. It can also be used for reference to see how another company has been set up. To access the sample company, click on *File* at the top of any of the master menus and then click on the *Change Company* option on the *File* menu.

DBA is a multi-company system. If after you've built up your files you want to practice on your own data, you can make a copy of your company to another company number and then experiment running the various programs in that company. Later on you can erase that company when it is no longer needed. For instructions on setting up additional companies, see UT-I, *Create/Delete Company*.

Phase 2 – Develop numbering schemes

You must develop numbering schemes for the following list of key field names used by the various data files throughout the accounting system. Once numbering schemes are established they are difficult to change, so give them careful thought before entering actual records.

Item Class codes

This is a four character, alphanumeric code, upper case only. Each inventory item must be assigned to an item class. *Sales Analysis* reports, the *Inventory Value* report, and the *Month End Inventory Costing* report subtotal by item class. Discount tables for pricing are established by item class and item class is a selection filter in many other reports. Each item class is accompanied by a 30-

character description.

Category codes

This is a four character, alphanumeric code, upper case only. Each inventory item can be assigned to a category. The category field is used as a selection criterion on reports and provides an alternate means of grouping products to the item class. Unlike *Item Class*, which is mandatory and must be established in advance, *Category* is an optional field and need not be pre-defined.

Item numbers

This is a 15-character, alphanumeric field, upper case only. Every item used in the system, including purchased parts, finished goods, and subassemblies (types R, F, and A) must have an item number. If you plan to purchase or invoice non inventory items such as shop supplies or if you need to invoice customers for non-stock items such as tooling or restocking charges, non inventory item numbers (type N) can be used. If you want to use a light manufacturing approach to handling labor, it can be set up with item numbers (type L) and can be included in bills of material. If you do not already have item numbers, it is best to establish short, non-significant numbers for ease and accuracy of data entry. You might consider grouping like items together with an alpha letter prefix so they appear in a logical order on reports and lookup windows. The inventory master record has a variety of fields to further describe each item, such as two description lines, type, category, class, and specifications, so it is not necessary to build in great significance to the item number itself.

We do not recommend using vendor item numbers because it is common to purchase the same item from more than one vendor, each of who could have a different item number for the same item. The inventory master allows entry of an unlimited number of vendor and manufacturers and their item numbers for cross-reference purposes and automatic printing on purchase orders.

Terms codes

20 user defined terms codes can be set up and are used by the *Accounts Receivable*, *Sales Orders*, and *Purchase Orders* modules to designate payment terms. Codes are numbered 1 through 20, and each code is accompanied by a 10 character, alphanumeric description. Try to number your most commonly used terms first so they appear at the top of lookup windows. Do not rearrange the terms table once you have begun processing orders because it is the Term Number that is actually stored in the orders and invoices.

Vendor codes

This is a 10-character, alphanumeric code, upper case only. Vendors are used in the *Sales Commissions*, *Contact Manager*, *Purchase Orders*, *Inventory*, *Bills of Material*, *Material Requirements*, and *Routings* modules. A common numbering method is to start the vendor code with the first three letters of the vendor's sort name, followed by the first three letters of his second name. For example, Acme Manufacturing Company would be coded ACMMAN. If the vendor only has one name, use the first six letters of that name. With this coding method you almost

never encounter duplicate codes for similarly named vendors.

Vendor Class codes

Vendors can be assigned an optional four character, alphanumeric class code in order to group or filter vendors on reports by class code.

Customer codes

This is a 10-character, alphanumeric code, upper case only. Customers are used in the *Accounts Receivable*, *Sales Orders*, *Sales Analysis*, *Inventory*, *Work Orders*, *Job Costing*, and *Estimating* modules. A commonly used method is to start the customer code with the first three letters of the customer's sort name, followed by the first three letters of his second name. For example, Ace Manufacturing Company would be coded ACEMAN. If the customer only has one name, use the first six letters of that name. With this coding method you will almost never encounter duplicate codes for similarly named customers.

If a bill-to customer has multiple shipping addresses, each one will be set up with a customer code. A good technique is to attach a three-letter code for the destination city on the end of the standard customer code. For example, if Ace Manufacturing Company has a ship-to address in Industry and another in Anaheim, the first one could be coded ACEMANIND and the second one ACEMANANA. This method of numbering ship-to's makes it easy to identify the correct customer code when viewing codes in a lookup window.

Customer Class codes

Customers can be assigned an optional four character, alphanumeric class code in order to group or filter customers on reports by class code.

Work Center codes

This is a 12-character, alphanumeric field. You will divide your factory into work centers and assign each routing sequence (operation) to a work center. Outside Processing is also set up as a work center or series of work centers.

Scrap codes

This is a 2-character, alphanumeric field, upper case only. Scrap codes are accompanied by a 30-character description. Scrap codes can be used to track unplanned scrap of raw materials or semi-finished products in the work order modules and rejected purchased items returned to vendor to specific reasons.

QC codes

This is a 2-character, alphanumeric field, upper case only. QC codes are accompanied by a 30-character description. QC codes can be used to track rework labor and "use as is" purchased items to specific reasons.

Machine codes

This is a 4-character, alphanumeric field, upper case only. Machines are set up in the machine file in the *Routings* module and are used to define specific machines in the shop.

Tool codes

This is a 15-character, alphanumeric field, upper case only. Tools are set up in the tool file in the *Routings* module and may be used to define tools, dies, fixtures, etc.

Operation Templates

Operation templates are used to define common operations and are identified with a three character numeric field. Templates are copied into routings as needed. Common operations should be numbered together so they display together when viewed in lookup windows.

Phase 3 – Enter master files

The following master files should be set up roughly in the order they are listed. Some files are mandatory and some are optional. The entry program for each master file is in parenthesis next to the name. Refer to the program documentation for details.

NOTE: Enter data in the default company that you see when the system first starts up. Only create companies in UT-I, *Create/Delete Company*, if you actually have additional companies or if you want to create companies for training or archival purposes.

Terms Table (SM-E)

The terms table is set up through SM-E, *Enter Terms Table*. You must enter at least one term type; we suggest “Net 30” for the first term. Terms will be used when entering sales orders, purchase orders, and AR vouchers.

Vendors (SM-B)

Vendors are set up in SM-B, *Enter Vendors*, and are used by the *Sales Commissions*, *Purchase Orders*, *Inventory*, *Bills of Material*, *Contact Manager*, and *Routings* modules. Before vendors are entered you should have your terms table set up (see above). You do not have to enter all your vendors at this time. At a minimum you should enter all vendors to whom you owe money, as well as vendors for sales tax authorities and payroll.

Employees (SM-G)

Employees, which are used in *Work Orders* and *Data Collection* for labor reporting and tracking, and in *Purchase Order Receiving* and *Inspection Buyoff* are set up in SM-G, *Enter Employees*. If you plan on charging labor to work orders using work center standard rates, it is not necessary to enter employee pay rates in SM-G.

Tax Codes (SM-E) & Tax Groups (SM-F)

You can skip setting these up if you will be invoicing within your accounting

system. Tax codes and tax groups are set up for sales tax processing on invoices and purchase orders. Each tax authority (state, provincial, local, etc.) is set up with a tax code, and then tax codes are combined into tax groups. Each taxable customer will be assigned to a tax group. Before entering tax codes you will need to set up vendor(s) to which the sales tax liability will be transferred when it comes time to pay the taxes through *Accounts Payable*.

Salespersons (CS-A)

You can skip this if you will be invoicing within your accounting system. DBA allows you to identify specific salespeople for a customer's sales order, either for reference purposes or for commissions tracking.

Customers (SM-A)

Customers are set up in SM-A, *Enter Customers*, and are used by the *Accounts Receivable*, *Sales Orders*, *Sales Analysis*, *Work Orders*, and *Estimating* modules. Before entering customers you should set up your terms table, salesmen, tax authorities, and discount codes (if applicable).

Item Classes (SM-C)

Item classes are set up in SM-C, *Enter Item Classes*. All inventory items will be assigned to an item class. You must set up at least one item class.

Items (IN-B)

Items are set up in IN-B, *Enter Inventory*, and are used by most modules in the system. Before entering inventory items you should set up your item classes and vendors (see above).

Item Prices (SO-Q-A thru H)

You can skip this if you will be invoicing within your accounting system. You can set up base prices or price codes (with quantity price breaks) or contract prices for individual customers. All these price options are used by the *Sales Orders* module. You must have item numbers created for the items you sell before you can assign prices.

Discount Codes (SO-Q-F)

You can skip this if you will be invoicing within your accounting system. Discount codes allow you to set up discount tables for each item class. If you are going to assign customers to discount codes, set them before entering customers.

Standard Costs (IN-L-A)

Standard costs are used by most manufacturers in order to establish accurate product costs. Job shops tend to use standard costs primarily for use with the *Estimating* module.

You can enter standard costs for materials in IN-L-A, *Enter Standard Costs*, or directly from IN-B, *Enter Inventory*, by clicking on the *Std Cost* button (or by pressing <PgDn>) from within the inventory entry screen.

An alternative to entering material standard costs is to enter average costs through

IN-K, *Adjust Physical Levels*, when entering your beginning inventory, then run IN-L-E, *Update Material Std Costs*, which will roll the average cost into the standard cost field for R (purchased parts) type items. After bills of material are entered, BM-G, *Print/Rollup Standard Costs*, can be used to roll up the costs for all assemblies instead of having to roll them up individually.

Bills of Material (BM-A)

Accurate bills of material are essential for inventory control, costing, and production planning. Bills of material are entered in BM-A, *Enter Bills of Material*. Before entering bills of material you should set up all your inventory items (see above).

Shop Calendar (SM-H)

The shop calendar determines which days are available for production. It is used by the *Sales Orders*, *Work Orders*, *Material Requirements*, and *Scheduling* modules and is set up through SM-H, *Enter Shop Calendar*.

Work Centers (RO-C)

Work centers are entered in RO-C, *Enter Work Centers*. You must set up at least one work center in order to use routings.

Machines (RO-D)

Machines are set up in RO-D, *Enter Machines*.

Tools (RO-E)

Tools are setup in RO-E, *Enter Tools*.

Operation Templates (RO-I)

Operation templates can be setup for standard operations and can be copied into routings. They are entered through RO-I, *Enter Operation Templates*. Before entering operation templates you should set up your work centers and, if applicable, your machines and tools (see above).

Specs Templates (RO-K)

Specs templates are user-defined templates that can be copied into routing sequences. They are entered through RO-K, *Enter Specifications Templates*. Specs templates can be set up as forms with column headings, field names, etc., and can be used to define setups, temperatures, inspection steps; whatever is needed for your specific operation that is not handled by the standard routing fields. Once specs templates are copied into routings, they can then be modified for a particular product/sequence, and they will print on the shop traveler.

Routings (RO-A)

Routings are entered through RO-A, *Enter Routings*. Before entering routings you should set up your work centers and, if applicable, your machines and tools (see above).

QC Codes (RO-F)

QC codes are entered through RO-F, *Enter QC Codes*.

Scrap Codes (RO-G)

Scrap codes are entered through RO-G, *Enter Scrap Codes*.

Phase 4 – Enter module defaults

Each module has default settings that allow you to specify various preferences within the application programs.

Company Name & Address (SD-A)

This establishes the company name that will be displayed on menus and reports and the name and address that can optionally print on forms such as invoices, purchase orders, etc.

Work Orders Defaults (SD-B)

Various set up parameters for the *Work Orders* module are entered via SD-B, *Work Orders Defaults*.

Purchase Orders Defaults (SD-C)

Various set up options for the *Purchase Orders* module are set up through SD-C, *Purchase Orders Defaults*.

Inventory Defaults (SD-H)

You can select your choice of either average, FIFO, or LIFO costing. Unless you have a specific need for FIFO or LIFO, select average costing.

Routings Defaults (SD-I)

Various set up options for the *Routings* module are established through SD-I, *Routings Defaults*.

Bills of Material Defaults (SD-J)

Various set up options for the *Bills of Material* module are established through SD-J, *Bills of Material Defaults*.

Sales Orders Defaults (SD-M)

Enter your *Sales Orders* default settings through SD-M, *Sales Orders Defaults*.

Sales Commissions Defaults (SD-N)

Enter your *Sales Commissions* default settings through SD-N, *Sales Commissions Defaults*.

Accounts Receivable Defaults (SD-P)

Enter your *Accounts Receivable* settings through SD-P, *Accounts Receivable Defaults*.

Phase 5 - Set up printing and order any forms

If you've not already done so, set up your printing for forms, labels, and reports. All printing set up issues are covered in appendix A, *Setting up Printing*.

All forms (acknowledgements, packing slips, shipping labels, invoices, purchase orders) are available in plain paper formats or formatted for pre-printed forms.

DBA forms are provided by *Fortune Forms, Inc* (800-884-8804). A universal form is available that offers the advantage of using a single form for all printed forms. Pre-printed forms are available for the above listed items plus statements and AP checks for DBA accounting. Call *Fortune Forms* directly for any questions related to DBA compatible printed products.

Phase 6 – Train personnel to use the programs

During phase 7 you should see that your personnel are trained to use the various programs. By this time most of your master file data has already been entered, which will provide you with familiar data to work with during the training period.

Make a copy of your company (UT-I)

Training is most effective if done with your own data. One way to use your own data for training without risk of adversely affecting your permanent files is to create a test company via UT-I, *Create/Delete Company*. When creating a company you are given the option of copying the data from another company. Copy your company's data to the test company and conduct your training exercises in the test company.

During training make sure people fully understand how to operate the programs they will be using. If using work orders, for example, go through at least one complete work order cycle – enter the work order, release it, print a traveler, issue material, enter labor, enter finished production, close the order – and all the while print the various related reports and make sure the data is correctly processed. Do the same with the purchase order cycle and the sales order cycle.

Do not advance to phase 8 until your people know how to use the programs. Phase 7 marks the starting point when the system actually gets used.

Phase 7 – Enter open orders and inventory on-hand

Final phase 7 requires careful planning. The normal practice is to establish a target startup date. Just prior to the startup date (such as the weekend before) you must gather information from your old system and enter your current inventory

counts and open work orders and purchase orders as of the startup date. This involves intensive data entry in a short period of time, so make sure the time is allocated and you have enough personnel on hand to get the job done.

You will need to have the following information from your existing accounting system prior to beginning phase 8.

- Open (unreceived) purchase orders.
- Open (unfinished) work orders.
- Open (unshipped) sales orders
- Outstanding invoices and credits for all customers.
- Current inventory on-hand balances.

Use UT-K-A (if necessary) to remove unwanted transactions

If for any reason you have made some transactions during the training phase that you now want to remove so that you have a “clean” system (while retaining all your master file data), you can run UT-K-A, *Clear Data*. The program presents you with an entry screen from which you will be able to select partial or complete deletion of data files. Before you make each selection, click the *Help* button (or press F1) for additional information on each choice. For each data category on the screen, enter a C if you want to keep the master file data, but want to clear all transactions such as sales information, manufacturing transactions, etc. Enter an N if you want to leave the data untouched.

Enter open purchase orders (PO-A)

When you are ready to begin using purchase orders, you must enter all open (unreceived) purchase orders so that all requirements for material and outside processing are in the system and existing purchase orders can be received.

Enter open work orders (WO-A)

When you are ready to begin using work orders, you must enter all open (uncompleted) work orders so that all requirements for material and labor are in the system and existing work orders can be processed.

Open work orders will already have some costs accumulated in them at the time you begin using DBA. You can either ignore these costs and enter an estimated cost when these work orders are completed, or you can enter costs-to-date as of the time you begin using DBA.

A simple way to get a total cost into work orders is to enter total costs-to-date through WO-H, *Enter Misc/Extra Costs*. If you want costs broken down into labor, setup, overhead, etc., an alternative approach would be to set up labor and outside processing item numbers as well as a dummy number for material and issue these item numbers through WO-G, *Issue Materials*. In doing so, all the appropriate cost buckets will be updated.

Enter open sales orders (SO-A)

When you are ready to begin using sales orders, you must enter all open sales orders. If you are using the manufacturing system, this step is essential in order to provide the production planning system with complete requirements.

Enter open accounts receivable aging (AR-B)

You can now enter invoice and credit amounts representing transactions up to the cutover date in *Accounts Receivable*.

To enter your opening aging balances, select the *Beginning Balance* option in AR-B, *Enter AR Vouchers*. This option will create records in the aging file, but will not post to the *General Ledger*.

At this point you can run AR-F, *Print Aging*, and compare the total on the report with the total amount in your accounting system. If they don't match, you will have to use AR-B, *Enter AR Vouchers*, to back out and re-enter information so that these figures will be brought into balance.

Enter physical inventory (IN-K)

When you are ready to begin processing sales orders, purchase orders, and work orders, you should enter a beginning physical inventory. From that point on the system will maintain a perpetual inventory through the system transactions.

You can print a list of items to count through IN-J, *Print Physical Check*.

The actual counts and beginning average costs are entered through IN-K, *Adjust Physical Levels*. You can verify balances and inventory value through IN-F, *Print Inventory Value*. The inventory value should agree with your general ledger account value(s); otherwise adjustments will have to be made either to the inventory or to the ledger accounts until the starting figures are in agreement.

After all the average costs are in, you might want to roll the average costs for purchased parts into the standard cost field to establish beginning standard costs for materials. This can be done through IN-L-E, *Update Material Standard Costs*.

For a more sophisticated physical inventory involving freezing the inventory, capturing counts on tags, rolling in costs, etc., DBA offers a complete *Physical Inventory* module. See chapter 12, *Physical Inventory*, for details.

Perform a cost rollup (BM-G)

You should make sure you have a standard cost established for all materials (type R items). You can automatically create standard costs for purchased (Type R) parts from your last or average costs by running IN-L-E, *Update Material Standard Costs*.

Once all this is done and all your routings are entered, you should run BM-G, *Print/Rollup Standard Costs*, to roll up the standard costs for all your cost categories: material, labor, setup, outside processing, fixed overhead, variable

overhead.

You will now have current standard costs for all products. Standard costs are used by the *Work Order* module to calculate estimated costs, the *Job Costing* module to compare actual performance with standard, the *Inventory* module to compare inventory value at standard cost versus average cost, the month end inventory reports to compare actual costs with standard costs, and the *Estimating* module for material costs if none are specified on RFQ's or through ES-H, *Enter Material Costs*.

This completes the implementation process. The system is now ready for daily processing of work order, purchase orders, and sales orders.

Implementation checklist

Phase 1 – System orientation

- ☐ Study the user manual, especially chapter 3
- ☐ Look at sample company 99

Phase 2 – Develop numbering schemes

- ☐ Item Class codes
- ☐ Category codes
- ☐ Item Numbers
- ☐ Terms codes
- ☐ Vendor codes
- ☐ Vendor Class codes
- ☐ Customer codes
- ☐ Customer Class codes
- ☐ Work Center codes
- ☐ Scrap codes
- ☐ QC codes
- ☐ Machine codes
- ☐ Tool codes
- ☐ Operation Templates

Phase 3 – Enter master files

- ☐ Terms Table (SM-E)
- ☐ Vendors (SM-B)
- ☐ Employees (SM-G)
- ☐ Tax Codes (SM-E) & Tax Groups (SM-F)
- ☐ Salespersons (CS-A)
- ☐ Customers (SM-A)
- ☐ Item Classes (SM-C)
- ☐ Items (IN-B)
- ☐ Item Prices (SO-Q-A thru H)

- ☐ Discount Codes (SO-Q-F)
- ☐ Standard Costs (IN-L-A)
- ☐ Bills of Material (BM-A)
- ☐ Shop Calendar (SM-H)
- ☐ Work Centers (RO-C)
- ☐ Machines (RO-D)
- ☐ Tools (RO-E)
- ☐ Operation Templates (RO-I)
- ☐ Specs Templates (RO-K)
- ☐ Routings (RO-A)
- ☐ QC Codes (RO-F)
- ☐ Scrap Codes (RO-G)

Phase 4 – Enter module defaults

- ☐ Company Name & Address (SD-A)
- ☐ Work Orders Defaults (SD-B)
- ☐ Purchase Order Defaults (SD-C)
- ☐ Inventory Defaults (SD-H)
- ☐ Routings Defaults (SD-I)
- ☐ Bills of Material Defaults (SD-J)
- ☐ Sales Orders Defaults (SD-M)
- ☐ Sales Commissions Defaults (SD-N)
- ☐ Accounts Receivable Defaults (SD-P)

Phase 5 – Set up printing and order any forms

- ☐ See appendix A for printing setup instructions
- ☐ Order any forms needed from *Fortune Forms* (800-884-8804)

Phase 6 – Train personnel to use the programs

- ☐ Make a copy of your company (UT-I)

Phase 7 – Enter open orders and inventory on-hand

- ☐ Run UT-K-A (if necessary) to remove unwanted transactions

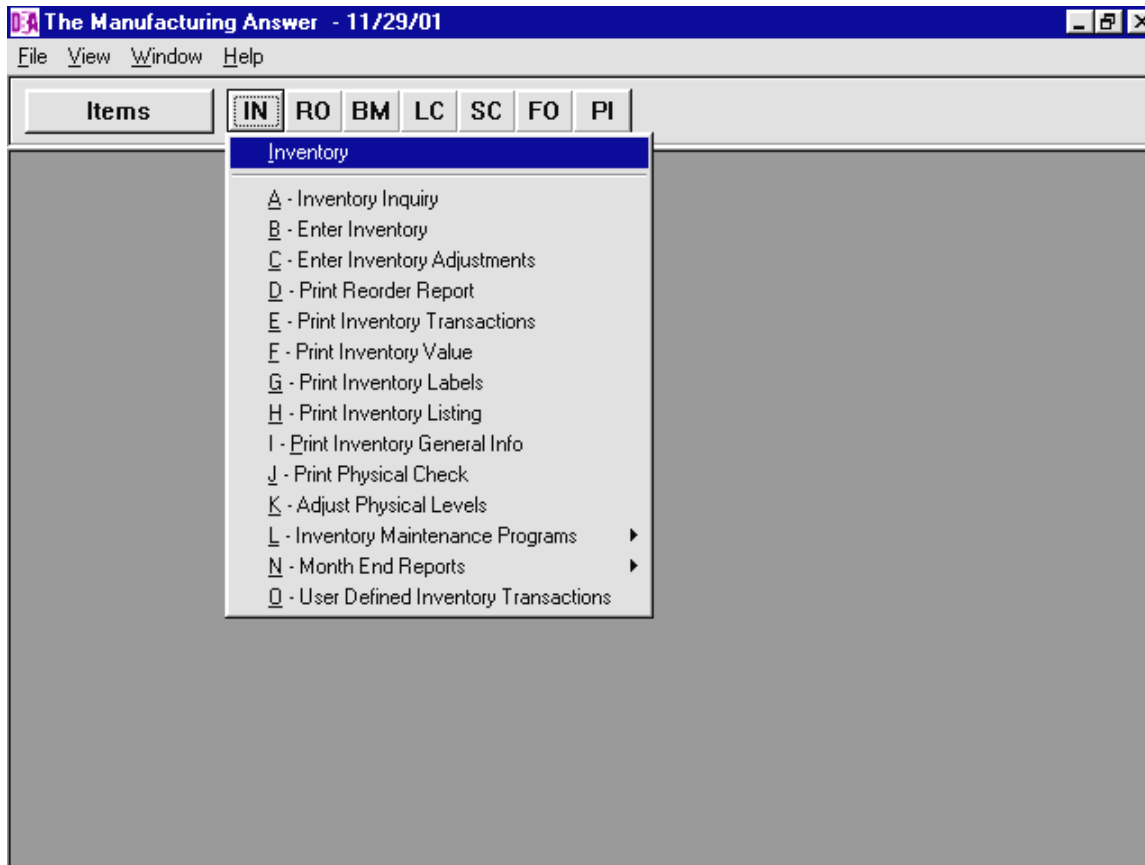
Implementation checklist

- ❑ Enter open purchase orders (PO-A)
- ❑ Enter open work orders (WO-A)
- ❑ Enter open sales orders (SO-A)
- ❑ Enter open accounts receivable aging (AR-B)
- ❑ Enter physical inventory, costs (IN-K)
- ❑ Perform a cost rollup (BM-G)

8. INVENTORY

- **Inventory overview**
- **Item costing**
- **Getting freight into the cost of items**
- **Using multiple locations**

Inventory overview



The *Inventory* module consists of the programs listed on the menus displayed above, plus additional program on the *Inventory Maintenance* and *Month End Reports* submenus. The *Inventory* module performs the following functions.

PRODUCT DATABASE

The *Inventory* module serves as the primary database for storing most product information.

Both inventory items and non-inventory items (such as service and repair) are set up in *Inventory*.

A variety of fields are available for specific manufacturing related functions. Products may be organized into classes and categories for reporting purposes.

ORDERING TOOL

The *Reorder Report* in combination with the *Inventory Inquiry* provides an excellent tool for determining what to purchase or manufacture. Information is displayed as to what's on hand, on sales order, on back order, on purchase order, on work order, allocated, and in work-in-process. Reorder levels and reorder

amounts may be established for each item.

INVENTORY TRACKING

All inventory transactions (purchase order receipts, shipments, adjustments, issues to work-in-process, manufacturing receipts) are tracked and available for review. Month end reports provide all information needed for month end inventory accounting.

PHYSICAL INVENTORY

The *Physical Check* report allows you to select items for counting, and the *Adjust Physical Levels* program provides a means for entering new balances and tracking the dollar effect of inventory adjustments. In addition to these simple physical inventory tools, DBA offers an entire *Physical Inventory* module (see chapter 12 for details).

MULTIPLE LOCATIONS

The system provides for multiple warehouses or manufacturing plants and can maintain separate stock balances and costs by location.

Item costing

DBA maintains three costs on all items

DBA automatically maintains an *Average Cost* on all items (except for non-stock items). This is the cost that is used to value your inventory and is the cost used for job costing when items are issued from inventory to work orders.

DBA optionally holds a *Standard Cost* on all items. Standard costs are maintained manually on individual components and are automatically calculated via cost rollups of your routings and bills of material. Standard costs do not change until you change them. As you shall see later in this section, standard costs can be highly useful for some types of manufacturing.

Finally, DBA maintains a *Last Cost* on all items for reference purposes and for use on purchase orders as a default cost when there are no costs in the vendor price file.

Using average costs

Many manufacturers speak of using "actual" costing. In reality, they are speaking of average costing, which is about as close as one can get to "actual" costing in a manufacturing system.

With purchased items and services, yes, you can track actual costs. Anything manufactured, however, gets overhead applied to it, which is not an actual cost, but an estimated cost. Also, direct labor is often applied to manufactured items at standard rates, not exact wage rates, so again, it is an estimated, not actual cost.

Therefore, on any manufactured item, be aware that the average cost you see is an estimated cost, not an actual cost.

How is the average cost calculated?

The average cost is a running average. Whenever new units of an item are added to existing units (such as when receiving a purchase order), the total cost of the new units is added to the total cost of the existing units, then is divided by the new total number of units.

The average cost is recalculated in the following situations.

- When you receive inventory items through purchase order receiving.
- When you receive work order finished production into inventory.
- When you make a positive adjustment to on-hand inventory.

What if a transaction causes the on-hand to go negative?

The above formula works fine as long as the on-hand quantity after the transaction is a positive amount. In actual practice, however, it is not uncommon for inventory items to have negative on-hand balances. This can be caused by

counting errors or by being behind or out of synch on reporting inventory transactions to the system.

If after an inventory transaction the resulting on-hand quantity will be a negative amount, the system ignores the standard formula (which can give distorted results when using negative numbers) and continues to use the average cost that was in effect before the transaction, rather than try to calculate a new average cost.

Inventory value is maintained in each item's *Book Value* field

Each item has a *Book Value* field that gets adjusted up or down with each inventory transaction by the total cost of the transaction. The total of all your item book value amounts represents the total value of your inventory.

On normal transactions that result in a zero or positive on-hand quantity, this cost will equal the transaction units multiplied by the average cost.

On transactions that result in a negative on-hand quantity, the total cost will not equal units multiplied by average cost.

The reason we are bringing this to your attention is that you may expect the total units on hand of an item multiplied by the item's average cost to equal the item's total book value when running the IN-F, *Print Inventory Value*, report. This would be the case if the item had always maintained a zero or positive on-hand balance. But if the item's on-hand balance had gone negative at some point, book value could differ from this equation.

What costs are used for non-stock items?

Non-stock items do not have on-hand inventory and therefore have no book value, but the system does maintain last and standard costs on these items in order to have a cost to apply in the event they are issued to work orders or shipped to customers.

For item types N (non-stock), L (labor) or T (outside processing), the system looks first to see if there is a standard cost. If it finds one, it uses it as the cost. If it doesn't find a standard cost, it looks for the last cost. If it finds one, it uses it as the cost.

What about FIFO or LIFO costing?

DBA offers FIFO or LIFO costing, in which case the *Average Cost* field is calculated using first-in, first-out cost buckets or last-in, first-out cost buckets.

In general there is no advantage to using either of these methods over straight average costing. Unless there is severe inflation in the economy, all three methods will produce very similar results. FIFO/LIFO costing involves more maintenance and is therefore not recommended over the simpler alternative of using straight average costing.

Your costing preference is set in IN-L-I, *Change Costing Method*.

Using standard costs

The screenshot shows a software window titled "IN-L-A Enter Standard Costs". At the top, there's a menu bar with "File", "View", "Window", and "Help". Below the menu is a toolbar with buttons: "Items", "IN", "RO", "BM", "LC", "SC", "FO", and "PI". The "IN" button is selected. The main area contains several input fields: "Part No", "Desc", "Lot Size" (with a value of 0), "Type", "Last Cost" (0.0000), "Avg Cost" (0.0000), and "Stock UM". Below these fields are two tables. The first table, "This Level Costs", lists: Material (0.000000), Freight (0.000000), Labor (0.000000), Setup (0.000000), Outside Proc (0.000000), Fixed Overhead (0.000000), and Var Overhead (0.000000). The second table, "Rolled-up Costs (All Levels)", lists: Material and Freight (0.000000), Labor (0.000000), Setup (0.000000), Outside Proc (0.000000), Fixed Overhead (0.000000), and Var Overhead (0.000000). At the bottom of the tables is a "Total Standard Cost" field with a value of 0.000000. At the very bottom of the window are navigation buttons: "Help", "Lookup", "Clear", "First", "Last", "Prev", "Next", and "Exit".

IN-L-A, Enter Standard Costs

Any manufacturer with a product line or any job shop that makes multiple level products on a repeat basis should use standard costs.

Instead of a single cost like the average cost and last cost, with standard costs you can break down the cost of each item into separate costs for material, setup, labor, outside processing, fixed overhead, and variable overhead.

Standard costs are used for the following:

- Standard costs are your best estimate of what your products actually cost to manufacture.
- Standard costs are the best costs to use when entering finished production of partially completed work orders.
- Standard costs give you a basis of comparison with actual costs as a measure of performance.

Use standard costs to estimate product costs

Standard costs can be rolled up through your routings and bills of material to establish total standard costs for manufactured items.

Each manufacturing item has "this level" costs, which are the direct costs that go into the item's final level, and "rolled up" costs, which are the "this level" costs plus all the costs for all the lower levels of the product structure.

Therefore, the first benefit of using standard costs is that it can give you a good estimate of what your existing products cost to make, and also what any new products should cost to make.

Standard costs are rolled up through BM-G, *Print/Rollup Standard Costs* or by clicking the ROLLUP Button in *IN-L-A Enter Standard Costs*.

Post partial finished production at standard cost

If you have multi-level products, it is difficult to record all your finished production at actual cost.

Quite often you will have partial completions where subassemblies are being issued to other work orders before all production is completed and all costs are reported. Therefore the cost that is attached to the partially completed production has to be an estimated cost.

Only when the work order is finally completed and all costs are reported can you know the actual cost of the work order.

DBA allows you to enter finished production at standard cost in WO-I, *Enter Finished Production*.

A good practice that we recommend is to enter all finished production for partial completions at standard cost. Only the last transaction is entered at actual cost.

On the last transaction the *Enter Finished Production* program will make a balancing entry that takes the total cost of the work order, deducts all previously reported costs, and uses the remainder as the cost of the last transaction. The costs of all transactions added together will exactly equal the total cost of the work order. You can also enter all finished production at standard cost and when the work order is closed, a variance posting will balance out the net cost into the work order and the finished production taken out.

Use standard costs as a comparison with actual costs

Standard costs for manufacturing items should constantly be reviewed. The more accurate your standard costs are, the more confidence you will have that they accurately represent the costs of your products.

A good report to review is IN-N-C, *Print Closed Work Orders Costing*. This report shows standard and actual costs and the variances between the two, broken down into material, labor, setup, outside processing, fixed and variable overhead.

If you notice big variances in any of the cost categories, it means that either your standard costs are inaccurate or something went wrong during production. In either case, corrective action should be taken so that the standard is corrected or the source of the production problem is addressed.

How are standard costs established for labor and overhead?

Standard costs for labor, setup, and overhead are established by the time-per-part or parts-per-hour production rates entered within RO-A, *Enter Routings*.

JC-H, *Print Work Order History*, is an excellent tool for reviewing the history of each routing sequence of an item across multiple work orders. This lets you use past history to help estimate the time-per-piece or parts-per-hour production rates for each sequence.

The alternative to using past history is to actually conduct time studies to determine production rates. You may not want to do this on every product you make, but it can be worthwhile on your more important products.

Standard rates for labor and overhead are maintained in RO-C, *Enter Work Centers*.

In the case of labor, you would normally set rates by calculating your total direct labor costs per month divided by the total number of production hours for the month. This will give you an average hourly rate. If you wish to get more precise, you can make the same calculation on a departmental or work center basis, since each work center can have its own rates.

Also be aware that for setup labor you must specify the typical production run quantity in the routing's *Lot Size* field so that the program can translate setup cost (which is the same fixed cost whether you run one part or thousands) into a per unit cost.

In the case of overhead, DBA lets you express overhead as a percent of labor or as an hourly rate. If you express it as a percentage of labor, you would compare your total overhead for a month with your total direct labor for the month in order to arrive at the percentage of labor. If you use the hourly rate method, take total overhead for the month and divide it by the total number of production hours.

DBA lets you organize your overhead into fixed and variable overhead categories. Some companies use both overhead categories, others will lump all their overhead into one and will typically use the fixed overhead category to hold the rates.

If you are using the "light manufacturing" method of handling labor, standard costs for labor and overhead are entered against the labor item numbers (item type L), which are then incorporated in your bills of material. To enter standard costs for type L items, use IN-L-A, *Enter Standard Costs*.

How are standard costs established for outside processing?

Standard costs for outside processing are established within RO-A, *Enter Routings*

In each outside processing routing sequence you can enter a cost. This is the cost that will be used in the standard cost rollup.

How are standard costs established for materials?

To enter standard costs for purchased materials, use IN-L-A, *Enter Standard Costs*.

A nice tool for periodically updating standard costs for materials is IN-L-E, *Update Material Standard Costs*. This program lets you replace the current standard cost for all or a range of items using the item's *Last Cost* or *Average Cost*.

Standard costs are also used by the *Estimating* module

ES-A, *Enter Estimates*, will use the standard cost for materials if an RFQ is not specified for a material or if there is no entry on file in the estimate *Material Costs* file.

Getting freight into the cost of items

Freight can be a significant portion of the total cost of many materials and therefore should be reflected in their inventory cost in the job cost when materials purchased directly for work orders.

Materials purchased for inventory

When materials are purchased for inventory, it is often the case that there are several materials purchased on the same purchase order. When these materials are received, there is only one freight charge, which must somehow be apportioned to each material. This is a complex undertaking that is difficult to put into a formula.

Instead, DBA offers a *Freight Pct* field that can be set up in advance for each item number. For example, if experience tells you that an item has a roughly 20% freight cost, you can enter 20.00 in its *Freight Pct* field. Whenever the item gets received, PO-C, *Receive Purchase Orders*, will automatically increase its cost by 20%

Materials purchased for work orders

When you purchase materials for a work order instead of for inventory, if all items on the purchase order are designated for the same work order number, the freight amount will be charged to the work order. AP-C, *Enter PO Invoices*, will add a special component to the work order bill of material called *Job Cost Freight* and will charge the cost to that component.

This special item number for *Job Cost Freight* is set up in advance through SD-C, *Purchase Orders Defaults*.

Using multiple locations

The screenshot shows a software window titled "The Manufacturing Answer - 11/29/01" with a menu bar (File, View, Window, Help) and a tabbed interface. The "Items" tab is active, showing sub-tabs for IN, RO, BM, LC, SC, FO, and PI. The main window is titled "IN-L-B Enter/Assign Locations". It contains three main sections: "FACTORY / WAREHOUSE LOCATION" with fields for Loc Name, Add1, Add2, CSZ, Resale #, Contact, Phone, and FAX; "SINGLE PRODUCT ENTRY" with fields for Part Number and Description; and "GENERATE LOCATION RECORDS" with fields for From Part Number, Thru Part Number, From Class, Thru Class, For Product, and Type(s). At the bottom are buttons for Help, Lookup, Print, and Exit.

IN-L-B, Enter/Assign Locations

DBA allows you to maintain independent inventory data for your items at up to 250 different locations.

A location is typically another factory or warehouse, but it could represent anything you want. For example, a location could be an area within a warehouse. Or, if you have trucks in the field with inventories, each truck could be a location.

Within each location you can track units on hand, on sales order, on purchase order, on work order, allocated, and the location's inventory value.

You can create customer orders, purchase orders, and work orders for a single location only. Items received or sent from more than one location require separate orders.

Work orders include a field for *Location* if you ever go to a second or third plant and need to schedule and track orders by location.

You should give careful consideration how you name your locations codes so that they are easily used and recognizable by those who will be working with them.

We will now review three possible scenarios for using locations.

- You have a single location for all your items.
- You have set up your items without any location information and then find it necessary to add location information for a few items that will be stored away from your original location, such as with a consignment inventory.
- You have multiple locations and all your items are subject to being stored at any of them.

In all examples, the item record is created without entering any information about units on hand, etc. Instead, you will use customer orders, purchase orders, and work orders to update your various inventory balances after the locations for each item are created.

Example 1: A single location.

If you plan on using only one location, it is not necessary to create any locations. The system automatically assign all items to a default blank “ ” location.

The default blank location is set up in SD-H, *Inventory Defaults*. When setting up your inventory defaults, leave the *Default Location* field blank.

When entering purchase orders and work orders, skip over or press <Enter> through the *Location* field.

Example 2: Adding a 2nd location after setting up your inventory.

In our second example, you've been using the system with a single location as described above. You now want to add a second location for a few items, perhaps in a satellite warehouse.

The first step is to go to IN-L-B, *Enter/Assign Locations*, and check the name of your first locations. If blank, it should now be given a name, otherwise users will mistake a blank name for meaning no location at all. To do so, run UT-K-E, *Consolidate Inventory Locations*, and reassign the location name. Once the first location has a name, proceed with creating the second location in IN-L-B, *Enter/Assign Locations*.

Next, using IN-L-B, individually assign item numbers to the new location, or assign ranges of item numbers by item class or item type.

The new location will now be listed whenever you perform a lookup on your locations.

Example 3: Setting up a multiple location inventory system.

In our third example you will be using a multiple location system.

Be sure and assign short, identifiable names for your locations that will be easily recognized by users of the system.

When you have multiple locations, it is not a good ideal to have the default location as a blank location. Users can accidentally forget to enter a location and

the system will automatically use the blank location by mistake. Therefore, you should set up the default location in *Inventory Defaults* with an actual code rather than a “blank” code. So, if you have a blank location, before setting up any additional locations, run UT-K-E, *Consolidate Inventory Transactions*, and assign the blank location to a name.

Then proceed with creating additional locations through IN-L-B, *Enter/Assign Locations*. As you do so, be sure and assign individual items or ranges of items to each location.

Using your multiple inventory locations

Now that your locations are set up and your items assigned to locations, you can use purchase orders and work orders to receive items into the new locations.

You cannot enter items on customer orders, purchase orders, and work orders unless they are assigned in advance to the location specified in the order header.

The system maintains stock status fields (on sales order, on purchase order, on work order, allocated, in WIP) for each item for each location.

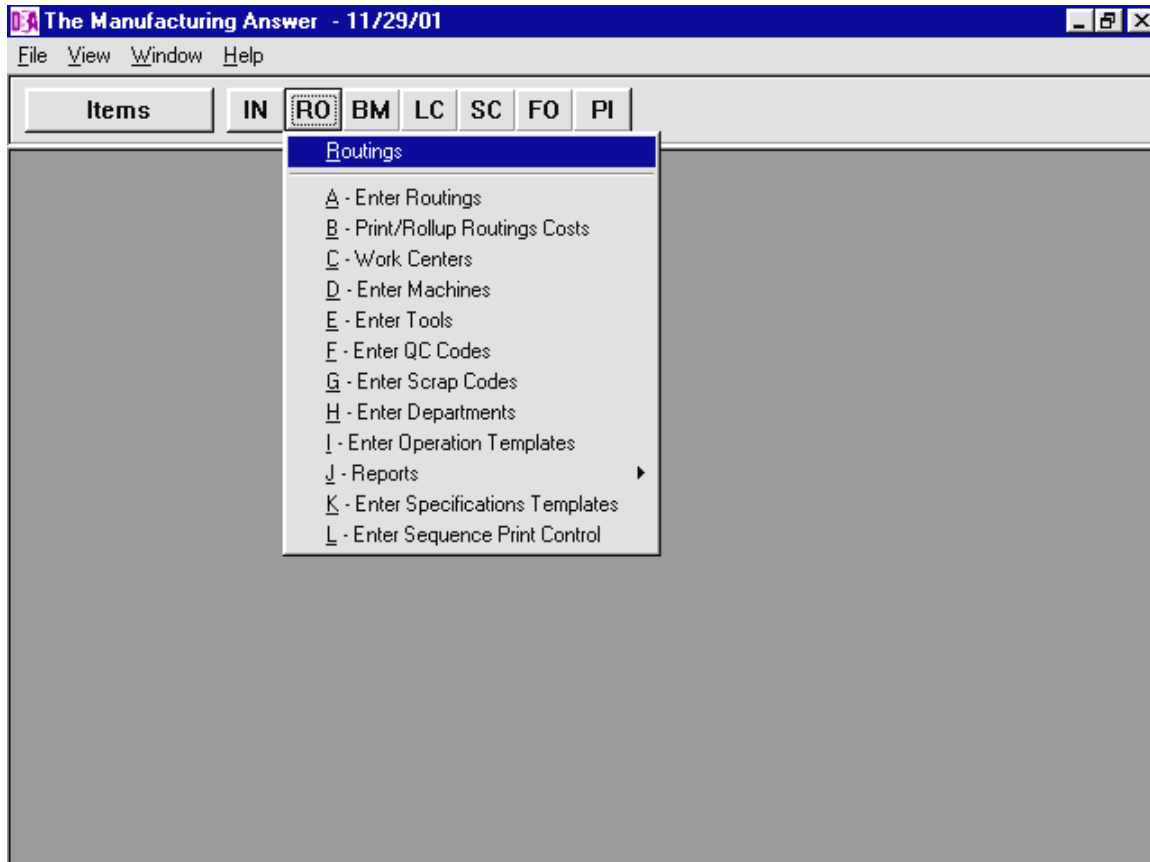
You can see stock status via IN-A, *Inventory Inquiry*, the IN-D, *Print Reorder Report*, the IN-J, *Print Physical Check*, report, or via IN-K, *Adjust Physical Levels*.

To move inventory from one location to another, use IN-L-J, *Transfer Inventory*.

9. ROUTINGS

- **Setting up work centers**
- **Entering routings**
- **Setting production rates**
- **Using operation templates**
- **Tracking machines and tools**

Routings overview



A routing refers to the way an assembly is “routed” through the factory. In the Routings module we define how and where assemblies are manufactured. The Routings module consists of a number of database files needed by the *Work Orders* module for the manufacturing process. These database files include:

ROUTINGS

Routings define each labor sequence (operation) or outside service. Production standards are set and work centers specified for each sequence. Extensive space is provided for free form text and specifications defining how each operation is to be performed.

WORK CENTERS

The factory can be divided into work centers. Rates for setup, labor, and overhead are established for each work center. Work centers are valuable for costing and scheduling purposes.

MACHINES

You may set up your machines in the machine file. Sequences and operations can

be specified to run on particular machines, in which case the machine time is tracked automatically so that preventive maintenance can be scheduled.

TOOLS

Tools, dies, fixtures, molds etc. may be set up in the tool file. Sequences and operations can be specified to use specific tools. The number of parts produced per tool can be automatically tracked by the system so that preventive maintenance can be scheduled.

QC CODES

User defined QC codes can be set up for tracking rework labor. These codes can be incorporated into an overall quality assurance program.

SCRAP CODES

User defined scrap codes can be set up for tracking unplanned scrap. These codes can be incorporated into an overall quality assurance program.

DEPARTMENTS

If work centers are to be grouped into broader departments, you may set up department codes and descriptions in the department file.

OPERATION TEMPLATES

Commonly repeated operations may be set up as templates which can be copied into routing sequences wherever specified.

SPECS TEMPLATES

Specs templates are user-defined templates that can be copied into routing sequences. Specs templates can be set up as forms with column headings, field names, etc., and can be used to define setups, temperatures, inspection steps; whatever is needed for your specific operation that is not handled by the standard routing fields. Once specs templates are copied into routings, they can then be modified for a particular product/sequence, and they will print on the shop traveler.

In addition to these database files, a *Print/Rollup Routings Costs* program calculates total setup, labor, overhead, and outside processing through all the sequences and passes the total standard costs to the inventory record for each parent product. Once in the inventory record, these costs can become part of the *Print/Rollup Standard Costs* in the *Bills of Material* module. This two-step cost rollup gives you precise control over product costing.

Setting up work centers

The Manufacturing Answer - 11/29/01

File View Window Help

Items IN RO BM LC SC FO PI

RO-C Work Centers

WC Code	WC Description	Dpt	Dpt Descrip	P Parent WC	% Util
05	CUTTING				0.00
10	ASSEMBLY				90.00
20	PRESS BRAKE				90.00
30	PAINT BOOTH				90.00
40	OUTSIDE PROCESSING				0.00
50	INSPECTION				100.00

Start typing to narrow search. Use top row <not keypad> for numeric searches.

Add Edit Delete

Chg Search Key Exit

RO-C, Enter Work Orders

Work centers are used to organize your factory and outside processes into specific work centers for costing, scheduling, and documentation purposes.

To set up work centers, use RO-C, *Enter Work Centers*.

Each routing sequence set up in RO-A, *Enter Routings*, gets assigned to a work center. Work centers can be set up for in-house labor or for outside processing.

Setting up in-house work centers

In-house work centers are set up for any labor or processes that are performed within your factory.

Setting work center capacity

When you set up in-house work centers you can specify the *Total Hours/Day*, the *Total Shift Hours*, and a *% Utilization* factor. These three parameters are used by the system for scheduling and capacity planning purposes.

The *Total/Hours Day* field represents the total number of hours per day available in this work center for production. For example, if you have five men working

eight hours per day each, you would have 40 hours per day available.

The *Total Shift Hours* field represents the total number of shift hours per day available for production. For example, a work center working a single shift would have eight shift hours, two shifts would have 16 hours, etc. This field value is used by SH-M, *Lead Time Estimator*, to determine how many hours in a day are available to routing sequences (it does not assign the entire work center capacity to one work order; only the shift hours amount).

Out of the total hours per day available for production, the *% Utilization* factor is the percentage of time truly available. The percent utilization accounts for normal down time due to maintenance, stoppages, etc. Ninety percent would be entered as 90.00. SH-E, *Finite Scheduling*, uses this field to determine how much run time to allocate to routing sequences.

Actual work center loads can be compared to work center capacity via SH-K, *View Work Center Load*, and SH-L, *Print Work Center Load*.

Entering standard rates for labor and overhead

Standard rates for labor, setup, and overhead are maintained in your work centers.

In the case of labor and setup, you would normally set rates by calculating your total direct labor costs per month divided by the total number of production hours for the month. This will give you an average hourly rate. If you wish to get more precise, you can make the same calculation on a departmental or work center basis, since each work center can have its own rates.

In the case of overhead, DBA lets you express overhead as a percent of labor or as an hourly rate. To express it as a percentage of labor, set the *Post Overhead as % of Labor?* field to *N* in SD-B, *Work Orders Defaults*.

If you express it as a percentage of labor, you would compare your total overhead for a month with your total direct labor for the month in order to arrive at the percentage of labor.

If you use the hourly rate method, take total overhead for the month and divide it by the total number of production hours.

DBA lets you organize your overhead into fixed and variable overhead categories. Some companies use both overhead categories, others will lump all their overhead into one and will typically use the fixed overhead category to hold the rates.

Entering overhead rates for estimates

RO-C, *Enter Work Centers*, allows you to set different fixed and variable overhead rates for use by the *Estimating* module.

The reason for separate rates is this. The standard rates for fixed and variable overhead are for accounting purposes. They are set to accurately absorb actual overhead into your WIP inventory.

These may not, however, be the rates you want to use for estimating purposes.

When the factory is not busy, for example, fixed overhead goes up as a percentage of your costs. If you used the accounting overhead rate for your estimates, they would calculate with higher prices, making you less competitive.

You should therefore set the estimating overhead rates at reasonable rates that represent your expected overhead costs under normal conditions.

Using parent-child work centers (finite scheduling only)

If you plan on using SH-E, *Finite Scheduling*, you can set up *parent* and *child* work centers. If you plan on using SH-F, *Infinite Scheduling*, or manual scheduling, you cannot use *parent-child* work centers.

Parent-child work centers work as follows.

It is common for work centers to be comprised of several identical machines or workstations, any one of which can be used interchangeably. Rather than treat them as one work center, each machine or workstation can be set up as its own work center. These will be known as *child* work centers. A *parent* work center will be set up separately and represents all its children.

When setting up routings, you typically assign the sequence to the *parent* work center rather than a *child* work center (you can make an exception where a particular operation has to run on a specific *child* work center). When the finite scheduling program goes to schedule a routing sequence assigned to a *parent* work center, it looks among its *child* work centers and schedules the sequence to the *child* with the lowest contention (waiting time).

If you are using infinite or manual scheduling

If you plan on using infinite or manual scheduling, in the above situation you would set up the group of interchangeable machines as one work center only. For example, if you had five machines each running eight hours per day, you would set up one work center with 40 total hours per day.

Setting up outside processing work centers

Outside processing work centers are set up for any outside processes such as painting, plating, heat treating, etc.

Outside processing default settings

In RO-C, *Enter Work Centers*, you can enter settings for outside processing work centers that will make default entries when creating outside processing routing sequences in RO-A, *Enter Routings*.

The *Lead Time* field sets the default lead-time for outside processing performed in this work center.

If there is a minimum charge for services performed in this outside processing work center, it can be entered in the *Minimum Charge* field.

If this is an outside processing work center that has a standard cost per pound, per inch, or per foot, such as with a plater or heat treater, that cost can be entered in the *Cost/lb/in/ft* field. This will be a default that will be multiplied by the manufactured item's *Weight* or *Foot Factor* field (depending on the *Surface or Wt?* field described below) to create an entry in the *Cost* field in the routing.

The *Surface/Weight* field determines whether the outside processing cost is determined by the weight of the item, using the *Weight* field from the item's inventory record, or the surface area of the item, determined by the item's *Foot Factor* field.

Generally speaking, work centers set up with these defaults are usually set up for one vendor rather than for multiple vendors. In other words, the outside services vendor is the work center.

Entering routings

The Manufacturing Answer - 11/29/01

File View Window Help

Items IN RO BM LC SC FO PI

RO-A Enter Routings

Part W-100 RED WAGON Lot Size 50

Seq	Op	T	Description	Work Center	Time/Pt	Parts/Hr	Setup
10	10	L	FORM	20	00:01:05	55.00	00:00:00
20	20	P	PAINT	40	00:00:00	0.00	00:00:00
30	30	L	ASSEMBLE WAGON	10	00:03:00	20.00	00:00:00
40	40	L	FINAL INSPECTION	50	00:00:18	200.00	00:00:00

Help Exit

RO-A, Enter Routings

Routings are used to define how and where products are manufactured. The manufacturing process is broken down into specific sequences (operations) for in-house direct labor and for outside processing (for services such as painting, plating, heat treating, etc).

To enter routings use RO-A, *Enter Routings*.

The following are some of the capabilities of DBA's routings and some of the different ways they can be used.

Notes and specifications templates help document your products

Up to 15 lines of free form text can be entered against any routing sequence to define how the sequence is to be performed.

Additionally, you can create what are called *Specs*, which are similar to notes except that they have predetermined fields and headings that are set up in advance using RO-K, *Enter Specifications Templates*.

Specifications templates allow you to customize your routings with specifications unique to your industry. For example, you might want to set up fields to hold information such as die settings, temperatures, inspection fields, etc. You can set

up any number of specifications templates you wish.

While in the *Specs* screen on any routing sequence, you can copy in a specifications template by clicking the *Copy Spec Templates* button.

Standard costs are rolled up from routings

Routings allow you to accurately estimate product costs for labor, setup, overhead, and outside processing.

BM-G, *Print/Roll up Standard Costs*, rolls up standard costs up through all your routing sequences to help establish total standard costs for your manufactured items.

In the case of direct labor sequences, the cost for labor and overhead is determined by the routing sequence's time-per-piece production rate multiplied by the work center's standard rates for labor, fixed overhead, and variable overhead.

The cost for setup is determined by the work center setup rate multiplied by the routing sequence's setup time divided by the item's *Lot Size*, which is specified in the header of RO-A, *Enter Routings*. The lot size is your estimate of the typical work order quantity for the item and is used by the cost rollup program to translate setup cost (which is the same for a work order quantity of one as it is for thousands) into a per-unit cost.

The cost for outside processing is the per-unit cost specified in your outside processing routing sequences.

How routings are used with work orders

When a work order is created, the routing for the item is copied to a work order routing, at which point the labor on each sequence is allocated against the corresponding work centers in the factory and against your outside processing work centers.

At this point the work order is now part of the scheduling system and is subject to start, finish, and due date changes as it gets scheduled through the factory.

The nice thing having a separate work order routing from the standard routing is that changes can be made at the work order level for custom manufacturing purposes without affecting the standard routing.

Changes to the work order routing can be made using WO-K-A, *Enter Work Order Routings*.

Routings are used for scheduling

The work centers and production rates specified in your routing sequences are integral elements of the scheduling system.

The work centers tell the scheduling system where and in what sequence the work order operations are to be performed. The production rates specified in each work order sequence tell SH-E, *Finite Scheduling*, how long each sequence will take to

complete.

If you are using manual or infinite scheduling, the production rates specified in each work order sequence create loads on each work center, which can then be compared against available capacity for capacity planning purposes. Work center loads can be viewed via SH-K, *View Work Center Load*, or printed out via SH-L, *Print Work Center Load*.

Copy feature

RO-A, *Enter Routings*, lets you copy all the sequences from one item number to another.

Products often share the same routing sequences with only minor differences between them. It is much faster to use a similar product's routing as a template to copy from and then edit than it is to enter a new routing from scratch.

To copy a routing, bring up the routing of the item you wish to copy from. While on the header screen, click the *Copy Routing* button.

Using outside processing work center default settings

Be aware that you can create defaults in your work centers for the *Cost*, *Minimum Charge*, and *Lead Time* fields that are entered in your outside processing sequences.

If you have an outside processing work center that has a standard cost per pound, per inch, or per foot (such as in the case of a plater or heat treater), the outside processing sequence *Cost* can be calculated by the program as follows.

The standard cost for the service is entered in the *Cost/lb/in/ft* field in RO-C, *Enter Work Centers*. When you enter the outside processing sequence in RO-A, *Enter Routings*, this standard cost will be multiplied by the manufactured item's *Weight* or *Foot Factor* field (in the item master record) to create the *Cost* entry in the routing.

The program knows to use either the item's *Weight* or *Foot Factor* based on the *Surface/Weight* field default in the work center (S for surface, W for weight).

The *Lead Time* work center default sets the lead-time for outside processing performed in this work center.

The *Minimum Charge* work center default sets a minimum charge for services. The *Cost* field in the routing cannot be less than the minimum charge.

Generally speaking, work centers set up with these defaults are usually set up for one vendor rather than for multiple vendors. In other words, the outside services vendor is the work center.

Backflushing work order materials by routing sequence

Even though routings deal with labor and outside processing and not material, there is one exception. DBA does allow backflushing of work order materials by

routing sequence if you assign bill of material components in BM-A, *Enter Bills of Material* to specific routing sequences.

Specifying sequences means that inventory will be backflushed as you complete various routing sequences instead of at time of finished production, which means your inventory will be relieved much closer to when it was physically issued.

Backflushing at time of finished production is good if your production runs are short in nature. If you have relatively long work production runs, however, backflushing by sequence is the preferred alternative.

Another nice feature of specifying a routing sequence at the component level is that you can optionally print the components on the shop traveler within that sequence, thus visually tying the material to the operation in which it is used.

Using multiple routings

RO-A, *Enter Routings*, allows you to assign each routing sequence to a routing number. Even though there is technically only one routing allowed per manufactured item, each routing number prints as a separate routing and thus effectively functions as a separate routing.

This feature can be useful with multi-level products where you want a single work order instead of separate work orders for all the subassemblies. This is accomplished by defining your subassemblies as phantom assemblies (item type B) instead of true subassemblies (item type A).

While only one routing is technically allowed per work order, you can get around this by defining all your sequences, both for the parent product and all its phantom assemblies, on one big routing and assigning each assembly's sequences to its own routing number. When you print the shop traveler, a separate routing will print for each assembly.

For more information on phantom assemblies, see *Using phantom assemblies* in chapter 10.

Using alternate operations

Routing sequences can be created for alternate operations. An alternate operation is one that is not standard, but could be used as an alternative.

For example, you might normally paint an item in-house, but if your paint facility was overbooked or broken down, you could send the part to an outside vendor for painting under the alternate operation.

Alternate operations are bypassed by MR-G, *Print/Rollup Standard Costs*, so that they do not add to the standard cost of a product. They are also bypassed by SH-E, *Finite Scheduling*, and SH-F, *Infinite Scheduling*.

Normally, alternate operations print on shop travelers for reference purposes only. They let factory personnel know that there is an alternate operation available.

If during the course of production it is determined that an alternate operation is to

Entering routings

be used, you can go into WO-K-A, *Enter Work Order Routings*, delete the standard operation, then change the operation type on the alternate from type A (alternate) to type L (direct labor) or type P (outside processing).

Setting production rates

Standard costs for labor, setup, and overhead are established by the time-per-part or parts-per-hour production rates entered within your routings in RO-A, *Enter Routings*.

You can enter production rates either using the time-per-part method or the parts-per-hour method; either way the program translates it into a time-per-part figure for standard costing purposes.

The time-per-part is multiplied by the work center standard rates for labor and overhead to arrive at the standard cost for labor and overhead.

Reviewing past performance using the *Work Order History* report

JC-H, *Print Work Order History*, is an excellent tool for reviewing the history of each routing sequence of an item across multiple work orders. This lets you use past history to help estimate the time-per-piece or parts-per-hour production rates for each sequence.

Conducting time studies

The alternative to using past history is to actually conduct time studies to determine production rates. You may not want to do this on every product you make, but it can be worthwhile on your more important products.

Entering setup

Setup labor is entered as the total setup time required.

For setup labor you must also specify the typical production run quantity in the routing's *Lot Size* field so that the program can translate setup cost (which is the same fixed cost whether you run one part or thousands) into a per unit cost.

Using operation templates

The screenshot shows the 'RO-I Enter Operation Templates' window. The title bar reads 'The Manufacturing Answer - 11/29/01'. The menu bar includes 'File', 'View', 'Window', and 'Help'. The toolbar has buttons for 'Items', 'IN', 'RO', 'BM', 'LC', 'SC', 'FO', and 'PI'. The 'RO' button is highlighted. The main form area contains the following fields and values:

Field	Value
Template No.	0
Oper Desc	
Work Center	
Vendor Name	
Lead Time	0 Days
Out Proc Cost	0.000000
No Processes	0
Time/Process	00:00:00 or Processes/Hr 0.00
Time/Part	00:00:00 or Parts/Hour 0.00
Setup Hours	0.000000
Use Std Time?	00:00:00
Routing No	0
No Persons	0.00
Machine No	
Tool Number	
Tool Desc	

At the bottom of the window, there are buttons for 'Help', 'Lookup', 'Notes', and 'Exit'.

RO-I, Enter Operation Templates

Operation templates allow you to pre-define your commonly used direct labor and outside processing operations in the form of templates (complete with notes and specifications) that can be copied into routings and estimates, then edited for any changes.

To enter operation templates, use RO-I, *Enter Operation Templates*.

Using the routing generator

Once you have operation templates set up, you can then use the *routings generator*.

Whenever you enter a routing for the first time, you are given the option of using the routing generator.

The routing generator lists your operation templates in a window and lets you click on them one at a time in the sequence that the operations are to be performed. As you click each template you will see its sequence order displayed.

Once you have finished selecting (and de-selecting, if necessary) all your operations, you click the *Save* button and the *Enter Routings* program will generate routing sequences for all the operations selected, in the order selected.

Once the sequences are created, you can then edit them for any needed changes. Using the routings generator can be a tremendous timesavings in setting up new products and entering estimates.

Tracking machines and tools

The screenshot shows a software window titled "The Manufacturing Answer - 11/29/01". Below the title bar is a menu bar with "File", "View", "Window", and "Help". A toolbar contains buttons for "Items", "IN", "RO", "BM", "LC", "SC", "FO", and "PI". The "RO" button is highlighted. Below the toolbar is a sub-window titled "RO-D Enter Machines". This sub-window contains a form with the following fields: "Machine No." (a small text box), "Description" (a larger text box), "Work Center" (a text box), "No. Hours between Service" (a text box with "0" entered), "Date Last Service" (a text box with "00/00/00" entered), "No. Hours Used to Date" (a text box with "0" entered), and "Notes" (a large text area). At the bottom of the sub-window are buttons for "Help", "Lookup", "Delete", "Save", and "Exit".

RO-D, Enter Machines

Machines and tools can be set up with their own numbers and can be specified on routing sequences.

To set up machines, use RO-D, *Enter Machines*. To set up tools, use RO-E, *Enter Tools*.

Preventive maintenance capability

Not only are machines and tools used as references on shop travelers, but they offer a simple preventative maintenance capability.

In the case of machines, the system automatically records the number of hours of production that get reported through WO-F, *Enter Labor*, WO-M, *Batch Labor Entry*, and DC-A, *Enter Labor & Production*.

In the case of tools, the system automatically records the number of parts that get reported through the labor entry programs.

At any time you can go to the *Machines* or *Tools* setup programs and you can see the number of hours or the number of parts that have been produced since the last time maintenance was performed.

Each time you perform maintenance, you can go into the program and reset the number of hours or number of parts back to zero and the count will begin again.

The screenshot shows a software window titled "The Manufacturing Answer - 11/29/01". Below the title bar is a menu bar with "File", "View", "Window", and "Help". A toolbar contains buttons for "Items", "IN", "RO", "BM", "LC", "SC", "FO", and "PI". The "RO-E Enter Tools" window is active, displaying a form with the following fields:

- Tool Number: [text input]
- Description: [text input]
- No Parts between Maint: [text input] 0
- Date Last Maintenance: [text input] 00/00/00
- Parts Produced to Date: [text input] 0
- Notes: [text area]

At the bottom of the window are buttons for "Help", "Lookup", and "Exit".

RO-E, Enter Tools

Machines can be manually scheduled

The machine you specify on a routing sequence is for reference only. You can make this a specific machine or give it a dummy machine number that represents a group of machines. If you make it a dummy number, it is a reminder that when it comes to production time, you must assign the sequence to an actual machine.

Once a work order is in actual production, you can manually assign work order sequences to the machines of your choice. To manually schedule machines, use SH-D, *Manually Schedule Machines*. To get a printout of your current machine schedule, use SH-J, *Print Machine Schedule*.

If you do not manually assign a machine in advance, you can wait and do so after-the-fact when reporting production through WO-F, *Enter Labor*. The program provides a *Machine* field that lets you enter the machine that was actually used.

Using machine trim sizes with the estimate yield calculator

ES-A, *Enter Estimates*, features a *yield calculator* within its bill of material entry, with which you can specify an X dimension and a Y dimension for a part and the

program will determine how much of a sheet or length of material will be required. Each machine can have a default *trim size*, which determines how much material is needed to be “gripped” by the machine. This trim material is not available for use by the part and is therefore an important factor in the yield equation.

Production reports

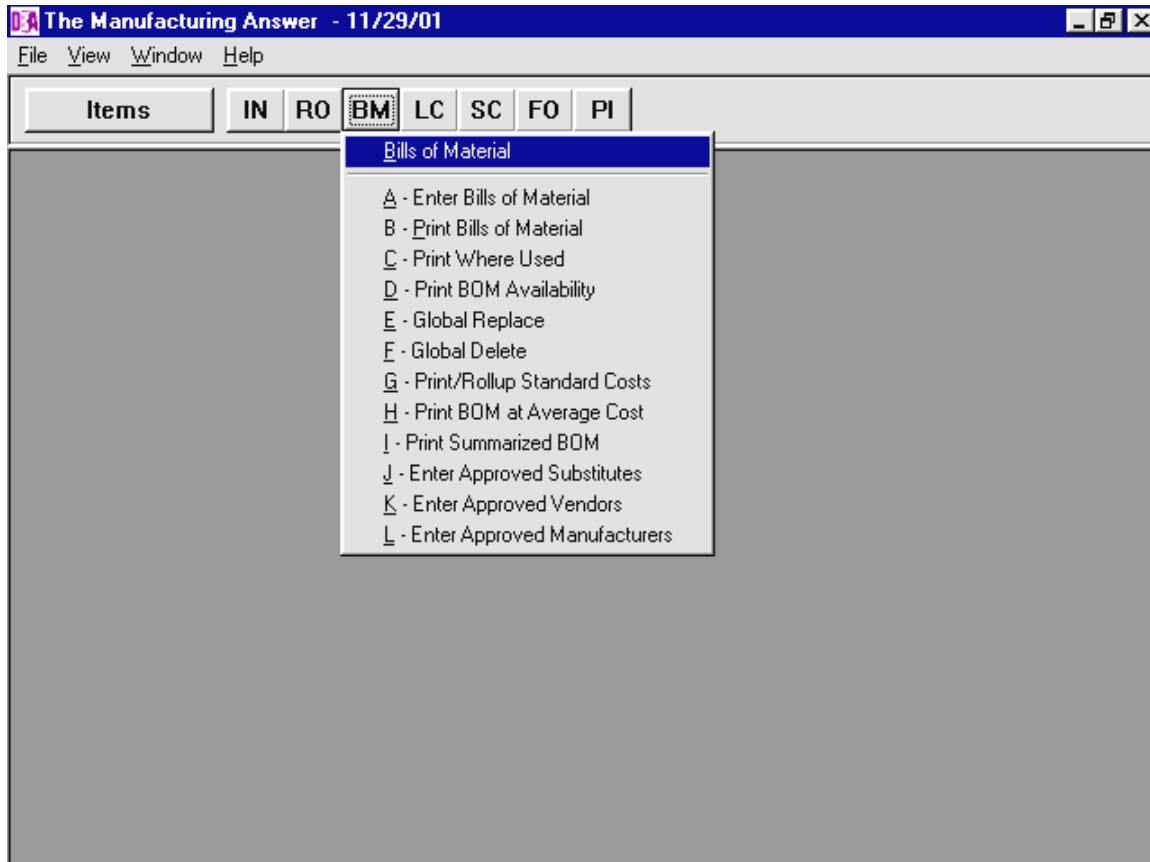
To get a listing of production by machine, use JC-J, *Print Production by Machine*.

To get a listing of production by tool, use JC-K, *Print Production by Tool*.

10. BILLS OF MATERIAL

- **Entering and using bills of material**
- **Rolling up standard costs**
- **Using approved substitutes, vendors, manufacturers**

Entering and using bills of material



A bill of materials is a list of items which comprise a finished product, subassembly, phantom assembly, selling kit, or make-from. Accurate bills of material are essential to a good manufacturing inventory system and serve as the foundation for the entire costing and planning process.

Bills of material are entered through BM-A, *Enter Bills of Material*.

The following are some of the capabilities of DBA's bills of material and ways they can be used.

Remarks and notes help document your products

The bill of material file allows extensive remarks to be maintained on each component and the entry of notes at the parent part level. If you are using the "light manufacturing" method of handling labor, labor (and outside processing) can be given item numbers (types L and T) and included in bills of material. These features allow extensive manufacturing documentation to be built into the bill of material for companies that do not wish to use routings and work centers.

Indented reports can handle up to 35 levels

The indented bill of material reports will handle up to 35 levels in a product

structure. The parent product is considered to be level zero and its components level one. If any of the components in level one are subassemblies, their components are considered level two, and so forth. You do not need to specify levels; the system automatically defines the relationships.

Cost rollup capability

The bill of material is an integral part of the costing system. Standard costs are maintained at the item level for material, setup, labor, outside processing, and overhead. BM-G, *Print/Roll up Standard Costs*, will calculate and add these costs up through all levels of the bill of material and establish a total standard cost for the top-level item.

The standard cost rollup can give you an accurate estimate of what it will cost to manufacture a product, even with complex products comprised of many components and subassemblies.

See the next section for more details on rolling up standard costs.

Copy feature

The copy feature makes it easy to copy entire bills of material or portions of bills of material from one assembly to another.

When you copy components, you can optionally include remarks and notes.

You are not restricted to one copy. You can copy components from multiple bills of material. After one copy procedure is completed, you can perform another. In this manner several bills of material can be merged into one bill of material.

To use the copy features, click the *Copy BOM* button while in the BM-A, *Enter Bills of Material*, component entry screen.

Global replace and delete capability

BM-E, *Global Replace*, allows you to replace a component with another component throughout all bills of material that contain that component.

For example, you may want to replace an aluminum washer with a stainless steel washer. The washer is used in hundreds of assemblies. Rather than having to change each bill of material, use *Global Replace* to automatically replace this component everywhere it is used.

BM-F, *Global Delete*, allows you to delete a component across all bills of material that contain that component.

How bills of material are used with work orders

When a work order is created, the bill of materials for the item is copied into a work order bill of material, at which point the required quantities for each component are allocated in inventory and compared with what's on hand and already on order. From this decisions can be made as to what needs to be purchased, ordered, or rescheduled to meet material requirements.

The nice thing about this feature is that the work order bill of material is separate from the standard bill of material. Therefore, changes can be made to the work order bill of material, if necessary, without affecting the standard bill of material. This is ideal for custom manufacturing where the standard bill of material serves as a master template, but on each work order various features and options are changed according to customer preferences.

There are times, especially on new products, where you don't always know what the final bill of material will be until you manufacture the product for the first time. During manufacturing you can modify the work order bill of material as much as you want. If you want the final work order bill of material to become your new standard bill of material, you can click the *Copy to BOM* button in WO-K-B, *Enter Work Order Bills of Material*, to do so.

Selling kits

Selling kits, used in sales orders and quotations, are defined as bills of material. A selling kit allows you to specify a kit of items with one ordering number and one price. The kit item number itself is never manufactured or stocked. When the order is shipped, inventory is relieved for each of the kit components.

Approved substitutes, vendors, manufacturers

You can define approved substitute parts (BM-J, *Enter Approved Substitutes*), approved vendors (BM-K, *Enter Approved Vendors*), and approved manufacturers (BM-L, *Enter Approved Manufacturers*)—globally for all situations, by specific parent item numbers, or only for specific customers. These can print on bill of material reports, purchase orders, and work order pick lists.

Backflushing by routing sequence

Backflushing, in which component inventory is relieved after-the-fact when recording finished production, can be done at the routing sequence level if you specify a sequence number in the *Seq* field while entering a bill of material component in BM-A, *Enter Bills of Material*.

Specifying sequences means that inventory will be backflushed as you complete various routing sequences instead of at time of finished production, which means your inventory will be relieved much closer to when it was physically issued.

Backflushing at time of finished production is good if your production runs are short in nature. If you have relatively long work production runs, however, backflushing by sequence is the preferred alternative.

Another nice feature of specifying a routing sequence at the component level is that you can optionally print the components on the shop traveler within that sequence, thus visually tying the material to the operation in which it is used.

Using phantom assemblies

Phantom assemblies represent a kit of items that is never manufactured on its own or stocked. The parent part on a phantom assembly is coded with an item type B

(for phantom assembly).

When a work order gets created, the phantom's components automatically are pulled into the work order bill of material and print in a separate section of the shop traveler so that they are identified as a separate assembly from other components.

One valuable use of phantom assemblies is for companies that make multi-level products who want to make those products on a single work order instead of having separate work orders for each subassembly.

To do this, instead of making each subassembly an actual subassembly, you set them up as phantom assemblies and include them in the top-level parent assembly as components. DBA allows phantom assemblies within phantom assemblies, so you can use phantoms in a multi-level structure.

When the work order gets created, all the phantom assembly components get pulled into a single work order bill of material, including phantoms within phantoms. The phantom assemblies still print separately on the shop traveler so that they appear as distinct assemblies, but they are all part of a common single level work order bill of material.

If you use this method of consolidating all your assemblies into a single work order, be aware that this affects your routing as well. Even though you can only have one routing per work order, DBA lets you assign routing sequences to different routing numbers (it's still one routing, but each routing number prints separately).

You could therefore split up your routing sequences on different routing numbers and then print what appears to be separate routings for each phantom assembly.

Another use of phantom assemblies is as templates for common sets of parts used by manufacturers such as custom machine builders. When creating new bills of material, you can bring in entire sets of parts by entering just the phantom item number, instead of entering each component individually.

Make-froms

Instead of tediously having to create a work order just for the purpose of getting outside processing performed on some inventory parts, *make-froms* are used instead.

For example, a part sits on the shelf in a non-plated state. You want to send some of the parts out for plating. When the parts come back they should have a different item number to differentiate between the plated and non-plated parts.

A *make-from* is an item that receives an outside process such that it becomes a different item number than the item without the process.

Make-from items are coded as type *M* items in IN-B, *Enter Inventory*, and are set up with a bill of material in BM-A, *Enter Bills of Material*, consisting of the raw part(s) (in this example the non-plated part) that go into the finished *make-from*.

When a *make-from* is received in purchase order receiving, inventory is increased for the *make-from* item and inventory is deducted for its component(s). The cost of the component(s) and the outside processing added together becomes the cost for the *make-from* item.

For more information on how makes froms are used in purchase order entry and receiving, see *Using make-froms* in chapter 14.

Importing items and bills of material from CAD systems

Some manufacturers generate new item numbers and bills of material within their CAD systems.

If your CAD system can create export files in ASCII text format, you can set up data import routines in DBA's *Data Exchange* module and import these files.

To set up the item import, use DE-B, *Import Inventory*. To set up the bill of material import, use DE-C, *Import Bills of Material*.

Rolling up standard costs

Any manufacturer with a product line or any job shop that makes multiple level products on a repeat basis should use standard costs.

Standard costs are used for the following:

- Standard costs are your best estimate of what your products actually cost to manufacture.
- Standard costs are the best costs to use when entering finished production of partially completed work orders.
- Standard costs give you a basis of comparison with actual costs as a measure of performance.

For a complete discussion on using standard costs, see the *Item costing* section chapter 8.

Standard costs are rolled up through BM-G, *Print/Roll up Standard Costs*.

When you make changes to standard costs for individual items, be aware that this will not affect any assemblies that use that item until you run a cost rollup. You should therefore run the cost rollup on a regular basis and incorporate it into your planning routine.

If you value your inventory at standard cost

The only exception to running cost rollups frequently would be for companies that value their inventory at standard cost. The vast majority of DBA customers value their inventory at average cost, so this would not affect them.

If you value your inventory at standard cost, each time you run a cost rollup you are revaluing your inventory. You therefore need to run the *Inventory Value* report (which shows your inventory at average and standard cost) before the cost rollup and after the cost rollup.

Companies that use a standard cost inventory typically run cost rollups once or twice a year, although it can be done quarterly or even more often, if needed.

Using approved substitutes, vendors, manufacturers

The screenshot shows a software window titled "The Manufacturing Answer - 11/29/01". Inside, there's a menu bar with "File", "View", "Window", and "Help". Below the menu is a toolbar with buttons: "Items", "IN", "RO", "BM", "LC", "SC", "FO", and "PI". The main window area is titled "BM-J Enter Approved Substitutes". It contains three input fields: "Std Part Number", "Parent Part No", and "Customer Code". Below these is a "Line #" field with the value "0". There is a table with two columns: "Substitute Part" and "Description". At the bottom of the window are three buttons: "Help", "Lookup", and "Exit".

BM-J, Enter Approved Substitutes

Some manufacturers are required to use components in the manufacturing process only from certain vendors or manufacturers, or they may be allowed to use substitute components when standard components are not available.

The use of approved substitutes, vendors, and manufacturers is common in the electrical assembly industry and others.

Approved substitute parts

Use BM-J, *Enter Substitute Parts*, to enter approved substitute parts.

A substitute part is another inventory item that can be used in place of a particular component should there be no stock available for that component.

You can define a substitute part globally, meaning that it can be substituted within any bill of material, by parent item number, meaning that it can only be substituted within a particular assembly, or by customer, meaning that it can only be substituted within a particular customer's products.

These substitute parts can optionally print within BM-B, *Print Bills of Material*,

BM-I, *Print Summarized BOM*, and the work order pick list (WO-D, *Print Pick Lists*).

BM-K, Enter Approved Vendors

If during the course of production substitute parts are used in place of standard components, they can be swapped into the work order bill of material through WO-K-E, *Swap Substitute Parts*.

Approved vendors

To enter approved vendors, use BM-K, *Enter Approved Vendors*.

You can designate approved vendors globally, meaning that the vendors are applicable within any bill of material, by parent item number, meaning that the vendors are designated only within a particular assembly, or by customer, meaning that the vendors are only approved for a particular customer's products.

You can also enter the vendor's item number as a cross-reference to your item number, if applicable. When entering a purchase order for the vendor, the vendor's item number will come into the purchase order as a comment line following the entry of your item number. Purchase Order Defaults can be set to allow, warn, or prohibit the use of an unapproved vendor when entering a purchase order.

Approved vendors can optionally print within BM-B, *Print Bills of Material*, and BM-I, *Print Summarized BOM*.

Approved manufacturers

To enter approved manufacturers, use BM-L, *Enter Approved Manufacturers*.

You can designate approved manufacturers globally, meaning that the manufacturers are applicable within any bill of material, by parent item number, meaning that the manufacturers are designated only within a particular assembly, or by customer, meaning that the manufacturers are only approved for a particular customer's products.

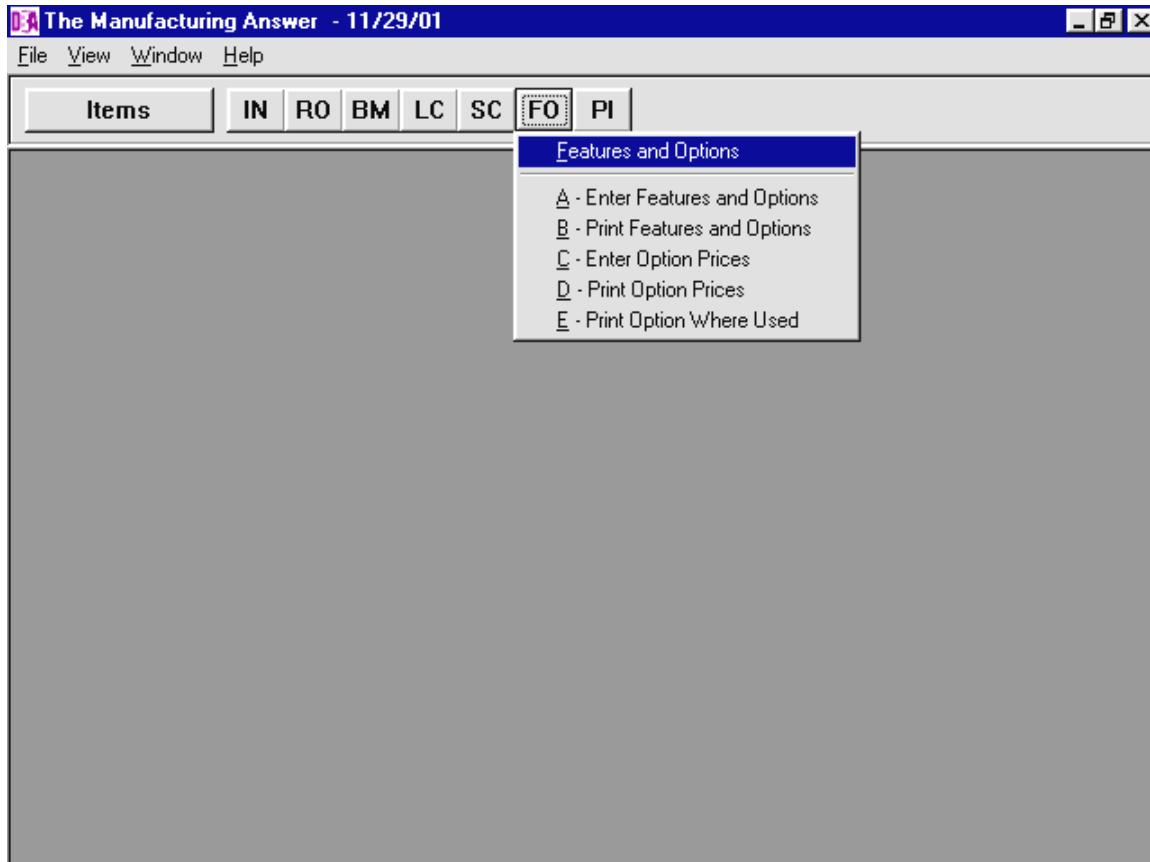
You can also enter the manufacturer's item number as a cross-reference to your item number, if applicable. When entering a purchase order, the manufacturer's item number will come into the purchase order as a comment line following the entry of your item number.

Approved manufacturers can optionally print within BM-B, *Print Bills of Material*, and BM-I, *Print Summarized BOM*.

11. FEATURES & OPTIONS

- **Features & Options overview**
- **Setting up Features & Options**
- **Using features and options in sales orders**

Features & Options overview



The *Features & Options* module provides a means of selecting a series of options from within a product through the sales order entry or sales quotations programs. Once a series of options has been selected, the options may be passed over to the work order system and will produce a custom bill of material for the manufacture of the item. The option system may also be used to configure selling kits within sales order entry.

This options capability is ideal for products that have more options than an item numbering scheme can accommodate. An office desk, for example, could be available with different types of wood, different sized desk tops, rounded or square edges, left or right return, two or three drawer pedestals, half or full height modesty panels, etc. It can be very difficult, if not impossible, to devise an item numbering scheme that can accommodate every mathematical combination of options.

The *Features & Options* module allows you to define an inventory type "O" item (which we refer to as a *Feature*) in the product's bill of material wherever there are multiple options for a particular feature. For example, if a desk is available in six different colors, you could create an inventory type O item in the inventory file with the product code "COLOR." You could then create a bill of material for

the COLOR feature which would contain all the actual color options available. In the bill of material for the desk you would enter the feature COLOR instead of indicating specific colors.

When entering a sales order for the desk, the system will automatically display all the color options in a window. You may then select a specific color, which will feed into the sales order as a line item following the desk. You may define any number of these features within a product, and you can have features within features and within options. Features can be defined in the bill of material as mandatory, meaning that an option must be selected, or non-required, meaning that the feature may be skipped.

You can attach prices to options directly within the bill of material, or you can specify that the customer's standard pricing (price code, contract, or base price) be used. Pricing can also be set up to be a percentage charge against the price of the parent product. You can specify whether the price is to be separately itemized or if it is to be added to the price of the parent product.

You are not limited to one choice per feature. For example, if 10 desks are ordered, you may select five desks with a black color option and five desks with a white color option.

Once the options are selected and the sales order is saved, you may run SO-N, *Convert Sales Orders to Work Orders*. A work order will automatically be created for the parent product in the *Work Orders* modules. All manufacturing type options that were selected in the sales order, including any comments or changed descriptions, will be passed over to the work order bill of materials, thus creating a custom bill of materials. At that point inventory is immediately allocated for all materials in the work order bill of material.

DBA's *Features & Options* module is ideal for custom manufacturers. Not only does it solve the item numbering and pricing dilemma, but it reduces ordering mistakes, sales order and work order processing time, facilitates materials planning, and increases turnaround time in the shop.

Setting up Features & Options

Step 1 - Set up features in *Inventory*

An *O* type inventory item is an item number that represents a group or set of options. We refer to this as a *Feature*. A feature is not a tangible item and is never stocked or ordered but represents the “node” at which a choice will be made. A bill of material is created for the feature in which the components themselves represent the actual options. We will refer to this as an *Option Bill*. As an example, an *O* type item could be setup for a feature using the product code “COLOR.” An option bill could then be entered for COLOR in which all the actual color options are entered as components. Create an *O* type item for all features that will be needed for your product line.

Step 2 - Create option bills of material

After *O* type inventory items are created for all the *Features* needed, you may create *Option Bills of Material* through FO-A, *Enter Features and Options* or BM-A, *Enter Bills of Material*. Enter the *O* type feature as the *Parent Product* and the actual options as components.

Step 3 - Add features to parent bills of material

Wherever a finished good or subassembly parent product has a set of options within its bill of material, add a *Feature* (*O* type item) as a bill of material component instead of the specific options. During sales order entry the options within that feature will be presented in a pop-up window for selection.

Step 4 - Review FO-B, *Print Features and Options*

You can print out the features and options for any parent product through FO-B, *Print Features and Options*. If you answer yes to *Print the Subassemblies?* you will see all levels of the product structure. This report is limited to features and options only and does not include the standard components.

Step 5 - Test options in SO-A, *Enter Sales Orders*

The best way to verify if your features and options are setup properly is to enter a sales order in SO-A, *Enter Sales Orders*. You can enter an order for the item or items you wish to test. You can press <Esc> and not save the order, thus allowing you to test as many products as you wish without creating live sales orders. You can also use SO-P-A, *Enter Sales Quotations* (which has identical functionality as SO-A), for testing purposes.

Features & Options defaults

Defaults are established in SD-L, *Features and Options Defaults*. These defaults are standard settings for various prompts used during features and options entry and allow you to tailor the system to your needs. All default settings can be overridden during actual features and options entry.

Using features and options in sales orders

The screenshot shows the 'SO-A Enter Sales Orders - Add New SO' window. The 'SO' tab is active. The 'Features of: DESK, OFFICE [Level 1]' dialog box is open, displaying a list of features with checkboxes. The features listed are: 1100 (checked) DESKTOP, 1200 (checked) COLOR, 1300 (checked) MODESTY PANEL, 1400 (checked) RETURN, and 1500 (checked) DRAWERS. To the right of the features list, there are fields for 'Line #', 'Tax?', 'Release?', 'Est Ship', and a value of '0.00'. At the bottom of the window, there are 'Select/Unselect' and 'Save' buttons.

Features Selection Window

Upon completing entry of a line item for which the product's *Has Options?* field is set to *Y* in FO-A, *Enter Features and Options*, a window will appear which lists all the first level *Features* (inventory Type O's) defined in the product's bill of materials. Mandatory features will be identified with a check mark. You can move the highlight bar to any non-mandatory feature and click on it (or press <Enter>) to mark it for selection. You can click (or press <Enter>) on any check-marked feature to remove the check mark. When you are satisfied that all features you want are checked, click on the *Save* button (or press F10). The program will then begin presenting options for each feature selected.

The *Options* available within the first feature selected are displayed in a window. Highlight the option desired and click on it (or press <Enter>). The option will come into the order as a line item. The description for the item is indented one character and contains a dot (period) in the first position. This serves to give the options an indented structure when printed out on sales order documents. You can change the description if you wish. (NOTE: if you wish to suppress the use of these dotted indentations, you can set the *Suppress Headers/Footers/Indents?* field in SD-L, *Features and Options Defaults*, to *Y*).

You can accept the *Ship Qty* or change it. If you override the quantity with a

smaller amount, the program will return you to this option window once all sub-options (if any) in the option structure are completed for the first option/quantity entered. Only when the original expected quantity is equaled or exceeded will the program move on to the next feature's options.

If the option has a price, it will either be entered on the line or it will be added to the price of the parent product, depending on how each option's pricing was defined in FO-A, *Enter Features and Options*. If the price is to be added to the parent, you will not see the parent price change until the sales order is saved and redisplayed.

Options that are to be passed over to work order bills of material are marked with an *M* in the *Lin* field. Options that are to remain as sales order line items only are marked with a *K*.

When the sales order is saved, inventory is designated as "On Sales Order" for *K* type items. Inventory is not allocated for *M* type items, for they will be passed over to the work order bill of material, at which point inventory will be allocated for all the work order bill of material components.

When all level one features for the sales order line item (parent product) have been fully selected through all levels, entry is completed and the line item entry area will be cleared and ready for entry of a new line item.

Skipping options

If you are in the middle of entering a series of options for an item and you wish to skip over a particular option, click on the *Exit* button (or press <Esc>). You will be asked *Exit without choosing any of these options?* If you click *Yes* or press <Enter>, you are then asked *Cancel the rest of the feature/option selection process?* Click *No* or press <Enter> and you can move on to the next choice.

Canceling the features/options process

If at any point during entry of a series of features and options you wish to cancel the process (rather than finish all the remaining features and options), click on the *Exit* button (or press <Esc>) while in the features or options selection window. First you will be asked if you want to exit without choosing any of the features or options. Click *Yes* or press <Enter>. Then you will be asked *Cancel the rest of the feature/option selection process?* Click *Yes* (or press *Y*) and you will be returned to the line item entry area, which is now cleared for entry of a new line item. The parent product and options entered up to this point remain in the sales order, however, and will have to be deleted manually.

The Manufacturing Answer - 5/15/2002

File View Window Help

Sales SO SA CS CM

SO-A Enter Sales Orders - Add New SO

SO No. 6031 SO Date 05/15/02 Ship to ADU-01-X Last Invc# 0
 Customer Name ADU-01 Entered by ADVANCED AUTOMATION

Options for: COLOR [Level 1]

5.00	1205	GREY COLOR	[R]
5.00	1210	BLUE COLOR	[R]
5.00	1215	WHITE COLOR	[R]
5.00	1220	BLACK COLOR	[R]
5.00	1225	BROWN COLOR	[R]

Li Sub

Select

Options Selection Window

Features within features

Features can be set up within features, allowing you to set up a chain of choices. For example, the first feature selected could be “fabric”. After fabric is chosen the next feature choices could be “patterns” or “solids.” Only after one of those features is selected would actual fabric options be displayed for selection. Each feature within a feature is considered a “level.” The order entry program can accommodate up to nine levels of features or options.

Features within options

A selected option can have its own bill of material which contains *Features* (inventory Type O’s). For example, you could select a particular desktop size, but then you need to know if it is to have a rounded or square edge. Within the desktop’s bill of material you would add a feature for edge detail. In order entry the desktop size option, once selected, will have its *Features* displayed in a window as if a new parent product is being entered, thus allowing a second level of options. In the same manner, any of the second level options can also contain features, thus providing a third level of options. Up to nine levels of sub-options can be processed within the sales order.

2nd description/specifications/customer cross-reference/comments

If an option has a second line of description, inventory specifications, or customer-cross reference item numbers or descriptions, these will automatically come into the sale order following the option. They will be indented (without being preceded by dots) the same number of characters as the option.

Once line item entry is completed for an option structure, you can edit the option lines or comment lines (or insert comment lines) or type over the standard *Description* with a customized description.

Converting sales orders to work orders

When sales orders are converted to work orders through SO-N, *Convert Sales Orders to Work Orders*, the program will automatically pass all options coded *M* in the sales order *Lin* column over to the newly created work order's bill of materials.

If an option is a phantom assembly (inventory type *B*), work order bill of material records are created for each component in the phantom's bill of material, thus pulling in a set of components by means of one item number.

When a shop traveler is printed for the work order, option components passed over from sales orders are listed in a separate section of the bill of materials entitled "OPTIONS SELECTED." Any phantom assemblies are labeled and listed separately. Any standard bill of material remarks attached to any component or option.

The work order bill of material is a marriage of the standard components found in the manufactured item's bill of material, plus the options and phantom components selected during sales order entry.

Once a work order bill of materials is created, inventory is then allocated (for production planning purposes) for each of its components.

12. PHYSICAL INVENTORY



The *Physical Inventory* module provides a complete physical inventory system, which provides an organized means for taking a physical inventory, increasing accuracy and minimizing disruption to operations.

Sequence of events

The physical inventory process is as follows.

Inventory will be "frozen" as of a particular date via PI-A, *Capture Frozen Inventory*. "Freezing" the inventory means that a snapshot of the current on-hand quantity and costs is taken and stored in the Physical Inventory module files for comparison with actual counts. You can freeze all or portions of the inventory so that complete or partial or cycle counting can be accommodated. The inventory counts and costs held in the DBA's data files are captured in a separate file where they will ultimately be compared with the new physical counts. Operations will cease while the inventory is counted.

The actual counts can be recorded on prenumbered tags or on the frozen inventory report. Typically tags are mounted on all the actual physical items, shelves, pallets, bins, etc. As items are counted, the item numbers and counts are entered

on the tags. When finished, all the tags are collected.

The tag numbers, item numbers, and counts are recorded in PI-C, *Enter Tag Counts*. The *Enter Tag Counts* program fully handles items set for lot control and/or serial control. A missing tag report can be printed to make sure that all tags are accounted for.

If you wish to change any costs before updating actual inventory, you can do so through PI-E, *Edit Frozen Inventory Costs*.

After all the counts and costs are entered and fully edited, actual inventory is updated with the physical inventory through PI-G, *Update Actual Inventory*].

Operations can resume as soon as tags are collected

A unique feature of the DBA physical inventory system is that operations can resume as soon as the tags are collected, even if it takes several days to enter the counts and complete the editing process. When the actual inventory gets updated, the program nets out any transactions that have occurred since the physical count was taken. You can therefore take as much time as needed for accurate processing of the physical inventory without the pressure of having to complete it as a requirement for resuming operations.

***Adjust Physical Levels* program provides a simple alternative**

If you simply want to adjust the counts on a selected range of items, IN-K, *Adjust Physical Levels*, provides a simple alternative to the formal physical inventory system described above.

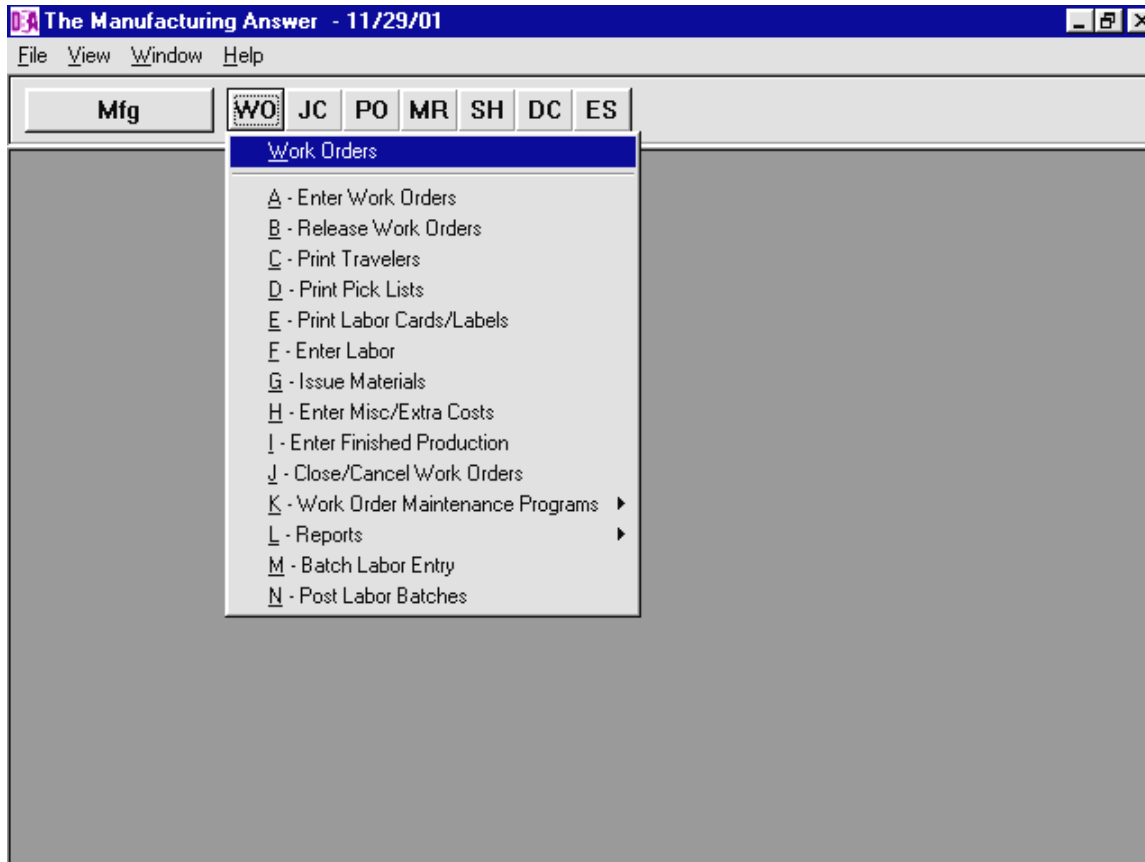
The program allows you to enter a new on-hand quantity (by location, if you have multiple locations) that overrides the existing on-hand quantity.

You can print a list of items to be counted beforehand using IN-J, *Print Physical Check*.

13. WORK ORDERS

- **Work Orders overview**
- **How work orders get created**
- **Creating multi-assembly work orders**
- **Creating multi-date work orders**
- **Work Orders flow chart**
- **Releasing work orders**
- **Printing shop paper**
- **Issuing material from stock**
- **Purchasing material for work orders**
- **Purchasing outside processing for work orders**
- **Reporting labor to work orders**
- **Using data collection to report labor**
- **The "light manufacturing" method of handling labor**
- **Entering miscellaneous and extra costs**
- **Recording finished production**
- **Closing work orders**
- **Job costing reports**

Work Orders overview



Work orders are the foundation of the manufacturing system. A work order tells the system that you intend to manufacture a specific quantity of some item at a particular time.

When you create the work order, any material or labor needed for that work order is allocated by the system against available inventory and labor resources.

During the course of production materials can be purchased or issued to the work order, outside processing (such as painting, heat treating, etc.) can be purchased for the work order, and setup and direct labor can be reported to the work order.

When items are finished, all these costs are rolled up into a total cost for the items and then those items can be transferred to inventory. Once in inventory the items can be sold (invoiced) or can be issued to other work orders (such as in the case of subassemblies).

How work orders get created

The Manufacturing Answer - 11/29/01

File View Window Help

Mfg WO JC PO MR SH DC ES

AW-A Enter Work Orders

Work Order	0-0	Status		Cust	
Location		Priority		Name	
Part#		Class		Attn	
Desc				PO #	
Quantity to Make		0.00		Job #	
Quantity Completed		0.00		Quote #	0
Start Date	00/00/00	Actual	00/00/00	SO #	0
Finish Dt	00/00/00	Actual	00/00/00	Price	0.0000 UM 0
Due Date	00/00/00			Change Order Number	

	Estimate	Actual	Variance	%
Setup	0.00	0.00	0.00	0.00
Material	0.00	0.00	0.00	0.00
Outside Proc	0.00	0.00	0.00	0.00
Labor	0.00	0.00	0.00	0.00
Fixed Overhead	0.00	0.00	0.00	0.00
Var Overhead	0.00	0.00	0.00	0.00
Misc	0.00	0.00	0.00	0.00
Extra	0.00	0.00	0.00	0.00
TOTAL COST	0.00	0.00	0.00	0.00

Help Lookup Clear Exit

WO-A, Enter Work Orders

Work orders are created four different ways.

- If you are a make-to-order manufacturer, such a job shop repeating a previous order or a product line manufacturer who builds to order, you can enter items on sales orders and convert them to work orders through SO-N, *Convert Sales Orders to Work Orders*.
- If you are using the *Materials Requirements* module to plan your requirements, MR-F, *Generate Material Requirements*, will suggest which items need work orders, which can then be automatically generated through MR-I, *Generate Work Orders*.
- If you want to manually enter work orders, you can do so directly using WO-A, *Enter Work Orders*.
- If you have quoted your customer using the *Estimating* module, you can convert your estimate to a work order and a customer order through ES-E, *Convert Estimates*.

Creating multi-assembly work orders

The Manufacturing Answer - 11/29/01

File View Window Help

Mfg WO JC PO MR SH DC ES

AW-K-D Create Multi-Assy Work Orders

From Work Order 0-0
Thru Work Order 0-0
Offset Lead Times [Y/N] N
Use Shop Calender for Start Date? N
Create Work Orders only for Products
below reorder level [Y/N] Y

Help Lookup Process Exit

WO-K-D, Create Multi-Assy Work Orders

There are make-to-order products that consist of multiple sub-assemblies, each with its own bill of material and routing, that need to be made on separate work orders. Machine builders, for example, manufacture in this manner.

Rather than have to manually enter a work order for each subassembly, DBA gives you the ability to enter just one work order for the parent part, then run WO-K-D, *Create Multi-Assy Work Orders*, to automatically create work orders for all the lower level assemblies.

The program can be restricted to creating work orders for a single level or it can create work orders down through all levels of the product structure.

Lead times can be offset so that subassemblies are scheduled for completion prior to parent work orders being started, and work order creation can be restricted to subassemblies that lack on-hand inventory.

Work order numbering

Each work order that gets created will share the same work order prefix number as the original work order number, but will have a different suffix number beginning with -002, -003, -004 etc.

Creating multi-date work orders

There are products for which you might wish to schedule production runs well into the future, perhaps to fulfill a contract with multiple delivery dates.

Rather than have to manually enter a work order for each date, DBA gives you the ability to enter just one work order, then run WO-K-C, *Create Multi-Date Work Orders*, to automatically create work orders for all dates.

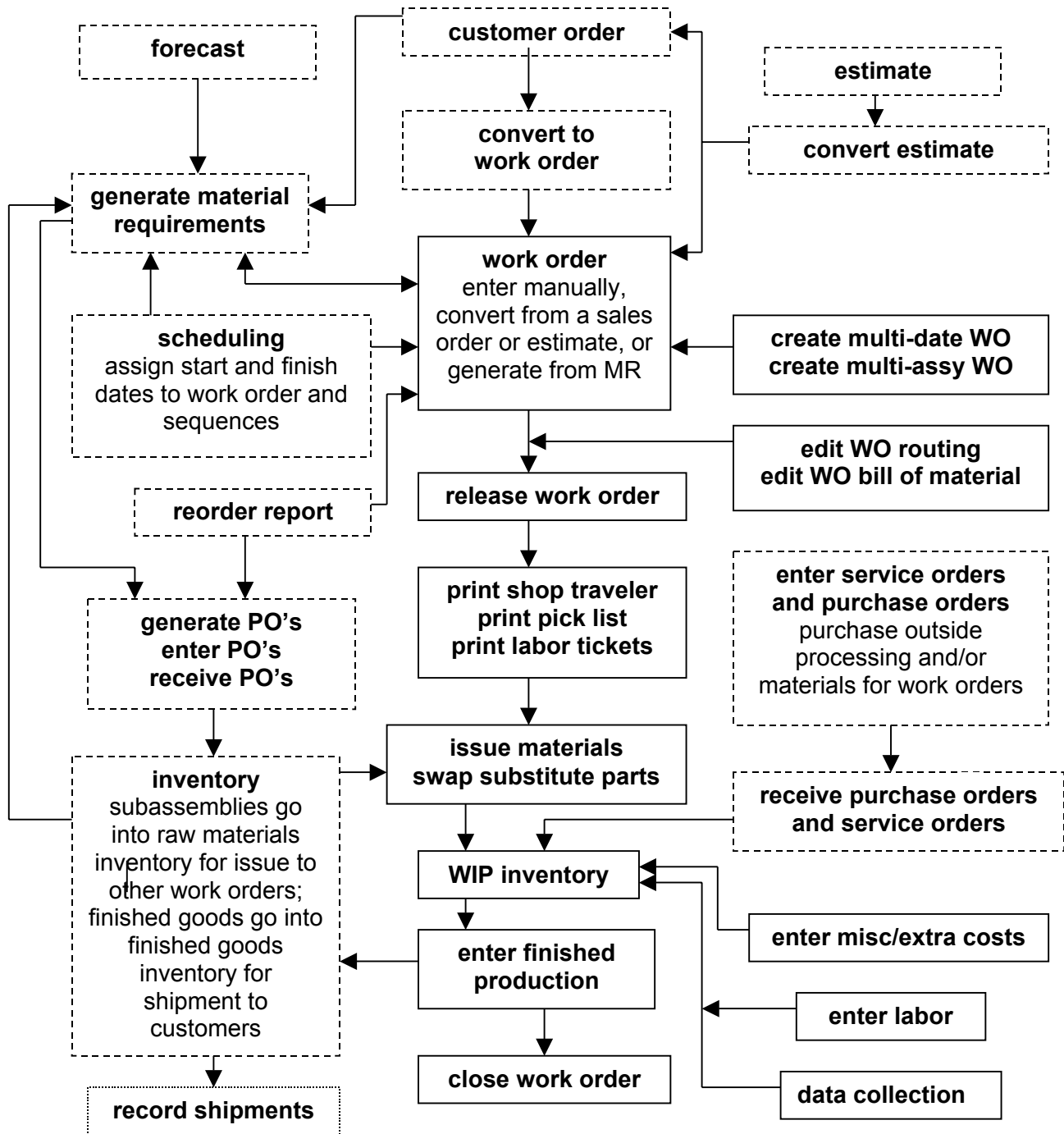
When you manually create a work order using WO-A, *Enter Work Orders*, you can enter multiple due dates. These due dates are for reference purposes only and print on the shop traveler in the work order schedule section of the header.

When you run WO-K-C, *Create-Multi-Date Work Orders*, a separate work order will be created for each delivery date and quantity.

Work order numbering

Each work order that gets created will share the same work order prefix number as the original work order number, but will have a different suffix number beginning with -002, -003, -004 etc.

Work Orders flow chart



Releasing work orders

DBA offers a work order release feature which allows you to distinguish between work orders that are scheduled versus those that are live and in production.

This release feature enables you to freely create work orders, even if they are scheduled to begin far into the future, yet keep them separate from work orders that are actually in production.

Work orders are released through WO-B, *Release Work Orders*.

Setting up work orders for automatic release

If you want to skip this step and have all your work orders automatically released upon creation, you can specify so in SD-B, *Work Orders Defaults*. This would be appropriate for situations where the majority of your work orders are all completed in a short time frame.

When you release a work order, its status gets changed from *F* (firmed) to *R* (released), and an *Actual Start Date* gets assigned to it.

Shortage report

The program also optionally produces a shortage report that checks the work order bill of material and lets you know if any components lack sufficient inventory to cover this order and other outstanding work order allocations for that component.

Printing shop paper

"Shop paper" is a term used to describe various documents and forms that accompany the work order as it moves around on the shop floor. Often shop paper is assembled into a plastic jacket, sometimes referred to as a "shop packet."

Shop paper can include the following items.

Shop traveler

The shop traveler, printed via WO-C, *Print Shop Traveler*, documents how the item is to be manufactured and consists of the following elements.

HEADER: The header section lists the work order number, the item being produced, the start and finish dates, the customer, sales order, purchase order, and other related order header information.

SPECIFICATIONS: If specifications are maintained on this item in the inventory file, they will print in this section.

INSTRUCTIONS: Any special instructions for this order are printed in this section.

WORK ORDER SCHEDULE: If multiple start/finish dates and quantities were specified for this order, they will be printed in this section.

JOB SCHEDULE: If other work orders share the same work order prefix number, their work order numbers, item numbers, start/finish dates, and quantities will print in this section for cross-reference purposes.

ROUTING: All sequences (operations) and related fields print in this section. If components were specified for specific sequences, they will print within those sequences as well as in the bill of materials section at the end of the traveler. The print formats optionally include fields for quality control sign-offs.

BILL OF MATERIAL: All components and total quantities required print in this section. If any components are part of a second level "phantom" assembly, they will print separately and will be labeled as phantom components.

OPTIONS SELECTED: If the work order was generated from a sales order with options in it (see the *Features & Options* module), the options selected for the item will print in this section.

Pick list

The pick list, printed via WO-D, *Print Pick List*, provides a listing of the components that must be pulled from stock for a work order or range of work orders.

If a component is repeated in several places within the work order bill of material, it will be consolidated into a single quantity for stock picking purposes.

The pick list prints the quantities remaining to be issued, not the original quantity required. Therefore you can print a pick list midway through a work order and it

will be restricted to components that have not yet been issued.

A range of work orders by work order number or by job number can be consolidated into a single pick list.

The pick list shows the component item number, description, quantity, unit of measure, bin location, and offers an entry column for recording quantities, lot numbers, serial numbers, etc.

Labor cards or labels

Labor cards or tickets are used as documents to help capture daily labor for reporting through WO-F, *Enter Labor*. You can print either 3 x 5 labor cards or 1" x 3-1/2" adhesive labels for each routing sequence.

Each routing sequence is printed on a separate card or label. The worker is given a card or label upon starting an operation. The start and stop times are either written in or punched on a time clock and the number of pieces produced are recorded.

Cards and labels are gathered each day and are entered through the *Enter Labor* program, either at the end of the day or first thing the next morning. Through this process work order status and job costing are updated in a timely fashion.

DBA labor cards are designed to print on 3-1/2" x 6" labels (Avery SKU #4167).

Drawing

Often a drawing of the item being manufactured is included with the shop paper. DBA references the drawing number in the header of the shop traveler, but the drawing itself is not stored in the system and must be furnished by you.

Issuing material from stock

The screenshot shows a software window titled "The Manufacturing Answer - 11/29/01". Below the title bar is a menu bar with "File", "View", "Window", and "Help". A toolbar contains buttons for "Mfg", "WO", "JC", "PO", "MR", "SH", "DC", and "ES". The main window area is titled "AW-G Issue Material". It contains a form with the following fields and values:

Field	Value
Work Order	3-0
Issue Date	00/00/00
Parent Part	
Description	
Reference	
Kit Issue?	
Item Number	
Description	
Serial Number	
Lot Number	
Qty Issued	0.0000
Qty Scrapped	0.00
Scrap Code	
Description	
Actual Cost	0.00

At the bottom of the window are buttons for "Help", "Lookup", "Dsp Material Trans", and "Exit".

WO-G, Issue Materials

Material can be issued to work orders from stock (inventory). You can issue material from stock three different ways: component issues, kit issues, or via backflushing.

Component issues

With component issues you issue one component at a time, usually based on pick lists or material tickets that come in from the shop floor as components are physically pulled from the shelf. Component issues are generally used when small numbers of components are involved that are picked at different times during the work order process. This method provides accurate inventory control but can be time consuming for long Bills of Material. To issue components, use WO-G, Issue Materials

Kit issues

With a kit issue you can issue entire sets of the work order bill of material in one transaction. Kit issues are ideal for large numbers of components that are picked all at once and allow you to remove them from inventory at the time they are pulled.

Kit issues are made via WO-G, *Issue Materials*. However, be aware that a kit issue will pull every component on the Bill of Materials regardless of inventory stock levels, thus allowing inventory on-hand to go negative.

Backflushing

With backflushing you issue complete sets of the work order bill of material after-the-fact when you report finished production. Backflushing involves the fewest number of steps and is appropriate where overall production time is relatively short and it is ok to deduct components from on-hand inventory after-the-fact.

Backflushing is performed in WO-I, *Enter Finished Production*. However, be aware that a backflush will pull every component on the Bill of Materials regardless of inventory stock levels, thus allowing inventory on-hand to go negative.

Backflushing by sequence

Backflushing by sequence is where components tied to a particular sequence are automatically relieved from on-hand inventory when completed parts are reported for that sequence in WO-F, *Enter Labor* or WO-M *Batch Labor Entry* or via *Data Collection*. This allows backflushing to take place incrementally during the course of the work order instead of after-the-fact when finished production gets reported.

To use this feature, you must assign components to sequences in BM-A, *Enter Bills of Material*. Be aware that a backflush will pull every component on the Bill of Materials regardless of inventory stock levels, thus allowing inventory on-hand to go negative.

When a component is assigned to a sequence, it will optionally print on the shop traveler (long form only) within that particular sequence.

Purchasing material for work orders

DBA lets you purchase materials directly for work orders, in which case inventory is bypassed and the materials are received to the work order.

This is accomplished by specifying a work order number on the purchase order line item. Items with work order numbers specified are received to work orders; items without work order numbers specified are received into inventory.

Being able to specify the work order number at the line item level (as opposed to the header level) allows you to purchase for multiple work orders on the same purchase order.

To purchase items for work orders, use PO-A, *Enter Purchase Orders*.

Purchasing outside processing for work orders

Outside processing (services such as heat treating, painting, etc.) is ordered via a special type of purchase order known as a *service order*.

With service orders you do not use item numbers for outside processing. Instead, you use the item number of the item being processed as the purchase order line item. When entering a service order you are required to specify the work order number and the appropriate outside processing sequence (operation) number from the work order routing.

When the processed items are received, they are charged directly to the work order and are recorded as completed.

To enter service orders, use PO-A, *Enter Purchase Orders*.

Reporting labor to work orders

The screenshot shows the 'AW-F Enter Labor' window. The title bar reads 'The Manufacturing Answer - 11/29/01'. The menu bar includes 'File', 'View', 'Window', and 'Help'. Below the menu bar is a toolbar with buttons: 'Mfg', 'WO', 'JC', 'PO', 'MR', 'SH', 'DC', and 'ES'. The main window area is divided into several sections. On the left, there are input fields for 'Work Date' (11/29/01), 'Employee' (0), 'Work Order' (0-0), 'Sequence' (0), and 'Machine'. On the right, there are input fields for 'N Setup?' (0.00), 'SETUP HOURS' (0.00), 'N Rework?' (0.00), 'QC Code/Desc', '0.00 RUN HOURS', '0.00 QTY COMPLETED', '1 No. in Team', '0.00 QTY SCRAPPED', 'Scrap Code/Desc', and a checkbox for 'Is this sequence now complete?'. At the bottom left, there is a summary box with the following text: 'R Reg/Over/Db1/Sick/Hol/Uac', '1 Shift', and '1 No. Jobs Worked'. At the bottom right, there are buttons for 'Help', 'Lookup', 'Clear', and 'Exit'.

WO-F, Enter Labor

Labor (setup and direct labor) is collected via time sheets or labor tickets and is reported to work orders via WO-F, *Enter Labor*, WO-M *Batch Labor Entry*, or if you are using the *Data Collection* module, it can be entered directly by production employees on the factory floor through DC-A, *Enter Labor & Production*.

When entering labor you must specify the work order number, employee, sequence (operation), setup time or production time, and the quantity completed and/or scrapped.

To enter labor, use WO-F, *Enter Labor*, or WO-M, *Batch Labor Entry*.

Labor rates can be work center rates or actual wage rates

With DBA you have a choice of using work center standard rates or actual wage rates (as maintained in the employee file).

As a general rule, using work center standard rates will provide more even and useful cost comparisons among work orders because it eliminates some work orders getting charged more than others for labor simply because they were assigned to higher paid employees.

There are, however, situations where exact job costing is required, in which case

you would use actual wage rates.

To use actual wage rates, set the *Use Actual Costs in Labor Entry?* field in SD-B, *Work Orders Defaults*, to *Y*. To use work center standard rates, set this field to *N*.

The Manufacturing Answer - 11/29/01

File View Window Help

Mfg WO JC PO MR SH DC ES

AW-M Batch Labor Entry

Date	11/29/01
Employee #	0
Work order #	0 - 0
Sequence #	0
Run/Setup Hrs	R
Start Time	00:00:00
Finish Time	00:00:00
Parts	0.00
Scrapped	0.00
Number of Jobs	1
Backflush by Seq	N
Run Hrs	0.00
Setup Hrs	0.00

Help Save Exit

WO-M, Batch Labor Entry

Using data collection to report labor

The *Data Collection* module allows production employees to report their labor directly to work orders and eliminates the need for after-the-fact labor entry from time cards or labor tickets and labels.

Employees can enter their labor via keypad entry or bar code scanning. Extra computers are located on the shop floor, and employees use the computer as a time clock for clocking in and out of work order routing sequences.

Unlike most other data collection systems, which use separate networks of dumb terminals with batch updating of the manufacturing system, DBA's *Data Collection* system interacts directly with the database and is part of the same network. The computers on the shop floor are simply extra nodes on the network.

For more details see *Data Collection*, chapter 17.

The "light manufacturing" method of handling labor

Manufacturers with simple product lines who don't want to track labor can use what is sometimes referred to as "light manufacturing" as a simple means of accounting for the cost of labor without having to use routings.

In light manufacturing labor is defined as a component(s) in the work order bill of material. Most companies using this approach do not track labor and instead are using it as rough estimate in order to get labor into the cost of the finished item. In this case the labor component would be included as part of kit issues or backflushes (see previous section *Issuing Material from Stock*) and would be treated the same as material components.

With light manufacturing you can still track actual labor, if necessary. In this case you can set up the system in SD-B, *Work Orders Defaults*, so that labor is excluded from kit issues and backflushes. You can then track your labor (on time sheets or time cards) and issue it in units of decimal hours as a single component issue via WO-G, *Issue Materials*.

To exclude labor components from kit issues and backflushes, set the *Labor Prompt in Kit Issues?* field in SD-B, *Work Orders Defaults*, to *Y* and answer *N* to include labor when issuing kits or backflushing.

To issue labor in units of decimal hours, use WO-G, *Issue Materials*.

Only use light manufacturing for ultra-simple products

DBA offers the light manufacturing alternative because many manufacturers coming from general accounting systems are accustomed to using that method.

Using routings to define labor, however, offer several advantages over the light manufacturing method of defining labor as bill of material components.

First, using routings can be kept just as simple as the bill of material method. At a minimum, all that is needed is a single work center, perhaps called "factory." You can enter a single sequence called "labor" and then enter a single hourly amount for the total labor that goes into the product. This simplified routing entry takes no more time than entering labor components in a bill of material.

Using routings, though, enables you to define multiple sequences (operations) and gives you a platform for entering documentation as to how your products are manufactured that for purposes of printing on shop travelers.

If you do any significant amount of outside processing (plating, heat treating, etc.), routings almost have to be used as the best and easiest way to handle the purchasing and costing of these services. Trying to use type T item numbers in your bills of material to account for outside processing is a clumsy process that requires unique item numbers for every process for each item you make, whereas using routings involves no item numbers at all.

Unless you have an ultra-simple product line with little or no outside processing, we do not recommend using the light manufacturing approach.

The light manufacturing method is not appropriate for job shops at all, because job shops need the detailed shop travelers and work center scheduling that can only be accomplished by using routings.

Entering miscellaneous and extra costs

The screenshot shows a software window titled "The Manufacturing Answer - 11/29/01". Below the title bar is a menu bar with "File", "View", "Window", and "Help". A toolbar contains buttons for "Mfg", "WO", "JC", "PO", "MR", "SH", "DC", and "ES". The main window is titled "AW-H Enter Misc/Extra Costs - Add New Entry". It contains a form with the following fields:

Cost Type	E E-Extra M-Miscellaneous
Date	00/00/00
Work Order	0-0
Sequence No	0
Cost Amount	0.000000
Description	
GL Account	

At the bottom left is a "Help" button, and at the bottom right is an "Opening Screen" button.

WO-H, Enter Misc/Extra Costs

There may be occasions where you want to charge extra costs to a work order that are not specifically for material, labor, or outside processing. For example, you may want to charge a work order for items such as engineering fees, tooling, crating, letters of credit fees, etc.

DBA offers two types of extra costs. *Extra* costs are costs made at the work order level and *miscellaneous* costs are extra costs tied to specific routing sequences.

Extra and *miscellaneous* costs can also be entered in estimates and can be compared to the actual costs entered in WO-H, *Enter Misc/Extra Costs*.

There is no limit to the number of *extra* and *miscellaneous* costs that can be entered.

Recording finished production

The Manufacturing Answer - 11/29/01

File View Window Help

Mfg WO JC PO MR SH DC ES

AW-I Enter Finished Production

		Estm Costs	Actual Costs
Date Received	11/30/01	Matl	0.00
Work Order No	0-0	Setup	0.00
Parent Part		Labor	0.00
Desc		OutPr	0.00
Reference		FixOH	0.00
Qty to Make	0.00	VarOH	0.00
Qty Completed TD	0.00	Misc	0.00
Qty Completed	0.00	Extra	0.00
Use Std Cost?	0.0000	Total	0.00
Actual Unit Cost			0.00
Lot Number		Current Costs	
Serial Number		Avg	0.0000
		Last	0.0000
		Std	0.000000

Help Lookup Exit

WO-I, Enter Finished Production

When manufactured items are completed, they are ready to be transferred to finished goods inventory, from which they can then be sold (invoiced) or issued to another work order (as in the case of a subassembly). Recording finished production is done through WO-I, *Enter Finished Production*.

Entering partial completions

It is not necessary for the entire work order quantity to be completed in order to enter finished production. In fact, it is common to enter partial completions before the entire work order is completed so that rush orders can be shipped or subassemblies can be issued to other work orders.

Using standard cost or actual cost

As you make transactions you are given the choice of using the standard or actual cost to update the inventory average cost. The inventory average cost is used as the cost of goods sold when the item gets shipped or is used as the cost that gets rolled up into a higher-level assembly if the item gets issued to another work order.

If you use choose actual cost and are entering a partial completion, the program

will calculate an estimated cost based on the percent completion and the costs accumulated to date. Be aware that this is only an estimate and may not be very accurate, depending on the quality and amount of cost information available.

For this reason companies that maintain standard costs often record all partial completions at standard cost and then make a balancing entry on the last transaction (see next).

Some companies (usually those with product lines, not job shops) perform all manufacturing transactions at standard cost and calculate variances between actual and standard cost at month end. If this is the case with your company, you would make all your entries at standard cost.

For more information on using standard costing, see *Using standard costs* in chapter 8.

Costs are balanced on the last transaction

On the very last transaction that is made when the work order is finally completed, the program optionally makes a balancing entry through adjustment of the last unit cost so that all costs that were charged to the work order equal the total cost of the finished goods that were transferred to inventory. In other words, this last balancing transaction ensures that all costs charged to the work order equal all costs that went into finished goods inventory -- or, simply put, "all dollars in equal all dollars out."

If your company makes all transactions at standard cost, you would not take the balancing entry option and a variance posting will be made when the work order is closed zeroing out the cost in the work order due to any difference between actual costs incurred and standard finished production removed from the work order.

Closing work orders

The screenshot shows a software window titled "AW-J Close/Cancel Work Orders". The window has a menu bar with "File", "View", "Window", and "Help". Below the menu bar is a toolbar with buttons for "Mfg", "WO", "JC", "PO", "MR", "SH", "DC", and "ES". The main area of the window contains a text box with the instruction: "<You may enter the work order prefix or the complete work order number.>". Below this text box are several input fields: "From Work Order" and "Thru Work Order" (both showing "01- 0"), "From Job Number" and "Thru Job Number" (both empty), "GL Post Date" (showing "00/00/00"), and "Close or Cancel [C/X]" (showing "C"). At the bottom of the window are buttons for "Help", "Lookup", "Process", and "Exit".

WO-J, Close/Cancel Work Orders

Work orders are not considered finished until they are formally closed.

Many companies close the work order as a separate step in conjunction with some sort of sign off procedure to insure that all costs are accounted for and all procedures completed.

Work orders can be closed two ways

There are two ways to close work orders.

If you want to save a step and close work orders automatically upon completion of production, you can set a default that prompts you to do so within WO-I, *Enter Finished Production*. To set this default, enter a Y in the *Close WO in Enter Finished Production?* field in SD-B, *Work Orders Defaults*.

If you want to close work orders as a separate step, use WO-J, *Close/Cancel Work Orders*.

Canceling work orders

You can also use WO-J, *Close/Cancel Work Orders*, to cancel a work order. If you cancel a work order, any materials that were issued are automatically returned

to inventory.

If for some reason these materials are not in a returnable state, you will have to make manual inventory adjustments to get them back out of inventory.

Job costing reports

The *Job Costing* includes a series of job costing reports that provide you with detailed analysis of your manufacturing activities.

JC-A, *Print Job Cost Report*, provides detail and summary breakdowns of all costs on a work order.

JC-B, *Print Profit Projection*, gives you a projected estimate of what your final job costs will be, based on costs reported to-date.

All your job cost transactions are available on separate reports: JC-C, *Print Labor Transactions*, JC-D, *Print Overhead Transactions*, JC-E, *Print Material Issues*, JC-F, *Print Outside Purchases* and JC-Q *Print Work Order Receipts*.

JC-G, *Print Labor Efficiency*, is a great tool for analyzing past performance within labor sequences for the purpose of setting time standards within your routings.

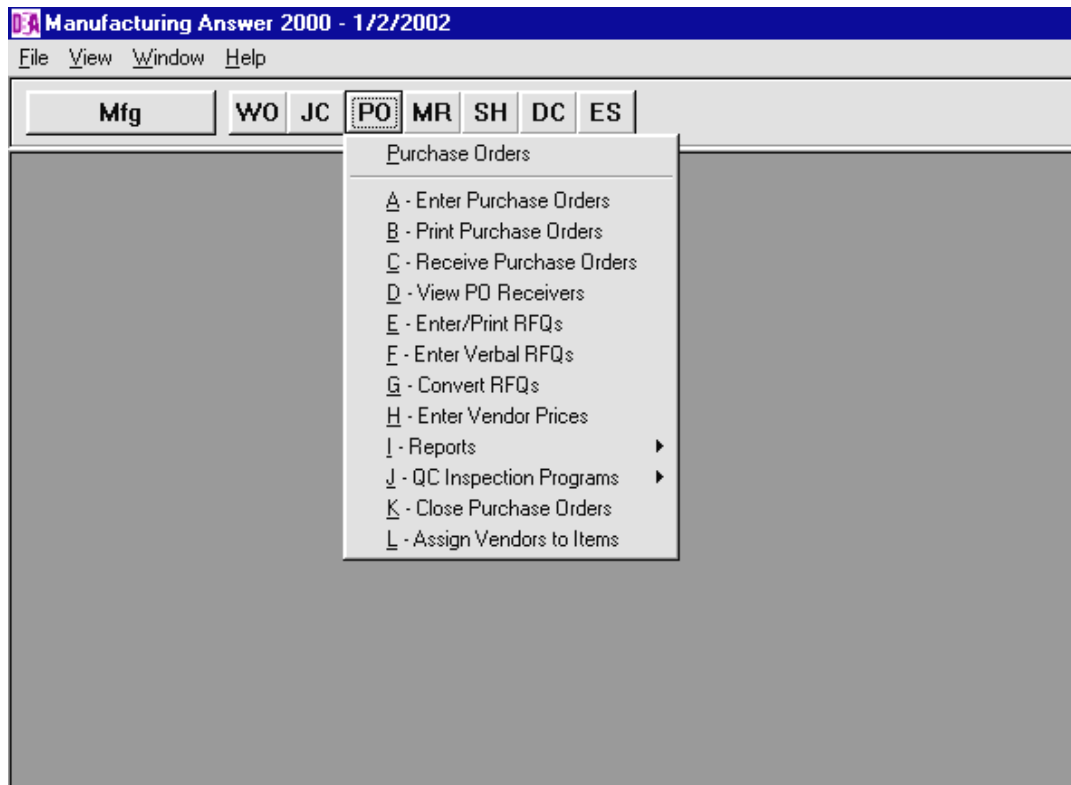
Production reports let you analyze your manufacturing resources: JC-I, *Print Production by Work Center*, JC-J, *Print Production by Machine*, JC-K, *Print Production by Tool*.

JC-M, *Print WIP Summary*, gives you a complete breakdown of your work-in-process inventory costs, summarized by work order.

14. PURCHASE ORDERS

- **Purchase Orders overview**
- **Purchase Orders flow chart**
- **How purchase get created**
- **Purchasing material for inventory**
- **Purchasing material for work orders**
- **Purchasing outside processing for work orders**
- **Purchase order prices**
- **Receiving purchase orders**
- **Reconciling and closing purchase orders**
- **Using the vendor price file**
- **Using make-froms**
- **Using RFQ's**

Purchase Orders overview



Purchasing is an integral part of the manufacturing process. As work orders get created, requirements are generated for the materials and outside processing services that will be needed to complete those work orders. Purchase orders are used to order and track these materials and services.

Purchasing materials

Purchase orders are used to order materials. Materials can include anything from raw materials to subassemblies to complete products.

You can purchase materials for general inventory from which they can then be issued to work orders or shipped to customers, or you can purchase materials for specific work orders. In the latter case, upon receipt the materials bypass general inventory and go directly to WIP (work-in-process) inventory.

On a single purchase order you can order a mixture of items for inventory and for multiple work orders.

Purchase orders can also be used for scheduled deliveries as far into the future as needed. You can enter an item multiple times, each with a different due date. The purchase order can stay open indefinitely and can be received against any number of times.

Purchasing outside processing services

In manufacturing it is common to order outside processing services for things such as plating, painting, heat treating, etc.

DBA offers a special type of purchase order known as a *service order*. When you designate a purchase order as a *service order*, each line item must be marked for a specific work order. No item numbers are required to identify the service being ordered; you can simply use the item number of the item being manufactured.

When the *service order* is received, its costs get charged directly to the work order(s) designated.

Quality control sub-system

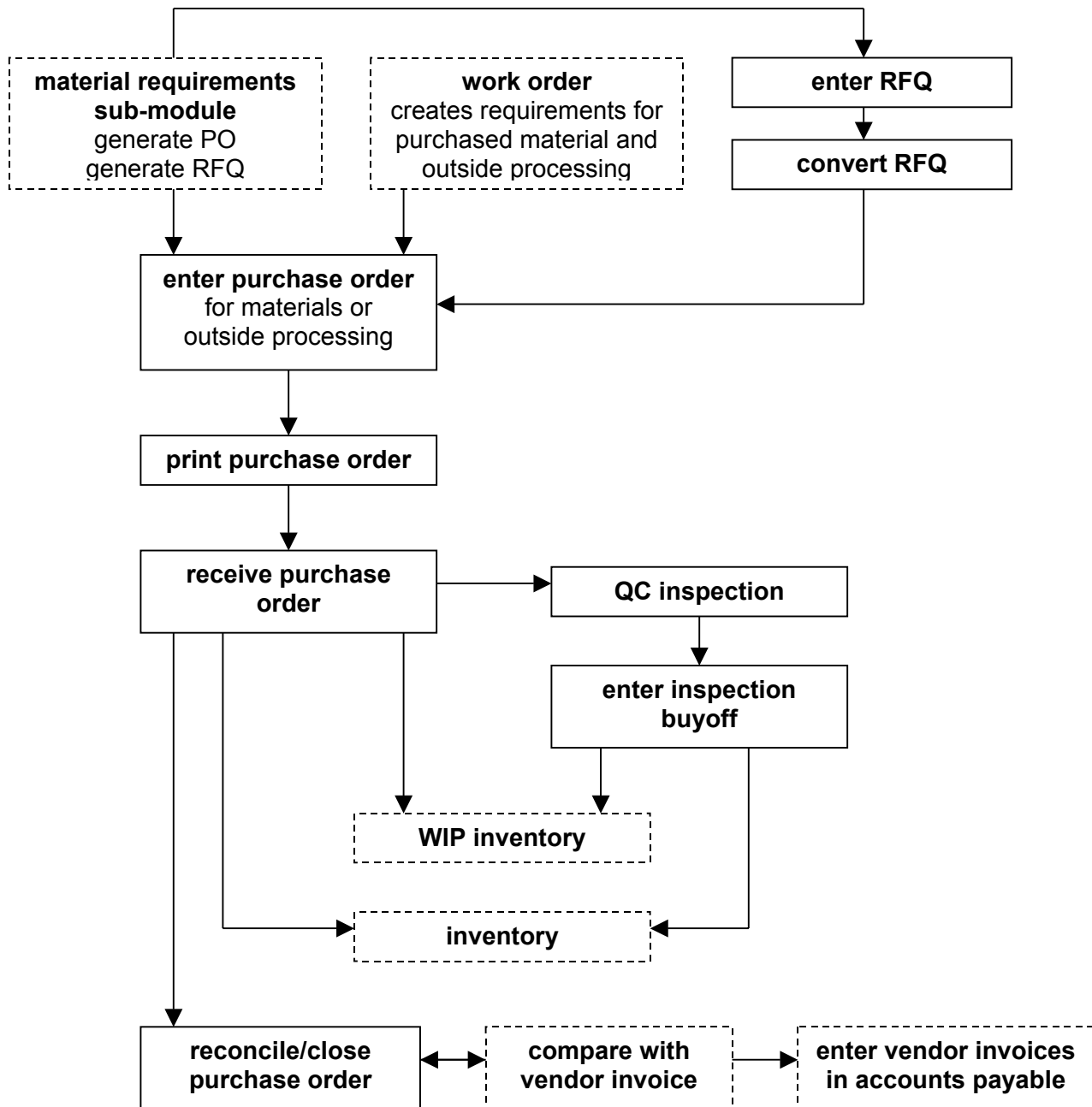
The *Purchase Orders* module offers an optional QC inspection system. You can receive items into QC inspection instead of into inventory. Only after items in QC are inspected and "bought off" are they then released to general inventory or WIP inventory.

RFQ's

Some manufacturers must get bids from multiple vendors for materials. The *Purchase Orders* module offers RFQ's (requests for quote) that lets you send RFQ's to multiple vendors. RFQ's that are accepted can be automatically converted to purchase orders without having to re-key any data.

RFQ's are also linked to the *Estimating* module. Once a vendor submits his bids, the RFQ costs can automatically be incorporated into the estimate's cost structure.

Purchase Orders flow chart



How purchase orders get created

The screenshot shows the 'PO-A Enter Purchase Orders' header screen. The window title is 'The Manufacturing Answer - 11/29/01'. The menu bar includes File, View, Window, and Help. The 'Mfg' tab is selected. The form contains the following data:

PO No	5000	PO Date	11/15/99	Ship-to Vendor or Customer?	
Vend Cd	STE-01	PO Type	P		
Name	STEEL WHEELS				
Address	444 Lemon Avenue				
	PO Box 4565				
City, St	Altamont, CA				
Zip	93412				
Country	USA				
Attn	Brian Jones				
Phone #	405-876-9898				
Fax					
PO Desc					Confirming?
Ship Via	BEST WAY				N
Terms Cd	5 NET 30				Taxable?
Job No					N
FOB					Tax Rate(%)
Ent By					0.000
Currency					
	GL Dept				
	Location				
Subtotal	3244.00	Tax	0.00	Total	3244.00

Navigation buttons: Help, Line Items, Exit.

PO-A, Enter Purchase Orders (header screen)

Purchase orders can be created three different ways.

- Purchase orders can be entered manually via PO-A, *Enter Purchase Orders*.
- Purchase orders can be generated automatically through MR-J, *Generate Purchase Orders*, based on recommendations made by MR-F, *Generate Material Requirements*.
- Purchase orders can be created when RFQ's are converted into purchase orders via PO-G, *Convert RFQ's*.

Service orders

In most companies service orders are created manually as needed when outside processing sequences come up during the manufacturing process.

MR-F, *Generate Material Requirements*, does not deal with services at all and can't be used to generate service orders like it can with material purchases.

While RFQ's are generally used for quotes on materials, they can be used for outside processing services if needed.

How purchase orders get created

The Manufacturing Answer - 5/15/2002 [Min] [Max] [Close]

File View Window Help

Mfg WO JC **PO** MR SH DC ES

PO-A Enter Purchase Orders [Min] [Max] [Close]

PO No	5015	Date	01/15/02	Ship-to Vendor or Customer?	☐
Vend Cd	STE-01	Type	P		
Name	STEEL WHEELS				

	Orig. Qty	Line#
	150.00	1

REF	Item Number	Description	Quantity	Due Date
	A-210	15" AXLE	50.00	02/07/02

Price	1.1500	UM	EA	Conv. Fact.	1.00000	Tax?	N	Disc%	0.00	Estimate	0	Work Order	0- 0	Seq	0
-------	--------	----	----	-------------	---------	------	---	-------	------	----------	---	------------	------	-----	---

Subtotal	57.50	Tax	0.00	Total	57.50
----------	-------	-----	------	-------	-------

Display Lines	First	Last	Prev	Next	1st Screen
Help	New Line	Ins Line	Del Line	Dupe Line	Notes
					Save
					Exit

Line Item Screen

Purchasing material for inventory

It is common in manufacturing for materials to be needed by multiple work orders at the same time. Instead of buying materials for each work order, it is more efficient to group requirements together and purchase for general inventory and then issue the material from there to the individual work orders. In the process you will get lower prices by making bigger purchases.

Planning tools

There are two tools in DBA that help you group your requirements and plan your purchases.

- The most powerful tool is MR-F, *Generate Material Requirements*, which suggests purchase orders based on comparing existing inventory with work order requirements, and staying within guidelines you've established for reorder levels and lead times.
- The other tool is the IN-D, *Print Reorder Report*, which shows you your existing inventory, sales, and work order requirements and gives you all the information you need to manually prepare your purchase orders.

A major advantage to using MR-F, *Generate Material Requirements*, especially if you have lots of purchase orders, is that you can automatically generate purchase orders based on its recommendations through the MR-J, *Generate Purchase Orders*, instead of having to enter them all manually.

Entering purchase orders

Purchase orders are entered through PO-A, *Enter Purchase Orders*.

To purchase items for inventory is simple. Any line item that is entered without specifying a work order number will get received into general inventory.

On the same purchase order you can freely order some items for inventory and some for specific work orders.

Purchasing material for work orders

Many manufacturing companies, especially job shops, strive to avoid maintaining any inventory and therefore purchase materials only on an as-needed basis for each work order. DBA easily accommodates this type of purchasing.

Purchase orders are entered via PO-A, *Enter Purchase Orders*.

Each line in the purchase order has an optional field for the work order number. If you specify a work order number, upon receipt the material will bypass general inventory and will be received into WIP inventory and its costs will be charged to the work order specified.

If you leave the work order number blank, the material will be received into general inventory.

On a single purchase order you can freely purchase material for any number of different work orders by entering multiple lines each assigned to a different work order.

Purchasing outside processing for work orders

In manufacturing it is common to order outside processing services for things such as plating, painting, heat treating, etc.

DBA offers a special type of purchase order known as a *service order*. When you designate a purchase order as a *service order*, each line item must be marked for a specific work order. No item numbers are required to identify the service being ordered; you simply use the item number of the item being manufactured.

When the *service order* is received, its costs get charged directly to the work order(s) designated.

Entering service orders

Service orders are entered through PO-A, *Enter Purchase Orders*.

At the top of the first screen is the *PO Type* field. Enter an *S* in this field to create a service order.

You must specify a work order number on all service order line items. You can freely purchase services for different work orders on the same service order.

Purchase order prices

When entering purchase order and service order line items, the *Price* field automatically displays a price according to the following formulas.

Materials

When entering a purchase order for materials, PO-A, *Enter Purchase Orders*, first looks in the vendor price file. If it finds a price for this item from this vendor, it will use that price. You can access the vendor price file through PO-H, *Enter Vendor Prices*.

If it does not find a price in the vendor price file, it automatically inserts the *Last Cost* value maintained in the inventory file in the *Price* field. The *Last Cost* field is automatically updated by the system each time an item is received.

You can then either accept the default price or override it.

Service orders

When entering a service order you must specify a work order number and routing sequence. If a vendor cost is maintained on that work order routing sequence, PO-A, *Enter Purchase Orders*, will display that cost in the *Price* field. You can then either accept this price or override it.

Vendor costs for outside processing sequences can be maintained in the manufactured item's permanent routing (as entered in RO-A, *Enter Routings*) or in the work order routing (as entered in WO-K-A, *Enter Work Order Routings*). Vendor costs for outside processing can also be maintained in PO-H, *Enter Vendor Prices*.

Receiving purchase orders

The Manufacturing Answer - 11/29/01

File View Window Help

Mfg WO JC **PO** MR SH DC ES

PO-C Receive Purchase Orders

Purchase Order No.	5009	Vendor	STEEL WHEELS
PO Type	Purchase	Receive all Lines?	N
Currency		Receive thru Due Date	00/00/00
Receipt Date	11/30/01	Display Comment Lines?	Y
Receive Into	Inventory	Display Fully Received Lines?	N
Packing Slip No			

Lin	Item Number	Description	Due Date	Work Order	Seq
	A-210	15" AXLE	07/18/00	0 - 0	0

PO Lines

Item Number	Description	Lin	Remaining	This Rcpt	Due Date
A-210	15" AXLE		5000.00	0.00	07/18/00
	Vend#:SW-210				
	1/2" diameter - chrom				
A-220	WHEEL		8832.00	0.00	07/18/00
	Vend#:SW-5000-RW				

Exit

PO-C, Receive Purchase Orders

Purchase order and service order items are received via PO-C, *Receive Purchase Orders*.

Materials that are designated for specific work orders will go directly to WIP inventory and their costs will be charged to those work orders.

Materials not designated for specific work orders will go to general inventory.

Outside services costs are added to WIP inventory and are charged to specific work orders.

You can receive into inventory or QC inspection

When receiving a purchase order you are given the choice of receiving into *inventory* (general inventory or WIP) or *QC inspection*.

If you wish to set a default for the *Receive into* field, you can do so in SD-C, *Purchase Orders Defaults*.

If you choose *QC inspection* the items will not go into inventory until they are transferred out of *QC inspection* via PO-J-C, *Enter Inspection Buyoffs*.

Changing prices during receiving

It is common to place purchase orders without always knowing the exact price. It is not always practical to verify prices in advance with every purchase order, especially for routing supplies or other small items.

Depending on the default set in SD-C, *Purchase Order Defaults*, when you receive each item, you may be given the opportunity of changing the unit price. If you do so, it will change the price on the purchase order itself, so if there are to be any future receipts, it will be at this new price.

If you do not know the price at time of receiving, you can still change the price at a later time when you receive your vendor invoice and process it using AP-C, *Enter Vendor Invoice* (see next section).

Reconciling and closing purchase orders

Manufacturing Answer 2000 - 1/2/2002

File View Window Help

Acct + Sales GL AP AR SO SA CS CM

AP-C Process Invoices for ACME PAINT COMPANY

Open Receivers

Excludes Receivers dated after 01/02/02

Date	PO#	Item	Qty	Price	Ext.	↓
11/27/01	5511	A-210	2.0000	1.5000	3.0000	
11/30/01	5506	RESISTOR	0.8000	500.0000	400.0000	
12/18/01	5510	A-210	200.0000	1.5000	300.0000	

Invoice Amount 746.23 Balance 746.23

Tag Untag Sort By PO Sort By Date Done Exit

PO-N, Enter Purchase Order Invoices

A purchase order gets closed when all items are fully received and reconciled with vendor invoices. Once closed the purchase order information is saved in the purchase history file and deleted from the open purchase order file.

Purchase orders are closed in a two-step process.

The first step is completely automatic. When all line items are fully received in PO-C, *Receive Purchase Orders*, the purchase order gets marked as closed for receiving purposes. This receiving close keeps fully received purchase orders from appearing in lookups that are meant to be confined to open purchase orders.

The second step is a formal close, normally accomplished via PO-N, *Reconcile Vendor Invoices*. It is a formal close in order to provide a procedure for insuring that the purchase order has been compared with the vendor invoice and is fully accurate and complete.

If you want to bypass PO-N and simply close your purchase orders without reconciling the purchase prices, you can do so via PO-K, *Close Purchase Orders*.

If you are using DBA accounting, use AP-C, *Enter PO Invoices*, instead of PO-N. AP-C creates entries in the *Accounts Payable* system for ultimate payment of vendor invoices.

Reconciling purchase orders with vendor invoices

Purchase orders need to be reconciled with vendor invoices to make sure that inventory and job costing and the vendor's billing are accurate and in agreement with one another. This is performed through PO-N, *Reconcile Vendor Invoices*.

It is common to place purchase orders without always knowing the exact price. It is not always practical to verify prices in advance with every purchase order, especially for routing supplies or other small items. Often you won't know the final price until you have the vendor invoice in hand.

PO-N, *Reconcile Vendor Invoices*, offers a price review function that allows you to compare your vendor's invoice with your purchase order line item prices. When you change a price after an item has already been received, the program will make adjusting entries to previously posted inventory and job costs to correct for the price change.

Use PO-I-F and PO-K as an alternative

An alternative to running all your vendor invoices through PO-N is to compare your vendor invoices with your purchases using PO-I-F, *Print Received Not Invoiced*. Any purchase orders that do not need price changes can simply be closed via PO-K, *Close Purchase Orders*. Purchase orders that need price changes can then be reconciled through PO-N, *Reconcile Vendor Invoices*.

Using the vendor price file

Manufacturing Answer 2000 - 12/17/01

File View Window Help

Mfg WO JC **PO** MR SH DC ES

PO-H Enter Vendor Prices

Part Number			
Description			
Vendor Name			
Purch U/M		Conv Factor	0.0000
Lead Time	0	Exp Date	00/00/00
Quantity 1	0.00	Cost	0.0000
Quantity 2	0.00	Cost	0.0000
Quantity 3	0.00	Cost	0.0000
Quantity 4	0.00	Cost	0.0000
Quantity 5	0.00	Cost	0.0000
Last Update	00/00/00		

Help Lookup Delete Display Quotes Exit

PO-H, Enter Vendor Prices

DBA offers a vendor price file that gives you a place to enter vendor prices and other vendor specific parameters. When a line item is entered in PO-A, *Enter Purchase Orders*, the program first checks the vendor price file first to see if there is a current entry. If so, it automatically pulls that price, unit of measure, PO Conversion factor and lead-time into the purchase order.

The vendor price file allows you to enter up to five quantity price breaks and also contains an expiration date so that older prices expire if not periodically reviewed.

To enter Parameters in the vendor price file, use PO-H, *Enter Vendor Prices*.

You can review vendor prices through the PO-I-D, *Print Vendor Prices*, report. A nice feature of the report is that you can limit it just to expired prices.

You certainly do not have to maintain vendor prices in this file for all items you purchase. On important items, however, that you use all the time, this file gives you a place to maintain and refer to current prices and quantity breaks by vendor.

Using make-froms

A *make-from* is an item that receives an outside process such that it becomes a different item number than the item without the process.

For example, a part sits on the shelf in a non-plated state. You want to send some of the parts out for plating. When the parts come back they should have a different item number to differentiate between the plated and non-plated parts.

Instead of tediously having to create a work order just for the purpose of getting outside processing performed on some inventory parts, *make-froms* can be used instead.

Setting up make-froms

Make-from items are coded as type *M* items in IN-B, *Enter Inventory*, and are set up with a bill of material in BM-A, *Enter Bills of Material*, consisting of the raw part(s) (in this example the non-plated part) that goes into the finished *make-from*.

Entering make-froms on purchase orders

Even though *make-froms* are used for purchasing services, they are entered on type *P* purchase orders because they are not tied to a work order sequence like regular type *S* service orders are.

When a *make-from* is entered as a line item, the component(s) in the *make-from*'s bill of material automatically come into the PO as line items following the *make-from* item; however, they have negative quantities. This properly creates a negative value against the *On PO* status field in inventory, which tells the MR-F, *Generate Material Requirements*, program and the IN-D, *Reorder Report*, that the components are allocated for a PO and are not available for other use.

Receiving make-froms

When *make-froms* are received in PO-C, *Receive Purchase Orders*, you not only receive the *make-from* item, but you also receive the item's components. Because they are negative quantities, by receiving them you actually deduct them from inventory. When the *make-from* is added to inventory, PO-C, *Receive Purchase Orders*, adds its PO unit cost to the cost of the component(s) so that the total cost for the *make-from* accurately reflects both the service rendered to it as well as the cost of the components that went into it.

The use of *make-froms* provides a convenient alternative to using the work order process for handling the outside processing costing and inventory properly for the *make-from* item and its components.

Using RFQ's

In some industries such as the electronic assembly business it is necessary to get price quotations from vendors prior to starting a work order. If so, RFQ's can be a valuable tool in this process.

An RFQ is a "request for quote" document that is sent out to a vendor or set of vendors. It gives you the ability to keep track of these requests, record the responses, and ultimately convert the "winning" RFQ to a live purchase order.

Two types of RFQ's are offered

DBA offers two types of RFQ's: printed RFQ's and verbal RFQ's.

Entering a printed RFQ is identical to entering a purchase order, except that you don't enter prices until they are sent back to you by the vendor. You would use a printed RFQ when you want a printed document to send to the vendor or usually when there are multiple items on the RFQ.

You would use a verbal RFQ when you make a verbal request to a vendor for a price on one item. It not only gives you the means to record the request, but it also holds up to five quantity price breaks for the item.

How are RFQ's created?

Printed RFQ's can be entered manually via PO-E, *Enter/Print RFQ's*.

Printed RFQ's can also be generated automatically by MR-K, *Generate RFQ's*, based on purchase order recommendations made by MR-F, *Generate Material Requirements*.

Verbal RFQ's are entered manually via PO-F, *Enter Verbal RFQ's*.

Tracking RFQ's

RFQ's are considered open if there are no prices entered and are considered closed once prices have been received and entered.

Open and closed Verbal RFQ's can be tracked through PO-I-C, *Print RFQ Status*.

Converting RFQ's

Printed RFQ's can be converted to live purchase orders via PO-G, *Convert RFQ's*.

Using printed RFQ's as purchase order templates

Another use of RFQ's is as purchase order templates. Often there are situations where you buy many line items from the same vendor with only minor variations from purchase order to purchase order.

Instead of having to enter a new purchase order from scratch each time, you can set up a permanent RFQ for that vendor containing all the line items you buy from him. Each time you need to place a purchase order, convert the RFQ to a

purchase order using the *Convert RFQ's* program, then go into the purchase order and delete any lines that may not apply to this particular order.

It is often much faster to edit a purchase order created from a template than it is to create a big, multiple line purchase order from scratch.

Using RFQ's in estimates

RFQ's can be a useful tool within ES-A, *Enter Estimates*, if you need to get costs from vendors for particular materials or services and then need to reflect those costs in the estimates calculations.

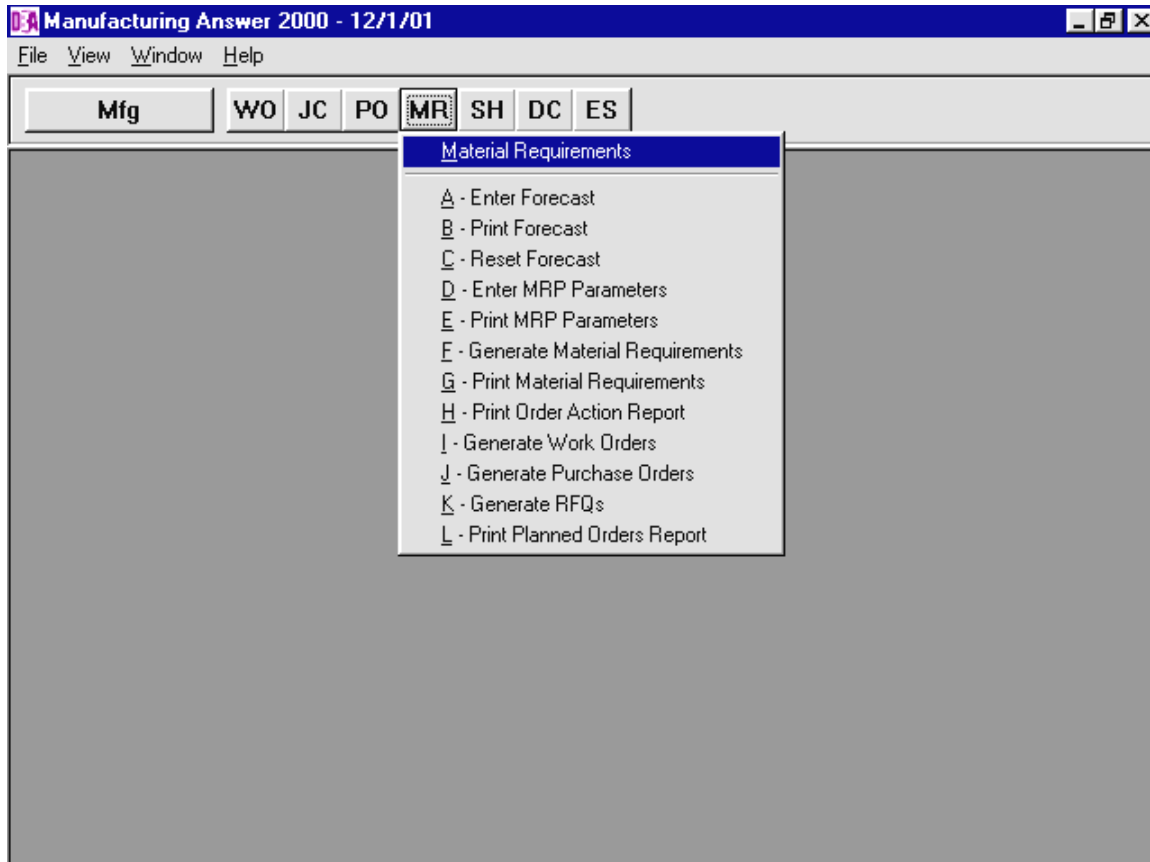
Both types of RFQ's allow you to tie the RFQ to a particular estimate number. If you have multiple RFQ's in the file for the same inventory item, you will only be allowed to tie one of them to a particular estimate. This one RFQ represents the "winning" RFQ.

If an RFQ from a vendor is tied to a particular estimate, the RFQ price(s) will be used in the estimate calculations instead of the normal cost sources.

15. MATERIAL REQUIREMENTS

- **Material Requirements overview**
- **Planning system flow chart**
- **Setting up material requirements parameters**
- **Generating material requirements**
- **Creating suggested work orders**
- **Creating suggested purchase orders**
- **Expediting and delaying existing orders**
- **Using forecasts to enter future demand**
- **Generating planning lead times**

Material Requirements overview



The *Material Requirements* module is a vital part of the DBA planning system, which provides a variety of tools that assist in the complex process of materials planning and work order scheduling.

A good planning system assures that materials are available at the time needed for production. Without material you cannot manufacture anything.

Short term materials planning tools

Companies with short term planning requirements where work orders are completed in relatively quick production cycles and material have short lead times tend to use either IN-D, *Print Reorder Report*, BM-D, *Print BOM Availability*, or WO-L-F, *Print Work Order Shortages*, to determine which items need to be purchased or manufactured.

These reports will list any items that are not on hand and will show you any other work orders that may also need the same items.

Generating material requirements

By far the most powerful materials planning and scheduling tool available to you is MR-F, *Generate Material Requirements*, which takes all requirements, whether

from existing sales orders, work orders, or forecasts, and then generates materials requirements down through all levels of the product structures by comparing projected on-hand inventory with material needed, suggesting specific purchase orders and work orders as it goes.

Besides recommending when to place purchase orders and create work orders, MR-F, *Generate Material Requirements*, functions as a powerful scheduling tool that advises you when to delay or expedite existing orders in response to changes elsewhere in the system. If you have multiple levels in your bills of materials and common subassemblies and components among different products, scheduling these interdependent items so that you don't end up bringing in material too early (tying up capital) or too late (delaying production) is fantastically complicated and almost impossible to keep track of manually. The *Generate Material Requirements* program keeps all these interdependent relationships synchronized with one another.

MR-F, *Generate Material Requirements*, is highly recommended as a planning tool. To use it effectively you must have accurate bills of material, lead times, and be dedicated to maintaining realistic and accurate due dates and on-hand inventory counts.

Scheduling

Working hand-in-hand with MR-F, *Generate Material Requirements*, is the *Scheduling* module. The *Scheduling* module offers two types of scheduling: finite and infinite.

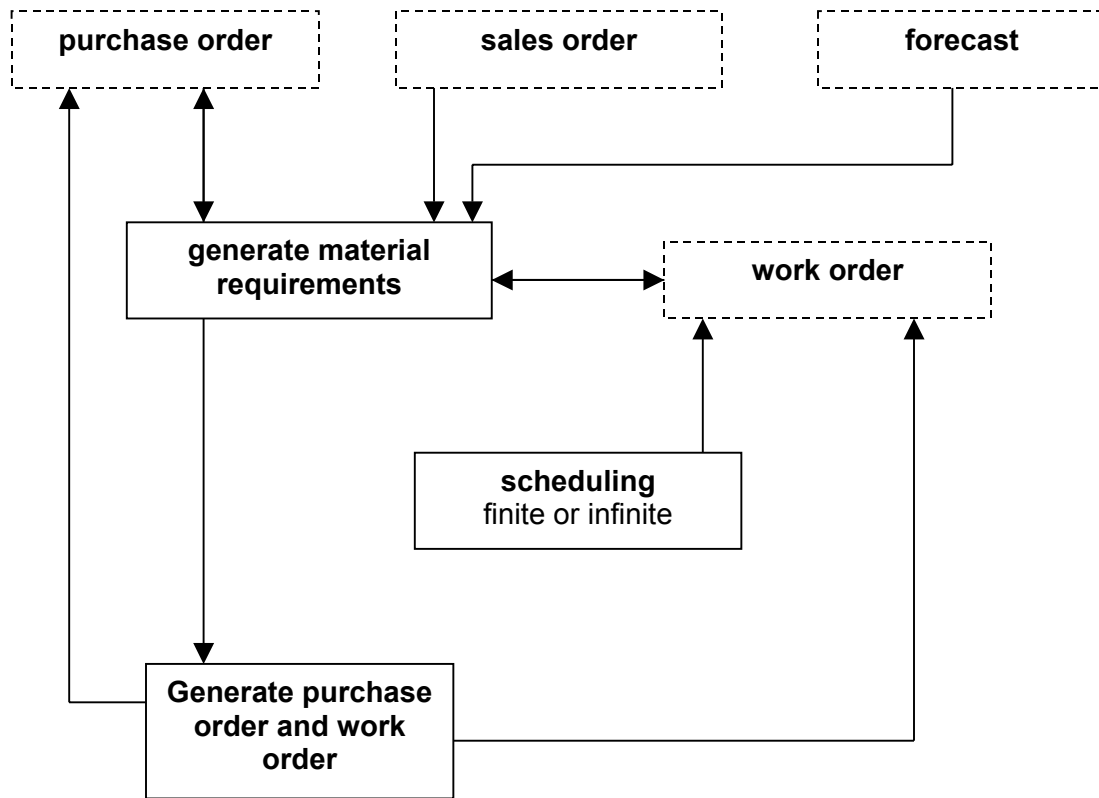
The finite scheduling program determines work order finish dates based on your available capacity. Open work orders are first sorted in priority order and then the factory work centers are loaded, one work order at a time in routing sequence order. The last routing sequence's finish date determines the overall scheduled finish date of each work order.

The infinite scheduling program leaves work order start and finish dates fixed and determines start and finish dates on the routing sequences without regard for plant capacity. Work center loads can be viewed in a graphical display so that you can compare available capacity against your commitments.

If your company has limited short-term ability to change capacity and quotes realistic due dates that reflect your current production capability, you would use finite scheduling. If your company quotes fixed due dates and meets those dates by adapting your capacity as needed (such as adding extra shifts or overtime), then you would use infinite scheduling.

As new work orders get scheduled and existing work orders get re-scheduled, the next time you run MR-F, *Generate Material Requirements*, it will reflect the new start and finish dates and will compare those against the interdependent demand of related subassemblies and materials and will let you know if related work orders and purchase orders need to be re-scheduled also.

Planning system flow chart



Setting up material requirements parameters

Manufacturing Answer 2000 - 12/1/01

File View Window Help

Mfg WO JC PO MR SH DC ES

MR-D Enter MRP Parameters

Part Number		Type	
Description			
Include in MRP Generation? [Y/N]	☐		
Reorder Level	0		
Reorder Amount	0		
Lead Time	0		
Round MRP quantities to the next whole number? [Y/N]	☐		
EXPEDITE	Buffer	0 Days	
	Sensitivity	0 Days	
DELAY	Buffer	0 Days	
	Sensitivity	0 Days	

Help Lookup Exit

MR-D, Enter MRP Parameters

Material requirements parameters are specific fields used by MR-F, *Generate Material Requirements*, as guidelines to determine when and how much to orders for each item.

Specifically these fields are: *include in MRP generation, reorder level, reorder amount, lead time, rounding to next whole number, expedite buffer, expedite sensitivity, delay buffer, and delay sensitivity.*

Material requirements parameters can be entered via MR-D, *Enter MRP Parameters* or through IN-B, *Enter Inventory*.

Setting default values

Many of your items will use the same parameters. You can set up default values for these parameters in SD-D, *Material Requirements Defaults*, so that as you create new item numbers the default values will be automatically assigned to them.

The following are discussions on some of the important parameters

Include in MRP generation? field

You must set this field to Y for an item to be included in the material requirements generation.

Sometimes items such as rivets, oil, labels, etc. are bought in bulk and are not necessary to include in the MRP program.

We suggest you set the default for this field to *Y* and then manually set this field to *N* for any items that are not to be part of the material requirements generation.

Reorder level

This is the minimum inventory on-hand quantity you want to keep at all times, also known as safety stock level. MR-F, *Generate Material Requirements*, will suggest work orders or purchase orders necessary to keep inventory at this reorder level.

You want to set the *reorder level* to a quantity that insures that shortages will not occur, which is a function of how much is normally needed, how long does it take to get new stock in from production or through purchasing, and how much safety stock is to be included to cover unexpected requirements.

If you do not wish to carry inventory on an item and you want MR-F, *Generate Material Requirements*, to suggest orders only when a requirement exists, set the *reorder level* to zero.

Reorder amount

When a work order or purchase order gets suggested, MR-F, *Generate Material Requirements*, will use this amount as the minimum order quantity. If more than the reorder amount is needed, MR-F will use the exact amount required.

The *reorder amount* is usually an economical order quantity that reflects factors such as vendor minimums, better price level, frequency of orders, carrying cost of inventory, etc.

If you do not wish to carry inventory on an item and you want MR-F, *Generate Material Requirements*, to suggest order quantities equivalent to just what is needed, then set the *reorder amount* to zero.

Lead time

With manufactured items this is the time it takes to start and finish a work order for the item once all materials and subassemblies are complete and available.

With purchased parts this is the time it takes from ordering from the vendor to receipt of the item.

MR-F, *Generate Material Requirements*, uses the lead-time to determine the start date for work orders and the order date for purchase orders, thus ensuring that items get ordered early enough to arrive when needed.

Manufacturing lead times will vary as the plant gets busier or slower. These lead

times can be changed manually or SH-N, *Generate Lead Times*, recalculates manufacturing lead times across the entire inventory using a formula that reflects current plant capacity.

Lead times for purchased parts are maintained manually.

Rounding quantities to the next whole number

Certain items by their physical nature cannot be ordered fractional quantities. For example, sheet steel may have to be ordered in complete sheets. By setting the *Round MRP quantities to the next whole number?* field to Y, MR-F, *Generate Material Requirements*, will always round the suggested order quantity to the next whole number.

Expedite buffer

This field controls the EXPEDITE message issued by MR-F, *Generate Material Requirements*. An EXPEDITE message is issued whenever enough material is on work order or purchase order, but is scheduled to arrive after it is actually needed. It alerts you to move up the finish date or expected receiving date so that the material will arrive on time. The expedite buffer refers to the number of days within which an order can arrive and be considered late, as opposed to being considered “pegged” to a later requirement.

For example, if a shortage occurs on March 1st and the expedite buffer is set for 20 days and the shortage is met by a purchase order arriving March 19th, then the system will not issue a BUY message (suggested purchase order) and will instead issue an EXPEDITE message. If, on the other hand, the purchase order was due to arrive March 21st, it would fall outside the expedite buffer and the MRP program will assume it is designated for a later requirement. It will then issue a BUY message and a recommended PO quantity and order date.

If work order dates are frequently changed, existing work orders at the lower levels, such as for subassemblies, are likely to require date changes as well, which is where the EXPEDITE and DELAY messages can be helpful to insure that schedules stay coordinated.

There are not specific guidelines for setting expedite buffers; it varies with the volatility of date changes, what type of reorder levels are maintained, if you build to order or to stock, etc. In general, the longer you set the buffer, the fewer planned orders will be suggested. You might start with a setting of 30-60 days, evaluate the EXPEDITE messages that result, then adjust the buffers to your needs.

Expedite sensitivity

Most material requirements dates are loosely planned, meaning that they are not so precise that an item arriving one or even a few days late actually needs attention. You can reduce these unnecessary EXPEDITE messages through use of the *Expedite Sensitivity* setting. Any requirement whose number of days late is equal to or less than the *Expedite Sensitivity* number of days will not receive an

EXPEDITE message on any reports.

Delay buffer

Unlike the *Expedite Buffer*, which controls how MR-F, *Generate Material Requirements*, handles late arrivals, the *Delay Buffer* controls how early arrivals are handled. Any order due to arrive within the delay buffer period prior to a requirement is assumed by MR-F to be designated for that requirement and will receive a DELAY action message on reports. Any order that arrives prior to the delay buffer period is assumed to be for some other requirement and will receive a REVIEW action message.

DELAY messages alert you when orders arrive earlier than needed so that you can reschedule them and reduce unneeded inventory.

You will have to experiment to see which setting best fits your needs. The *Delay Buffer* works in tandem with the *Delay Sensitivity* setting (see next). You might start with a setting of 30-60 days and then adjust it to suit your needs as you gain experience. The shorter you set the buffer, the more REVIEW messages you will receive.

Delay sensitivity

This field controls the printing of DELAY messages. It should be set equal to the number of days an item can arrive early before it needs attention.

For example, if your policy is that items can arrive up to 10 days early without it causing any problems, you can set the Delay Sensitivity for 10 days. Only if an item arrives 11 days earlier than needed will you receive a DELAY or REVIEW action message.

Generating material requirements

MR-F, *Generate Material Requirements*, generates material requirements down through all levels of the bills of material, suggests work orders and purchase orders, and issues expedite and delay messages on existing orders.

You can run the program as often as needed. Some companies run it once a week, some run it once a day, and some run it more often than that. It depends on how often your requirements change and how often you do your planning. If you sit down to do your purchasing and scheduling twice a week, then you might run the program twice a week.

The program performs a complete generation from scratch each time and can take hours to complete where large files are involved, so it is best run during off hours.

How the program works

The program begins by identifying existing material requirements needed by customer orders, forecasts, and work order bills of material. It then compares requirements with existing work orders, purchase orders, and inventory. Wherever shortages are encountered, work orders and purchase orders are suggested (referred to as *planned* orders). Wherever finish dates and expected receiving dates are to be moved up or moved back, expedite and delay messages are issued.

As work orders get suggested, new requirements are created by the planned work orders' bills of material. Again, work orders and purchase orders are suggested. This process continues down through all levels of the bills of material.

Shortening generation time

You can limit the program to necessary requirements by optionally excluding forecasts, sales orders, and scheduled work orders (S types) from the generation, and you can limit the generation up through a particular estimated ship date. Limiting the generation can significantly reduce the run time of the program. You can also reduce the generation time by using SM-J-J, *Purge Closed Sales Orders* to clear closed sales orders from the open order file.

If you have not set up any reorder levels or reorder amounts on your inventory items, MR-F, *Generate Material Requirements*, will respond to each individual requirement and create many material requirements records. This is all right for a small job shop that purchases material for each order, but for a make-to-inventory company it is not advisable. Establishing reorder levels and reorder amounts will reduce the number of material requirements records and will shorten generation time.

Creating suggested work orders

Manufacturing Answer 2000 - 12/1/01

File View Window Help

Mfg WO JC PO MR SH DC ES

MR-I Generate Work Orders

Item Number From Thru

Start Date From 00/00/00 Thru 00/00/00

Finish Date From 00/00/00 Thru 00/00/00

Item Class From

Item Number: A-200
Description: WHEEL ASSEMBLY
Start Date : 01/17/02
Finish Date: 01/31/02
Quantity : 4000.00

Help Exit

MR-I, Generate Work Orders

MR-F, *Generate Material Requirements*, suggests work orders be created for specific quantities on specific start dates and due dates. These suggested work orders are given a sequential number beginning with the prefix "PL" followed by an up to five digit number, and are listed on the MR-G, *Print Material Requirements*, and MR-H, *Print Order Action*, reports in the *Order No* column.

Each time MR-F, *Generate Material Requirements*, is run, these planned work order numbers are reassigned.

At this point actual work orders have not been created -- these are merely suggested work orders.

You can either create actual work orders manually or you can generate them automatically.

Manually creating work orders

Work orders can be created manually based on suggestions made by MR-F, *Generate Material Requirements*. There are two reports that give you the information needed for making work order decisions.

MR-G, *Print Material Requirements*, shows you all current and future

transactions for an item, with a projected on-hand quantity next to each. This report shows you the whole picture on an item and may cause you to make changes to suggested work orders due to things you can see using your judgment that go beyond what the program can do, which is to simply follow the parameters it is given.

If you find that the program's suggested work orders are fine and there's no need to scrutinize MR-G, *Print Material Requirements*, the MR-H, *Print Order Action*, report can be filtered to give you a listing of all the suggested work orders that need attention.

Or, you can use both reports. Study MR-G, *Print Material Requirements*, to verify whether you want to follow the suggested work orders or make any changes, then use MR-H, *Print Order Action*, report as a worksheet to mark the work orders you wish to confirm and those that are to be changed.

Another good report is MR-L, *Print Planned Orders Report*. Use this report to help track the relationships among planned orders -- including the requirements that generated them and the requirements they in turn generate.

The actual work orders can then be created using WO-A, *Enter Work Orders*.

Generating work orders automatically

MR-I, *Generate Work Orders*, automatically converts planned work orders suggested by MR-F, *Generate Material Requirements*, into actual work orders.

The program gives you two run options.

The automatic option automatically generates work orders, meaning that all work orders within the selection range will be created automatically with no changes to the suggested quantities and dates.

The review option presents each suggested work order in a window in succession. You are given the opportunity to revise the start date, finish date, and quantity, or enter a zero quantity to skip creating the work order altogether. The work order gets created, then the next suggested work order is presented. The process continues until all suggested work orders within the selection range are reviewed.

Creating suggested purchase orders and RFQ's

Manufacturing Answer 2000 - 12/1/01

File View Window Help

Mfg WO JC PO MR SH DC ES

MR-J Generate Purchase Orders

Part Num Range 	Product Class Range
Start Date Range 00/00/00 00/00/00	Category Range
Due Date Range 00/00/00 00/00/00	Vendor Code Range
Make PO's for Location Include Product Specifications? Y Include Approved Manufacturers? Y Purchase Order Date Default 12/01/01 Due Date Default 00/00/00 Automatically Generate or Allow Review? [A/R] A	

Help Lookup Print Exit

MR-J, Generate Purchase Orders

MR-F, *Generate Material Requirements*, suggests purchase orders be created for specific quantities with specific purchase order dates and due dates. These suggested purchase orders are identified on the MR-G, *Print Material Requirements*, and MR-H, *Print Order Action*, reports with the message BUY in the *Order No* column.

At this point actual purchase orders have not been created -- these are merely suggested purchase orders.

You can either create actual purchase orders manually or you can generate them automatically.

Manually creating purchase orders

Purchase orders can be created manually based on suggestions made by MR-F, *Generate Material Requirements*. There are two reports that give you the information needed for making purchase order decisions.

MR-G, *Print Material Requirements* shows you all current and future transactions for an item, with a projected on-hand quantity next to each. This report shows you the whole picture on an item and may cause you to make changes to

suggested purchase orders due to things you can see using your judgment that go beyond what the program can do, which is to simply follow the parameters it is given.

If you find that the program's suggested purchase orders are fine and there's no need to scrutinize MR-G, *Print Material Requirements*, the MR-H, *Print Order Action* report can be filtered to give you a listing of all the suggested purchase orders that need attention.

Or, you can use both reports. Study MR-G, *Print Material Requirements*, to verify whether you want to follow the suggested purchase orders or make any changes, then use the MR-H, *Print Order Action*, report as a worksheet to mark the purchase orders you wish to confirm and those that are to be changed.

The actual purchase orders can then be created using PO-A, *Enter Purchase Orders*.

Generating purchase orders automatically

MR-J, *Generate Purchase Orders*, automatically converts purchase orders suggested by MR-F, *Generate Material Requirements*, into actual purchase orders.

The program gives you two run options.

The automatic option automatically generates purchase orders, meaning that all purchase orders within the selection range will be created automatically with no changes to the suggested quantities and dates. The program will automatically assign the vendor to the primary vendor specified in IN-B, *Enter Inventory*.

The review option allows each suggested purchase to be viewed and edited before being converted. The review function is especially useful with purchase orders because you may wish to override the vendor and buy from someone else.

The purchase orders that are generated will use prices from the vendor price file and, if none are found there, will use the *Last Cost* maintained in each item's master record.

Generate in date range batches

To avoid having many individual purchase orders for the same vendor, within a vendor the program groups all items selected into a single purchase order, all with the same due date, regardless of the suggested due dates.

Therefore, you should select items in groups of date ranges.

For example, let's say you want to purchase all items needed for due dates from February 1-8. Some items will have a four week lead time, some three weeks, some two weeks, etc. Start with the items with a four-week lead-time. Generate purchase orders by entering a start date range of 12/28/XX thru 1/3/XX (4-5 weeks prior to the due date) and a due date range of 2/1/XX thru 2/8/XX. Any BUY records that are to be ordered within the start date range that are needed within the due date range will be selected for purchase order generation. Set the

Purchase Order Date Default to 12/28/XX and the *Due Date Default* to 2/1/XX prior to generating the purchase orders. This tells you to send the purchase orders out on 12/28/XX and all line items on the purchase orders will have due dates of 2/1/XX.

Continuing our example, after generating the first date range, now generate another, this time for the three-week lead-time items. Keep the due dates the same, but now enter a start date range of 1/4/XX thru 1/11/XX. This will generate all the three-week lead-time items. Your *Purchase Order Date Default* will be set to 1/4/XX and the *Due Date Default* will remain at 2/1/XX. This tells you to send this batch of purchase orders out on 1/4/XX, with all line items having due dates of 2/1/XX. Continue in this manner with the two-week lead-time items and then the one-week lead-time items.

Using MR-J, *Generate Purchase Orders*, requires some planning. Unlike work orders, which always are for one item only, purchase orders can contain many line items, and purchased parts have a variety of different lead times. Therefore you have to devise a batch purchasing method similar to that described above, where batches of purchase orders are generated according to lead time/due date combinations. For some companies, thinking in terms of weekly batches makes sense, for others monthly batches are appropriate, for others semi-weekly or even daily batches are better.

MR-J, *Generate Purchase Orders*, is not suitable for entering scheduled orders with multiple delivery dates, which must be manually entered. Once entered, though, they are reflected in the MRP generation and, assuming all your requirements are covered, will not generate any suggested purchase orders.

Generating RFQ's

Any BUY recommendations can be converted into RFQ's instead of purchase orders if you use MR-K, *Generate RFQ's*.

The only difference between MR-K, *Generate RFQ's*, and MR-J, *Generate Purchase Orders*, is that RFQ's are generated without any prices.

RFQ's can be automatically converted to purchase orders using PO-G, *Convert RFQ's*.

Expediting and delaying existing orders

Equally important as MR-F, *Generate Material Requirements*, ability to suggest work orders and purchase orders is its scheduling function.

Whenever changes occur to dates at any level, be it a customer order, work order, purchase order, or forecast, MR-F reschedules all planned order dates of all items affected by the change.

On actual work orders and purchase orders, the program will issue EXPEDITE messages on items that are arriving too late and should be moved up, and will issue DELAY messages on items that are arriving too soon. Items that the program cannot relate to specific requirements will receive a REVIEW message.

Responding to these messages by rescheduling dates is an essential element of the planning system.

Responding to DELAY messages is just as important as expediting

Particularly important is to push dates out in response to DELAY messages. We always tend to focus on the need for expediting orders (an all too common practice in manufacturing companies) but we rarely think about de-expediting (delaying) orders.

In everyday practice, however, delays happen all the time. A machine may go down, a key employee might not show up, a critical material was not delivered on time, etc.

When one work order gets delayed, it can affect lots of other orders that are interdependent with it. Why manufacture or bring in material earlier than needed, when all it does is tie up money and factory resources that could be used for more urgent orders?

Using forecasts to enter future demand

Manufacturing Answer 2000 - 12/1/01

File View Window Help

Mfg WO JC PO MR SH DC ES

MR-A Enter Forecast

Item Number		Type	
Description			
Quantity	0.00		
Due Date	00/00/00		

Help Lookup Save Opening Screen

MR-A, Enter Forecasts

Many manufacturers have predictable order patterns for many products. They may not have actual customer orders in hand, but they know from previous experience that a certain number of orders will materialize.

DBA lets you enter these unconfirmed orders as forecasts so that the planning system can incorporate them with firm customer sales orders as it plans what to build and order.

The ability to incorporate forecasts in your planning can be especially helpful with product lines that have long planning horizons, as is the case when raw materials have long lead times and must be ordered well in advance of actual need.

Forecasts are entered via MR-A, *Enter Forecasts*. They can be manually entered or imported from an ASCII file containing item number, quantity and ship date.

Normally you enter forecasts for finished goods only. The planning system will automatically handle the lower level requirements for subassemblies and raw materials.

You can enter an unlimited number of forecasts for each finished good as far into the future as you want. Most typically you will enter a single forecast per finished

good for each month's projected sales.

Resetting forecasts

If you use forecasts in your planning, you need to reset them on a regularly scheduled basis so that they don't get out of date as time moves forward and forecasts become actual orders.

There are three ways to reset your forecasts.

- Consuming the forecast means that you can have actual customer orders or shipments deducted from the forecast within a specified date range. When the forecast was originally entered, actual customer orders did not exist. As actual orders materialize, you can have them deducted from the forecast to avoid overstating your actual requirements.
- Erasing the forecast means you can delete forecasts within a specified date range. As an example, at the end of the month you might want to erase the earliest month within your forecast and then add a new forecast on the end for one more month.
- Rolling over the forecast means that each forecast date will be moved forward one month. If you have a relatively stable, non-seasonal type forecast, this function allows you to move the forecast forward without having to re-enter new forecast records.

Forecasts are reset through MR-C, *Reset Forecast*.

Generating planning lead times

MR-F, *Generate Material Requirements*, uses the *Lead Time* field in each item's master record to establish a start date for each work order and purchase order early enough to meet the required due date.

For purchased parts the lead-time must be maintained manually based on past experience or your current vendor lead-time quotes.

For manufactured items, lead times will vary depending on current conditions in the factory. If you are busy and manufacturing at full capacity, lead times will grow longer. If business is slow and you have excess capacity, lead times will be shorter.

Generate Lead Times program

Rather than manually enter all these lead times, you can use SH-N, *Generate Lead Times*, to calculate these for you.

The program calculates the projected total run time days (using the item's *Lot Size* as the quantity), which includes production run time, outside processing lead-time, and any work center queue time. The *Lot Size* represents the typical production quantity for this item. The *Lot Size* can be entered via IN-L-A, *Enter Standard Costs*, or via RO-A, *Enter Routings*.

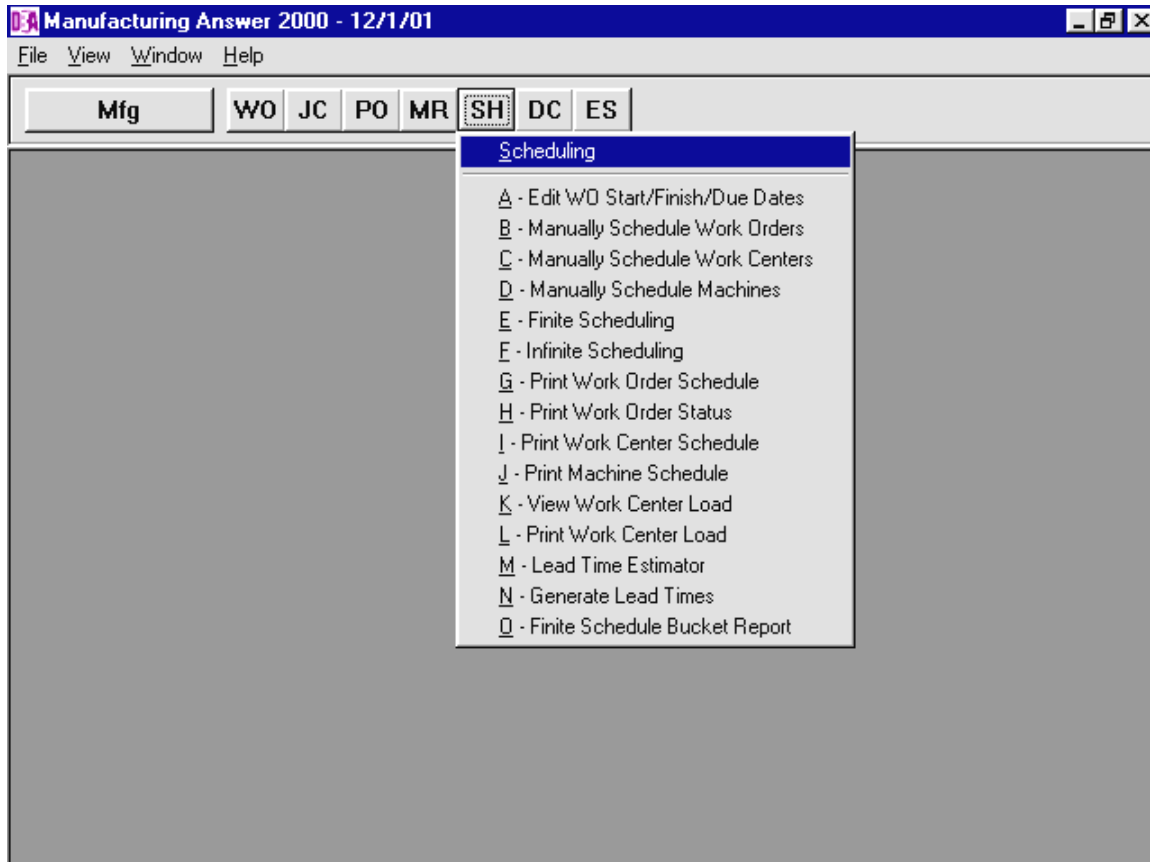
Work center queue time represents a guess as to how much contention (waiting time) there may be at each work center. The program uses the *Average Queue Time* field in each work center, which can be entered in RO-C, *Enter Work Centers*.

When the program finishes its calculations it produces a report that lists any lead-time changes that were made.

16. SCHEDULING

- **Scheduling overview**
- **Quoting lead times to customers or prospects**
- **Generating planning lead times**
- **Capacity planning**

Scheduling overview



The *Scheduling* module provides a variety of programs to assist with scheduling work orders through the factory. To use the *Scheduling* module you must use routings and collect and report labor.

Three methods of scheduling are available: *finite scheduling*, *infinite scheduling*, and *manual scheduling*. You cannot mix scheduling methods - you must use one method system-wide. Each of the three methods of scheduling is briefly described as follows.

Finite scheduling

Finite scheduling is the most automatic of the three methods. The only date you assign is a start date when the work order gets created. The finite scheduling program will then prioritize all your work orders based on the number of days from today to the due date divided by remaining production time. It then takes each work order in succession and schedules each of its routing sequences in the next available time slot in each of its work centers. The work order finish date will equal the finish date of the last routing sequence. Work order finish dates are therefore dependent on work center capacity and must contend with other work orders for the capacity that is available. Finite scheduling is generally used when accurate and realistic due dates are needed and where products are complex and

order volume is such that it overwhelms the ability to schedule manually or semi-manually.

Infinite scheduling

Infinite scheduling is semi-manual in the sense that you manually assign a start date and finish date to each work order, then the infinite scheduling program automatically assigns the start and finish dates to the routing sequences. It is called “infinite” scheduling because it assigns routing sequence dates without regard to available capacity in the work centers; in other words, it assumes that there is infinite capacity available. Infinite scheduling is generally used where due dates are relatively inflexible and you adapt your capacity (through overtime or hiring temps, for example) to meet your commitments to customers

Manual scheduling

Manual scheduling means that you manually assign all dates, including the start and finish date of the work order as well as optionally on each routing sequence. Manual scheduling is generally used when relatively few work orders are involved and products are simple. DBA provides a handful of entry programs that make it easy to manually schedule work orders.

For details on how each scheduling method is used, see the next three sections of this chapter: *How Finite Scheduling Works*, *How Infinite Scheduling Works*, and *How Manual Scheduling Works*.

Quoting lead times to customers or prospects

Queue Times	Est Finish	Lead Time
None	00/00/00	0
Priority 1	00/00/00	0
Priority 2	00/00/00	0
Priority 3	00/00/00	0

SH-M, Lead Time Estimator

DBA provides a *Lead Time Estimator* program (SH-M) that helps you quote realistic delivery dates to customers or prospects.

The program calculates the projected total run time days for the quantity entered, which includes production run time, outside processing lead time, and any work center queue time.

Work center queue time

Work center queue time represents a guess as to how much contention (waiting time) there may be at each work center. You can establish different queue times for each work order priority code; in other words in a given work center priority 1 work orders may only have to wait a day, but priority 3 work orders may have to wait 3 days. Work center queue times are entered in RO-C, *Enter Work Centers*.

The program then takes the *Start Date* you have entered, adds the projected total run time days, skips over any non-working days (as defined on the shop calendar), and projects an *Estimated Finish Date* for each work order priority code. For comparison purposes, the program also displays a projected *Finish Date* with no queue times included so that you can see the relative effect of queue time on the projections.

Generating planning lead times

Manufacturing Answer 2000 - 12/1/01

File View Window Help

Mfg WO JC PO MR SH DC ES

SH-M Lead Time Estimator

Part Number

Description

Quantity

Start Date

Queue Times	Est Finish	Lead Time
None	00/00/00	0
Priority 1	00/00/00	0
Priority 2	00/00/00	0
Priority 3	00/00/00	0

Help Lookup Exit

SH-N, Generate Lead Times

MR-F, *Generate Material Requirements*, uses the *Lead Time* field in each item's master record to establish a start date for each work order and purchase order early enough to meet the required due date.

For purchased parts the lead-time must be maintained manually based on past experience or your current vendor lead-time quotes.

For manufactured items, lead times will vary depending on current conditions in the factory. If you are busy and manufacturing at full capacity, lead times will grow longer. If business is slow and you have excess capacity, lead times will be shorter.

Generate Lead Times program

Rather than manually enter all these lead times, you can use SH-N, *Generate Lead Times*, to calculate these for you.

The program calculates the projected total run time days (using the item's *Lot Size* as the quantity), which includes production run time, outside processing lead-time, and any work center queue time. The *Lot Size* represents the typical production quantity for this item. The *Lot Size* can be entered via IN-L-A, *Enter*

Generating planning lead times

Standard Costs, or via RO-A, *Enter Routings*.

Work center queue time represents a guess as to how much contention (waiting time) there may be at each work center. The program uses the *Average Queue Time* field in each work center, which can be entered in RO-C, *Enter Work Centers*.

When the program finishes its calculations it produces a report that lists any lead-time changes that were made.

Capacity planning

Capacity planning is the process of periodically reviewing your manufacturing capacity to make sure you have enough resources -- machinery and labor -- to meet current and projected demand.

DBA offers some simple tools that help with this process. SH-K, *View Work Center Load*, and SH-L, *Print Work Center Load*, both display the load on your work centers, compared with available capacity.

By reviewing work center loads you can quickly see which work centers are bottlenecks and can respond by adding labor with extra manpower (more workers, additional shifts, overtime) or adding more machinery.

If you are using manual or infinite scheduling

If you are using manual scheduling or infinite scheduling and capacity is overloaded or underloaded, you have two courses of action.

One, you can move due dates in or out (and communicate so with your customers) until the overload or underload is corrected.

Two, you can increase or decrease capacity. In the short term this is almost always done with labor, either through overtime or adding temporary labor or by layoffs. Long term you can add machines, floor space, automation, etc. to boost capacity.

If you are using finite scheduling

If you are using finite scheduling your capacity never gets exceeded because the finite capacity scheduling program fits the schedule within your available capacity. As you get busier, lead times will grow longer and longer. Conversely, as you slow down, lead times will get shorter.

One danger in using finite scheduling is that it is easy to get complacent and let the schedule conform to your capacity instead of continually reviewing your capacity and adjusting it to meet your customers' needs.

Long term capacity planning

If you wish to do long term capacity planning, you must enter forecasts through MR-A, *Enter Forecasts*, then run MR-F, *Generate Material Requirements*, then you must generate actual work orders based on its suggestions via MR-I, *Generate Work Orders*.

At that point you will have created loads on your work centers that can be viewed on screen or on a report.

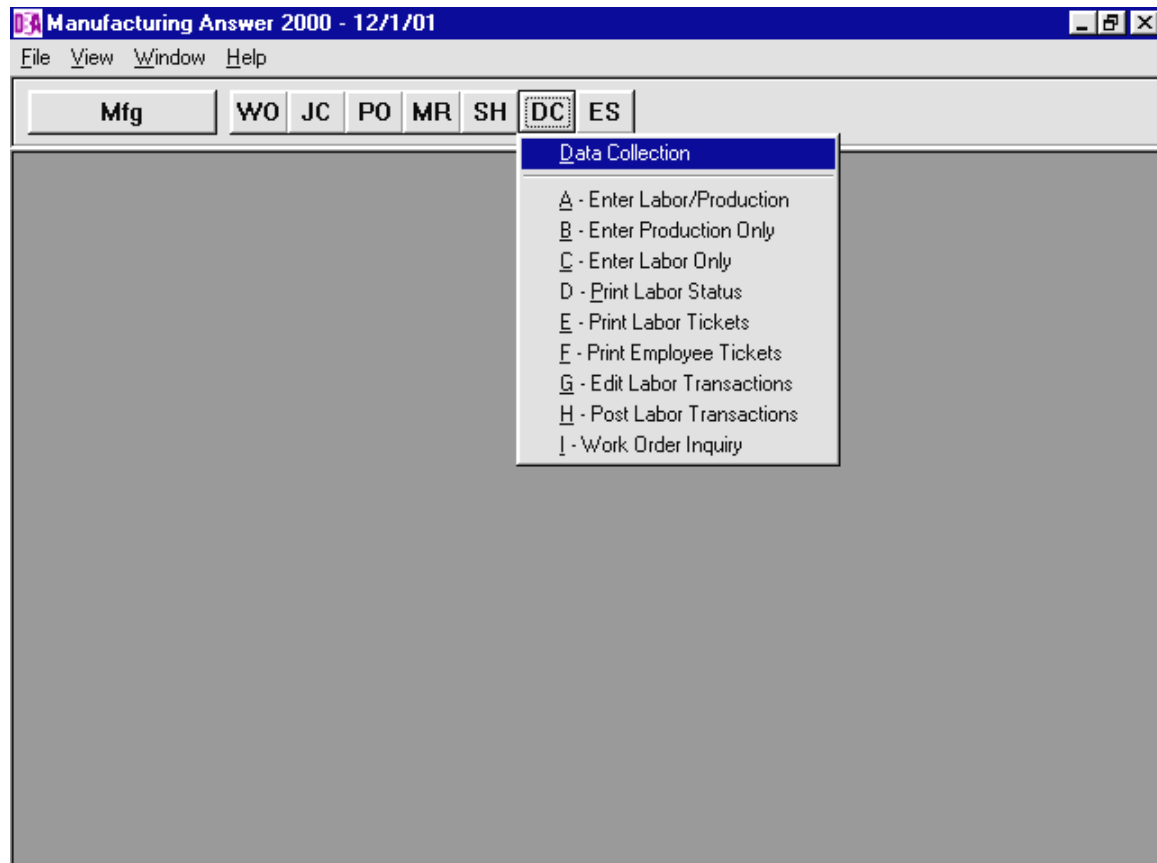
Using planning bills

If you have many products, entering long term forecasts for each would be a tremendous amount of work. One way to shortcut this process is to create

planning bills of material. Planning bills of material are dummy products that consist of families of products. You can create a dummy item number and then create a bill of material consisting of all or a group of your finished goods products. If you will be making monthly forecasts, the "quantity per" for each product would be its projected monthly sales.

Now all you have to do is enter forecasts for the planning bill dummy item number(s). MR-F, *Generate Material Requirements*, will explode this bill into separate bills for all your products and you will get the same effect as if you had entered separate forecasts for all your products.

17. DATA COLLECTION



Overview

The *Data Collection* module allows production employees to report their labor directly to work orders and eliminates the need for after-the-fact labor entry from time cards or labor tickets or labels.

Employees can enter their labor via keypad entry or bar code scanning. Extra computers are located on the shop floor that the employees use as time clocks for clocking in and out of work order routing sequences.

Labor transactions are stored in the data collection transactions file and are eventually batch posted to the work order files. This batch posting can be performed once a day or several times a day, depending on how timely your work order status information needs to be.

You can use one of three data collection modes: labor and production, in which both time and parts counts are recorded, production only, in which time is ignored and only parts counts are recorded, or labor only, in which only time is collected and parts counts are not entered.

Your labor shifts can be defined in SD-F, *Data Collection Defaults*. Buffer

periods can be defined so that employees can clock in before and after shifts begin, yet job costing only occurs when the shift starts and stops when the shift ends. Lunches and breaks can be defined so that job costing stops without requiring employees to clock out.

NOTE: The employee does not designate a shift when using the data collection programs. Instead, each employee is assigned to a shift in SM-G, *Enter Employees*.

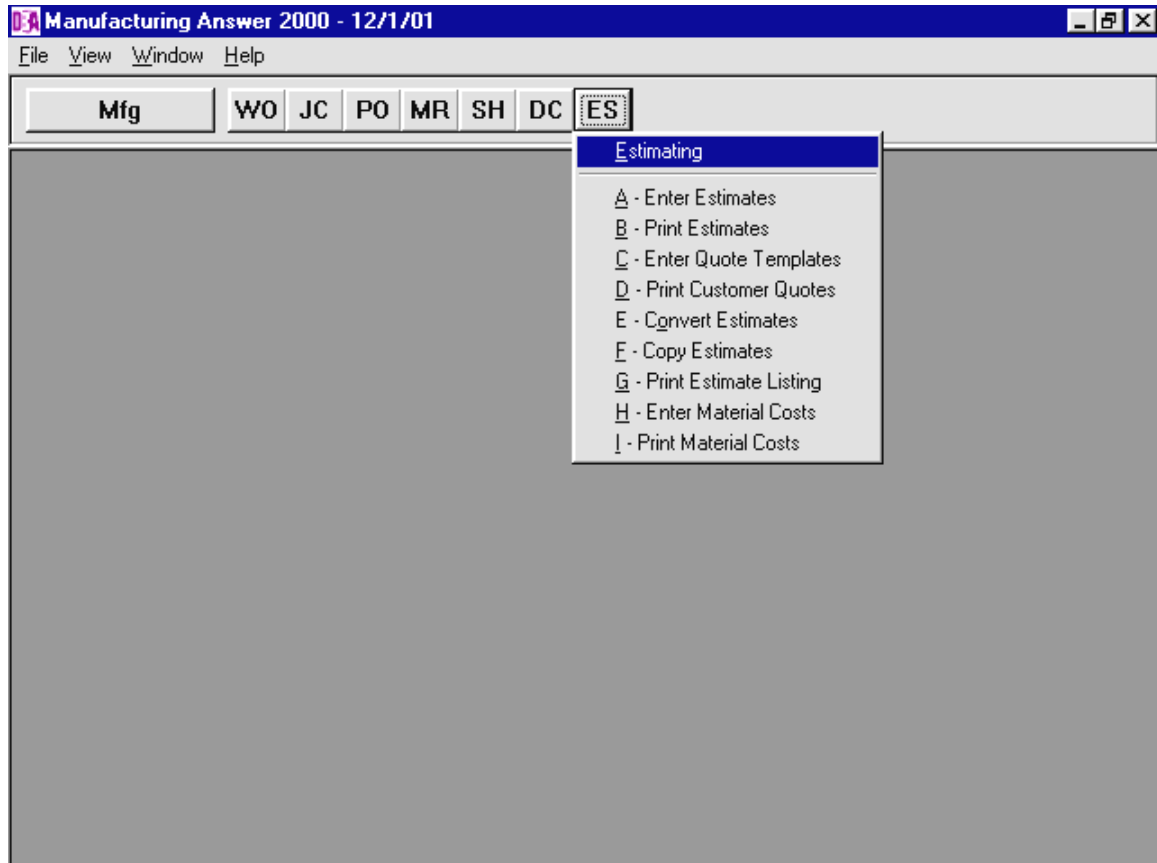
There are three other settings you can make in SD-F, *Data Collection Defaults*. First, you can specify whether you want the data collection entry programs (DC-A, DC-B, DC-C) to operate in full screen mode or in a two-line format used by some data collection terminals. If you using standard PC monitors, you should select full screen mode. The second setting determines whether employees are allowed to enter multiple work orders at the same time or whether they are restricted to working on only one work order at a time. Third, you can turn on Employee shift start/stop that will create a single shift transaction for each employee per day for payroll purposes in addition to the multiple work order transactions. This eliminates the need to use Indirect work orders to collect down time.

Hardware

If you wish to use bar coding, standard PC's can be equipped with "wedge" bar code readers that fit between the keyboard and the PC. The most commonly used wedge readers in factory environments are the wand type. For a supplier of wedge bar code readers, call *American Microsystems*, Euless, Texas at 800-648-4452.

Whatever you use for bar code readers, make sure they can handle a 3-character bar code. Some devices have a 4-character minimum and therefore cannot read DBA's 3-character work order suffix and sequence number.

18. ESTIMATING



Overview

The *Estimating* module is especially useful for job shops that have to bid for orders and is also valuable for manufacturers as a costing and pricing tool.

How job shops use Estimating

Job shops use *Estimating* for two purposes: calculating customer quotes and for setting up jobs.

ES-A, *Enter Estimates*, lets you enter estimated bills of material and routings and lets you apply profit margins to each of your cost categories (labor, material, outside processing, overhead). It can produce detailed and accurate internal estimates and summary type customer quotes on your stationary or on a form.

Many job shop veterans, however, have been quoting for years and don't want to conform to a different system. Even if you continue to do your own estimating, the *Estimating* module still has value as a great tool for setting up jobs, once you have a confirmed customer order.

In one program (ES-A, *Enter Estimates*) you can set up the customers, items, the

bill of material, and a routing. Once the estimate is completed, ES-E, *Convert Estimates*, automatically creates a work order and sales order without any additional entry.

The *Estimating* module is also useful if you ever repeat a job. You can retrieve an estimate, make any changes that may have occurred since the last time you ran the job, and then you can convert it once again into a new work order and new customer order.

Manufacturers use estimates as a costing and pricing tool

You may not directly bid for jobs, but you can still use the estimating program as a costing tool. The program allows you to see your overall cost broken down into separate costs for labor, setup, material, outside processing, and overhead. You can apply profit margins to each of these costs and perform sophisticated costing and margin analyses that can help determine product prices.

Copy Estimates program can be a big time saver

An estimate tied to one item number can be copied to another, meaning that the inventory, bill of material, routing, profit margins, etc. from one item number will all be transferred to the new item number.

If you have products or past jobs that are similar to new ones, you can save tremendous time by using the copy feature and then modifying the new estimate for any changes from the original.

To copy an estimate, use ES-F, *Copy Estimates*.

Converting estimates to work orders and customer orders

When an estimate becomes a firm order you can run ES-E, *Convert Estimates*, to automatically create a work order and sales order.

Many companies take shortcuts when originally entering the estimate, fudging on materials and entering minimal routing information. This is understandable in situations where many bids are given and only a few become actual orders.

When one of these "short cut" estimates become a confirmed order, though, you should take the time to go back into the estimate and add more detail before you convert the estimate to a work order. The bill of material and routing information in your estimate will become the actual work order information that prints on the shop traveler, so be sure to complete the estimate before converting it.

Setting up Estimating

One big advantage to using the *Estimating* module is that it is part of a common database, meaning that all items and costs are in one system and only have to be entered or changed in one place to be used by all programs and users.

Before actual estimates are created, you need to establish costs on materials, labor, setup, overhead, and outside processing. These are created as follows.

Materials

Material costs can optionally be entered in ES-H, *Enter Material Costs*. Use the material costs file if you wish to reflect quantity price breaks in your material costs. Up to five quantity breaks may be specified. The material cost file is only used by the *Estimating* module.

If you issue material from inventory such that there is only one cost and no quantity breaks, you can either use ES-H, *Enter Material Costs*, and only enter a quantity of 1.00 and a single cost, or you can ignore the material cost file altogether and use IN-L-A, *Enter Standard Costs*.

ES-A, *Enter Estimates*, always looks in the material cost file first, and, if no cost is found, it then reverts to the inventory standard cost file.

If you periodically want your *Last Cost* or *Average Cost* to become your new *Standard Cost*, you can run IN-L-E, *Update Material Standard Costs*, which rolls either the *Last* or *Average Cost* into the *Standard Cost* fields for purchase parts only.

Labor

Labor can be costed either through work center labor rates multiplied by production rates established within routings, or by defining labor as an item (item type *L*) and inserting it into the bill of material, using the item's standard cost for the hourly rate.

If you use routings to define each labor sequence, you will be specifying production rates in terms of parts/hour or time/part for each labor sequence. ES-A, *Enter Estimates*, allows different production rates for each quote quantity within the same sequence. The production rates will be multiplied by the labor rate specified in the work center for each labor sequence.

If you define labor in the bill of material, you can create one or a series of item numbers for labor in inventory, assigning each item number an item type *L*. You can enter a standard hourly rate for each labor item number in IN-L-A, *Enter Standard Costs*. Insert labor in bills of material, expressed in terms of decimal hours.

Work center labor rates are entered through RO-C, *Enter Work Centers*.

Standard costs for labor item numbers are entered through IN-L-A, *Enter Standard Costs*.

Setup

For *Setup* costing, routings provide more accuracy than the "light manufacturing" method of defining labor in the bill of material. Each work center can hold a *Setup* rate and setup time can be entered in each labor sequence within ES-A, *Enter Estimates*. If the bill of material approach is preferred, you could create an item number for setup and enter costs in ES-H, *Enter Material Costs*, which would allow different setup costs to be reflected for five different quantity levels.

Overhead

If you use the routings approach, hourly rates for overhead are specified in each work center and are applied to the labor and setup times defined in each labor sequence in the estimate routing. Work centers have special fields for *Estimating Fixed Overhead* and *Estimating Variable Overhead*, thus allowing you to have separate overhead rates for *Estimating* as opposed to the fixed and variable overhead rates used for actual job costing and inventory accounting.

If you use the bill of material approach, you can include an overhead rate within the standard cost for labor as defined in IN-L-A, *Enter Standard Costs*. You do not have to set up separate item numbers for overhead. Enter the labor rate under *Labor* and the overhead rate under either *Fixed Overhead* or *Variable Overhead* or both. Both the labor and overhead rates will be applied to each unit of labor specified in the bill of material.

Outside Processing

If you use the routings approach, outside processing is defined as specific sequences in the estimate routing, where you can specify a vendor, a unit cost, and a minimum cost if applicable.

If you use the bill of material approach, item numbers can be set up for outside processing with an inventory type *T*. You can then set up a standard cost for this item in either IN-L-A, *Enter Standard Costs*, or, if multiple quantity breaks are applicable, you can set up costs in

ES-H, *Enter Material Costs*. Include the item number for outside processing in the bill of material as required.

Estimating defaults

Defaults can be established in SD-G, *Estimating Defaults*, for setting field values in ES-A, *Enter Estimates*. Among the defaults that can be set are the days to expiration date, class, universal form or letterhead, and profit margins for material, labor, outside processing, overhead, and total margin.

19. HOW THE SALES SYSTEM WORKS

Using the sales system is optional

DBA Manufacturing comes with a complete sales system that includes sales orders, invoicing, and tracking of customer payments through Accounts Receivable. While these activities can also be performed in your accounting system, doing them within DBA is more convenient.

To use DBA's planning system effectively, sales orders are needed to compare orders with existing inventory. Invoicing within DBA automatically relieves finished goods inventory. And, tracking customer payments within DBA eliminates having to enter customers and AR details in your accounting system.

Some companies, however, especially job shops, do not have much inventory and only generate a few invoices a day. In this case, you might want to continue generating invoices and tracking customer payments within your accounting system.

To do this, go to AI-A, *Accounting Configuration*, and deactivate Accounts Receivable processing. Then go to PS-D, *Activate/Deactivate Modules*, and deactivate the *Sales Orders*, *Sales Analysis*, *Sales Commissions*, and *EDI Interface* modules. Do your invoicing in your accounting system. To relieve finished goods inventory in accordance with your invoices, do so using IN-C, *Enter Inventory Adjustments*.

What is a sales order system?

DBA uses a sales order system rather than an invoice-only system for entering customer orders.

An invoice-only system is typically used in retail or service situations where an order is entered and immediately invoiced at the time merchandise is purchased or when services are completed.

In manufacturing we use a sales order system instead, which involves a two-step process whereby the order is entered immediately, but usually invoiced at a later time after goods are manufactured.

Even when items are shipped immediately from stock, often there are backordered items that are to be shipped later that remain on the sales order until the order is fully shipped.

The next sections describe the many useful features of a sales order system and how it is used by different types of manufacturers.

Sales order features

Sales orders drive the planning system

The manufacturing planning system is driven by sales orders because they provide the requirements as to which items are needed and when they are needed. The planning system then compares those requirements with available inventory and suggests work orders and purchase orders that will fulfill those requirements.

Scheduled orders handle multiple deliveries

In manufacturing it is not uncommon to receive scheduled orders where a customer wants scheduled deliveries of an item at different dates in the future. DBA's sales order system allows each of these delivery dates to be entered as separate line items on the same sales order. Separate invoices will be produced as each shipment is made.

Multiple invoices can be produced from a single sales order

The sales order system allows you to enter a customer's order on one sales order, even if there are multiple delivery dates and backordered shipments. This provides much better control and tracking than having to enter each delivery or backorder on a separate order. From a single sales order, though, you can produce multiple invoices, one for each shipment.

Backorders are separately tracked

Having items on backorder is common among make-to-stock manufacturers. Backorder amounts are created when some items on a shipment are not in stock and have to get shipped at a later date.

DBA's sales order system holds these backorder amounts in a *Qty Backordered* field, which is listed on the IN-D, *Print Reorder Report*, and tracked via the SO-O-B, *Print Backorder Listing* report. This backorder tracking capability lets you focus on your backorders so you can get them fulfilled as soon as you can.

Instant invoice capability

If you set the *Ready to Invoice?* flag to Y when entering a sales order, the program checks stock on all line items and places all available units in a *Ship Qty* field and all items without stock in a *Backord Qty* field. All lines are flagged by the program as ready to invoice. You can then go directly to the print invoice program and bypass having to release the line items that were shipped.

The instant invoice capability can be useful for point-of-sale situations or for spare parts type orders that are to be shipped out of stock.

Sales order line items can be converted to work orders

SO-N, *Convert Sales Orders to Work Orders*, automatically converts sales order

line items to work orders. The program will run through the sales order line items and will produce a work order for each item that is a finished good, subassembly, or manufactured item. Multiple shipping dates per item can be accommodated.

For build-to-order manufacturers this conversion capability eliminates the tedious task of entering the order twice: once for billing purposes and once for manufacturing purposes.

Sales orders can be used to configure custom products

If you have products with various features and options such that each order is unique, you can use the *Features & Options* module to design bills of material that accommodate multiple options within products. During sales order entry these options are presented in windows for selection. As each option is selected, it feeds into the sales order underneath the parent product. Options may be defined within options, and you can specify option prices to be itemized or to be added to the price of the parent product.

When the sales order gets converted to a work order, the various option components are combined with the items standard components to create a custom work order bill of material.

Selling kits bring in groups of line items with one item number

A selling kit is a group of items that are sold for a single price and are assembled at time of shipping, not in advance through a work order. A selling kit is set up with a special item number (item type *K*) and the kit components are entered as the item's bill of material.

When one enters a selling kit item during sales order entry, the kit components automatically come into the sales order as separate line items that can then be gathered and shipped.

How sales orders are used by different manufacturers

How sales orders are used by job shops

Job shops that build everything to order can theoretically skip entering a sales order and can directly enter a work order and wait until time of shipment to prepare a sales order and invoice. However, it actually involves less work to enter a sales order first and convert it automatically to a work order, and it offers some advantages.

First of all, by entering a sales order first, you already have your invoice "pre-prepared." At time of invoicing all you have to do is release the order and print it instead of entering all the information from scratch.

Second, there may be times when your manufacturing schedule might differ from your shipping schedule. For example, your customer might request scheduled deliveries on certain dates, but you might choose to run the entire job, or certain portions of the job, all at once. You can schedule your work orders separately to conform to your own schedule, but maintain your sales orders schedule the way the customer wants it.

Third, the *Sales Orders* module allows you to use features such as contract pricing and automatic conversion from estimates.

Fourth, if you receive orders for several items all on one purchase order from a customer, they can be combined on one sales order, yet manufactured on separate work orders.

For all these reasons, we recommend that job shops initiate each order with a sales order and use SO-N, *Convert Sales Orders to Work Orders*, to automatically create work orders.

How sales orders are used by make-to-stock manufacturers

Make-to-stock manufacturers attempt to maintain an inventory and ship the majority of orders out of inventory. DBA sales orders can be set up to check available stock as each item is entered and to automatically backorder any amounts that are not available.

When automatic stock checking is used, the program automatically releases each line for invoicing as it is entered and thus allows you to produce an instant invoice.

How sales orders are used by mixed mode companies

Many companies perform a combination of make-to-stock and build-to-order manufacturing. DBA easily handles both requirements.

When entering an order to be shipped from stock, you simply set the *Ready to Invoice?* field to *Y* and the order will perform stock checking and allow instant

invoicing.

For build-to-order situations, set the *Ready to Invoice?* field to *N* and as a second step run SO-N, *Convert Sales Orders to Work Orders*, to convert the sales order items to work orders.

You can even mix and match these modes on the same sales order. Items that are in stock can be automatically selected for invoicing, while selected items that need to be manufactured can be converted to work orders or can be backordered and automatically scheduled for work orders via the planning system.

How sales orders are used by custom manufacturers

Custom manufacturers not only build-to-order, but their products consist of different features and options such that each order is unique.

The *Features & Options* module allows bills of material to be designed to accommodate multiple options within products. During sales order entry options are presented in windows for selection. As each option is selected, it feeds into the sales order underneath the parent product. Options may be defined within options, and you can specify option prices to be itemized or to be added to the price of the parent product.

After the options are selected and the sales order is saved, the sales order can be converted to a work order, during which all the options are passed over to the work order's bill of material where they are combined with the parent product's standard components.

The *Features & Options* capability allow custom bills of material to be built on-the-fly within sales order entry and is a useful tool for custom manufacturers whose products are subject to a variety of options. This capability eliminates complex item numbering, results in more accurate ordering and pricing, and gets orders to the factory floor more quickly.

How the sales system works

The sales process begins with contacts

The DBA sales system includes a *Contact Manager* module that can help you make and manage the contacts with prospective customers that will ultimately lead to actual sales.

The advantage of using DBA's *Contact Manager* module is that it is integrated with the rest of the system so that you don't have to manage two separate systems and don't have to enter data twice when prospects become customers.

Furthermore, the *Contact Manager's* follow-up and history notes capabilities are just as useful with customers as they are for prospects and therefore are best handled within an integrated system.

Using estimates and sales quotes to bid for jobs

Once contacts are made, the next step for many manufacturers and most job shops is to bid for jobs via a quotation process. DBA offers two types of quotes.

The *Estimating* module produces detailed customer quotes using information maintained in the various files of the manufacturing system, including material costs, RFQ's, and work center rates for labor and overhead.

Sales quotations in the *Sales Orders* module produce printed customer quotes, but does not perform any calculations. The quote prices can be entered manually or can be automatically generated from the price code and contract price files.

When quotes generated in the *Estimating* module become firm orders, they can be automatically converted to sales orders via ES-E, *Convert Estimates*. Quotes generated from sales quotations can be converted to sales orders via SO-P-C, *Convert Sales Quotes*.

Entering sales orders

Orders that do not come from converting customer quotes are entered manually through SO-A, *Enter Sales Orders*.

First the header information is entered -- the bill to and ship to accounts, customer PO number, salesperson(s), tax group (if applicable), payment terms, etc.

Then the line items are entered, each with its own estimated ship date (due date). You can enter the same item multiple times, each with its own estimated ship date, if multiple deliveries will be required.

Prices can be automatically assigned by the program based on price codes, discount tables, or contract prices, or you can manually enter prices.

How the *Sales Commissions* module interacts with sales order entry

When you enter a sales order, you can specify one or even two salespersons if you wish to credit them with sales commissions for the order.

The actual commission percentages and amounts can be automatically assigned by the program based on pre-defined rates set up at the salesperson, customer, or item levels, or commissions can be manually entered.

How the *Features & Options* module interacts with sales order entry

If a product is defined with features and options in the *Features & Options* module, during sales order entry its various features and options will automatically be presented in a succession of selection windows. Line items will be created for each option.

How else are sales orders created?

Besides manual entry, sales orders can be created automatically by ES-E, *Convert Estimates*, in the *Estimating* module and by SO-P-C, *Convert Sales Quotes*, in the *Sales Orders* module.

Saving the sales order generates requirements for manufacturing

When a sales order gets saved, the *On Sales Order* and *On Backorder* stock status fields for each of its line items get updated by the order quantities.

The line items, their quantities, and estimated ship dates are automatically reflected in the manufacturing planning system and immediately show up on the IN-D, *Print Reorder Report*, and on the MR-G, *Print Material Requirements*, report the next time MR-F, *Generate Material Requirements*, is run.

The planning system will now compare these new requirements with existing and projected inventory and will suggest the work orders and purchase orders needed to fulfill them.

Acknowledgements can be sent to customers for review

A common practice, especially when complex orders are involved, is to send a printed acknowledgements of sales orders to customers so that orders can be reviewed for accuracy in quantities, pricing, and due dates.

SO-B, *Print Acknowledgements*, will produce acknowledgements in your choice of text or graphical print formats on pre-printed forms or plain paper.

Packing slips can be printed in advance or when needed

Packing slips are ultimately needed at time of shipment for insertion with the goods as a listing of the shipment's contents. A copy is normally kept and sent to the office for use in preparing the customer's invoice.

Some companies prefer to print packing slips in advance and keep them in the shipping department in a file drawer or pinned to display boards for a form of visual scheduling of shipments.

Other companies prefer to plan their daily shipments from reports and then print packing slips on an as needed basis.

When packing slips are printed in advance, they are usually marked by hand with checkmarks against items that ship complete and handwritten amounts for backordered items.

When you print packing slips when needed you can use SO-E, *Release Sales Orders*, to release (mark) the line items that are shipping and thus produce a clean packing slip that is not marked up by hand. Another advantage to using *Release Sales Orders* is that it prepares the orders for invoicing at the same time and eliminates this having to be done as a separate step.

There is no right or wrong way to use packing slips; however, printing them in advance and viewing them on a display board is an older style method that harkens back to using pegboards and other manual scheduling tools.

The *Sales Orders* module offers excellent reports such as the SO-O-F, *Print Available to Ship*, and SO-O-E, *Print Shipping Schedule*, reports that are superior planning tools than shuffling through pre-printed packing slips.

If you choose to release sales orders and print packing slips on an as needed basis, be aware that you will have to equip your shipping department with a computer workstation and a printer.

Shipping labels

SO-D, *Print Shipping Labels*, produces shipping labels that can be affixed to cartons and packages. Like packing slips, shipping labels can be printed in advance or when needed.

One advantage to printing shipping labels on an as needed basis is that you know the carton count and can print the exact number of labels needed. When printing in advance, you don't know the carton count and usually end up wasting labels by printing more than you need.

If you print shipping labels on an as needed basis, be aware that you will have to equip your shipping department with a computer workstation and a printer. Shipping labels should be printed on a dedicated thermal or dot matrix printer using continuous form labels so that you can print one label at a time, if necessary, which is not possible with laser or ink jet printers.

Sales orders can be converted to work orders, if applicable

Build-to-order manufacturers can automatically convert sales orders to work orders using SO-N, *Convert Sales Orders to Work Orders*. This program would be run either immediately after entering the sales order, or perhaps once a day where all the day's orders can be processed at once.

A single sales order can contain multiple line items, but a work order can only be for one item at a time. Therefore, *Convert Sales Orders to Work Orders* will create separate work orders for each manufactured line item.

The program can be run in automatic mode where it will automatically create all the work orders without any review, or it can be run in review mode where each

planned work order is displayed in succession for review and changes, if needed.

Converting items with features and options to work orders

If you are converting line items that have features and options, SO-N, *Convert Sales Orders to Work Orders*, will automatically combine these option components with the item's standard components and will thus create a custom work order bill of material.

Releasing sales orders

SO-E, *Release Sales Orders*, is used for two purposes: to release line items for invoicing and for inclusion on packing slips (please see the discussion above on packing slips). Most companies prepare invoices at the end of each day or at the beginning of the next day from copies of the day's packing slips. The packing slips indicate which items were shipped and what the freight charges were.

Using *Release Sales Orders* you can tag those line items that were shipped complete and you can backorder any items that were shipped incomplete or not at all.

Printing and processing invoices

The printing and processing of invoices completes the process of shipping an order.

It is important to print and process invoices as soon as possible, not only for the cash-flow benefits of getting paid by the customer as early as possible, but because inventory and sales history files do not get updated until you do so.

Invoices are printed through the *Print Invoices* program. As they are printed they are marked as being printed so that they cannot be printed twice by accident.

Invoices are processed through SO-G, *Post Invoices*. This program reduces shipped line items on the sales order, updates the invoice history file, relieves inventory, makes entries in the sales commissions file, and posts entries to the accounts receivable aging file and to the general ledger.

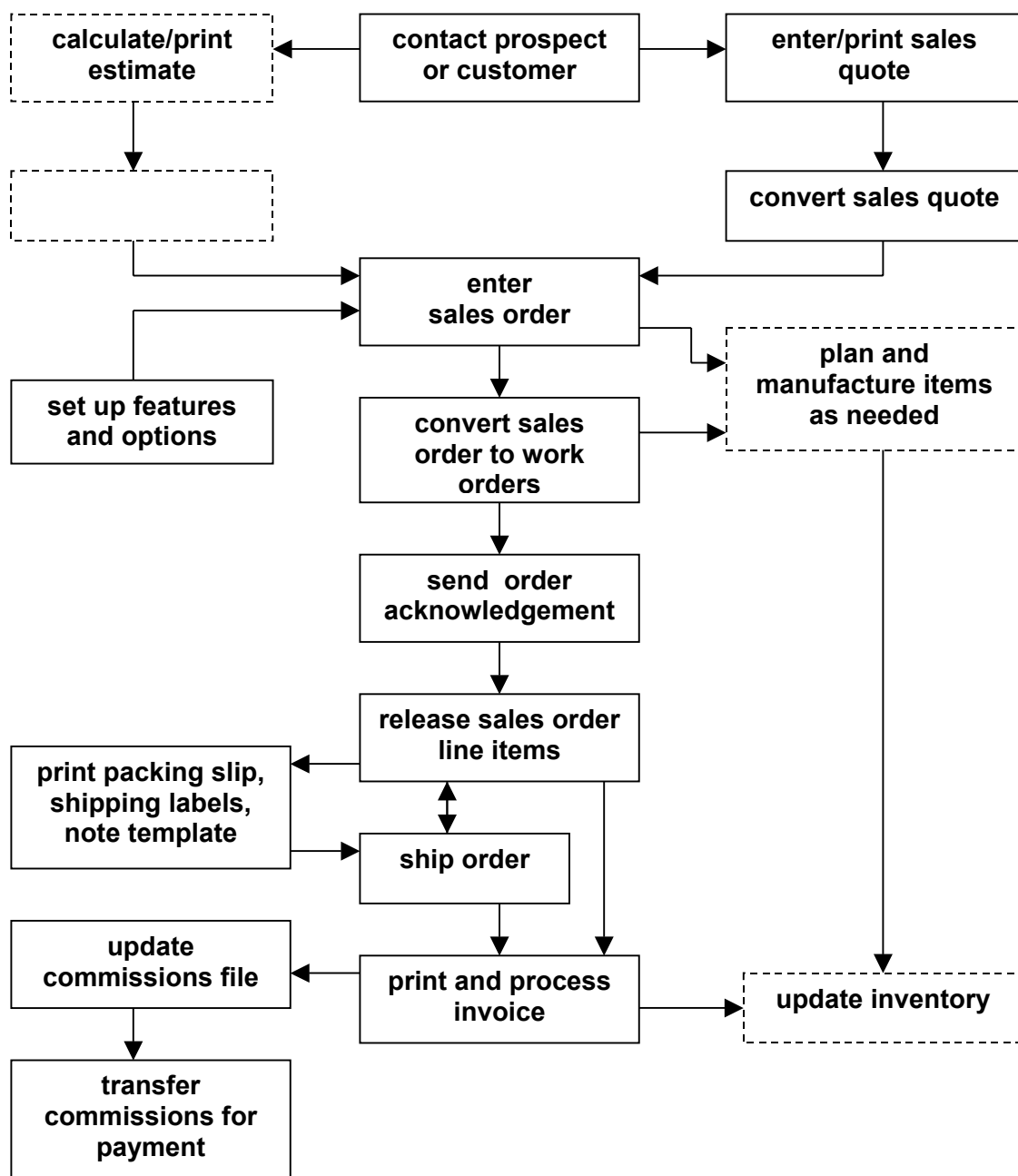
Using note templates for special forms

Some manufacturers must prepare special forms such as certifications that must accompany shipments. You can create your own custom forms through SO-L, *Enter/Print Note Templates*, which combines your customized wording with sales order information, thus saving you from having to type the information from scratch.

Summary

As you can see, the *Contact Manager*, *Sales Orders*, *Sales Commissions*, and *Features & Options* modules all contribute to the sales cycle and interact with one another, as well as with the manufacturing system.

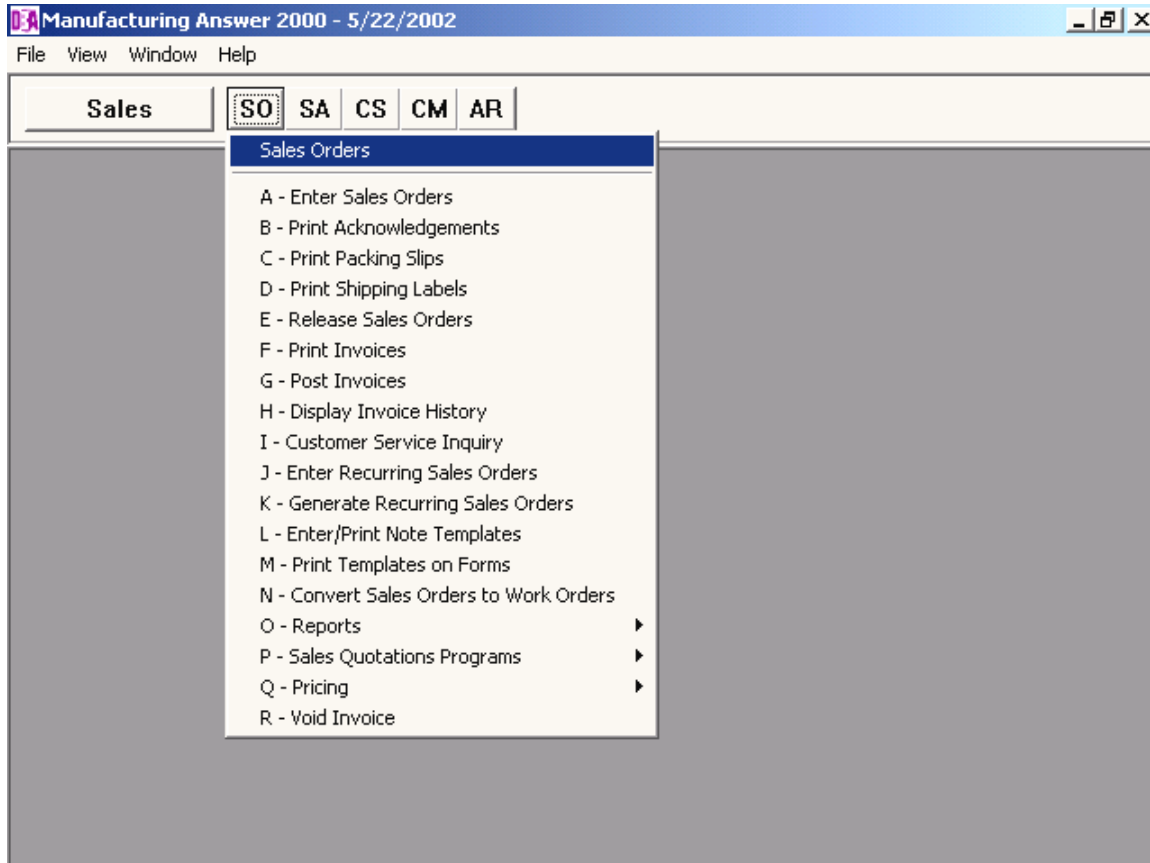
Sales system flow chart



20. SALES ORDERS

- **Introduction**
- **Entering sales orders**
- **Printing acknowledgements**
- **Printing packing slips**
- **Printing shipping labels**
- **Converting sales orders to work orders**
- **Releasing sales orders for printing and invoicing**
- **Printing and processing invoices**
- **Using note templates for special forms**
- **Product pricing**
- **Printing forms and labels**

Introduction



For an overview of the *Sales Orders* module and how it interacts with the rest of DBA, please read chapter 3, *How the Sales System Works*.

The *How the Sales System Works* chapter covers the following topics:

- What is a sales order system?
- Sales order features
- How sales orders are used by different manufacturers
- How to use a sales orders system
- Sales system flow chart

This chapter goes over many of the same topics, but in more detail. We suggest you read the *How the Sales System Works* chapter first and then use this chapter for reference on specific topics.

Entering sales orders

The Manufacturing Answer - 5/15/2002

File View Window Help

Sales **SO** SA CS CM

SO-A Enter Sales Orders - Edit Existing SO

SO No.	6027	SO Date	12/28/01	Ship to		Last Invc#	0
Customer	MSS-01	Entered by		MSS SYSTEMS			
Name	MSS SYSTEMS			18652 Alabama St.			
Address	18652 Alabama St.			Suite 150			
	Suite 150			Costa Mesa		93213	CA
City, St	Costa Mesa		CA	USA			
Zip			93213	John Gault			
Country	USA						
Attention	JOHN GAULT						

Ord Desc		Discont Cd	RETAIL
Cust PO		Slsp1 Cn%	101ULL 10.0000
Location		Slsp2 Cn%	0 0.0000
Terms Cd	1 NET 10	Ship Via	BEST WAY
		Taxable?	Y Rate 7.7500 %
FOB	PLANT	By	CAOR
Job No		GL Dept	
		Price Cd	1
		Ready to Invoice?	

Subtotal	14685.00	Tax	1138.09	Freight	0.00	Total	15823.09
----------	----------	-----	---------	---------	------	-------	----------

Help SO Notes Line Item Screen Save SO Exit SO

SO-A, Enter Sales Orders (header screen)

How are sales orders created?

To manually enter sales orders, use SO-A, *Enter Sales Orders*.

Besides manual entry, sales orders can be created automatically by ES-E, *Convert Estimates*, and SO-P-C, *Convert Sales Quotes*.

Entering a sales order

First the header information is entered -- the bill to and ship to accounts, customer PO number, salesperson(s), tax group (if applicable), payment terms, etc.

Then the line items are entered, each with its own estimated ship date (due date). You can enter the same item multiple times, each with its own estimated ship date, if multiple deliveries will be required.

Prices can be automatically assigned by the program based on price codes, discount tables, or contract prices, or you can manually enter prices. See the next section, *Product pricing*, for more details.

Stock status inquiry

Inventory stock status (on sales order, on backorder, on purchase order, in QC inspection, on work order, allocated) is available from three places during line item entry. While performing a lookup in the *Item Number* field, you can highlight an item and then click on the *Stock Status* button. You can also click on the *Stock Status* button while in the *Ship Qty* and *Backord Qty* fields.

Customer cross-reference capability

If an entry exists in the customer cross-reference file for an item, then the customer's item number and description for this item will automatically come into the sales order as comment lines immediately following the item. In this way both you and the customer see each other's item number and description. You can view these comment lines in the line item display window, which is accessed by clicking on the *Display Lines* button while in the *Lin* field.

To enter customer cross-reference items, use IN-L-C, *Enter Customer Cross-Reference*.

Product specifications

The program will also check to see if an item has specifications defined in its inventory master record. If so, the specifications will automatically display in a window where you are asked *Do you want to include specifications for this part?* You are given the choice of *Yes*, *No*, and *Edit*. If *Yes*, the specifications will come into the order as comment lines and will appear as extra lines of description. If you choose *Edit*, you can edit the specifications on-the-fly (just for this particular sales order) before the specifications are brought into the sales order.

Duplicating lines (for scheduled orders)

DBA accommodates scheduled or blanket orders by allowing you to enter the same item on multiple lines, each with its own estimated ship date. A *Duplicate* function allows you to repeat previously entered line items and comments without having to manually re-enter all the data fields.

After you enter a line item to be duplicated, along with any subsequent comment lines, you can click on the *Duplicate* button. The previous line item's information fields will be duplicated and the cursor moves to the *Est Shp* field. You can change the date and save the newly created line.

Entering sales order notes

You can click the *SO Notes* button from any point when you are in the line item entry area or from any point on the header screen to bring up the notes screen. Notes allow you to type in free form text using the entire width of the screen as an alternative to the 30-character width of comment lines. The notes will optionally print on acknowledgments, packing slips, invoices, and templates following the last line item.

The Manufacturing Answer - 5/15/2002

File View Window Help

Sales SO SA CS CM

SO-A Enter Sales Orders - Edit Existing SO

SO No. 6027 SO Date 12/28/01 Ship to Last Inv# 0
 Customer MSS-01 Entered by MSS SYSTEMS

Original Qty 140.00 Line # 1

Lin	Item Number	Description	Tax?	Y
1	W-100	RED WAGON	Release?	N

Ship Qty	Backord Qty	Price	UM	Disc%	Est Ship
140.00	0.00	99.0000	EA	0.00	01/31/02

Subtotal	14685.00	Tax	1138.09	Freight	0.00	Total	15823.09
----------	----------	-----	---------	---------	------	-------	----------

Display Lines Add Insert Delete Duplicate SO Notes Header Screen

First Last Prev Next Save SO Exit SO

SO-A, Enter Sales Orders (line item entry screen)

Editing sales orders marked as printed

Once an invoice from a sales order has been printed in via *Print Invoices* and has been marked in that program as having been printed, the sales order entry program does not allow changes to that sales order until it has either been posted or un-marked as having been printed. This is a protection against accidental changes being made to a sales order between the time the invoice has been printed and the time it gets posted (DBA allows invoice posting to take place after invoice printing) that could result in the invoice posting disagreeing with the printed invoice that may have already been sent to the customer.

If you attempt to edit a sales order marked as printed, you will receive a message indicating that an invoice from this sales order has been printed but not posted. You are further advised that if you want to edit the sales order anyway, you must reprint the invoice (or list it to the screen) through SO-F, *Print Invoices*, and after printing indicate that you don't want it marked as printed. This process "un-marks" the sales order and allows you to then edit it.

Reopening a closed sales order

A sales order automatically gets marked as closed when all line item quantities are invoiced and posted. Once closed, the sales order is subject to being purged via SM-J-J, *Purge Closed Sales Orders*. You can re-open a closed sales order,

however, if you need to change quantities, add more lines, etc.

Go to the opening screen in SO-A, Enter Sales Orders. Select the order you wish to reopen. You will be asked *This sales order has been closed. Do you want to re-open it?* Click *Yes* or press <Enter> and you can then edit the sales order for any changes or additions. When you save the sales order it will be saved as an open sales order.

Using sales orders as credit memos

The sales order allows entry of negative quantities. This can be useful as a means of producing printed credit memos. If you want the item being credited to be returned to inventory, enter the item's item number and a negative quantity. When the invoice gets posted, inventory will be increased. If you wish to credit a customer but not return inventory, enter a non-inventory item number and modify the description to identify the item. Use the sales order comments capability to describe the reason for the credit memo. When invoices are printed *in Print Invoices* for orders with a negative total, the title of the form gets changed from *INVOICE* to *CREDIT MEMO*.

Absorbing sales tax into line item prices

There are industries such as construction in which sales tax is often not itemized and is absorbed into the line item prices. The tax, however, is still owed and must be paid to the appropriate tax authorities, and the tax amount must be backed out from the sales amount of each line item or else sales will be overstated.

DBA does this type of sales tax processing. In SD-M, *Sales Orders Defaults*, set the *Prompt for Itemized Sales Tax?* switch to *Y*. Now each time you save an order you will be asked *Do you wish to itemize sales tax?* If you answer *N*, no tax will be printed on the invoice. If you answer *Y*, sales tax will be shown in the subtotal section of the invoice in the conventional manner.

The following applies only if you're using the DBA accounting. When the invoice is posted and sales tax is not itemized, instead of each item's sales GL account being credited for the full line item amount, it will be reduced by the amount of sales tax and a separate credit will be made to the default sales tax GL account. The tax authority file will also be updated by the sales tax amount.

Changing sales tax amounts

There are some situations where sales tax is not applied to the entire sales amount of a line item. Part of the price may have a labor content, for example, that is not taxable.

If you wish to have the capability of seeing and changing each line item's taxable amount, set the *Prompt for Taxable Line Item Amount?* switch to *Y* in SD-M, *Sales Orders Defaults*. If so, whenever a taxable line item is entered, the taxable amount (the amount the tax will be applied to) will be displayed in a pop-up window, where it can be manually overridden.

Printing acknowledgements

R Print Preview

50 % 5 Close

ACKNOWLEDGEMENT

Manufacturing Answer 2000
999 Success Avenue
Suite 100
Industry, CA 91789

Sales Order No. 0015
Order Date. 01/02/01
Page. 1
Customer Phone. 714.777.9294
Customer Fax.

B	ADVANCED AUTOMATION 1920 Susan St. Santa Ana, CA 92704 USA ATTN: Francisco d'Antonio	S	ADVANCED AUTOMATION 2000 Industry Road Unit F Cerritos, CA 92545 USA ATTN: Frederick W. Taylor
----------	--	----------	---

Facile. IN	Purchase Order.
Print Format. NLF 10	Ship Via. BEST WAY
Account Ctl. ADV-01	FOB. INDUSTRY
Salesperson. 10	Job Number.

Line	Order Qty	Part / Description	DISC	Price	UM	Extended Price	Est Ship
45	14-100	PSD MAGON Second Description line Comp XL-422 Robot File These are test specifications to show how they feed into various documents.	0.00	\$20,000	EA	\$4,400.00	03/15/01

Page 5 of 5

A common practice, especially when complex orders are involved, is to send a printed acknowledgements of sales orders to customers so that orders can be reviewed for accuracy in quantities, pricing, and due dates.

SO-B, *Print Acknowledgements*, will produce acknowledgements in your choice of text or graphical print formats on pre-printed forms or plain paper.

Printing packing slips

Print Preview

50% 1 Close

PACKING LIST

Manufacturing Answer 2000
999 Success Avenue
Suite 100
Industry, CA 91789

Shipment No. 9999
Ship Date. 03/28/01
Page. 1

B	KISS SYSTEMS 12002 Alabama St. Suite 100 Costa Mesa, CA 92626 USA ATTN: John Gault	S	KISS SYSTEMS 12002 Alabama St. Suite 100 Costa Mesa, CA 92626 USA ATTN: John Gault
---	---	---	---

Sales Order No.	8012	Purchase Order	
Order Date	12/04/00	Ship Via	BEST WAY
Cost Code	KISS-01	FOB	PLANT
Job Number		Salesperson	10

Line	Part Number	Description	Bin Loc	Qty This Shipment	Qty Actually Shipped
11-100		RED WAGON Second Description line These are test specifications to show how they feed into various documents.	A1000	100	

Page 1

Packing slips are ultimately needed at time of shipment for insertion with the goods as a listing of the shipment's contents. A copy is normally kept and sent to the office for use in preparing the customer's invoice.

Some companies prefer to print packing slips in advance and keep them in the shipping department in a file drawer or pinned to display boards for a form of visual scheduling of shipments.

Other companies prefer to plan their daily shipments from reports and then print packing slips on an as needed basis.

When packing slips are printed in advance, they are usually marked by hand with checkmarks against items that ship complete and handwritten amounts for backordered items.

When you print packing slips when needed you can use SO-E, *Release Sales Orders*, to release (mark) the line items that are shipping and thus produce a clean packing slip that is not marked up by hand. Another advantage to using *Release Sales Orders* is that it prepares the orders for invoicing at the same time and eliminates this having to be done as a separate step.

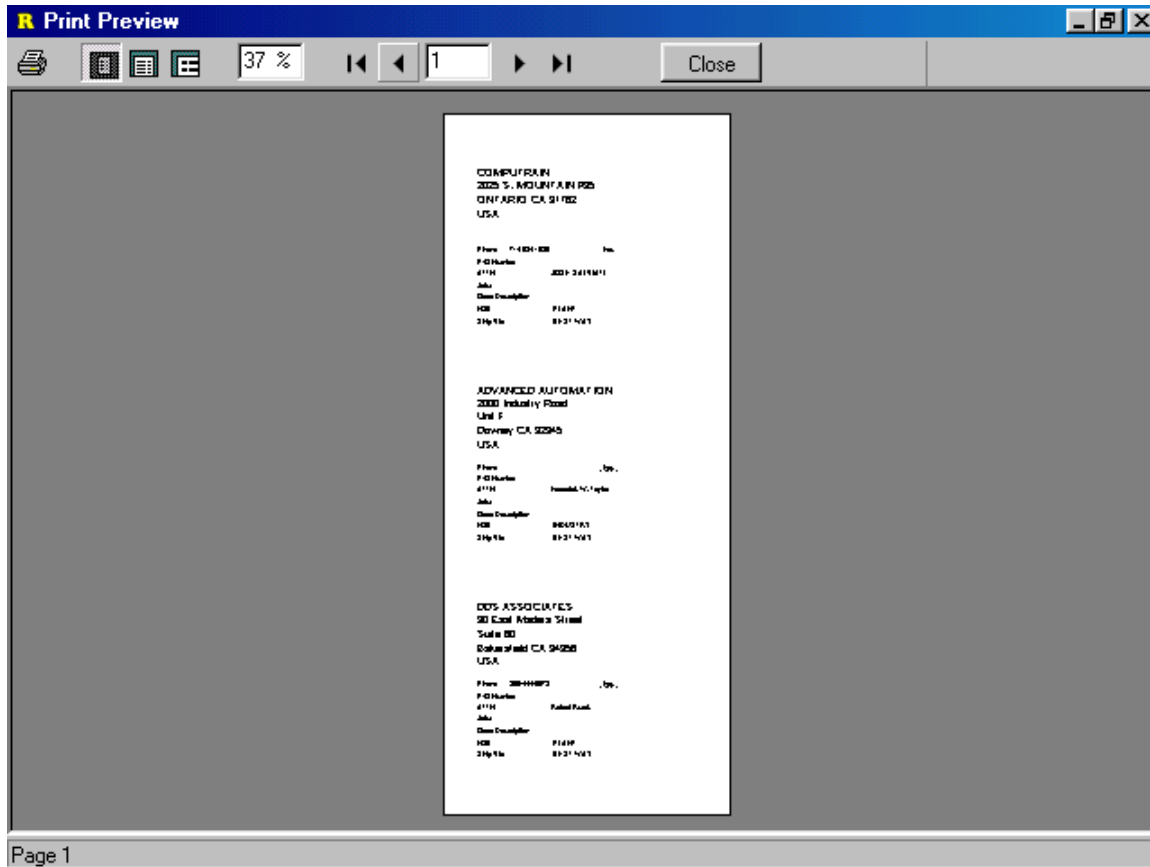
There is no right or wrong way to use packing slips; however, printing them in

advance and viewing them on a display board is an older style method that harkens back to using pegboards and other manual scheduling tools.

The *Sales Orders* module offers excellent reports such as the SO-O-F, *Print Available to Ship*, and SO-O-E, *Print Shipping Schedule*, reports that are superior planning tools than shuffling through pre-printed packing slips.

If you choose to release sales orders and print packing slips on an as needed basis, be aware that you will have to equip your shipping department with a computer workstation and a printer.

Printing shipping labels



SO-D, *Print Shipping Labels*, produces shipping labels that can be affixed to cartons and packages. Like packing slips, shipping labels can be printed in advance or when needed.

One advantage to printing shipping labels on an as needed basis is that you know the carton count and can print the exact number of labels needed. When printing in advance, you don't know the carton count and usually end up wasting labels by printing more than you need.

If you print shipping labels on an as needed basis, be aware that you will have to equip your shipping department with a computer workstation and a printer. Shipping labels should be printed on a dedicated dot matrix or label printer using continuous form labels so that you can print one label at a time, if necessary, which is not possible with laser or ink jet printers.

Converting sales orders to work orders

The Manufacturing Answer - 5/15/2002

File View Window Help

Sales **SO** SA CS CM

SO-N Convert Sales Orders to Work Orders

From Sales Order 6027
Thru Sales Order 6027

Combine Duplicate Items into Single Work Orders? N

Prompt for Work Order Quantity? Y

S/O Num: 6027 Item Num: W-100 RED WAGON

Please Enter a Quantity to Convert: 140.00 Est Ship Date: 01/31/02

On-Hand:	840.00	On Purchase Order:	0.00
On Sales Order:	160.00	On Work Order:	255.00
On Back Order:	190.00	Allocated	0.00
Available:	745.00	In Work-in-Process:	0.00

Increment Work-Order Suffix by? 1

Routing File is created

Exit

SO-N, Convert Sales Orders to Work Orders

Build-to-order manufacturers can automatically convert sales orders to work orders using SO-N, *Convert Sales Orders to Work Orders*. This program would be run either immediately after entering the sales order, or perhaps once a day where all the day's orders can be processed at once.

A single sales order can contain multiple line items, but a work order can only be for one item at a time. Therefore, *Convert Sales Orders to Work Orders* will create separate work orders for each manufactured line item.

The program can be run in automatic mode where it will automatically create all the work orders without any review, or it can be run in review mode where each planned work order is displayed in succession for review and changes, if needed.

Releasing sales orders for printing and invoicing

The Manufacturing Answer - 5/15/2002

File View Window Help

Sales **SO** SA CS CM

SO-E Release Sales Orders

SO No 6015 Cust Code ADU-01 Name ADVANCED AUTOMATION Addr1 1920 Susan St. Addr2 Addr3 Cty/St Santa Ana , CA Zip 92704		Release all Lines? Include Backorders? Release thru Est Date 00/00/00 Display Comment Lines? Auto Release Comments? Display Shipped Lines?	
Customer PO N Release? N Taxable? SO Date 01/05/01 Last Invc 10013 Slsp 10 GL Dept Loca Terms NET 10		Ship Via BEST WAY Shipper# 0 Ship Dt 00/00/00 Subtot 0.00 Tax 0.00 Freight 0.00 Total 0.00	
		Release line? ☐ Lin Est Ship 00/00/00 Item No / Desc Ship Qty 0.00 Backord Qty 0.00 Unit Price 0.0000 UM Disc% 0.00 Tax Extension 0.00	

Help Exit

SO-E, Release Sales Orders

SO-E, *Release Sales Orders*, is used for two purposes: to release line items for invoicing and for inclusion on packing slips (please see the discussion above on packing slips).

Most companies prepare invoices at the end of each day or at the beginning of the next day from copies of the day's packing slips. The packing slips indicate which items were shipped and what the freight charges were.

Using *Release Sales Orders* you can tag those line items that were shipped complete and you can backorder any items that were shipped incomplete or not at all.

Printing and processing invoices

R Print Preview

50% 1 Close

Manufacturing Answer 2000
999 Success Avenue
Suite 100
Industry, CA 91789

INVOICE
Invoice Number: 10000
Invoice Date: 11/20/00
Page: 1

B COMPUTRAIN
2025 S. MOUNTAIN RD
ONTARIO, CA 91762
USA
ATTN: JOSE SALVATI

S COMPUTRAIN
2025 S. MOUNTAIN RD
ONTARIO, CA 91762
USA
ATTN: JOSE SALVATI

Sales Ord No. 0000	Facable. N	Purchase Order
Order Date. 10/22/00	Print Terms. NET 00	Ship Via. BEST WAY
Account Ctl. COMPAQ	Shipped No. 0	FOB. PLANT
Subsequence. 0	Ship Date. 11/20/00	Job Number.

Line	Qty Shipped	Shipped Date	Part Number/Description	Amount	Price	Unit	Extended Price
	2		0 1000 DESK, OFFICE	0.00	\$500.0000	EA	\$1,000.00
			Options:				
M	2		0 1105 DESKTOP, 4 X 5	0.00	\$0.0000	EA	\$0.00
M	2		0 1106 JAGGED EDGE	0.00	\$0.0000	EA	\$0.00
M	2		0 1210 BLUE COLOR	0.00	\$0.0000	GL	\$0.00
M	2		0 1210 JAGGED EY PANEL, FULL HEIGHT	0.00	\$0.0000	EA	\$0.00
M	2		0 1410 RETURN, RIGHT HAND	0.00	\$0.0000	EA	\$0.00
M	2		0 1510 THREE DRAWER ASSY	0.00	\$0.0000	EA	\$0.00
M	2		0 1810 J. BUREAU, BRASS	0.00	\$0.0000	EA	\$0.00
			End Options				

Page 1

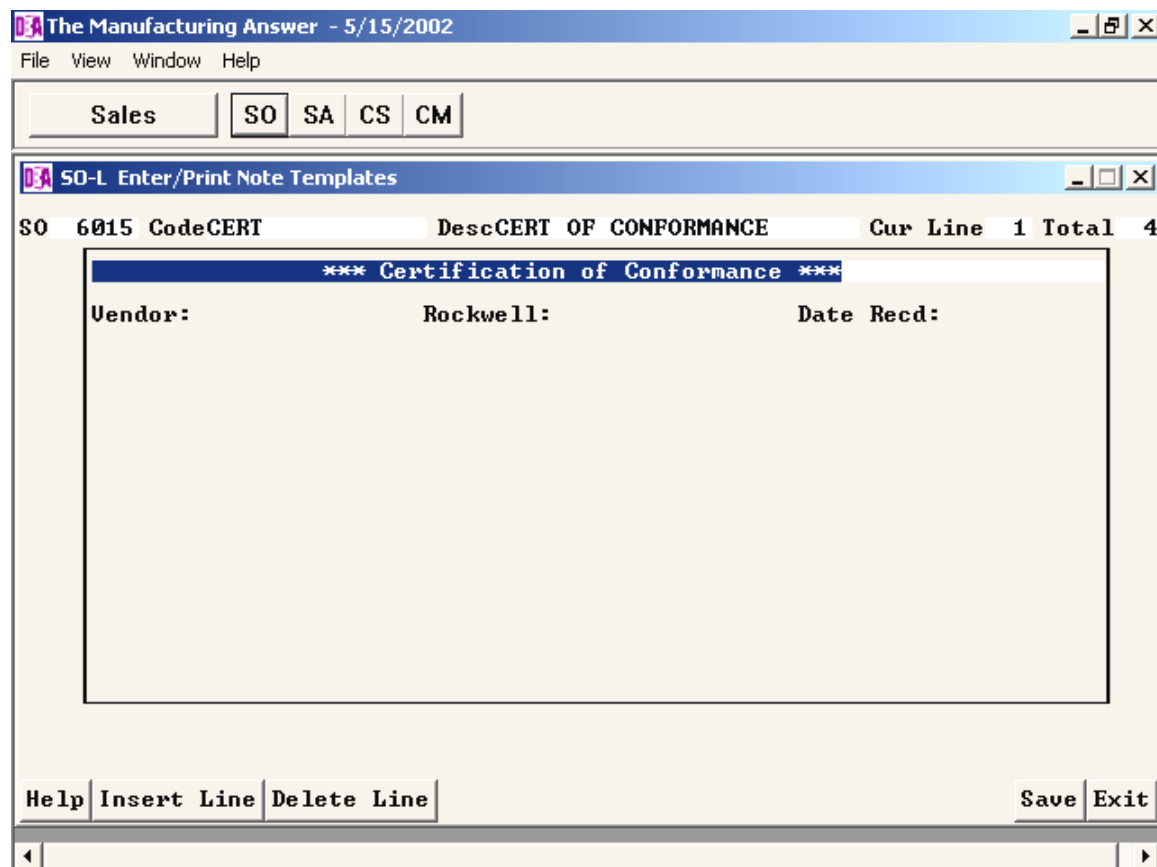
The printing and processing of invoices completes the process of shipping an order.

It is important to print and process invoices as soon as possible, not only for the cash-flow benefits of getting paid by the customer as early as possible, but because inventory and sales history files do not get updated until you do so.

Invoices are printed through the *Print Invoices* program. As they are printed they are marked as being printed so that they cannot be printed twice by accident.

Invoices are processed through the SO-G, *Post Invoices*. This program reduces shipped line items on the sales order, updates the invoice history file, relieves inventory, makes entries in the sales commissions file, and posts entries to the accounts receivable aging file and to the general ledger.

Using note templates for special forms



The screenshot displays the 'SO-L Enter/Print Note Templates' window. At the top, the title bar reads 'The Manufacturing Answer - 5/15/2002'. Below it is a menu bar with 'File', 'View', 'Window', and 'Help'. A toolbar contains buttons for 'Sales', 'SO', 'SA', 'CS', and 'CM'. The main window area shows a form for 'SO 6015 CodeCERT DescCERT OF CONFORMANCE Cur Line 1 Total 4'. The form has a header '*** Certification of Conformance ***' and three fields: 'Vendor:', 'Rockwell:', and 'Date Recd:'. At the bottom of the window, there is a status bar with buttons for 'Help', 'Insert Line', 'Delete Line', 'Save', and 'Exit'.

SO-L, Enter/Print Note Templates

Some manufacturers must prepare special forms such as certifications that must accompany shipments. You can create your own custom forms through SO-L, *Enter/Print Note Templates*, which combines your customized wording with sales order information, thus saving you from having to type the information from scratch.

price code at time of order entry.

The screenshot shows a software window titled "SO-Q-H Enter Contract Prices - Add New Contract Price". The window has a menu bar with "File", "View", "Window", and "Help". Below the menu bar is a toolbar with buttons for "Sales", "SO", "SA", "CS", and "CM". The main area contains a form with the following fields:

- Customer Code: []
- Name: []
- Item Number: []
- Description: []
- Base Price: 0.0000
- Expiration Date: 00/00/00

Below the form is a table with 10 rows and 3 columns:

Quantity	Price	Disc %
0.00	0.0000	0.0000
0.00	0.0000	0.0000
0.00	0.0000	0.0000
0.00	0.0000	0.0000
0.00	0.0000	0.0000
0.00	0.0000	0.0000
0.00	0.0000	0.0000
0.00	0.0000	0.0000
0.00	0.0000	0.0000
0.00	0.0000	0.0000

At the bottom of the window are buttons for "Help", "Lookup", and "Opening Screen".

SO-Q-H, Enter Contract Prices

Think of each price code as a price list. You can establish up to 999 separate price codes (price lists). How are these used? One price list might be for retailers, another for wholesalers, perhaps another for offshore accounts, etc.

Ten quantity price breaks may be defined for each product within each price code.

Price code prices can be entered through SO-Q-D, *Enter Price Codes*.

Base prices

The simplest form of pricing is the use of a base price on each item. The base price is a reference price or "list" price.

If the sales order entry program does not find a contract price or a price code price, it then uses the base price.

If you have a single price list for all your customers and don't offer quantity price breaks, use base prices for all your pricing. Exceptions for specific customers or items can be handled with contract prices.

The base price is also used as a reference by SO-Q-H, *Enter Contract Prices*, and SO-Q-D, *Enter Price Codes*. Prices entered in those programs are displayed as a

percentage discounted from the base price for comparison purposes.

Base prices are entered through SO-Q-A, *Enter Base Prices*.

The screenshot shows a software window titled "The Manufacturing Answer - 5/15/2002". Inside, there's a menu bar with "File", "View", "Window", and "Help". Below that is a main menu with buttons for "Sales", "SO", "SA", "CS", and "CM". The "SO" button is selected, leading to a sub-menu titled "SO-Q-F Enter Discount Codes - Add New Price". The main area of the window displays a table with the following structure:

Discount Code	Inventory Class	Dollar Amount	Disc %
		0.0000	0.0000
		0.0000	0.0000
		0.0000	0.0000
		0.0000	0.0000
		0.0000	0.0000
		0.0000	0.0000
		0.0000	0.0000
		0.0000	0.0000
		0.0000	0.0000
		0.0000	0.0000

At the bottom of the window, there are buttons for "Help" and "Opening Screen".

SO-Q-F, Enter Discount Codes

Global price change

If you want to globally change a range of prices rather than having to change each price individually, use SO-Q-C, *Global Price Change*. The program will change base prices and will optionally change price code prices and contract prices if they are tied to the base price by percentage amounts.

Generating base prices

If you establish your base prices as a markup over cost, you can use SO-Q-J, *Generate Base Prices*, to do this automatically for you.

If existing price code prices and contract prices are tied to the base price by percentage amounts, they can also be optionally updated by the new base prices.

Once base prices are generated, they can be edited through SO-Q-A, *Enter Base Prices*.

Discount codes

If you price your products with discounts, DBA offers discount codes that accommodate a variety of discount schemes.

SO-Q-F, *Enter Discount Codes*, can be used to establish different discount tables by item class. Each customer can be assigned a discount code in addition to a price code. The customer's default discount code can be overridden during sales order entry.

Each item class within a discount code can have its own discounts. This allows you to mix and match different discounts on the same invoice for different classes of product.

As an example, a discount code can be set up called DEALER. Within DEALER we can assign a discount of 50% for an item class consisting of complete systems, 35% for an item class consisting of system elements, and 25% for spare parts.

A customer given a discount code of DEALER can therefore be discounted three different ways, depending on the class of merchandise.

Within each item class up to ten different discounts may be established according to the dollar quantity of the item being purchased.

How to print forms and labels

The following forms and labels are used within the *Sales Orders* module.

- Acknowledgements
- Packing slips
- Shipping labels
- Invoices
- Note templates

With each printed form (acknowledgements, packing slips, invoices, note templates) you have three options: a graphical format on plain paper, a graphical format on a pre-printed form, or a text format on a pre-printed multi-part form.

Use SD-M, *Sales Orders Defaults*, to specify the desired format for each form.

Plain paper printing

With the advent of laser printers, many companies print forms on plain paper rather than pre-printed forms or multi-part forms. The graphical layouts make for nice looking forms and you can use the *Modify Forms* program to insert your company logo or change fonts and fields [*click File-Modify Forms*].

If you need multiple copies, you simply specify the number of copies at time of printing. One downside to this for some companies is that each copy will be the same color, whereas with pre-printed forms each copy can have a different color. Some companies load their laser printer trays with paper that is pre-sorted in different colors, but this takes coordination and care to make sure the colors stay synchronized.

An advantage of plain paper forms over multi-part forms is that they can be printed on high-speed laser printers instead of slower, dot matrix printers.

Graphical format on pre-printed forms

As an alternative to printing on plain paper, some companies prefer to print on pre-printed forms. Usually this is done when companies want color or some type of engraving or other printed effects that can't be achieved with plain paper laser printing.

This is still single sheet printing where the form is loaded into the printer tray and multiple copies are specified at the time of printing. For a multi-part form it is possible to purchase the forms in pre-collated multiple colors.

Text format on multi-part forms

Multi-part forms printed on impact type dot matrix printers are still popular. For this type of printing, DBA offers text formats that provide more consistent alignment and printing speed than graphical formats.

If you will be using text formats, we recommend using a *Generic / Text Only* printer driver (see further below).

Shipping labels

Shipping labels can print in graphical mode only either on laser jet or ink jet printers or on continuous form dot matrix printers.

Installing the *Generic / Text Only* driver for dot matrix printing

The *Generic / Text Only* printer driver bypasses the graphical Windows print drivers and uses the standard 10-character per inch font that comes with your dot matrix printer.

To install the *Generic / Text Only* printer driver, click *Windows Start*, then *Settings*, then *Printers*, then *Add Printer*. After indicating whether the printer is to be a local or network printer, you can then select a *Manufacturer*. Click on *Generic*. In the *Printers* list box you will then see *Generic / Text Only* as the only listing. Click *Next* and finish answering the rest of the prompts. You can give the printer a unique name if you wish, and you can also indicate whether you want the printer to be your default printer or not.

Whenever you print forms, be sure and select the *Generic / Text Only* printer (or whatever name you may have assigned it). If you use your dot matrix printer primarily for forms printing, it is advisable to make it your default printer so that it always displays as your first choice. To make a printer your default printer, go to *Start/Settings/Printers*. Click and highlight your printer icon. While highlighted, click on *File* and then click on *Set as Default* so that it is check marked.

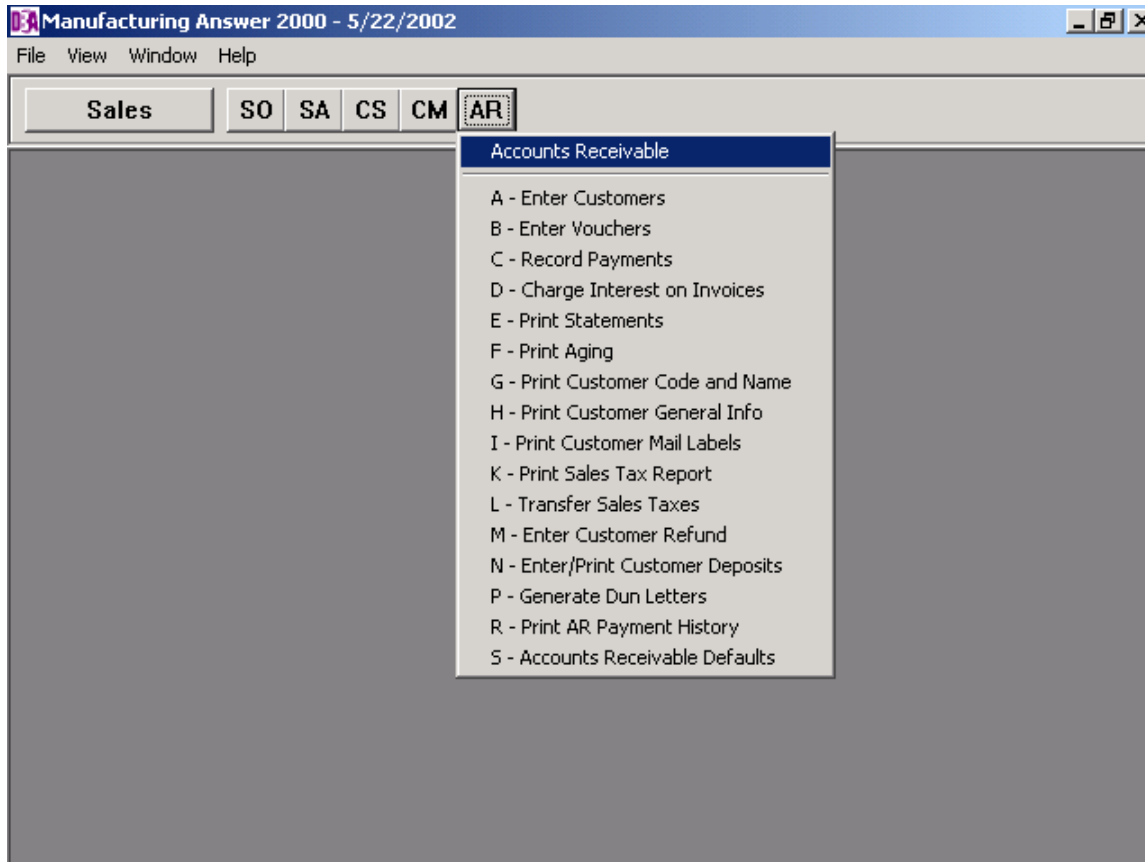
Ordering forms

DBA forms are provided by *Fortune Forms, Inc* (800-884-8804 or www.fortuneforms.com). Contact *Fortune Forms* directly for any questions related to DBA compatible printed products.

21. ACCOUNTS RECEIVABLE

- **Accounts Receivable overview**
- **Accounts Receivable flow chart**
- **Collection tools**
- **Handling customer deposits (prepayments)**
- **Handling customer refunds**
- **Sales tax processing**
- **Sales taxes flow chart**

Accounts Receivable overview



Activities involving the tracking and processing of customer payments are performed in the *Accounts Receivable* module.

See the next page for a flow chart that illustrates how receivables get processed.

As you can see from the flow chart, receivables either originate from invoices or from AR vouchers. Invoices are used for customer orders and credit memos; AR vouchers are generally used only for credit adjustments to accounts where a formal credit memo is not needed.

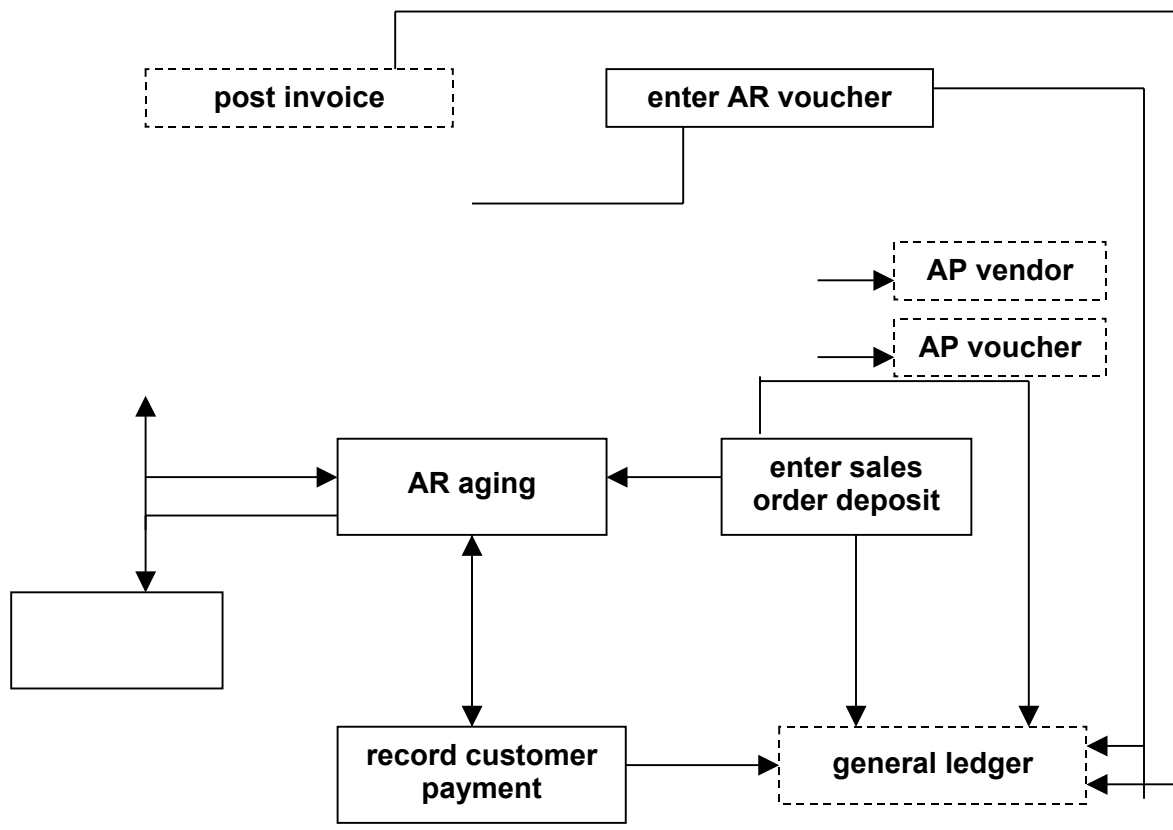
The *Accounts Receivable* module includes various tools for managing the receivables process, including account statements, dun letters, a credit status report, and the ability to use follow-ups within the aging report.

Besides tracking receivables and recording customer payments, you can also enter customer deposits (advance payments) and you can process customer refunds through the accounts payable system.

If you do not want to use the *Accounts Receivable* module

If you prefer to enter your receivables in another system, you can deactivate the *Accounts Receivable* module in AI-A, *Accounting Configuration*.

Accounts receivable flow chart



Collection tools

The *Accounts Receivable* module offers several tools that will help you with the collections process.

Credit assessment reports

Two reports can help you assess the credit worthiness of your customers.

The AR-F, *Print Aging*, report can be run to include past due items only and include either customer totals only or a detail list of overdue invoices.

The AR-R, *Print AR Payment History*, report lists customer payment transactions and can help you assess payment patterns, etc.

Statements of account

Statements of account can be produced and sent to customers as a reminder of payments that are due (AR-E, *Print Statements*).

If you wish, you can charge interest on overdue payments via AR-D, *Charge Interest on Invoices*.

Follow-ups

Follow-up codes and their associated dates and remarks can be entered on customers and optionally printed on the AR-F, *Print AR Aging*, and AR-Q, *Print Credit Status*, reports. Follow-ups serve as reminders to perform certain actions such as to telephone a customer requesting payment or to follow up on a payment promise. Follow-ups are entered off the main customer entry screen in AR-A, *Enter Customers*.

Dun letters

Dun letters can be an effective tool for reminding customers that they are late on their payments. In DBA you can set up a series of progressive dun letters that each correspond with a different range of days that payments are overdue. When dun letters are generated, the program automatically sends each customer the appropriate dun letter. (AR-P, *Generate Dun Letters*)

Credit hold status

Customers can be put on credit hold status as a means of signifying that they should not receive open account credit. This is done through the customer entry program by clicking on the *Credit Info* button and setting the *Credit Hold* field to *Y*. (AR-A, *Enter Customers*) Sales orders can be entered for customers on credit hold; however, you will be warned that they are on credit hold. You will be unable to print packing slips until the credit hold status is removed. This is a precaution designed to prevent accidental shipment to a customer on credit hold. If you need a packing slip anyway, you will have to temporarily remove the credit hold status, print the packing slip, then restore the credit hold status afterwards.

Handling customer deposits (prepayments)

Manufacturing Answer 2000 - 5/22/2002

File View Window Help

Sales SO SA CS CM AR

AR-N Enter New Customer Deposit

Customer Name

Deposit Date 05/22/02

Check Number 0

Open Deposit Amount 0.00

Orig Deposit Amount 0.00

Currency

Bank Account

Link to SO# 0

Description <for AR aging & statements>

Help Lookup Exit

AR-N, Enter/Print Customer Deposits

It is common for job shops and custom manufacturers to require advance deposits before running special orders. DBA handles the processing of customer deposits through AR-N, *Enter/Print Customer Deposits*, which does the following.

First, when you print a sales order acknowledgement or an invoice the deposit is shown in the subtotal section and is deducted from the total order amount. This way the customer clearly sees his deposit amount.

Second, if you are using DBA accounting, the general ledger posting for the deposit is automatically handled correctly. The initial deposit is first credited to your default *Customer Deposits* account, as set up in AD-A, *General Ledger Defaults*. This is a liability account, because for the time being you are holding someone else's money and are therefore liable for it. The offsetting entry is to your cash account.

When the order ships and the invoice is posted, the invoice posting program debits the deposit amount from your *Customer Deposits* account and credits it to your default *Accounts Receivable* account and automatically applies the deposit against the invoice in the aging file.

Handling customer refunds

This program is most useful if your using DBA accounting because it allows you to automatically process customer refunds through the Accounts Payable system.

Enter Customer Refund will actually create a record in the vendor file for the customer so that the refund can be processed through the normal accounts payable system like any other payment.

At time of entry you can choose to create a manual check or an AP voucher that can be selected and paid with the rest of your accounts payable.

If you are not using DBA accounting, this program will create a credit entry in the customer's aging file, but nothing more.

Sales tax processing

Manufacturing Answer 2000 - 5/22/2002

File View Window Help

System Mgr UT SM SD AI IM PS DE

AM-F Enter Tax Codes - Edit Existing Tax Code

Tax Code	Description	Tax ID Number
LA COUNTY	LOS ANGELES COUNTY	

Vendor FRA-01

SO TAXES

Tax Rate
2.125 %

PO TAXES

Tax Rate
0.000 %

THE SYSTEM IS SET TO NOT USE TAX CODES WITH POs.
No entry to the PO Taxes section is allowed here.

Help Save Opening Screen

SM-E, Enter Tax Codes

Sales tax processing in the United States is straightforward and usually involves a single sales tax rate applied to selective line items with a single tax total at the bottom of the invoice. Sometimes, however, multiple tax authorities are involved, such as state and local tax authorities.

In other countries multiple taxes (such as federal taxes, provincial taxes, local taxes, value added taxes, excise taxes) often apply and sometimes must be itemized on the invoice. In some cases, taxes are applied in a hierarchical fashion in which some taxes are “taxes on taxes.”

Setting up tax codes and tax groups

DBA allows you to set up tax codes for each type of tax via SM-E, *Enter Tax Codes*.

You can then group tax codes together via SM-F, *Enter Tax Groups*.

Any number of these tax groupings can be set up and assigned to particular customers and vendors. You can specify whether freight is to be taxed, whether a tax applies to other taxes, and whether the tax is to be itemized and printed out on the invoice.

Manufacturing Answer 2000 - 5/22/2002

File View Window Help

System Mgr UT SM SD AI IM PS DE

AM-G Enter Tax Groups

Group		Description				
CALA		LOS ANGELES COUNTY				
Tax Code	Description	%	ID	On	Frgt	Print
CA	STATE OF CALIFORNIA	6.000		N	N	N
LA COUNTY	LOS ANGELES COUNTY	2.125		N	N	N
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		Total %			Frgt %	
		8.125			0.000	

Help Lookup Exit

SM-F, Enter Tax Groups

Sales tax reporting

Even though tax codes are organized into tax groups, the tax itself is owed and reported to the tax code, which is usually some sort of tax authority such as a state tax board.

All the information needed for reporting sales taxes to tax authorities is found on the AR-K, *Print Sales Tax Report*. When printed in detail mode the report shows the taxable sales, non-taxable sales, sales tax, and freight for each invoice within a specified date range, grouped and subtotaled by tax code.

Transferring sales taxes for payment

If you are using DBA accounting, taxes are transferred for payment in the Accounts Payable system via AR-L, *Transfer Sales Taxes*. If you are not using DBA accounting, you can still use AR-L as a control mechanism to let you know what you've transferred or not transferred to your accounting system in regards to paying the appropriate tax authorities.

The program allows you to transfer all or portions of the total amount due for each tax code. When you do so, your default *Sales Taxes Liable* GL account gets debited and your default *Accounts Payable* account gets credited. AP vouchers are created for the appropriate tax code vendors and the selected sales tax transactions get marked as having been transferred. The AP vouchers can now be

selected and processed for payment through the accounts payable system.

Manufacturing Answer 2000 - 5/22/2002

File View Window Help

Sales SO SA CS CM **AR**

AR-L Transfer Sales Taxes

Tax Code	Description	Tax Rate	Post Date
KERN	KERN COUNTY	1.750	05/22/02

Tax Collected Tag	Code	Date	Number	Amount	Cust/Vend
	KERN	12/22/00	10007	14.88	DDS-01

Outstanding Amount	14.88	Ready to post (Y/N)	N
Amount Tagged	0.00	A/P Invoice Number	0

Tag One Tag Group Tag All Exit

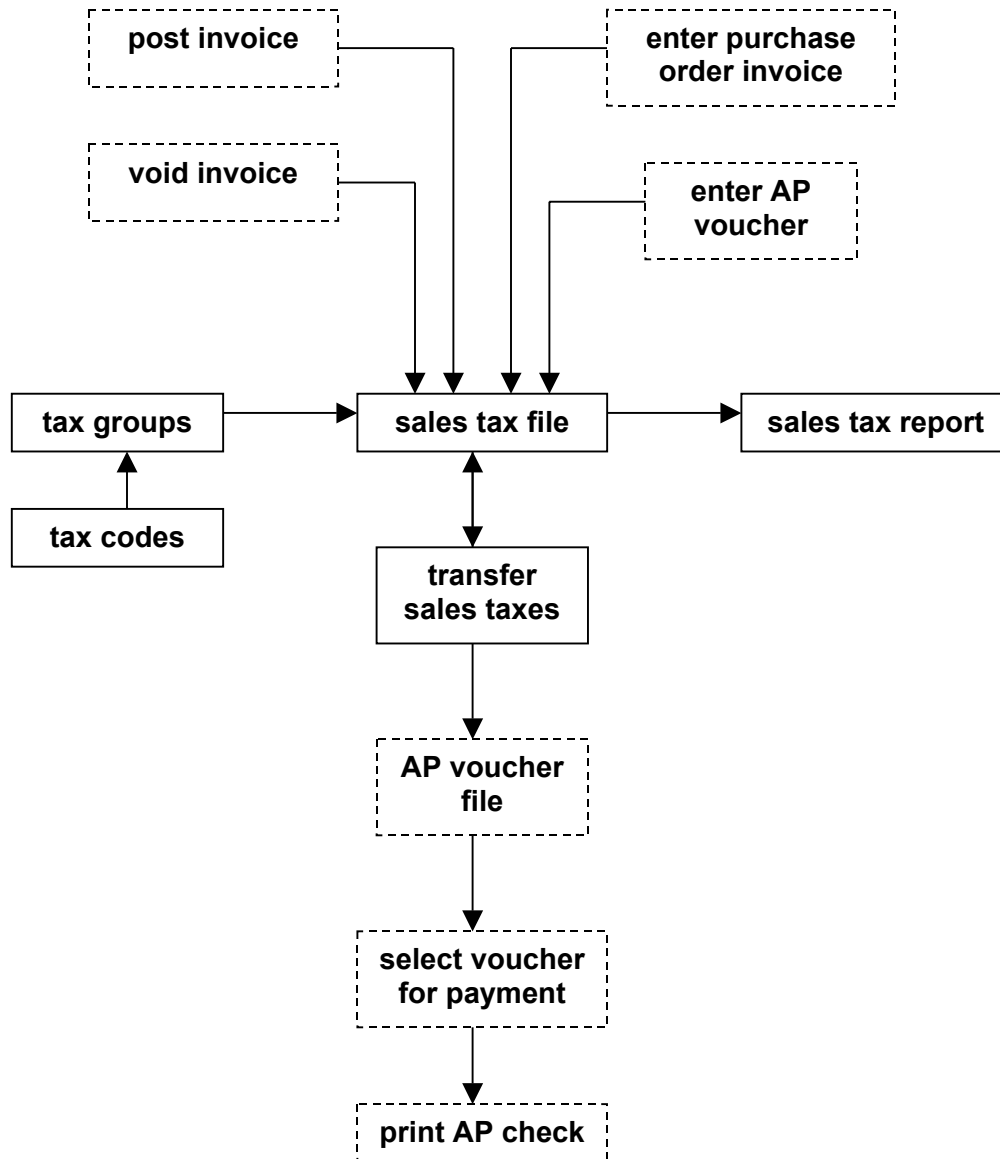
AR-L, Transfer Sales Taxes

Purchase tax processing

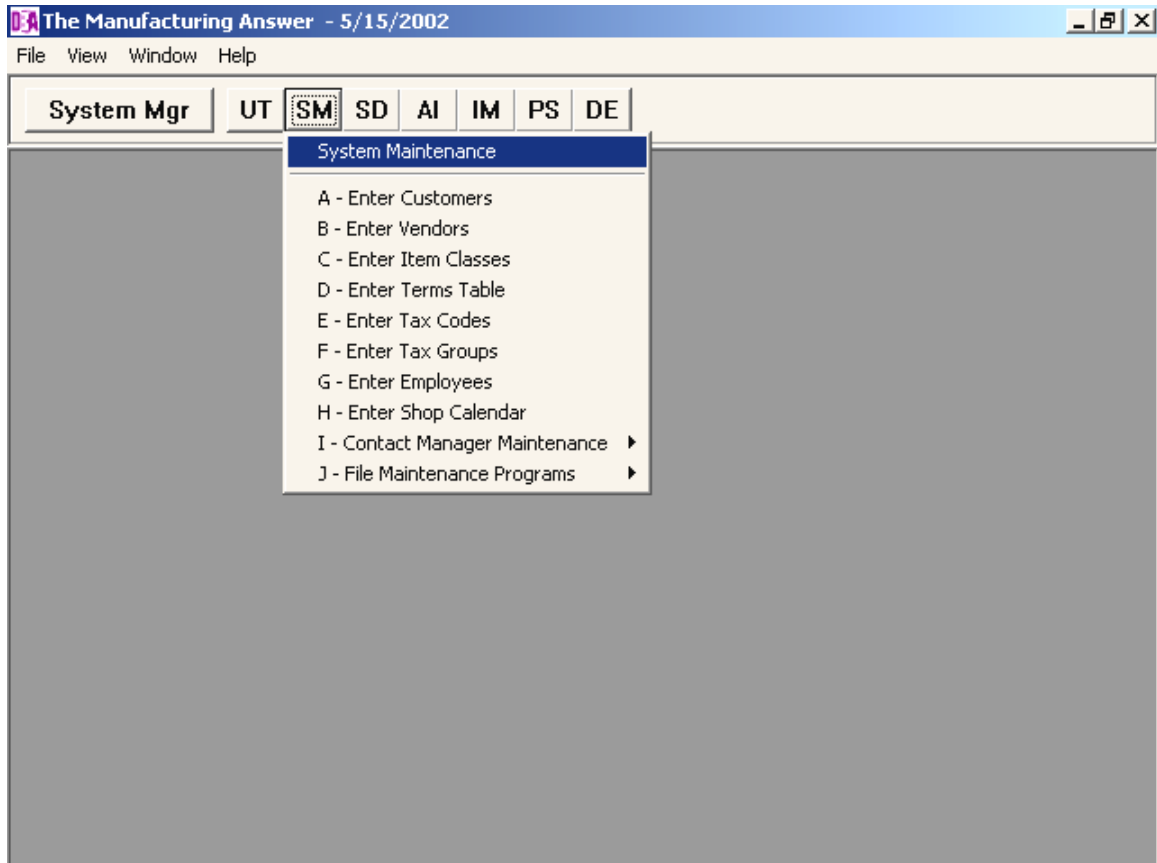
In the United States the collection and reporting of tax on purchases is the responsibility of the vendor, in which case no reporting and processing of taxes on purchases is required. If this is the case, set the *Track PO taxes using tax groups?* prompt to N in LD-B, *Purchase Orders Defaults*.

In some other countries, however, this is not the case and the tax must be accounted for and reported in the same fashion as sales tax. The same tax codes and tax groups used for sales taxes can also be set up for purchase orders in the SM-E, *Enter Tax Codes*, and SM-F, *Enter Tax Groups*, programs described above. Purchase order taxes can also be transferred for payment using AR-L, *Transfer Sales Taxes*.

Sales taxes flow chart



22. SYSTEM MAINTENANCE



The *System Maintenance* menu includes certain maintenance functions that you will perform from time-to-time, as well as options you will use when initially setting up your system. The programs involved with system startup – enter item classes, setting up the terms table, and setting up tax codes and tax groups – are detailed in this chapter.

SM-C Enter Item Classes

If you are using your own accounting system

If you are using your own accounting system, item classes are used to organize your items into meaningful groups for various reports. Each item set up in IN-B, Enter Items, must be assigned to an item class.

If you are using DBA accounting

If you are using DBA accounting, the item class is not only used for grouping items on reports, it also allow you to optionally control GL posting by item class.

System-wide defaults for Inventory, COGS, Taxable Sales, Non-Taxable Sales, WIP, Absorbed Labor, Absorbed Fixed Overhead, and Absorbed Variable Overhead are maintained in AD-A, *General Ledger Defaults*.

If you wish to control posting to these accounts by item class, you can do so in SM-C, *Enter Item Classes*. You are not obligated to specify GL accounts for each item class. Only those that are to differ from the system-wide defaults need to be assigned GL accounts. If GL accounts are not assigned at the item class level, posting programs will use the system-wide defaults.

GL accounts are assigned to item classes for each inventory location you are using. If you have a lot of item classes and/or multiple locations, SM-C, *Enter Item Classes*, offers a variety of options for copying and making mass changes to your item class records without having to do each one individually. See the on-line *Help* file for details.

SM-D Enter Terms Table

The Manufacturing Answer - 5/15/2002

File View Window Help

System Mgr UT **SM** SD AI IM PS DE

AM-E Enter Terms Table

Term Num	Description	Disc Amt	Typ	Days	Max Days til Due
1	NET 10	0.00		10	10
2	AMEX	0.00	A	0	0
3	CASH	0.00	C	0	0
4	NET 15	0.00		0	15
5	NET 30	0.00		0	30
6	NET 60	0.00		0	60
7	NET 90	0.00		0	90
8	NET 120	0.00		0	120
9	2x10,NET30	2.00	%	10	30
10	COD	0.00		0	0
11	LTR/CREDIT	0.00		0	0
12	SPECIAL	0.00		0	160
13		0.00		0	0
14		0.00		0	0
15		0.00		0	0
16		0.00		0	0
17		0.00		0	0
18		0.00		0	0
19		0.00		0	0
20		0.00		0	0

Help Save Exit

Use this program to enter or change the terms that you want to use for payment on invoices. What you enter here will be used when creating vouchers, invoices, purchase orders, or recording payments.

Field explanations

Term Num

You are allowed to have up to 20 different terms types.

NOTE: The terms that you create in the first position of your term choices is the default that will be used for NSF checks, interest charges, PR tax transfers, and voided AP checks. You should use a non-specific term in this position, such as NET 30.

Description

This is the terms specifier that will print in the *Terms* section on invoices, PO's, etc. This is a 10 character alphanumeric field. It is also used as the reference on pop-up menus when you are selecting a terms type.

Disc Amt

The amount of discount allowed if paid within the discount days. This is a 2 digit numeric field.

Typ

If the discount you entered above is a percentage, enter %. If it is a dollar amount, enter \$. If this is a cash type term, then enter a C. When SO-G, *Post Invoices*, encounters a cash item, it posts the amount directly to the default AR cash account instead of creating a receivable for it. If you want to be prompted for which account to use, enter an A (“ask”) for the type. If you enter this type, the invoicing programs will allow you to select from all bank accounts at the time you are posting the invoice.

Days

The number of discount days allowed. The invoice must be paid within this limit to qualify for the discount. This is a 3 character numeric field.

Max Days till Due

The maximum number of days allowed before the invoice is overdue. This is a 3 character numeric field.

SM-E Enter Tax Codes

Manufacturing Answer 2000 - 5/22/2002

File View Window Help

System Mgr UT **SM** SD AI IM PS DE

AM-F Enter Tax Codes - Add New Tax Code

Tax Code	Description	Tax ID Number

Vendor

SO TAXES Tax Rate 0.000 %	PO TAXES Tax Rate 0.000 %
--	--

THE SYSTEM IS SET TO NOT USE TAX CODES WITH POs.
No entry to the PO Taxes section is allowed here.

Help Opening Screen

Use this program to create tax codes for each tax authority to which sales tax is owed on sales and (optionally) on purchases.

Once each tax authority is set up with a tax code, tax codes are then grouped together in SM-F, *Enter Tax Groups*, and thus can handle tax processing on invoices and purchase orders where multiple tax authorities (state, local, etc.) are involved. Taxable customers and vendors are assigned to the tax groups that are created through SM-F.

All of these taxes due can be listed via AR-K, *Print Sales Tax Report*, and can be transferred to payment via AR-L, *Transfer Sales Taxes*.

Sales tax processing

Sales tax processing in the United States is straightforward and usually involves a single sales tax rate applied to selective line items with a single tax total at the bottom of the invoice. Sometimes, however, multiple tax authorities are involved, such as state and local tax authorities.

In other countries multiple taxes (such as federal taxes, provincial taxes, local taxes, value added taxes, excise taxes) often apply and sometimes must be

itemized on the invoice. In some cases, taxes are applied in a hierarchical fashion in which some taxes are “taxes on taxes.”

DBA allows you to set up tax codes for each type of tax via this program and then allows you to group these taxes together via SM-F, *Enter Tax Groups*. Any number of these tax groupings can be set up and assigned to particular customers and vendors. You can specify whether freight is to be taxed, whether a tax applies to other taxes, and whether the tax is to be itemized and printed out on the invoice.

Purchase tax processing

In the United States the collection and reporting of tax on purchases is the responsibility of the vendor, in which case no reporting and processing of taxes on purchases is required. If this is the case, set the *Track PO taxes using tax groups?* prompt to N in SD-C, *Purchase Orders Defaults*.

In some other countries, however, this is not the case and the tax must be accounted for and reported in the same fashion as sales tax. The same type of tax codes and tax groups used for sales taxes can also be set up in SM-E and SM-F for taxes on purchases.

Field explanations

Tax Code

This is the code that identifies the tax code and is used in SM-F, *Enter Tax Groups*, to create tax groups.

Description

The tax code’s description.

Tax Identification Number

This is optional and usually would be for a governmental identification number. This number prints on sales and purchase order documents, if specified so in SM-F, *Enter Tax Groups*.

Vendor Code

This is a valid vendor code the system will use to transfer the accumulated tax amount when processed via AR-L, *Transfer Sales Taxes*.

Percent %

The tax percentage used in calculating the amount of tax. The tax is based on the sales/purchase order line amount.

GL-Account (visible if you are using DBA accounting)

The *General Ledger* account to which the taxes will be posted.

SM-F Enter Tax Groups

Manufacturing Answer 2000 - 5/22/2002

File View Window Help

System Mgr UT **SM** SD AI IM PS DE

AM-G Enter Tax Groups

Group		Description				
Tax Code	Description	%	ID	On	Frgt	Print
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		0.000				
		Total %			Frgt %	
		0.000			0.000	

Help Lookup Exit

Use this program to create *tax group* combinations of the different *tax codes* that were created in SM-E, *Enter Tax Codes*. The *tax group* determines the total tax that will be charged to each customer or by each vendor.

A *tax group* can consist of just one *tax code*, if only one *tax code* applies to particular customers or vendors.

Customers are assigned to tax groups via SM-A, *Enter Customers*, and vendors are assigned to tax groups via SM-B, *Enter Vendors*.

If you plan on processing taxes on purchase order transactions, be sure and set the *Track PO taxes using tax groups?* prompt to Y in SD-C, *Purchase Order Defaults*.

Field explanations

Group

The code you will use to identify the *tax group*.

Description

The description of the *tax group*.

Tax Code

This is the tax code that was created in SM-E, *Enter Tax Codes*. Group the *tax codes* in the order that applies to this particular grouping. A *tax group* can consist of a single *tax code* or up to nine *tax codes*.

Description

The description of the tax code will automatically be displayed in this field.

% Rate

The tax code rate as set up in SM-E, *Enter Tax Codes*, will automatically display in this field.

I.D.

The identification number of the tax code will be automatically displayed in this field.

On

Set this field to Y if this tax is to be calculated on preceding tax code tax amounts. If the tax is independent of the previous tax code amounts, enter an N in this field.

Freight

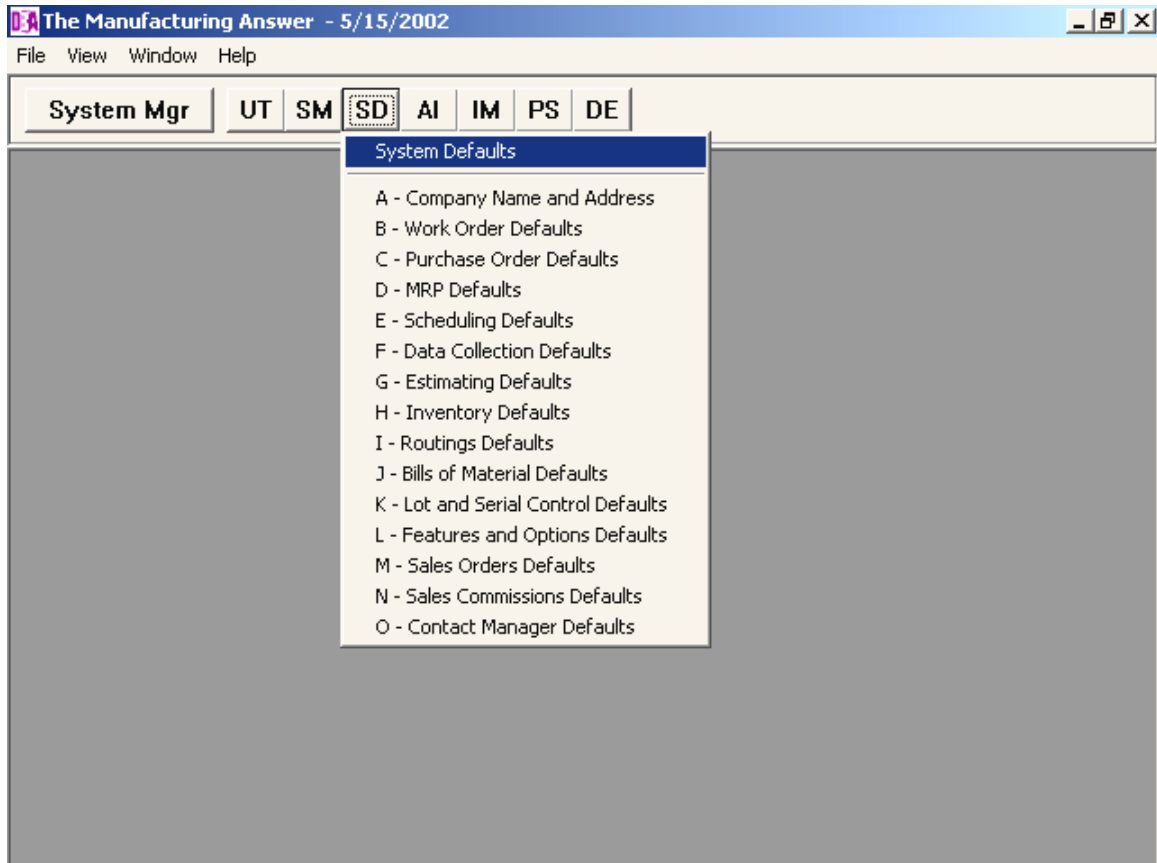
If the tax is to be calculated on freight, enter a Y in this field.

Print

If set to Y, the *Identification Number* entered in SM-E, *Enter Tax Codes*, will print to the left of the tax code on invoices, acknowledgements, sales quotations, purchase orders, and RFQ's.

In order for this feature to work, *Multi-Tax Forms* in IM-A, *International Configuration*, must be set to Y.

23. SYSTEM DEFAULTS



The *System Defaults* menu consists of a number of programs that provide default settings for various DBA Manufacturing modules.

DBA offers many features throughout, some of which are not applicable to every company. Through default settings you can turn various features on or off so that you can tailor the system to your needs. In this manner DBA can be kept very simple or it can offer sophisticated functions and features.

You should carefully go through all the default settings prior to using the system.

SD-A Company Name and Address

Use this program to enter the name of the company that will appear on the system main menus, on various reports, and the address that will print with the name on plain paper format purchase orders, acknowledgments, packing lists, invoices, statements, and quotations.

If your license expires

If you allow your DBA software license to expire, the information on this screen reverts to the following each time a user logs into the system.

LICENSE EXPIRED-See SD-A
Without a current license
this message will restore
itself w/each user logon.

If you let your license expire, you do not have the legal right to use the product, even though the product remains fully functional. It may be the case that you let your license expire via an oversight and the expiration is only temporary. If so, you can go to this program and change your name and address back to what it was, if needed for printing forms, financial statements, or anything else that uses the information on this screen. Be aware, however, that as soon as any user logs into the system, the expiration message will restore itself. Once you install a valid license, you need to use SD-A to re-enter the name and address information, it does not automatically reset when the new license is installed.

SD-B Work Orders Defaults

Field explanations

Work Order Status Code

Each work order is assigned a status code when it is created. Status code *F* is the most common default setting. The values are:

S = Scheduled. The work order is scheduled for the future and no labor and materials have been allocated.

F = Firmed. Labor and material are allocated.

R = Released. Labor and material are allocated and the work order is released to the shop floor with an actual start date.

Default Priority Code [1/2/3]

Each work order is given a priority code for scheduling purposes. Open work orders will be sorted by scheduled finish date within priority code. This field sets the default priority code. Priority code 2 is the most commonly used default.

Default Class Code

This is a user-defined classification code for any purpose you may have. It is used as a filter on various reports. It is a one character alphanumeric field.

Next Work Order Number

This is the next work order number that will be assigned by the system. This number increments each time a new work order is automatically assigned a number. This number will not increment when work orders are manually assigned numbers out of sequence.

Labor Prompt in Kit Issues?

If you are using the "light manufacturing" method of handling labor, you can include labor item numbers (inventory type L) as part of kit issues in order to charge the work order the standard labor defined in the work order bill of material. If you enter an *N* in this field, labor will automatically be part of kit issues within WO-G, *Issue Materials*. If you enter *Y* to this default, a pop-up window will ask you if you wish to include the labor, giving you the opportunity to answer no and report the labor separately for a simple form of labor job costing.

Backflush in Enter Finished Prod?

If you wish to use backflushing of bill of material components in WO-I, *Enter Finished Production*, enter a *Y* here so that a pop-up window will ask if you want

to backflush the components. If you enter an *N*, the window will not be presented and backflushing cannot be performed.

Close WO in Enter Finished Prod?

Normally work order closing is a separate step performed in WO-J, *Close/Cancel Orders*. You can have a pop-up window present the option of closing the work order during WO-I, *Enter Finished Production*, by entering a *Y* in this field. If you answer *N*, the window will not be presented and you will have to close the work order as a separate step.

View only in WO Bills of Mat?

Set this field to *Y* if you don't want users to have the ability to make changes to the work order bill of material after it is created. They will be able to view bill of material detail, but will not be able to edit the screen.

The following five fields set default switches for WO-C, Print Travelers. These settings can be overridden as shop travelers are selected for printing.

Print BOM Remarks? - Traveler

If *Y*, any remarks attached to bill of material components will print underneath the component in the bill of material section of the shop traveler.

Print BOM Comments? - Traveler

If *Y*, any comments added to the work order bill of material will print underneath the component in the bill of material section of the shop traveler.

Print Job Schedule? - Traveler

If there are multiple work orders sharing the same work order prefix and you want them printed in the header section of the shop traveler, enter a *Y* in this field. These would represent related work orders that are to be completed as a group. If this information is not needed, enter an *N*.

Print Short Form? - Traveler

Enter a *Y* if you want the short format traveler, enter an *N* if you prefer the long format traveler.

Print Bill of Mat? - Traveler

If you don't want the bill of materials printed on the shop traveler, enter an *N* in this field.

Use Std Cost in Ent Fin Prod?

When entering completed production through WO-I, *Enter Finished Production*, you have a choice of putting the items into finished goods inventory at standard cost, or at an automatically calculated or manually entered actual cost. If you

want to always enter finished goods at standard cost, enter a *Y*. If you ever want to use actual costs, enter *N*.

Print Mat in Seqs? - Traveler

If work order bills of material components are tied to specific routing sequences and you want them printed on the shop traveler within the sequences in which they are used or consumed, enter a *Y* in this field.

Print Machine and Tool - Traveler

If you are assigning machines and tools in your routings and want them printed on shop travelers, enter a *Y*. If you don't use machines and tools, enter an *N*.

Print Inspection Fields - Traveler

Optional inspection fields can print on the shop traveler which allow entry of sign-offs for *quantity*, *first article*, *last article*, *accepted*, *rejected*. If you don't want these fields, enter an *N*.

Use Actual Costs in Labor Entry?

If set to *Y*, the system will use labor rates from the employee file for job costing in the *Work Orders* and *Job Costing* modules. If set to *N*, the system will use standard labor rates from the work centers the labor takes place in.

Post Overhead as % of Labor?

If set to *Y*, the overhead rates (fixed and variable overhead) for each work center will be calculated as a percentage of labor (rather than as an hourly rate) and will be applied accordingly to each dollar of labor entered in the *Work Orders* module. If set to *N*, the overhead rates defined in the work centers are hourly rates that will be applied to each hour of labor reported.

Calculate Labor from Bills of Mat?

If during ES-E, *Convert Estimates*, you wish to have bill of material labor item numbers (type L) which are tied to routing sequences totaled and dumped into the work order routing, enter a *Y* in this field. This will cause a pop-up window to be presented within ES-E that will ask whether you want the calculation made or not.

Delete Labor after BOM Calc?

If you entered *N* in the previous default, enter an *N* in this field also. If you entered a *Y*, this default results in a second pop-up window during ES-E, *Convert Estimates*, which will ask if you want to delete the labor item numbers from the bill of material after they have been converted to time within the routing sequences. Generally you would answer *Y* to this default if you had entered *Y* to the previous default.

View only in Enter Routings?

Set this field to *Y* if you don't want users to have the ability to make changes to the work order routing after it is created. They will be able to view routing detail, but will not be able to edit the screen.

Backflush by Sequence in Enter Labor?

If you plan on using the backflush by routing sequence feature in which bill of material components are tied to specific sequences, enter a *Y* in this field and you will be prompted for backflushing of components via a pop-up window during WO-F, *Enter Labor*. If you do not plan on using this feature, enter an *N*.

Divide Labor Cost by # Jobs Worked

If an employee is working two work orders at the same time, by entering a *Y* to this default you can have the employee's labor cost divided in half (or thirds, or fourths, etc.). If you enter *N*, each work order will be charged the full labor rate. This default affects both WO-F, *Enter Labor*, and DC-A, *Enter Labor/Production*, and DC-C, *Enter Labor Only*.

Print Multi Routings? - Traveler

In RO-A, *Enter Routings*, you can specify different sequences to print on different routing numbers. When you print the shop traveler it will produce multiple routings within the same work order number instead of a single routing. If you intend to use this feature, enter a *Y* in this field.

SD-C Purchase Orders Defaults

Field explanations

Print Co. Name/Address on forms?

If you are printing PO's and RFQ's on plain paper or on forms without your name and address printed on them, answer Y to this prompt and your company name and address, as entered through SD-A, *Company Name and Address*, will print in the upper left hand corner of the forms.

Default Entered by

If one person generates the majority of your purchase orders, type in that person's initials. Otherwise, you can leave this field blank. You can override this default when entering a purchase order.

Default Ship Via

Type in the method you normally use for shipping. You can override this default when entering a purchase order.

RFQ DEFAULTS:

RFQ Print Format Number

You are offered four print format choices in a lookup window. Formats 1 and 2 are graphical formats. The format identification number, which is used by UT-G, *Enter Graphical Print Settings*, is located in parenthesis to the right of the description. Formats 3 and 4 are text-based formats normally used for dot matrix printing. Highlight the format you want and click on it or press <Enter>.

Print Title on RFQ?

This option allows you to control whether the forms title ("REQUEST FOR QUOTE") will print or not. Suppressing the title allows the title to be pre-printed so that different forms are readily identifiable when pulled from the shelf to be used.

Next RFQ Number

This shows the number that will be assigned to the next RFQ, which gets updated by an increment of one whenever an RFQ is saved through PO-E, *Enter/Print RFQ's*, or whenever an RFQ record is created in PO-F, *Enter Verbal RFQ's*.

PO RECEIPTS:

Require Pack Slip Info?

This sets a default for this prompt in the header section of PO-C, *Receive*

Purchase Orders.

Receive Into

This sets a default for this prompt in the header section of PO-C, *Receive Purchase Orders*. You are presented with a choice (via a pop-up window) of receiving directly into *Inventory* or into *QC Inspection*.

Receive all Lines?

This sets a default for this prompt in the header section of PO-C, *Receive Purchase Orders*.

Display Comment Lines?

This sets a default for this prompt in the header section of PO-C, *Receive Purchase Orders*.

Display Fully Rec'd Lines?

This sets a default for this prompt in the header section of PO-C, *Receive Purchase Orders*.

Next QC Receiver Number

If you receive to QC inspection, each item gets assigned a *QC Receiver Number*. This field shows the next number that will be used.

PO APPROVED VENDORS:

Force PO to use approved vendors?

This option sets the default for the *Approved Vendors in PO?* field in IN-B, *Enter Inventory*, and also controls how approved vendors are handled during purchase order entry. You are given three choices.

- If you do not wish to use approved vendors, meaning that you can purchase from any vendor without restriction, enter an X in the *Do not check vendor approval status* field.
- If you want a warning when using an unapproved vendor, but with the ability to go ahead and use an unapproved vendor, enter an X in the *Warn if unapproved but allow use* field.
- If you want only approved vendors to be used, enter an X in the *Prohibit use of unapproved vendors* field.

Be aware that any of the above defaults can be overridden at the item number level via the *Approved Vendors in PO?* field in IN-B, *Enter Inventory*.

PURCHASE ORDERS:

PO Print Format Number

You are offered six print format choices in a lookup window. Formats 1-4 are graphical formats. The format identification number, which is used by UT-G, *Enter Graphical Report Settings*, is located in parenthesis to the right of the description. Formats 1 and 3 offer a *Condensed* print format, which prints one line per line item using a small font. Formats 2 and 4 print two lines per line item using a larger, easier-to-read font. Formats 5 and 6 are text-based formats normally used for dot matrix printing. Highlight the format you want and click on it or press <Enter>.

Print Title on PO?

This option allows you to control whether the forms title (“PURCHASE ORDER”) will print or not. Suppressing the title allows the title to be pre-printed so that different forms are readily identifiable when pulled from the shelf to be used.

Next PO Number

This shows the number that will be assigned to the next purchase order. It automatically increases by an increment of one as each new purchase order gets saved. You can reset this number at any time.

Track PO Taxes using Tax Groups

If you need to track incoming sales tax on POs by tax group, enter Y

Default PO Tax Rate

If you do not track incoming sales tax on POs by tax group, enter the tax rate to be used on PO's that are designated taxable.

Item Number for Job Cost Freight

If all the items on a purchase order are for the same work order number and you wish to charge the freight to the work order, enter an *Item Number* in this field to which the freight cost will be charged (you will have to set the item number up in inventory first). The freight is entered and charged to the work order through AP-C, *Enter Purchase Order Invoices*. When freight gets charged to the work order, this item number is added to the work order bill of material and the cost is charged to it.

Copy in customer PO's from Work Orders as comment lines?

If within a purchase order line item you specify a work order number and this default is set to Y, you will be asked if you wish to copy in your customer's purchase order number from the work order to a comment line for this purchased

item, providing a cross-reference. If you do not need such a cross-reference, set this default to *N*.

Invoice PO receipts thru AP?

If you are using a non-DBA accounts payable system, enter an *N*. If so, you will be prompted to close purchase orders when they are fully received. If set to *Y*, you will not be prompted to close purchase orders until they are fully received and invoiced through AP-C, *Enter Purchase Order Invoices*.

Print PO Ending Lines?

If you want your printed purchase orders to include ending lines (see below), then enter *Y* here. If you answer yes, you will see a prompt when you are saving the purchase order that will allow you to choose if you want to print these lines or not. Ending lines can be used for any type of message you want printed in the lower left corner of your purchase orders. If you don't want to use ending lines, enter *N*.

PO Ending lines

You may include up to 5 description lines at the end of any purchase order. If you have answered *Y* to the *Print PO Ending Lines* question above, PO-A, *Enter Purchase Orders*, will then ask if you want to include the lines as each purchase order is saved.

RECEIVERS:

Next PO Receiver Number

Enter the tracking number to be used for the next receipt.

Allow Closed PO's to be reopened?

Enter *Y* or *N* to establish whether closed Purchase Orders can be reopened in PO-D, *View PO Receivers*.

Require the entry of Scrap Codes?

Indicate whether scrap codes will be required when rejecting items in PO-J-C, *Enter Inspection Buyoffs*

Require the entry of QC Codes?

Indicate whether QC codes will be required when accepting items designated "Use as is" in PO-J-C, *Enter Inspection Buyoffs*

Access to Prices on Receivers?

Enter 0 if receiving is to have access to change prices in PO-C, *Receive PO*; enter 1 if receiving can view prices but not change them; enter 2 to remove the price from the receiving screen.

Percentage of Overreceipt allowed?

Enter the percentage of overreceipt to be allowed without a warning message.

SD-D Material Requirements Defaults

Field explanations

Include in MRP Generation?

If set to *Y*, whenever new items are created in IN-B, *Enter Inventory*, the *Include in MRP Generation?* switch will be set to *Y*, which means this inventory item will be included in the MR-F, *Generate Material Requirements*, program's calculations. Almost always this default is set to *Y*. Individual items can be excluded from the MRP Generation through MR-D, *Enter MRP Parameters*.

Expedite Buffer (days)

This field controls the EXPEDITE message issued by the MRP program. An EXPEDITE message is issued whenever enough material is on work order or purchase order, but is scheduled to arrive after it is actually needed. The expedite buffer refers to the number of days within which an order can arrive and be considered late, as opposed to being "pegged" to a later requirement. This default value is assigned whenever new items are created through IN-B, *Enter Inventory*. You can override the default on selected inventory items through MR-D, *Enter MRP Parameters*.

Expedite Sensitivity (days)

This field reduces unnecessary EXPEDITE message issued by the MRP program. If a receipt is expected later than needed but within the expedite sensitivity buffer, no expedite message is issued.

Delay Buffer (days)

This field controls the DELAY message issued by the MRP program. A DELAY message is issued whenever material on work order or purchase order is scheduled to arrive earlier than it is actually needed. The delay buffer refers to the number of days beyond which an early arrival will get a DELAY message. This default value is assigned whenever new items are created through IN-B, *Enter Inventory*. You can override the default on selected inventory items through MR-D, *Enter MRP Parameters*.

Delay Sensitivity (days)

This field reduces unnecessary DELAY message issued by the MRP program. If a receipt is expected sooner than needed but within the delay sensitivity buffer, no delay message is issued.

SD-E Scheduling Defaults

Field explanations

Should PO entry and receipt [PO-A, PO-C], labor entry [WO-F], and Data Collection update the actual start/finish date of sequences?

If you are using finite scheduling, answer *Y* to this prompt. If you do so, in the above programs you will be asked “*Is this sequence now complete?*” when making entries. This enables the finite scheduling program to know when sequences are finished so that they will no longer be scheduled. If you are not using finite scheduling, enter an *N* in this field.

Allow entry to overlap settings in routings?

This default only applies to finite scheduling. If you enter *Y* to this default, when entering routings you will have access to the *forward OVERLAP* field. Some operations require extra hours of non-production before the next sequence can begin. For example, you might have a painting sequence that requires 24 hours of drying before they can move on to the next sequence. If you enter 24 in the *forward OVERLAP* field, the finite scheduling program (SH-E) will add 24 hours before setting the next sequence’s starting date.

Display Machine prompt in Enter Labor?

This default only applies to infinite or manual scheduling. If you are using finite scheduling, you can set this default to *N*. Sometimes machine assignments are changed or not known until the last minute. If you set this field to *Y*, during WO-F, Enter Labor, the default machine assigned to the work order routing will be presented in a pop-up window and can be overridden if the work was performed on a different machine. This keeps machine tracking accurate and avoids having to change machine assignments through SH-D, *Manually Schedule Machines*, before entering labor.

SD-F Data Collection Defaults

Field explanations

Allow clocking in/out on multiple jobs?

If you allow employees to work on two or more routing sequences at the same time, set this field to Y. If set to N, they will be required to clock out of one sequence before clocking into another.

Be aware that on each clock-in or clock-out, all open sequences will be automatically clocked out and clocked back in so that the labor cost can accurately be distributed among the sequences open during each time segment between clock-ins and clock-outs.

The result of this is that there can be multiple transaction records for a single segment of time an employee reports to a particular sequence. Also be aware that when a sequence gets split into several transactions as described above, the parts produced will only be reported to the last transaction.

Use full screen?

If set to Y, all transactions for an employee's shift are displayed in the lower portion of the screen. Displayed will be the work order number, sequence, start time, finish time, posting status, parts produced, parts scrapped, and run time.

If set to N, the program prompts will be confined to a two-line area in the center of the screen and no other information will be displayed.

We recommend using full screen mode unless you have some type of monitor that cannot display the complete screen.

Enable Employee Shift Start/Stop?

If set to Y then in addition to the work order transactions, each employee will have a shift record for each day from the first clock in until the last clock out that can be used for payroll.

Shift Schedule Button (Define Shifts)

Click the *Shift Schedule* button to access a screen where you can define your labor shifts so that labor reported through DC-A, Enter Labor & Production, or DC-C, Enter Labor Only, is correctly calculated for actual working time and overtime.

NOTE: The employee does not designate a shift when using the data collection programs. Instead, each employee is assigned to a shift in SM-G, *Enter Employees*.

Enter the *Shift Name* and give it a description. You may define up to three shifts.

Create a *Buffer* period by defining when the buffer begins and when the actual

shift starts. A buffer is a period within which employees may clock in, but actual job costing will not start until the shift *Start* time. The buffer period eliminates crowding at the data collection terminals at the beginning and end of the shift. Make the buffer period as large as is necessary.

You may define breaks and lunch times. If you do so, the data collection programs will automatically stop posting time for these periods without the employees having to clock in and out.

All time is entered in military time. Three thirty in the afternoon would be entered as *15:30:00*.

Define a buffer period at the end of the shift for clocking out. If an employee clocks out after the buffer period, the data collection programs will consider all his buffer time to have been working time. Define up to three shifts in the same manner. When you are completed, click on the *Save* button (or press F10) to save the record.

SD-G Estimating Defaults

Field explanations

Starting Quote Number

The next available quote number to be used by ES-A, *Enter Estimates*.

Default Status Code

All quotes are assigned a status code. The values are A=Active, C=Converted, I=Inactive, X=Canceled. Normally the default status code will be set to *A*.

Default Class Code

An optional class (classification) code is available to help categorize estimates for reporting and retrieval purposes. You may set a default value through this field. This is a four character alphanumeric field.

Cust Quote Print Format Number

You are offered four print format choices in a lookup window. Formats 1 and 2 are graphical formats. The format identification number, which is used by UT-G, *Enter Graphical Report Settings*, is located in parenthesis to the right of the description. Formats 3 and 4 are text-based formats normally used for dot matrix printing. Highlight the format you want and click on it or press <Enter>.

Print Co. Name/Address on Quote?

Enter *Y* if you want your name and address printed in the upper left corner of the universal form. This would not apply to the letterhead versions.

Print Title on Quote?

This option allows you to control whether the forms title ("QUOTATION") will print or not. Suppressing the title allows the title to be pre-printed so that different forms are readily identifiable when pulled from the shelf to be used.

Num Days to Expiration Date

Enter the default number of days the customer quote is good for. The estimate program will automatically calculate the quote expiration date by adding the number of days to the quotation date.

Material Margin

The default profit margin (not markup) for the material portion of the estimate.

Labor Margin

The default profit margin (not markup) for the labor and setup portion of the

estimate.

Outs Proc Margin

The default profit margin (not markup) for the outside processing portion of the estimate.

Overhead Margin

The default profit margin (not markup) for the overhead portion of the estimate.

Total Margin

The default profit margin (not markup) that will be applied to all costs and profit margins that precede it. The total margin is not applied to miscellaneous and extra costs and charges.

Default Machine for Trim Size

If you plan on using the *Yield Calculator* (see ES-A, *Enter Estimates*, for details), you must specify a default machine for trim size, as defined in RO-D, Enter Machines.

Use Yield Calculator in BOM entry

If you intend to use the *Yield Calculator* during bill of material entry within the *Estimating* program, enter a Y in this field. The *Yield Calculator* allows entry of an X and Y dimension and calculates how much sheet will be required to make a particular part.

SD-H Inventory Defaults

Field explanations

Default Inventory Location

If you have a single inventory *Location* (plant or warehouse), this location is normally left blank so that *Locations* do not have to be entered in sales orders, purchase orders, and work orders. *Location* “blank” is actually a valid *Location*. If you have multiple *Locations*, it is good practice to give your primary *Location* a name so that all *Locations* have names.

Average, FIFO, LIFO Costing? [A,F,L]

This setting determines the method that will be used to calculate the inventory *Average Cost* field whenever inventory is increased with a purchase receipt or adjustment, and it determines the cost that will be used whenever inventory is issued or sold. For a weighted, running average, enter an *A*. For first-in-first-out (FIFO) costing, enter an *F*. For last-in-first-out costing (LIFO), enter an *L*. Consult with your accountant as to which method is best for your company. Once you make a setting in this field, it can only be changed via IN-L-I, *Change Costing Method*.

SD-I Routings Defaults

Field explanations

Multiply or Divide by Num Processes [M,D]

This sets the default for the prompt of the same name within the *#Proc* window found in each of the three routings entry programs. See RO-A, *Enter Routings*, for a detailed explanation of how the number of processes can be divided into or multiplied by the processes/hour.

Use Standard Time [Y,N]

Sets the *Std Time?* default in routings entry. If set to yes, the system will apply standard time to any units reported through WO-F, *Enter Labor*, rather than require actual time to be entered.

Make Seq equal Template Number [Y,N]

If you copy in operation templates, setting this default to *Y* causes the program to assign the routing sequence the same number as the template, wherever possible. Use this if you want sequences and operations to have the same numbering.

Default Seq increment

When copying in multiple operation templates into routings, the program automatically assigns sequence numbers. This field sets the default increment. For example, if you like your routing sequences numbered 10, 20, 30, etc., enter a 10. If you want your numbering 1, 2, 3, etc., enter a 1.

Display Long Time Prompt

Normally time is entered in hours, minutes, and seconds. If you prefer entering decimal time, or if the number of hours per part is ever to exceed 99 hours, or one second per part is not precise enough, you can have an optional decimal field display in a pop-up window that will allow entry of up to 9,999 hours and seven decimal places. Answer *Y* to this prompt if you want the *Long Time* field displayed during routings entry.

SD-J Bills of Material Defaults

Use this program to set defaults for the *Bills of Material* module. All these defaults pertain to the *Seq* field within BM-A, *Enter Bills of Material*.

Components can be tied to the particular sequence within the routing in which they are actually used or consumed. With these defaults you can make entry of the *Seq* field mandatory for selected inventory types so that entry of this field cannot be inadvertently skipped.

Tying bill of material components to specific routing sequences can have two uses. One, materials tied to sequences can optionally print on the shop traveler within the sequence they are used. Two, backflushing of components can be tied to routing sequences as production gets recorded through WO-F, *Enter Labor*. This allows inventory to be backflushed as the work order progresses, rather than having to wait and backflush component inventory when finished goods are completed.

If neither of these features apply to your company, set these defaults to *N*.

Field explanations

Require Sequence Entry - Type N (Non Inventory)?

Enter *Y* if you want the *Seq* field mandatory for type N (non-inventory) items.

Require Sequence Entry - Type L (Labor)?

Enter *Y* if you want the *Seq* field mandatory for type L (labor) inventory items.

Require Sequence Entry - Type T (Out Process)?

Enter *Y* if you want the *Seq* field mandatory for type T (outside processing) inventory items.

Require Sequence Entry - Type R, M, F, A?

Enter *Y* if you want the *Seq* field mandatory for types R (purchased), M (make-from), F (finished goods), and A (subassemblies).

SD-K Lot and Serial Control Defaults

Field explanations

Print Lot Listing after Invoices/Packing Slips?

If you enter *Y* to this default, any time you print a packing slip or invoice that has items with lot control entered, you will be asked if you want to print a lot number listing. If you answer yes, a follow-up page will be printed (repeating the order header section) with a listing of the item numbers, quantities, and corresponding lot numbers.

Print Serial Listing after Invoices/Packing Slips?

If you enter *Y* to this default, any time you print a packing slip or invoice that has items with serial control entered, you will be asked if you want to print a serial number listing. If you answer yes, a follow-up page will be printed (repeating the order header section) with a listing of the item numbers and corresponding serial numbers.

SD-L Features & Options Defaults

Field explanations

Mandatory Feature?

If the majority of features will be mandatory, set this default to *Y*.

Display Duplicate Option prompt?

If you have options such that when one option is chosen, such as color, where all subsequent features containing color would use the same color, you can have the *Duplicate Option* prompt display that presents three rules to determine how you want duplicate options handled. If duplicate options are not a factor for your products, set this to *N* and the duplicate options prompt will not be displayed at all.

Duplicate Option Code?

If you plan on using the duplicate options capability, you can select a default duplicate option code. The values are.

- 1 = Display options - regardless of duplicates.
- 2 = Do not display - add duplicate to order.
- 3 = Do not display - do not add duplicate to order.

For more detailed explanations of these codes, see chapter 11, *Features & Options*.

Manufactured or Kit type?

Enter an *M* if the majority of options are manufactured and are to be passed over to the work order system, *K* if they are kit types that will not be passed over to the work order system.

Include in cost rollup?

If the majority of options are to be included in the *Print/Rollup Standard Costs*, enter a *Y*. Generally only one out of each group of options would be included in the cost rollup, so normally this field would be set to *N*.

Use std customer pricing?

If the majority of your options carry prices that will be maintained in the standard price code and contract price files, enter a *Y*. If option prices are universally applied to all customers and are feature specific, enter an *N*, in which case the option prices are maintained within the options bills of material.

Add price to parent product?

If the option prices are to be itemized on the sales order, enter an *N*. If the option prices are to be added to the price of the parent product, enter a *Y*.

Percentage pricing?

If the majority of your option prices are to be a percentage of the price of the parent product, enter a *Y* for this default. If you do not use percentage pricing, enter an *N*.

Display Option Code prompt?

In a future version an option code will allow each option to be defined in a string of codes on the sales order. This currently has no use and can be set to *N*.

Suppress Option Headers, Footers and Indents?

During entry of options in SO-A, *Enter Sales Orders*, the program outputs comment lines which say *Options selected:* and *Suboptions* and structures the listing of options selected as an indented bill of material. If you want these comments suppressed in order to save order lines, enter a *Y* in this field.

SD-M Sales Orders Defaults

Field explanations

Acknowledgment Print Format No.

You are offered six print format choices in a lookup window. Formats 1-4 are graphical formats. The format identification number, which is used by UT-G, *Enter Graphical Print Settings*, is located in parenthesis to the right of the description. Within the universal form or blank paper options you can choose a *Condensed* print format, which prints one line per line item using a small font, or you can choose a non-condensed format, which prints two lines per line item using a larger, easier-to-read font. Formats 5 and 6 are text-based formats normally used for dot matrix printing. They both print two lines per line item. Highlight the format you want and click on it or press <Enter>.

Packing Slip Print Format Number

You are offered six print format choices in a lookup window. Formats 1-4 are graphical formats. The format identification number, which is used by UT-G, *Enter Graphical Print Settings*, is located in parenthesis to the right of the description. Within the universal form or blank paper options you can choose a *Condensed* print format (which lists separate columns for *Qty Ordered*, *Qty Ship to Date*, *Ship Qty*, and *Backord Qty* and uses a smaller font) or a *Single Qty Format* (which combines the *Ship Qty* and *Backord Qty* into a single *Qty This Shipment* column and uses a larger, easier-to-read font). Formats 5 and 6 are text-based formats normally used for dot matrix printing; only the *Single Qty Format* is available. Highlight the format you want and click on it or press <Enter>.

Invoice Print Format Number

All DBA forms printing programs print either on the universal form (which is an all-purpose form without pre-printed titles, field names, or column headings) or on blank paper. Invoices, however, can be printed on an invoice form for customers who prefer a more formal looking invoice than the universal form. DBA forms are supplied by Fortune Forms, Inc. (see *Forms* in chapter 1).

You are offered nine print format choices in a lookup window. Formats 1-6 are graphical formats. The format identification number, which is used by UT-G, *Enter Graphical Print Settings*, is located in parenthesis to the right of the description. Within the universal form, blank paper, or invoice form options you can choose a *Condensed* print format, which prints one line per line item using a small font, or you can choose a non-condensed format, which prints two lines per line item using a larger, easier-to-read font. Formats 7-9 are text-based formats normally used for dot matrix printing. They both print two lines per line item. Highlight the format you want and click on it or press <Enter>.

Sales Quote Print Format Number

You are offered six print format choices in a lookup window. Formats 1-4 are graphical formats. The format identification number, which is used by UT-G, *Enter Graphical Print Settings*, is located in parenthesis to the right of the description. Within the universal form or blank paper options you can choose a *Condensed* print format, which prints one line per line item using a small font, or you can choose a non-condensed format, which prints two lines per line item using a larger, easier-to-read font. Formats 5 and 6 are text-based formats normally used for dot matrix printing. They both print two lines per line item. Highlight the format you want and click on it or press <Enter>.

Print Co. Name & Address on Forms?

Enter *Y* if you want your company name and address (as defined in SD-A, *Company Name and Address*) to print in the upper left corner on acknowledgments, invoices, packing slips, and sales quotations.

Print Discount Column on Forms?

If you want the *Discount* heading and column to print on forms, enter a *Y*. If your company doesn't offer discounts or if you only want the net discounted price to print on invoices and acknowledgements, enter *N* and the discount heading and column will not print.

Decimalized Quantities on Forms?

DBA features two decimal places on all inventory quantities, including quantities shipped, and backordered on invoices, acknowledgments, packing slips, and sales quotations. If you never need Decimalized quantities on these order documents, enter a *Y* in this field and the decimals will be suppressed on the printouts.

Next Sales Order Number

Sales orders are automatically numbered by the system when saved through SO-A, *Enter Sales Orders*. The next available number is shown here.

Next Invoice Number

The number of the next invoice generated when running SO-F, *Print Invoices*. When setting up DBA for the first time, this number will have a default value of 1. The number is automatically incremented whenever new invoices are printed.

Next Packing Slip Number

Packing slips are printed with a sequential number for tracking purposes and for reference on invoices. Enter the starting number here.

Next Recurring Sales Order No

DBA allows you to set up "templates" of invoices that are sent on a regular basis.

If you have set up recurring invoices in SO-J, *Enter Recurring Sales Orders*, you can convert them into live sales orders through SO-K, *Generate Recurring Sales Orders*. Each template created will increment the recurring invoice number displayed here.

Next Sales Quote Number

Sales quotations are automatically numbered by the system when created through SO-P-A, *Enter Sales Quotations*. The next available number is shown here.

Default Taxable

If any of your sales orders are ever taxable, enter *Y*. If they are never taxable, enter *N*. You can override this default when entering a sales order. The *Taxable?* field in SM-A, *Enter Customers*, overrides this default.

Ready to Ship Default

If you take advantage of the ready-to-ship feature in the sales order system, you will be able to generate instant invoices for any items that have on-hand inventory without having to use SO-E, *Release Sales Orders*. This default sets the *Rdy?* flag in the sales order header. The default can be overridden within sales order entry on any given order.

Release Qtys > On Hand

This setting provides control over whether items can be released in SO-E, *Release Sales Orders*, if there is not sufficient quantity on-hand. Set this field to *0* if you want no control. Set it to *1* if you want a warning, but still allow release. Enter a *2* if you want to prohibit shipping items that are not on-hand.

Salesperson

You can use this field if you have one salesperson that handles a majority of your sales. The salesperson number should be set up in CS-A, *Enter Salespersons*. If you have a specific salesperson associated with a set of customers, you should assign the salesperson number in SM-A, *Enter Customers*. You can override either default salesperson assignment when entering a sales order.

Entered by

If you have one person who enters the majority of your sales orders, enter the name/initials of the person in this field. If several people enter orders, leave this field blank.

Default Ship Via

Type in the method you normally use to ship your orders. You can override this default when entering a sales order. The *Ship Via* field in SM-A, *Enter Customers*, overrides this default.

Default FOB

If all your sales orders use the same FOB designation, enter the default here. This default can be overridden when entering a sales order.

Turn the Credit Limit Message off?

DBA can display a message in the sales order program to notify you if your customer exceeds his credit limit or is on credit hold. The program checks the customer for credit hold as soon as the Bill To customer code is entered and checks the credit limit as each line item is entered. If you don't want to see this message during the sales order entry process, enter *Y* here. If you want to see the message, enter *N*.

Prompt for Taxable Line Item Amt?

If you are in an industry (such as construction) where the tax is not always applied to the full amount of the invoice line item, you can answer *Y* to this prompt and the taxable amount will display in a pop-up window during sales order entry. You can then accept or override the taxable amount. Most users should set this to *N* as you can always mix taxable and nontaxable lines on the same sales order.

Prompt for Itemized Sales Tax?

In some industries (such as construction) the sales tax is often not itemized and is instead included as part of the line item price. The tax must still be tracked and paid, however. If you answer *Y* to this prompt, whenever a sales order is saved you will be asked if you want to itemize sales tax. If you say no, the tax will not be itemized at the bottom of the invoice. When the invoice is posted, the sales amount will be reduced by the amount of the sales tax, and the sales tax liability will be posted to your default sales tax GL account and will post to the tax authority file.

Prompt for Retention Billing?

Retention billing is common in certain industries such as construction. If you answer *Y* to this prompt, whenever you release a sales order for invoicing SO-E, *Release Sales Orders*, you will be asked if you wish to bill for retention. If you indicate yes, you will be asked to enter a percentage. The program will then create a separate sales order for the retention amount and will handle all the GL postings properly. For more information, see SO-E, *Release Sales Orders*.

Retention Part No

If you answer *Y* to the *Prompt for Retention Billing?* prompt, enter an item number that will be used by the system on the retention invoice. This item number should be set up in IN-B, *Enter Inventory*, as a non-inventory item (Type N), and should be given a general description that is appropriate when appearing on invoices.

Suppress Non-Tax Warning Message?

Normally if you do not enter a *Tax Group* on a sales order, the program gives you a warning that alerts you to this fact. Many manufacturing companies do not process sales taxes and find this warning an annoyance. If you enter a Y in this field, the warning will be suppressed.

Print Title on:

This option allows you to control whether the forms title (the main title in the upper right corner of each form) will print on acknowledgments, packing slips, invoices, or sales quotations. Even though the universal form can be used for all DBA forms, some companies print special information (such as terms and conditions) on the back of some forms, some have different numbers of copies on multi-part forms, and some color code different sheets within multi-copy laser forms. Suppressing the title allows the title to be pre-printed so that different forms are readily identifiable when pulled from the shelf to be used.

Allow Entry to Location

This option allows you to control the entry of *Locations* (warehouses) in SO-A, *Enter Sales Orders*. See *Using Multiple Locations* in chapter 8 of the *Manufacturing User Guide* for more information on using *Locations* within DBA. If your company only uses a single *Location*, you should set up a blank (no name) *Default Inventory Location* in SD-H, *Inventory Defaults*. You should then set the default on this screen to *N* so that in SO-A the user will *not* have access to the *Location* field in the sales order header screen. If you use multiple *Locations*, you can set this default to *Y* or *R*. If set to *Y*, the user will have access to the *Location* field in SO-A; however, he can enter a blank (no name) *Location* as well as *Locations* with actual names. If you want to allow access to the *Location* field but require entry of an actual *Location* name (to avoid entry of a blank name by mistake), you can set this default to *R* (for required). When using multiple *Locations* it is advisable to give all *Locations* actual names.

Allow Entry to Department

This setting only applies if you are using DBA accounting. This option allows you to control the entry of GL *Departments* in SO-A, *Enter Sales Orders*. See AM-D, *Enter General Ledger Departments*, for more information on using GL *Departments*. If your company does not use departmental accounting (the majority of companies do not), you should set this default to *N* so that in SO-A the user will *not* have access to the *Department* field in the sales order header screen. If you use departmental accounting, you can set this default to *Y* or *R*. If set to *Y*, the user will have access to the *Department* field in SO-A; however, entry of a department code is not required and the field can be left blank. If set to *R* (for required), the user must enter a department code and cannot leave the field blank.

Print Ending Lines?

If you want to set your default to include ending lines (see below), then enter *Y* here. If you answer 'yes', you will see a prompt when you are saving the sales order that will allow you to choose if you want to print these lines or not. Ending lines can be used for any type of message that you want printed in the lower left corner of the invoice, acknowledgment, or sales quotation. If you don't want to use the ending lines, enter *N*.

Sales Order Ending Lines

You may include up to 5 description lines at the end of any invoice. If you answered *Y* to the *Print Ending Lines?* field above, *SO-A, Enter Sales Orders*, will ask if you want to include these lines on each sales order.

Sales Quote Ending Lines

If you answered *Y* to the *Print Ending Lines?* field above, *SO-P-A, Enter Sales Quotations*, will ask if you want to include these lines on each sales order.

SD-N Sales Commissions Defaults

Field explanations

Enter Commissions at Sales Order Entry?

If you answer yes to this prompt, the salesperson number and commission percentage will be displayed for salesperson 1 and salesperson 2 (if applicable) in a pop-up window while entering the sales order header. You can accept the salespersons and percentages shown or change them on-the-fly. The suggested salesperson will be the default salesperson assigned to the Ship To customer, or if none are assigned to the ship to account, then the salesperson(s) assigned to the bill to customer will be used. If you answer no, the salesperson will be displayed and changes allowed but the system will use the default commission rate assigned to the sales rep and not allow entry or viewing of the field for commission rate. If commission percentages are not assigned at the customer level, the program will use the salesperson percentages defined in CS-A, *Enter Salespersons*.

Enter Commissions at Line Item Entry?

If you want to vary commissions at the line item level and have the ability to override them during sales order entry, enter *Y* to this prompt. As each line item is entered, the default commission will be shown in a pop-up window for salesperson 1 and salesperson 2 (if applicable). Line item commissions can automatically override the default percentages entered in the sales order header on an exception basis if commissions are defined in either CS-K, *Enter Price Code Commissions*, or CS-M, *Enter Contract Commissions*. If you answer *N* to this prompt, any commissions defined in those programs will be automatically used, but if you want the ability to see the commissions and override them, then answer *Y* to this prompt.

Enter Commissions for 2 Salespersons?

If you never split commissions between two salespersons, answer *N* to this prompt; otherwise answer *Y*. When two salesperson fields are displayed in sales order entry, you are not required to use the second salesperson field.

SD-P Accounts Receivable Defaults

Field explanations

Statement Print Format Number

You are offered five print format choices in a lookup window. Formats 1-3 are graphical formats. The format identification number, which is used by UT-G, *Graphical Print Settings*, is located in parenthesis to the right of the description. Highlight the format you want and click on it or press <Enter>.

Print Co. Name/Addr on Statements?

Enter *Y* if you want your company name and address to print in the upper left corner on statements. If you have universal forms with your company name already printed, enter *N*.

Print Title on Statements?

This option allows you to control whether the forms title ("STATEMENT") will print or not. Suppressing the title allows the title to be pre-printed so that different forms are readily identifiable when pulled from the shelf to be used.

Interest Rate on Overdue Accounts

The monthly interest rate percentage to be charged on overdue receivables. For example, if you are charging 18% per year, then this amount would be 1.5. (1.5 percent x 12 months = 18 percent)

#Days Past Due before Interest is Charged

The number of days an invoice must be past due before interest is charged for that customer.

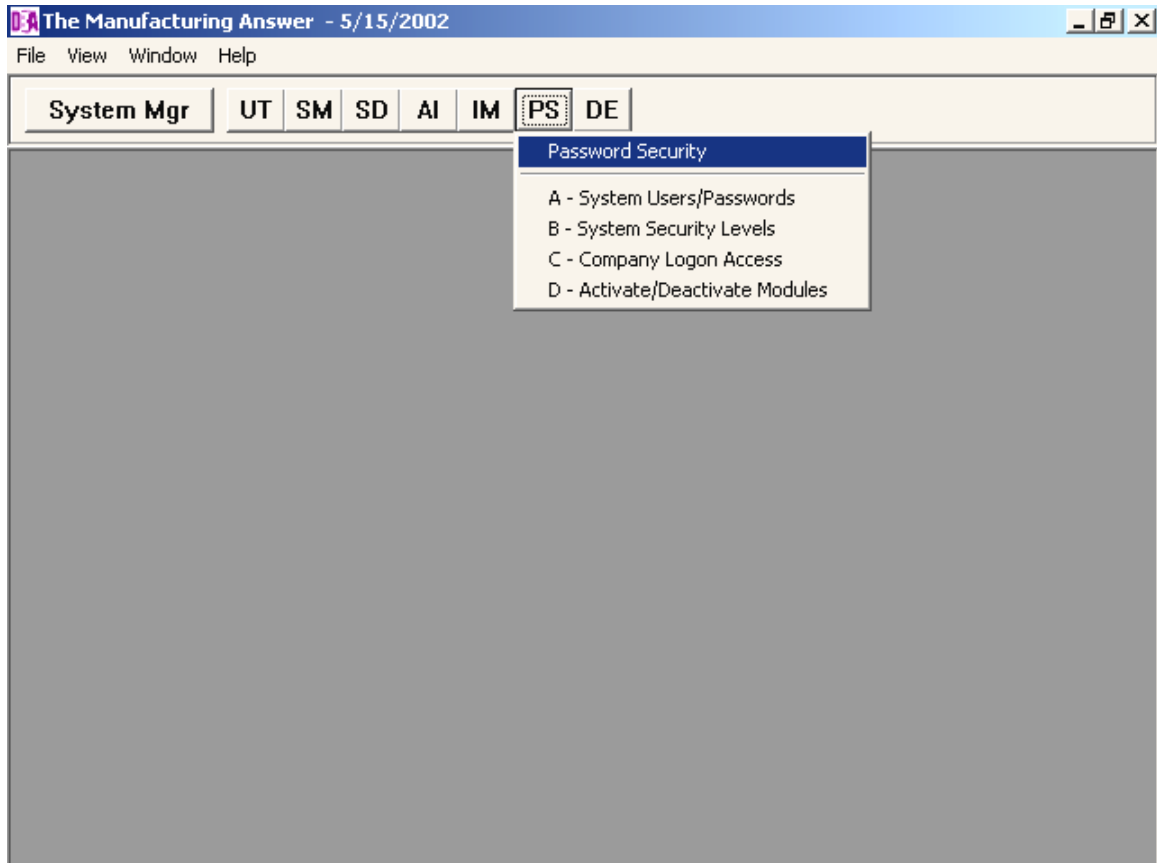
Aging Periods

You can establish up to five different aging periods, used when printing aging reports through AR-F, *Print Aging*, and on statements through AR-E, *Print Statements*. Defining these aging periods will allow you to view, sort, and print out invoices that are over 30 days old, etc. The first aging period should always be set to 0 days in order to include all invoices. These defaults can be changed on-the-fly when printing aging reports.

Customer Defaults

To quicken entry of new customers through SM-A, *Enter Customers*, you can set up defaults for the following fields: *Class*, *Price Code*, *Discount Code*, *Salesperson #1*, *Terms* and *Taxable code*. A new customer record will default to these values when saved, unless they are overridden.

24. PASSWORD SECURITY



The *Password Security* system lets you set up user *Logon ID*'s, passwords, and security levels in order to control access to individual programs within modules. You can also specify the menu and company each user is to be taken to at system startup. Finally you can disable unused modules system-wide to simplify menus.

The *Logon ID*'s and passwords you create in PS-A, *System Users/Passwords*, and the security levels you establish in PS-B, *System Security Levels*, are universal entries that apply to all companies on the system. In versions prior to 97.3 these entries were separately maintained within each company. In PS-C, *Company Logon Access*, you assign each user to security levels for each company to which the user is to have access. PS-D, *Activate/Deactivate Modules* allows you to completely eliminate unused modules (such as Lot or Serial Control if they do not apply to you).

Because the security settings made in these four programs are system-wide, you can set up all your security in any company of your choice and the settings will be saved to a common set of files in your DBAMFG subdirectory that apply to all companies.

You can set up as many as 99 security levels. Within each security level you

specify at the program level the access settings that will apply to each user assigned to that security level. For example, one security level might be called *System Administrator* and may be given access to all programs. Another level might be called *Order Entry* and might have access limited to just *Sales Orders*, *Accounts Receivable*, and *Inventory* programs.

Each time the user starts DBA he will either be taken directly to a startup company and menu (if none or only one *Logon ID* is created) or will be asked to enter a *Logon ID* and a *Password* (if security is activated). Once in the system there will be no further requests for passwords. If the user attempts access to a menu choice for which his security level is not authorized, he will receive a message indicating he is not authorized to use that program.

Sequence of events

The first thing to do is set up *Logon ID*'s and *Passwords* in PS-A, *System Users/Passwords*. If you are not immediately planning to activate security, you can still create *Logon ID*'s through PS-A to control startup menu and company preferences either by user or by a single *Logon ID* for all users.

Next, carefully consider how many security levels are needed to accommodate different categories of users and then create them through PS-B, *System Security Levels*.

Finally, through PS-C, *Company Logon Access*, assign users to specific security levels for each company to which the user is to have access. As soon as one user gets assigned to a security level within a company, security is then activated for that company and no user will be given access from that point on without a password.

Independent of the first three programs, PS-D, *Activate/Deactivate Modules* can remove or reinstate entire modules from the menu system-wide for all users.

PS-A System Users/Passwords

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File View Window Help

System Mgr UT AM AD MM MD IM PS DE

PS-A System Users/Passwords

Logon ID |
Password

Startup Settings for this Logon ID:

Startup Menu [0-4] 0

1 - Manufacturing
2 - Items
3 - Accounting
4 - System Manager

Startup Company

Can Use Maintain Database and Check File Integrity?

Help Save Exit

Use this program to create *Logon ID*'s and *Passwords* for your users. Whatever entries you make in the three security programs (PS-A, PS-B, PS-C) apply to all companies on your system, so you do not need to create separate settings within each company as was the case in versions prior to 97.3.

Against each *Logon ID* you can set preferences as to which menu is to be displayed and which company the user is to be taken to when starting the system.

You can create *Logon ID*'s without passwords if you don't need security but want to establish menu and company preferences by user. If you want all users to start with a company other than the default (blank) company, you can create one *Logon ID* that will be shared by everyone and set the menu and startup company against it. If only one *Logon ID* is set up, the system assumes everyone is sharing it and will automatically take ALL users to the menu and startup company specified against that one *Logon ID*.

Once more than one *Logon ID* is created, ALL users will be asked to enter a *Logon ID* when starting the system. If password security is not activated yet, users can share *Logon ID*'s, but once password security is activated for a particular company, each user will be asked to enter a unique *Logon ID* and *Password* from that point on.

You can set up *Logon ID*'s and *Passwords* immediately for setting menu and company preferences, but security itself won't be activated until security levels are defined through PS-B, *System Security Levels*, and users are assigned security levels by company through PS-C, *Company Logon Access*.

General program operation

You are first presented with a *System Users* lookup window that will display any existing *Logon ID* records. In the bottom portion of the screen there is help text explaining how the menu works upon system startup, depending on how many *Logon ID*'s you set up. If you set up no *Logon ID*'s, the system will always start in the default (blank) company at the *Acct + Sales* master menu. If you set up only one *Logon ID*, the menu assumes that all users will share that *Logon ID* and will take ALL users to the menu and company specified against that *Logon ID*. If two or more *Logon ID*'s are created, all users will be asked for a *Logon ID* upon startup and will be taken to the menu and company pertaining to the *Logon ID* entered.

To set up a new user, click on the *Add* button or press the <Insert> key. You are taken to a second screen. Enter the *Logon ID* that identifies the user and an optional *Password*. Next, enter the *Startup Menu* desired by entering the number opposite the four choices listed. Enter the *Startup Company* code. Answer Y to *Display Month-End Reminder?* if you want a message to display upon system startup that reminds you when it is the end of the month so that you won't forget to perform various calendar end of month activities.

When you've completed the last field the record is automatically saved and you are returned to the *System Users* lookup window where you will see your entry displayed. You can change any existing record by highlighting it in the window and then clicking the *Edit* button or pressing <Enter> to bring up the entry screen. Make whatever changes you wish and then click on the *Save* button (or press F10) to return to the *System Users* lookup window.

Continue this process until all *Logon ID*'s needed are created, then click *Exit* or press <Esc> to return to the *Password Security* menu.

To delete a *Logon ID* record, highlight the record desired and then click on *Delete* or press <Delete>.

PS-B System Security Levels

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File View Window Help

System Mgr UT AM AD MM MD IM PS DE

PS-B4 Manufacturing Security

Security Level 01 Supervisor

MODULE	YNC	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
Work Orders	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		
WO Maintenance Reports	Y	Y	Y	Y	Y	Y	Y	Y											
Job Costing	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Purchase Orders	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y						
PO Reports	Y	Y	Y	Y	Y	Y	Y	Y											
QC Programs	Y	Y	Y	Y															
MRP	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y						
Scheduling	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y			
Data Collection	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y									
Estimating	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y									

Help Save Exit

Use this program to create security levels for different categories of users. Each user will then be assigned security levels by company in PS-C, *Company Logon Access*.

Within each security level you can specify at the program level within each module whether users assigned to that security level are to have access to the program or not.

You may create as many as 99 security levels and there is no limit to the number of users that can be assigned through PS-C to each security level. Beginning with version 97.3 you can assign a 45-character description to each security level for reference purposes.

The security levels defined in this program are system-wide, meaning they apply universally to all companies, which was not the case prior to version 97.3.

General program operation

You are first presented with a *Security Levels* lookup window that will display any existing security levels. To create a new security level, click on *Add* or press <Insert>. You are swapped to another screen where you can enter a *Security*

Level numeric code and a *Description*. Once these two fields are completed the record is saved and you are returned to the *Security Levels* lookup window where your new security level record will now be displayed.

To define the security settings within each security level, highlight the *Level* desired, then click on *Menu Access* or press F3. You are taken to another screen where you can click or highlight and press <Enter> on the menu within which you wish to set access options. You will be taken to another screen where you see a grid of menu options.

The grid on the screen represents the menus within each master menu. From left to right, choices A through S correspond with each menu's letter choices.

In the left hand column titled "Y/N/C", if you enter a *Y* or *N*, all items on that menu will be set to *Y* or *N* and the cursor will move on to the next menu. If you enter a *C*, the cursor moves to right and allows you to individually set each menu option to a *Y* or *N*.

When you are finished setting the security for this level, click on the *Save* button (or press F10) to save the record. You will be returned to the *Menu Access* selection screen where you can select another menu. When finished, click on *Exit* (or press <Esc>) and you will be taken back to the main *Security Levels* lookup window.

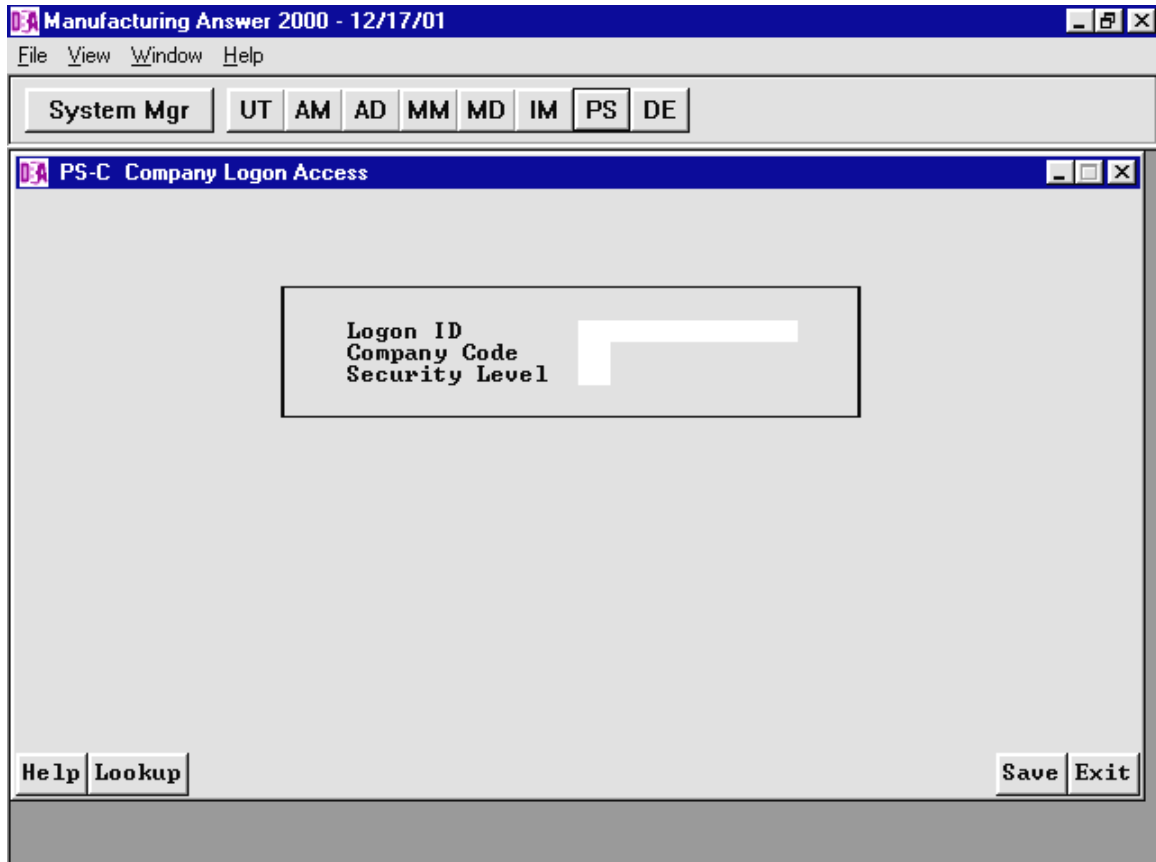
To change an existing security level description, highlight the *Level* desired and click on *Edit* or press <Enter>, make your change, then click on *Exit* or press <Esc> to return to the *Security Levels* lookup window.

To change an existing security level's menu access settings, highlight the *Level* desired, click on *Menu Access* (or press F3), make your changes, then click on *Exit* (or press <Esc>) to return to the *Security Levels* lookup window.

To delete a security level, highlight the *Level* desired and click on *Delete* or press <Delete>. You will be warned that any users assigned to this level will lose their access rights and that all access settings for this level will be lost. Click *OK* or enter *Y* and the security level and all its settings will be deleted.

When finished making entries, click on *Exit* (or press <Esc>) to return to the *Password Security* menu.

PS-C Company Logon Access



Use this program to assign users to security levels by company. Once one user is assigned to a security level in a particular company, security for that company is activated and all users from that moment on will be required to enter a password to get into that company.

Security levels are assigned in this program for all companies in the system. You no longer have to go into each company to assign security as was the case prior to version 97.3. For example, if you have three companies, all with security activated, and you want a user to have security access to all three companies, you would create three records for that user, each designated for one of the three companies.

General program operation

You are first presented with a *Logon Access by Company* lookup window that will display any existing company security level assignments.

To create a new record click on *Add* or press <Insert>. You are taken to another screen where you may enter the user's *Logon ID*, the *Company Code*, and then the appropriate *Security Level*.

Click on the *Save* button (or press F10) to save your entry and you will be returned to the *Logon Access by Company* lookup window where you can make another entry or edit an existing entry.

When you have finished making your entries, click on *Exit* or press <Esc> to return to the *Password Security* menu.

PS-D Activate/Deactivate Modules

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File View Window Help

System Mgr UT AM AD MM MD IM PS DE

PS-D Activate/Deactivate Modules

MENU MODULES AND SUBMODULES

Orig Status	Change To	Can Chg?	Code	Module Name
ON			WO	Work Orders
ON		no		Work Order Maintenance Programs
ON		no		Reports
ON			JC	Job Costing
ON			PO	Purchase Orders
ON		no		Reports
ON				QC Inspection Programs
ON			MR	Material Requirements
ON			SH	Scheduling
ON			DC	Data Collection
ON			ES	Estimating
ON			IN	Inventory
ON		no		Inventory Maintenance Programs
ON		no		Month End Reports
ON			RO	Routings

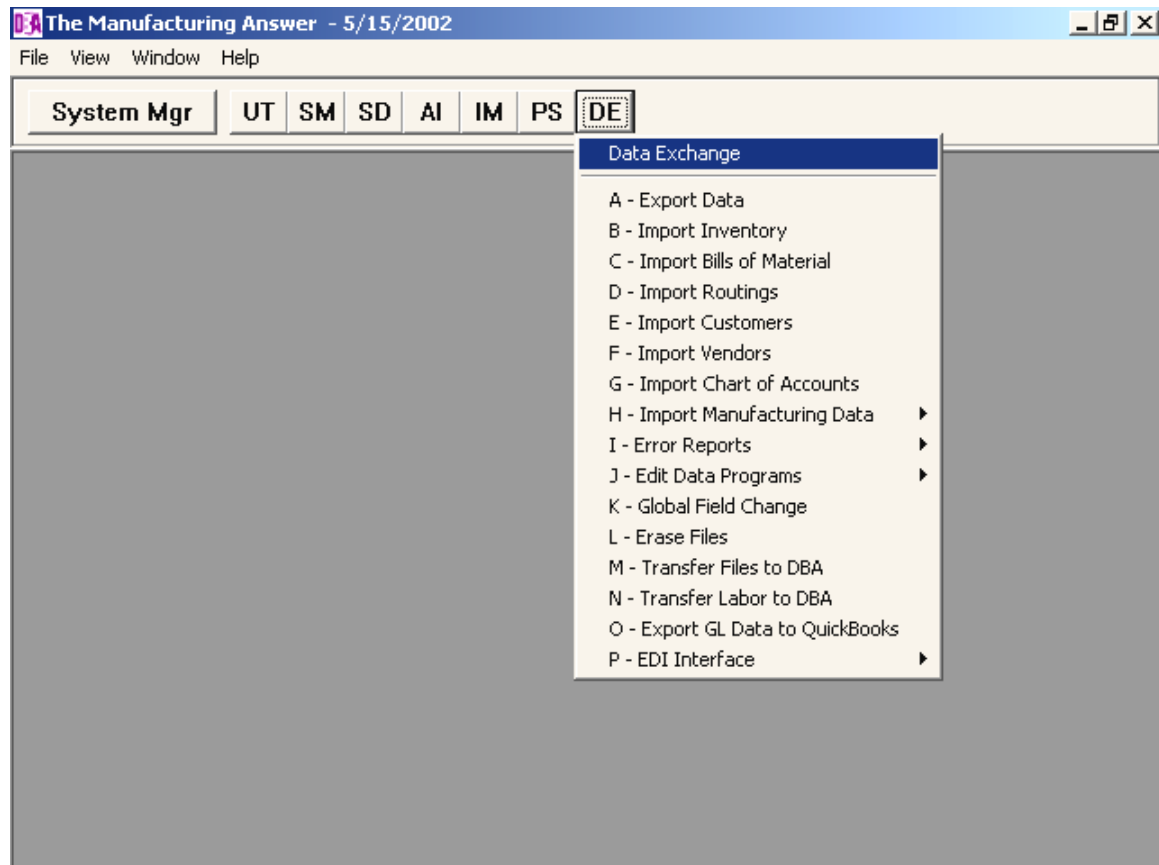
Toggle ON/OFF

Apply Cancel

General program operation

You are first presented with a list of modules and their current status. Select any module and click Toggle On/Off to change the status. Once all selections are made, click Apply. You will need to exit and restart DBA for the changes made to take effect. Some modules such as Password Security and Utilities are essential to system function and cannot be turned off.

25. DATA EXCHANGE



The *Data Exchange* module consists of a series of programs that enable you to import and export data into and out of other programs and systems. DE-A, *Export Data*, allows you to create export files from DBA data in various ASCII formats.

The DE-B thru DE-M programs are generally used to import master file data from another system. These master files are: inventory, bills of material, routings, customers, vendors, chart of accounts.

DE-H-A, *Import Labor*, and DE-N, *Transfer Labor to DBA*, are often used to transfer job cost hours collected from non-DBA data collection systems into DBA. DE-H-B, *Import Material Issues* and DE-H-C, *Import Finished Production* are used to collect manufacturing data regarding material movement on a portable bar code reader and import the data for posting to the work order and inventory files. DE-J-H and DE-J-I allow for editing of imported material issues and finished production, and for posting to work orders once the review is complete. Reports are available for review of the data prior to posting.

Finally, DE-O, *Export Data to Quick Books*, allows DBA *General Ledger* data to be exported into a *Quick Books* format. Generally this is used to send DBA GL data to outside CPA's for quarterly and year end purposes who do not know DBA accounting and are more familiar with the widely used *Quick Books* program

Importing master files (DE-B thru DE-M)

The *Data Exchange* module allows you to import master file data from your existing accounting system into DBA, thus greatly reducing the need for re-keying information by hand. You can import data into the following master files: customers, vendors, inventory, bills of material, routings, and GL chart of accounts.

You must be able to export your accounting master files into an ASCII text format, either comma delimited or fixed length. Most accounting systems have a data export capability that can do this.

DBA's data import design prevents bad data from getting into DBA's permanent files. Data is first imported into temporary holding files where they can be edited for errors. A transfer program then moves the data from the temporary files to the permanent files. Only valid records are transferred.

The data import is a three-step process. In step one you import your data into the temporary files. In step two you edit the data for any changes or additions that might be needed. In step three you transfer the data to the permanent files.

Step 1 - IMPORT DATA INTO TEMPORARY FILES

Export accounting data

First, you must export your accounting data for any of the following files: inventory, bills of material, routings, customers, vendors. You must export the data to ASCII text, and it can either be in a comma delimited or fixed length format. Most accounting packages have data export programs that will output your data to these formats. If not, you will have to get a programmer to create the ASCII files for you.

Comma delimited is slightly easier to work with, but if you have quotation marks within any of your data fields, those fields will have to be hand edited from your ASCII file before the import can be run. Otherwise, the import program gets confused because it looks for quotation marks to identify alphanumeric fields. Quotation marks are commonly found within inventory description fields because they are used to indicate inches. To edit your descriptions for quotation marks, you will have to edit the ASCII file through a text editor, such as DOS Edit. If you have massive incidence of quotation marks, you should go with a fixed length format.

Run import programs

DBA's import programs (DE-B thru DE-H) all function identically. You are first asked to enter the name of the ASCII file to import, including its data path. For example, if your ASCII file was named INVENT.TXT and was located in a

subdirectory named ACCT on drive F, you would enter the following:

F:\ACCT\INVENT.TXT

Next, you are asked if you want to *Skip* or *Replace* existing item numbers. If you have already imported some records and have already edited them, you can choose the *Skip* option so that those records are bypassed and won't be accidentally replaced during another import.

The *Replace* option can be very useful. Inventory data, for example, often comes from more than one file. Master data might be in one file, costs in another, and on-hand balances in another. If you select the *Replace* option, the program will replace just the field values for each field you select. Therefore, in the example of inventory, you could make three separate imports, each bringing in its own set of field values.

NOTE: Be aware that the *Skip* and *Replace* functions only apply to records that are still in the *Data Import* module and have not yet been transferred to the permanent files. These functions do not work with the permanent data files. When you actually transfer the data to the permanent files, the transfer program will always replace existing records for all master files except for bills of material, which, because DBA allows duplicate bill of material components, always appends records rather than replaces them.

IF YOU ARE IMPORTING BILLS OF MATERIAL: If you have duplicate components within your bills of material, do not use the *Replace* option, which will cause your last duplicate to replace the previous duplicates. Instead, use the *Allow Duplicate Components* option. Also, see the previous paragraph above in regards to how the data transfer program handles bills of material components.

Next, indicate whether your ASCII file is in comma delimited or fixed length format.

The rest of the screen consists of a listing of all the field names within the DBA master file, with two entry fields to the left of each field name.

Comma delimited files

If your ASCII file is comma delimited, you are only allowed entry of one field next to each DBA field name. In a comma delimited ASCII file, each field is separated on either side by a comma. Alphanumeric fields are further enclosed by quotation marks. A comma or commas between two quotation marks do not count as field separators. Simply identify your ASCII file field position numbers consecutively from left to right, then enter those numbers next to the appropriate DBA field.

Fixed length files

If your ASCII file is fixed length, you must enter two values to the left of each DBA field name. The first entry is the character position of the first character in the field, and the second entry is the number of characters in the field. It is easy to identify each field's starting character position and length by pulling up the file in a text editor that indicates the character location of the cursor. Simply move

the cursor to each field position and write down the location, then count the length of the field. Then enter the starting position and length next to the appropriate DBA field.

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File View Window Help

System Mgr UT AM AD MM MD IM PS DE

DE-B Import Inventory

Name of File to Import (with path):

Skip or Replace existing Part Numbers? IS,R1: Comma or Fixed Length? (C,F):

☐	☐	Part Number	☐	☐	Drawing No	☐	☐	Freight Cost
☐	☐	Desc 1	☐	☐	Std Pack	☐	☐	Labor Cost
☐	☐	Desc 2	☐	☐	Cycle Code	☐	☐	Setup Cost
☐	☐	Stock U/M	☐	☐	Freight Pct	☐	☐	Outside Proc
☐	☐	Price U/M	☐	☐	#1 Vendor Code	☐	☐	Fixed Overhd
☐	☐	Active?	☐	☐	#1 Vendor Part#	☐	☐	Var Overhead
☐	☐	Type	☐	☐	#2 Vendor Code	☐	☐	Standard Cost
☐	☐	Category	☐	☐	#2 Vendor Part#	☐	☐	Base Price
☐	☐	Class	☐	☐	#3 Vendor Code	Specifications		
☐	☐	Taxable	☐	☐	#3 Vendor Part#	☐	☐	#1
☐	☐	Purch U/M	☐	☐	Mfgr Name	☐	☐	#2
☐	☐	Conv Factor	☐	☐	Mfgr Part#	☐	☐	#3
☐	☐	Lead Time	☐	☐	Customer Code	☐	☐	#4
☐	☐	Reord Level	☐	☐	Units on Hand	☐	☐	#5
☐	☐	Reord Amt	☐	☐	Lot Size	☐	☐	#6
☐	☐	Bin Loc	☐	☐	Last Cost	☐	☐	#7
☐	☐	Weight	☐	☐	Average Cost	☐	☐	#8
☐	☐	Foot Factor	☐	☐	Material Cost	☐	☐	#9
☐	☐	Rev Level	☐	☐		☐	☐	#10
						☐	☐	#11
						☐	☐	#12
						☐	☐	Tax IN
						☐	☐	Duty Code
						☐	☐	Lot Control?
						☐	☐	Serial Control?
						☐	☐	UserDefined Sort

Help Exit

Import Inventory Screen

If any of your fields are larger than DBA's fields, you can crop your field length by specifying a starting position and length that reduces the size of the field being brought in. For example, you might have a unit of measure field of four characters, while DBA's unit of measure is three characters. Simply reduce the length of the field from four to three and the program will cut off the fourth character when importing the field data. On a right justified field, such as many numeric fields, specify a starting position to the right of the actual starting position and shorten the length, thus cropping the field.

Date fields

All imported date fields need to be in ISO YYYYMMDD format. If your data is different, you can define a custom format in Excel and convert your dates to the correct format.

Time fields

In the routings file there are a couple of time fields. DBA's time fields are in a HH:MM:SS format, meaning hours, minutes, and seconds. If your time fields are different, you may have to edit them after they are imported so they conform to the DBA format.

1000 character limit

Be aware that the DBA import programs only accept ASCII records of 1000 characters or less. Most accounting packages' data export programs allow you to export selected fields rather than the entire record, so if your ASCII record exceeds 1000 characters, export it again and leave out any unnecessary fields to get it under the 1000 character limit.

When all the field positions are entered, start the import by pressing F10. If the import stops and you get an error message indicating a field specification error, there is a problem with the ASCII record indicated on the screen. This usually is caused by a comma delimited record that has a quotation mark between two quotation marks, such as with an inventory description using a quotation mark to stand for inches. You will have to pull up your ASCII file in a text editor (such as DOS edit) and fix the bad field. Run the import again until it completes its run without any errors.

Each time you run an import, the field positions stay on the screen in the event you need to rerun the import again.

Erase files program

If for any reason any of your imports goes so badly that you want to start over, you can erase any of the temporary files and start over again by running DE-L, *Erase Files*. Simply enter *Y* next to each file to be erased. After entering *Y* or *N* next to each file, you are asked if you want to begin the program. If you answer yes, the appropriate temporary files will be erased.

STEP 2 - EDIT THE DATA

Print error reports

Now that your data has been imported into DBA's temporary master files, you should run error reports to check for missing or invalid fields needed by DBA's master files. Run the appropriate error reports (DE-I-A thru DE-I-E) as required.

The error reports will list all records that need to be edited, as well as any missing or invalid values needed by DBA.

Edit programs

You can edit individual records one-by-one through the edit programs (DE-J-A thru DE-J-G). These screens are identical to the master file entry screens within DBA, except that they access the temporary data rather than DBA's permanent

records.

For details on field explanations and general program operation, use the on-line Help to study the following:

- IN-B, Enter Inventory
- BM-A, Enter Bills of Material
- RO-A, Enter Routings
- SM-A, Enter Customers
- SM-B, Enter Vendors
- WO-F, Enter Labor

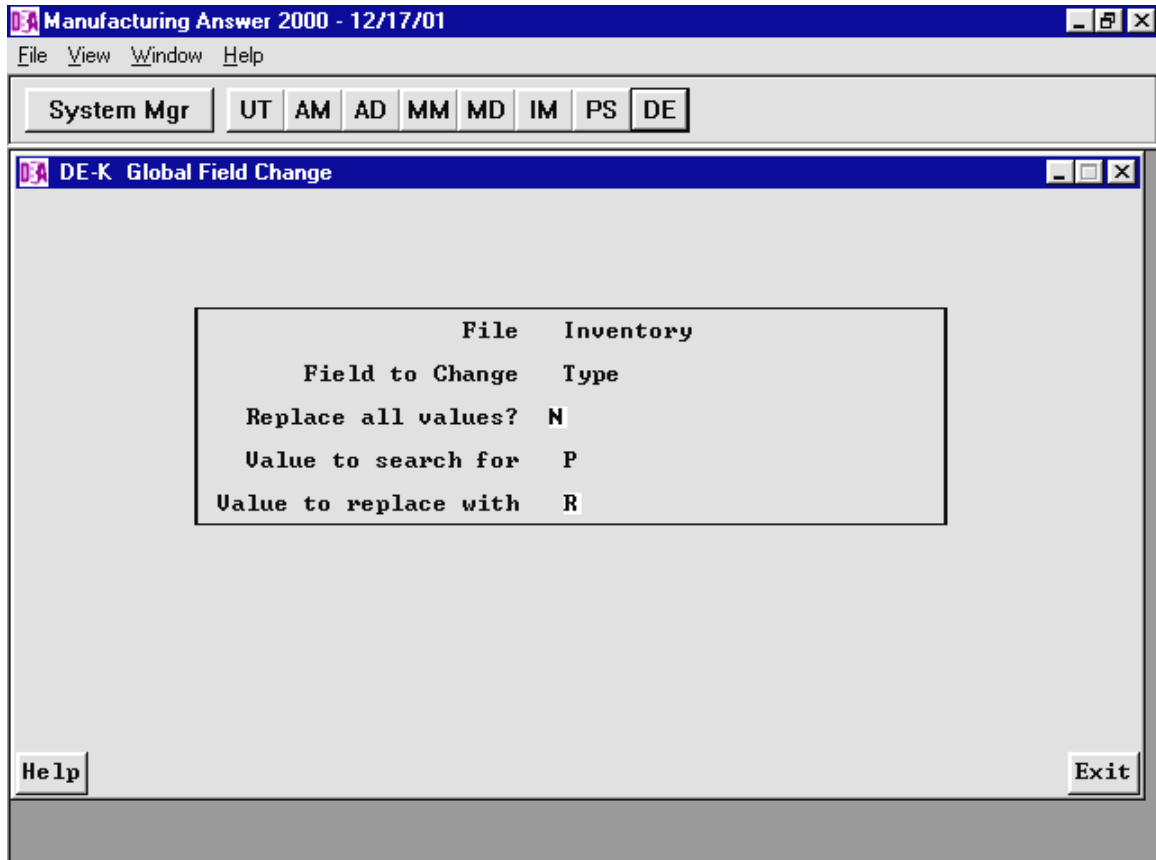
At a minimum you must edit the missing or invalid fields listed on the error reports. You can also add or change any other fields at this point, or you can wait until the temporary files are transferred into DBA and make further changes within the permanent files.

Global Field Change (DE-K)

This program can be very useful for entering or changing field values globally across many records. For example, in your accounting system a purchased item might be coded with an inventory type “P.” In DBA the inventory type code for purchased items is an “R.” Rather than have to individually change every inventory record, you can use the global field change program to change all records with a type “P” to a type “R.” The program only changes critical fields listed on the error reports.

First you are presented with a choice of four files with which the program can be used. Highlight the file you want to work with and press <Enter>.

Next you are presented with a window listing the critical fields within that file that can be globally changed. Highlight the field you want to work with and press <Enter>.



Global Field Change

If you want to superimpose a field value on all records in the file, answer Y to *Replace all values?* For example, you might want all items to have a selling unit of measure of “EA.” The cursor then moves to the *Value to replace with?* field. You would enter “EA” and press <Enter>. The program will then superimpose the value “EA” in the selling unit of measure field for all records within the file.

If you want to replace a particular value only within specific records, answer N. You are then asked for the value to search for. In our inventory *Type* example, you would enter a “P.” Enter the value to replace with, which in our example would be an “R.” The program will then automatically replace all “P” values in the inventory field with an “R.”

Erase files program

If for any reason any of your imports do not initially go well on the first try, you can erase any of the temporary files and start over again by running DE-L, *Erase Files*. Simply enter Y next to each file to be erased. After entering Y or N next to each file, you are asked if you want to begin the program. If you indicate yes, the appropriate temporary files will be erased.

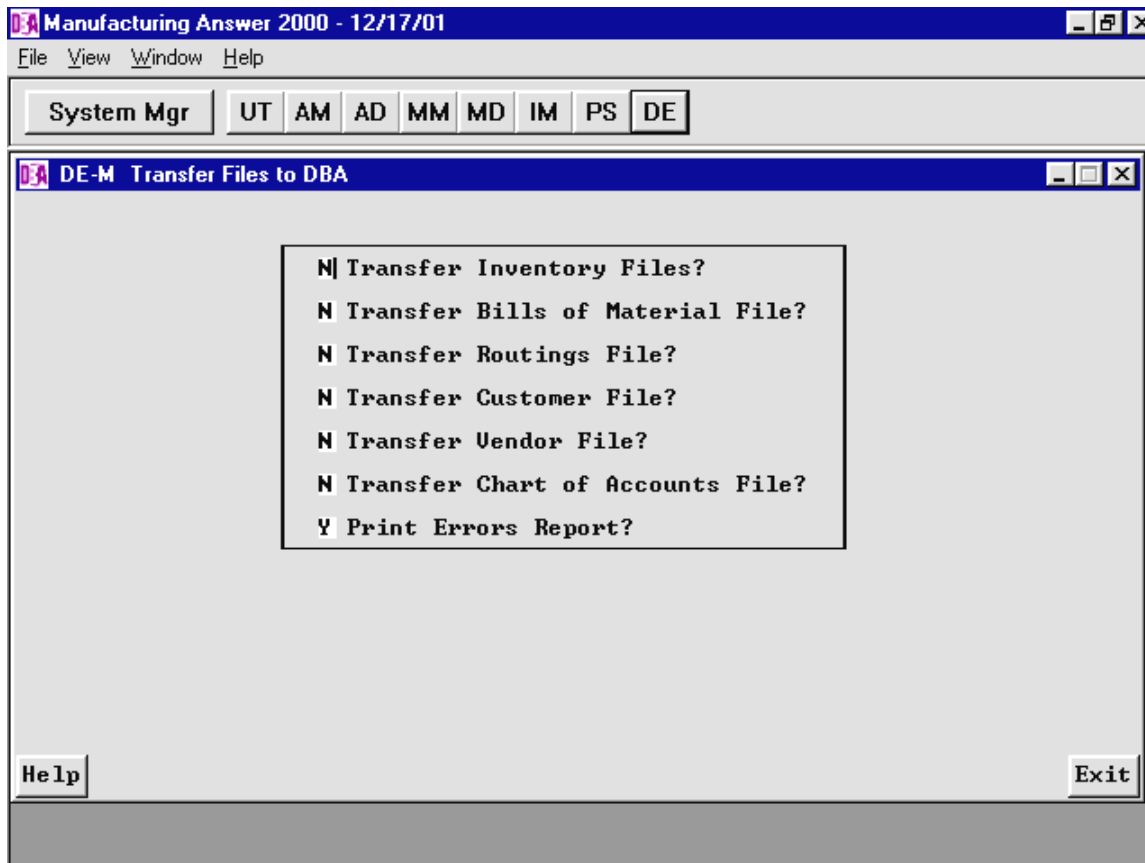
STEP 3 - TRANSFER DATA TO PERMANENT FILES

The final step is to transfer the temporary data to DBA's permanent master files through DE-M, *Transfer Files to DBA*. Only records without missing or invalid fields will be transferred by the program. Run the appropriate error reports before running this program to make sure all records to be transferred are without errors.

Enter a Y next to each file you wish to transfer. If you want an error report which lists any records that did not transfer, enter Y to *Print Errors Report?* The program will then begin the transfer.

The transfer completes the import process and insures that only good records get into DBA's permanent files.

NOTE: When you are importing data from multiple files into one DBA file, do not run this program until all your data has been imported and gathered together. You cannot transfer data in segments into the permanent data files.



Transfer Files to DBA

Importing Labor

DE-H-A, *Import Labor*, can be used to bring in labor data gathered by another (non-DBA) shop floor data collection system. Most data collection systems automatically produce ASCII files for transfer to other accounting/manufacturing systems.

Importing labor works the same as the other import programs. The only difference is that the other import programs are normally used just once during system start-up, while the labor import will be used daily as a permanent interface with the data collection system.

The data path and fields entered in DE-H-A, *Import Labor*, stay on the screen until changed, so they only have to be set up once. If your ASCII source file's name changes each time a new batch of labor is to be transferred, then you will have to go into the program and change the file name each time the program is run.

Name of File to Import (with path)					
Comma Delimited or Fixed Length? [C,F]					
☐	☐	Date	☐	☐	Setup Hours
☐	☐	Employee Num	☐	☐	Rework?
☐	☐	Work Order Prefix	☐	☐	Run Hours (XXX.XX)
☐	☐	Work Order Suffix	☐	☐	Qty Complete
☐	☐	Assembly	☐	☐	# Persons/Team
☐	☐	Description	☐	☐	Qty Scrapped
☐	☐	Sequence Number	☐	☐	Run Hours (XX)
☐	☐	Reg/Over/Dbl	☐	☐	Run Minutes (XX)
☐	☐	Shift	☐	☐	Run Seconds (XX)
☐	☐	# Jobs Worked			

Import Labor Screen

26. USING MULTIPLE COMPANIES

The number of companies you can maintain within your DBA system is for all practical purposes unlimited, with over 1200 logical possibilities for company code designations. The user-defined company code is a two-character, alphanumeric code.

Each company has its own set of data files, but only one set of programs is required and is shared by all the companies. Each user can perform transactions in only one company at a time.

The default company, which does not have a company code (and is therefore often referred to as the default or “blank” company) is always located in the primary DBAMFG subdirectory along with all the run programs. Additional companies are located in separate subdirectories one level down from the primary DBAMFG subdirectory.

Data files for the default company in DBAMFG all end with an extension of ***.B**. Data files for additional companies in the other subdirectories end with the same “B” extension followed by the two-character company code. For example, the data files that are included with the sample company 99, *The Manufacturing Answer*, are located in a subdirectory named TESTDATA and have an extension of ***.B99**.

Not only are the programs located in DBAMFG (ending with extensions of *.RUN and *.EXE), but so are the data dictionary files (all files beginning with FILE*.B) and the password security files, which are common data files shared by all companies.

Use default company for single company

If you only have a single company, it is best to use the default or “blank” company rather than create an additional company. The default company already exists when you install your DBA system and is the company that you are in when first starting the system.

Some users mistakenly think that they must create a new company in order to start with empty files for system implementation. This is not the case. If you are using DBA accounting, the only data in the default company are a handful of GL accounts in the chart of accounts that can be cleared using UT-K-A, *Clear Data* (see further below), or can be deleted manually through AM-C, *Enter General Ledger Accounts*.

Use UT-I if you need additional companies

When installed the system provides you with the default (blank) company in the DBAMFG subdirectory, which you should use as your primary company, and with company 99, *The Manufacturing Answer*, which is located in the TESTDATA subdirectory one level down from DBAMFG.

If you need one or more additional companies, they are created through UT-I,

Create/Delete Company. UT-I creates a subdirectory for the new company and sets up the locations in the data dictionary to enable the *Change Company* function (see below) to find the new company's data files. UT-I allows you to copy the data from one company to another, or it allows you to start with a set of initialized (empty) data files. If you copy data from another company but only want to retain master file data and not transactions, you can selectively clear transaction data using UT-K-A, *Clear Data* (see further below).

Sometimes additional companies are created for temporary purposes such as for training or experimentation when first learning the system. You can always create a temporary company through UT-I (copying your actual data if you wish) and then experiment all you want without affecting your real company data. Once you are finished using the temporary company, run UT-I again, this time using its delete function to delete the temporary company.

Changing companies

To change companies, click on *File* in the upper left corner of the Windows menu, then on *Change Company*. You will be presented with a list of companies in a window. Click on your choice and then click *OK*. The current company name is always displayed on the blue banner at the very top of the screen.

Using UT-K-A to clear files

UT-K-A, *Clear Data*, is a program that can selectively clear various sets of data files of either transaction data (leaving master file data intact) or of all data. This program is typically used on system start-up in one of two ways:

One, you have a single company and are installing it in the default (blank) company and you want to clear out the handful of GL accounts that come with the system or any other data that may have been entered by people experimenting with the system before implementation started. In this case you would have UT-K-A delete all data in the default company so you can start with completely clean files.

Two, you have two or more companies that share some common master file data, such as the same GL chart of accounts. You enter this data in the default company first and then use UT-I, *Create/Delete Company*, to create the second company, in the process specifying that all the files from the default company are to be copied over to the second company. You can then use UT-K-A in the second company to clear out any transaction data that may have occurred, but retain the master files such as the chart of accounts. This process eliminates needless double entry.

Running UT-K-A, *Clear Data*

When you start the program you are presented with an entry screen with a choice of the following six data categories (file groupings):

General Ledger Chart of Accounts
Accounts Receivable Customers

Accounts Payable Vendors
Inventory Items
Payroll Employees
Manufacturing Systems

Opposite each data category enter a *C* if you want to keep the master file data, but you want to clear all transactions (such as GL detail entries, aging detail, sales information, etc.) Enter a *D* if you want to clear all data completely, and enter an *N* if you want to leave the data category untouched.

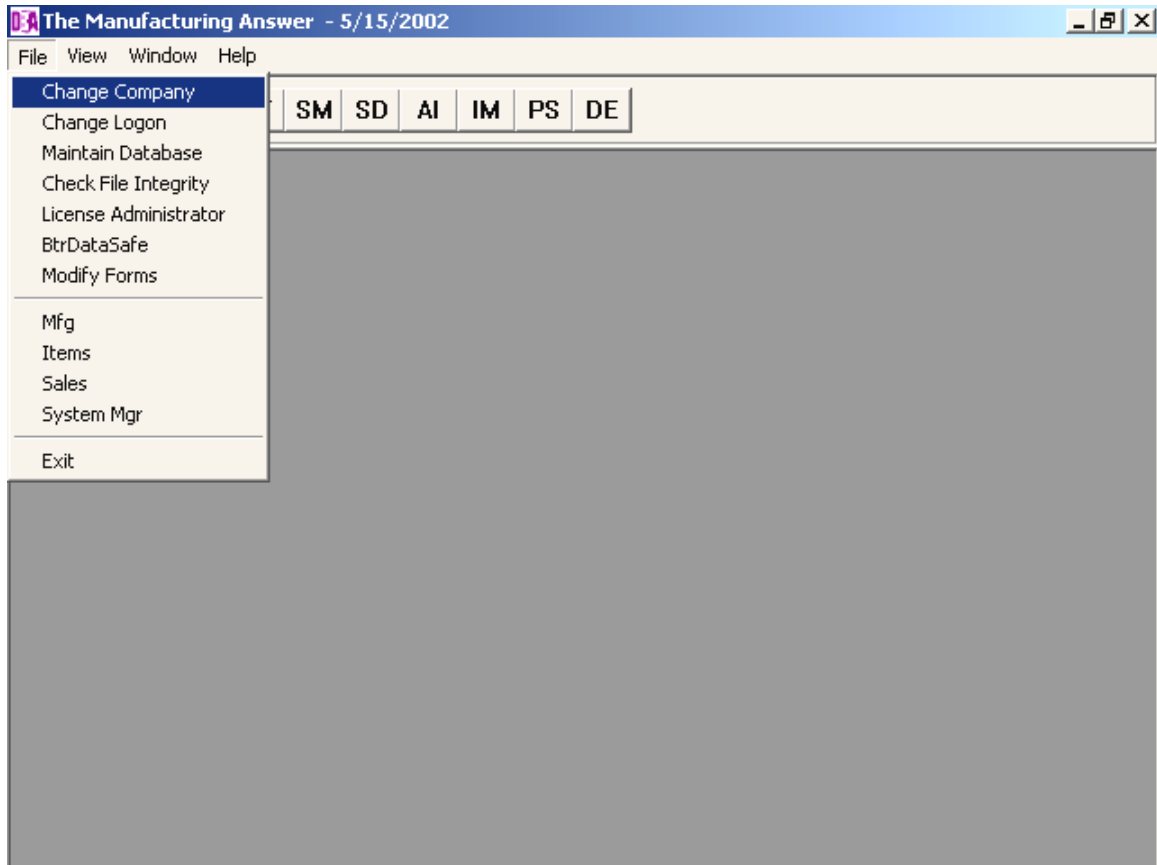
UT-K-A can take considerable time to run, depending on the sizes of the files being cleared.

Start-up preferences for company & menu

If you are using multiple companies you can set up each user with the start-up company and start-up menu of his choice. If a particular user tends to work primarily in company two rather than in the default company, he can be set up to bypass the default company and go directly to company two when starting the system. He can also be set up to always start at a particular menu.

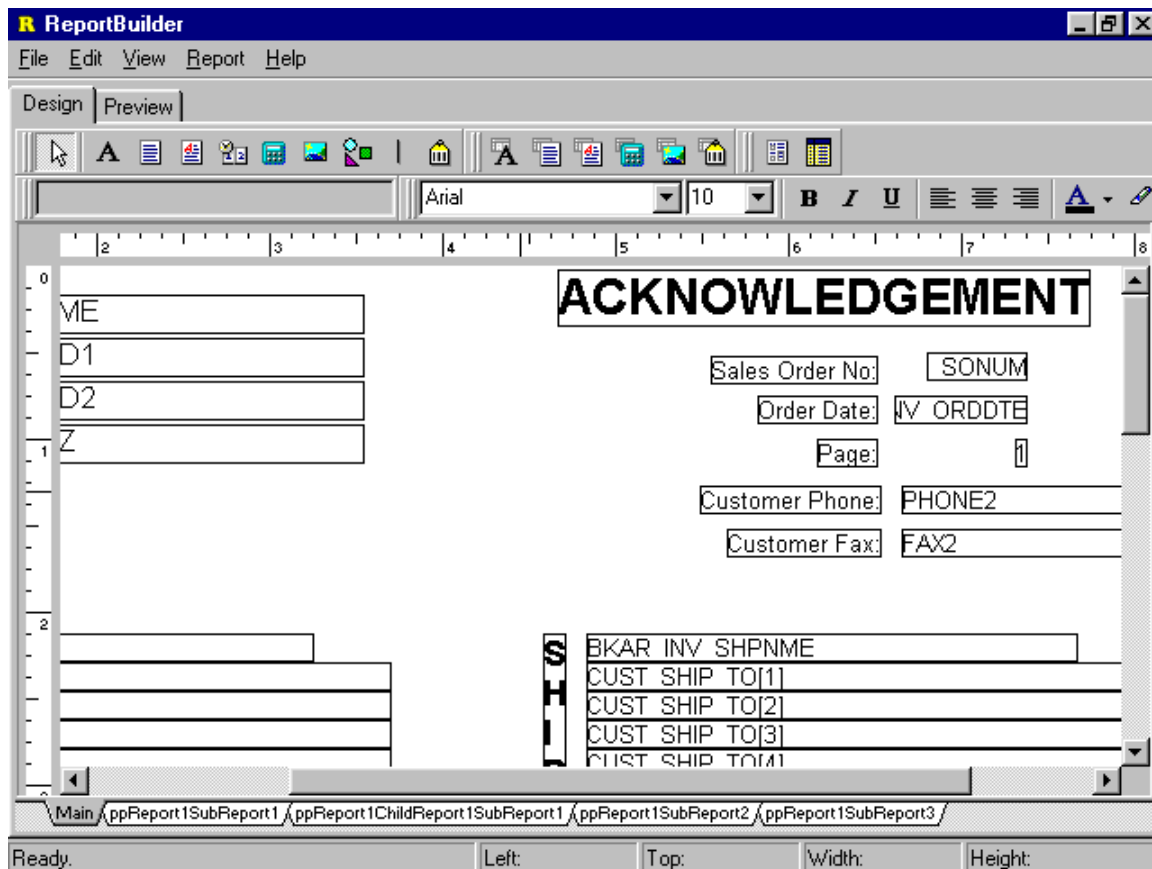
These preferences are set up through PS-A, *System Users/Passwords*. See chapter 24, *Password Security*, for details.

27. MAIN MENU (FILE) PROGRAMS



This chapter covers some of the programs that are found under *File* on the main menu.

Modifying forms



The *Modify Forms* program, found on the *File* menu at upper left on the DBA master menu, allows you to change the layout of graphical forms such as invoices, checks and labels. You can move and delete both text and data fields and you can add new text fields. The program does not have the capability to insert a data field.

General program operation

First, determine which RTM layout file you want to modify. Most of the RTM files are listed on the default screen where print formats are selected. For example, a lookup on the Acknowledgement print format field in SD-M indicates that format #2 Graphical Universal Form is BKS0B2.RTM. Only formats with an RTM designation can be modified. It is a good idea to make a copy of the RTM you want to modify and rename back to the original when you are happy with the results.

For a listing of RTM's, see the on-line Help.

To start the program, go to the main menu, click *File*, then *Modify Forms*. A box will come up with a button "Click here for list of RTM files". Click and select the

RTM you want to modify.

A format screen will come up showing the layout of the form you are modifying. Each item, or object, is enclosed in a box. Objects are either *Text* or *Data*. A text object is something that always prints on the form such as the word "ACKNOWLEDGEMENT". Data objects call the data from within DBA such as item number or customer name and the box contains the name of the field in the DBA file being called.

Double clicking on any object will select it. Once selected, the *Delete* key will remove it from the form. The font and size are shown at the top of the screen and can be changed for both text and data objects. For text objects, the actual text is shown on the upper left and can be changed as well. Selected objects can be moved or resized by dragging with the mouse.

A new text object can be created by copying an existing object. Select an object, click *Edit, Copy*, then move the cursor, then click *Edit, Paste*. The copy of the object can then be moved and the contents can be edited, thus effectively creating a new text object.

An existing data field can be bar coded by copying the object and changing the font to *CIA Code 39 Bar Code*.

For forms such as acknowledgements or purchase orders that contain multiple sections, there will be tabs at the bottom of the screen for selection of the sections to edit. Header, line items and notes are examples of different sections that could exist.

Once you have moved the fields where you want them, you can click the *Preview* tab to check your work. However, only text objects will be shown because no data files are being accessed. You must save the RTM file with the original name in order to use DBA to test the actual output.

RTX backup files

Be aware that copies of all original RTM layouts are maintained in a second set of files with an extension of *.RTX. If for any reason you wish to restore the original RTM and start again with a clean layout, simply copy and rename the appropriate RTX file to the RTM extension.

Inserting an image (such as a logo) into your layouts

Modify Forms provides the capability of inserting an image such as a logo into an RTM form. Select the desired RTM and, in the *Design* screen, from the toolbar select the image icon, which is the seventh icon from the left. Click on it, then click on the form where you want the image. Drag to relocate and stretch the frame as needed. Then right click inside the frame and choose *Picture*. Select the image file (BMP, ICO, EMF, and WMF formats are supported).

Be aware that your image must already be the correct size when selected. It cannot be enlarged or made smaller within this program.

Program descriptions

Check File Integrity

Use this program to check your data files for possible corruption.

If a data file is found to be corrupt, sometimes it can be corrected via UT-C, *Re-Index File*. Re-indexing may be helpful in recovering a file if the file key structure is corrupted, which can be caused by a power failure or damage to the hard drive.

If you are in an entry program and data does not respond in a normal fashion (such as when entering an item number you get the wrong description or data appears to be missing), this often can be corrected through re-indexing.

If a file cannot be repaired, the only alternative is to restore data from backup.

Maintain Database

Use this program to view or directly edit data file records.

CAUTION: be extremely careful when using this program. It generally is used as a support tool to repair data records. Whenever possible, you should use the actual programs themselves to work with data, because the programs properly update related files, whereas *Maintain Database* does not.

License Administrator

Use this program to install the DBA registration diskette at time of annual renewal or when upgrading your system for extra users or modules. You can also use the program to see how many users for which you are licensed, your serial number, and your license expiration date.

When you first start the program you will see your current registration information (serial number, expiration date, licensed number of users).

Place the DBA registration diskette in floppy drive A or B and make sure the corresponding drive number is shown in the *Registration File Location* field.

Click *Update Registration* and your registration information will be re-displayed with any changes.

BtrDataSafe

BtrDataSafe is a product from *Telistics Software Inc.* that checks your *Pervasive* database for file, field & index corruption and, if possible, repairs damaged files. *BtrDataSafe* can be set up to run on a timer during off hours and can also perform automatic backups. This product is available from *DBA Software* at a special price for DBA users.

A 30-day trial version is available and is located in your DBAMFG folder. To install it, double click on a file named *BtrDataSafe.exe*. The trial version will only scan files -- the re-index repair functions have been disabled. We suggest you install *BtrDataSafe* on a workstation rather than on the server. Once installed, it runs independently from DBA with its own startup icons.

To get a full working version, place your order with our sales department and we will immediately notify *Telistics Software* of your purchase. You can then contact *Telistics* directly to receive an unlocking code that converts the trial version into a full working version.

Modify Forms

Modify Forms allows you to change the layout of graphical forms such as invoices, checks and labels. You can move and delete both text and data fields and you can add new text fields. The program does not have the capability to insert a data field.

28. OFF-MENU PROGRAMS

The following are programs that are not located on any of the four master menus (*Mfg, Items, Sales, System Manager*).

BKCVUTIL Recalculate MTD/ YTD Totals

This program is used by companies that have written custom reports that use the *Month-to-Date* and *year-to-date* fields maintained in the customer and vendor files. These fields are recalculated by the SM-A, Enter Customers, and SM-B, Enter Vendors, programs when you click on the *Statistics* button. In a custom report, however, this recalculation routine does not get preformed, in which case you can run this program prior to running your custom report.

DBAMS.EXE Multiple Session Control

The DBA runtime engine/menu (TP5WDBA.EXE) has a control within it that prevents users from running multiple sessions of DBA. This is a protection against inexperienced users who mistakenly load up multiple sessions without realizing it and unnecessarily tie up system resources.

DBAMS.EXE allows you to turn off this control via a change in the Windows registry setting. It can therefore be run at the individual workstation level and can be confined to more experienced users.

Once the multiple session control is turned off, a message will be displayed each time DBA is started that reminds the user that the control is not activated.

ODBCDDF Make DDF Files for ODBC

DDF files are used by many programming tools as a means of connecting with the Btrieve database. An initial set of DDF files (FILE.DDF, FIELD.DDF, INDEX.DDF) are supplied with the DBA system and are located in the main DBAMFG directory. These DDF's are multi-company files and can be used by tools such as *Crystal Reports*.

Some programming tools such as *Microsoft Access*, which uses the ODBC driver that comes with *Pervasive.SQL* to connect with *Btrieve* via DDF's, have a limitation in that the DDF files must be in the same directory as the data files and therefore they cannot use the multi-company DDF's found in the DBAMFG directory. ODBCDDF is used to create separate DDF files for each of your DBA companies to get around this limitation.

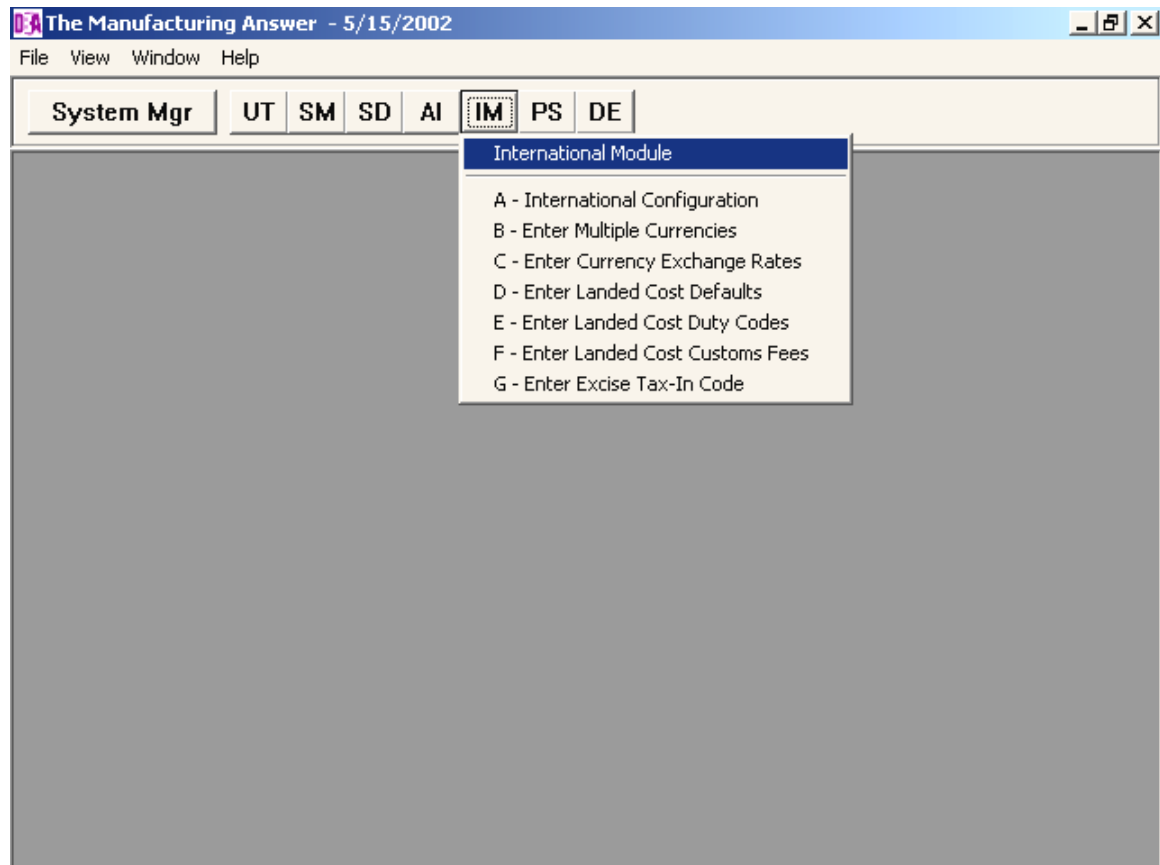
To use this program, each company on your system must have its data files located in a separate subdirectory than DBAMFG. If your default company files are located in DBAMFG, they must be relocated to a separate directory. See the Help file for this program for instructions on doing this.

The multi-company DDF files that are in DBAMFG and that can be re-created by

Off-menu programs

running UT-J, *Generate DDF Files*, will remain in the system. These DDF's are used by the *Maintain Database* program on the main menu.

29. INTERNATIONAL MODULE



Refer to *Accounting User Guide* if you are using DBA accounting

This chapter in this user guide is for companies who are using a non-DBA accounting system. If you are using DBA accounting, refer to chapter 9 in the *Accounting User Guide* for *International Module* details, which are different due to various general ledger settings.

International Module capabilities

The *International Module* includes two major capabilities, multi-currency processing and landed cost. In addition, the system has been equipped with a sales tax system that is flexibly designed to handle international requirements. These three international capabilities are described below.

Multi-currency processing

Multiple currency processing allows you to enter transactions in up to ten different currencies (which we refer to as *source currencies*) while keeping your financial reporting in your own currency (which we refer to as the *base currency*).

Each currency is set up via IM-B, *Enter Multiple Currencies*, and currency exchange rates are maintained in a table via IM-C, *Enter Currency Exchange*

Rates. Each time you update exchange rates, a record is saved of the previous exchange rates so that transactions, no matter how far backdated they are entered, can be matched to the closest historical exchange rate.

Landed cost processing

In many countries the landed cost of importing goods and materials (freight, duty, customs fees, broker fees) often represents a significant portion of cost that must be reflected in the inventory cost of items.

DBA provides a table of duty codes and percentages via IM-E, *Enter Landed Cost Duty Codes*, and a table of customs brokers and fees via IM-F, *Enter Landed Cost Customs Fees*.

Duty codes can be assigned to inventory items via IN-B, *Enter Inventory*, and customs brokers can be assigned during PO-C, *Receive Purchase Orders*. When items are received through PO-C, the landed costs (freight, duty, customs fees, broker fees) are calculated and added to each item's inventory cost.

Sales & purchase tax processing

Sales tax processing in the United States is straightforward and usually involves a single sales tax rate applied to selective line items with a single tax total at the bottom of the invoice. In other countries, however, multiple taxes (such as federal taxes, provincial taxes, local taxes, value added taxes, excise taxes) often apply and sometimes must be itemized on the invoice. In some cases, taxes are applied in a hierarchical fashion in which some taxes are "taxes on taxes."

DBA allows you to set up tax codes for each type of tax via SM-E, *Enter Tax Codes*, and then allows you to group these taxes together via SM-F, *Enter Tax Groups*. Any number of these tax groupings can be set up and assigned to particular customers and vendors. You can specify whether freight is to be taxed and whether a tax applies to other taxes. If you want the taxes itemized and printed out on the invoice, there is a setting in IM-A, *International Configuration*, which allows you to activate this feature.

In regards to sales taxes on purchases, in the United States the collection and reporting of such taxes is the responsibility of the vendor. In some other countries, however, this is not the case and the tax must be accounted for and reported in the same fashion as sales tax. The same type of tax codes and tax groups used for sales taxes can also be set up in SM-E and SM-F for taxes on purchases.

Finally, in some countries excise taxes are applied on both sales and purchases in which the tax is imbedded in the price of the product. DBA lets you set these type of taxes up via IM-G, *Enter Tax-In Codes*. When invoices get posted, the sales amounts that get reflected in sales reports are reduced by the amount of the excise tax so as not to be overstated. All of the taxes described above can be listed via AR-K, *Print Sales Tax Report*, and can be transferred for payment via AR-L, *Transfer Sales Taxes*.

Multiple currency startup

When you set the *Multi-Currency* field to *Y* in IM-A, *International Configuration*, the entire DBA system is set for multiple currency processing. Many programs will not let you use them until you get your system properly set up for multiple currency processing.

System setup includes the following:

- Enter currency codes
- Enter currency exchange rates
- Assign currencies to vendors (the program defaults all to *base currency*)
- Assign currencies to customers (the program defaults all to *base currency*)
- Assign the *Currency* field in all open and closed sales orders, purchase orders, and vouchers to *base currency*

All of this can be done via a single startup program named MCSTART. You must run this program before you can begin multi-currency processing.

MCSTART *Multi-Currency Startup* program

To run the multi-currency startup program, go to UT-A, *Run a DBA Program*, and type MCSTART and press <Enter>.

The first thing the program will do is take you to IM-B, *Enter Multiple Currencies*, where you can set up the various currencies you will be using. You can set up a minimum of one currency and return to IM-B later if you want to add the rest. At a minimum you must set up your *base currency*.

After escaping from that program, MCSTART automatically takes you to IM-C, *Enter Currency Exchange Rates*, where you can enter current exchange rates for each of your currencies.

After escaping from that program MCSTART takes you to a listing of vendors. The program assigns all vendors a default of *base currency*. You can then select just those vendors that transact in foreign currency and change them from *base currency* to *source currency*.

After completing vendor currency assignments, MCSTART takes you to a listing of customers. The program assigns all customers a default of *base currency*. You can then select just those customers that transact in foreign currency and change them from *base currency* to *source currency*.

After completing customer currency assignments, MCSTART asks you, as a first time multi-currency user, if you wish to assign the *Currency* field in all your open, and closed, and historical sales orders, purchase orders, and vouchers to base currency. You should do so, because at the time all these items were entered you were on a single currency system, which was your *base currency*. If you wish to change any open sales orders, purchase orders, and vouchers to *source*

Multiple currency startup

currency, you will have to manually change their *Currency* fields and adjust the amounts on the line items.

After this last routine is finished processing, you will be told that the multiple currency conversion is complete.

IM-A International Configuration

Field explanations

Multi-Tax Forms

If multiple tax codes apply to a sales order or purchase order and you want each tax code to be separately itemized and printed out in the bottom portion of the invoice, enter a Y in this field.

If you want only a single total for all taxes to print, enter an N in this field.

Excise Tax

If you will be using excise tax processing in which the tax is embedded in the item's selling or purchase price, enter a Y in this field.

Multi-Currency

If you will be using multi-currency processing, enter a Y in this field. Be aware that when you do so, you are switching the multi-currency system on and that you will not be able to run certain programs until you first complete your multi-currency setup. See *Multiple Currency Startup* at the beginning of the chapter for details.

Pay

This setting only applies if you are using DBA accounting. This option determines how currency gains or losses are handled on payments -- meaning payments you make to vendors and payments you receive from customers. If your country allows you to recognize the currency gain or loss at the time of the transaction, enter a Y in this field.

In some countries (such as Canada) you are required to recognize currency adjustments for payments at month end, rather than at the time payments are made or received. If so, enter an N in the *Pay* field.

Landed Costs

Enter a Y in this field if you will be using landed cost processing.

Auto-Tax Calc

This setting only applies if you are using DBA accounting. Set this to Y if you wish to have the capability of processing sales tax within AP-B, Enter Vouchers. Processing sales tax means that it will be posted to the *General Ledger* and also to the sales tax file where it can be ultimately be transferred for payment to the appropriate tax authorities.

30. USING DBA WITH YOUR ACCOUNTING SYSTEM

If you are using DBA's accounting system, daily and period end accounting procedures are covered in the *DBA Accounting* user guide.

DBA Manufacturing is designed to work independently from accounting (GL and AP), thus allowing you to use your existing accounting system or any accounting system of your choice. Interaction between the two systems is as follows.

GL accounts

You will set up approximately nine GL accounts in your accounting system that will be needed for making period end journal entries to reflect the activities of the DBA system.

Customer payments are entered in DBA and deposited in accounting

On a daily basis you will enter customer payments in DBA through AR-C, *Record Payments*, but you will make your bank deposits within your accounting system.

Payables can optionally be reconciled with DBA purchase orders

All your payables will be entered within your accounting system, including vendor invoices that pertain to purchase orders generated within DBA. If you want to match up DBA purchase orders with vendor invoices to check for accuracy and to make cost corrections to work orders and inventory, you can do so through PO-N, *Reconcile Vendor Invoices*.

Period-end journal entries

Once per period you will make a dozen or so journal entries in your accounting system to reflect the activities of DBA Manufacturing. All the information you need to make these entries is found on a single report: AI-C, *Period End Accounting Worksheet*.

Each time you run the report you can archive the data so that it doesn't get accidentally printed again and updated twice in your accounting system.

Once these dozen or so journal entries have been made, you can then run your financial statements (income statement, balance sheet, etc.) from within your accounting system.

Reconciliation reports

A recommended period end procedure is to reconcile the totals maintained in DBA for inventory, WIP, and PO's received not invoiced with those of your accounting system's general ledger.

Reconciling these balances is a good safety check to insure that all entries made in DBA are fully and accurately reflected in your general ledger.

Period end closing date

Another control over your data is provided by AI-B, *Reset Period End Close Date*, which lets you establish a closing date that prevents any transactions within DBA from being posted prior to that date. Therefore, you can coordinate DBA's closing date with that of your accounting system and prevent entries being made to closed periods.

Is it necessary to invoice and manage AR within DBA?

Ideally, invoicing and the management of accounts receivable should take place within DBA, because invoicing completes the work order in many cases and automatically relieves inventory. Because invoicing generates accounts receivable, it also makes sense to manage customer payments in the same system that generates the invoices.

You can, however, continue to invoice and track customer payments within your accounting system, as long as you don't mind the following.

- You will have to maintain two sets of customer records, unless you manufacture everything in DBA to stock.
- You will possibly have to set up item numbers for the items you sell in order to get them listed on invoices.
- You will have to relieve finished goods inventory in DBA through IN-C, *Enter Inventory Adjustments*, for all your shipments.

Generally speaking, it is much easier to invoice and track payments within DBA than it is to re-key that information into your accounting system. As you can see from the chart on the next page, there are no duplicate functions between DBA and your accounting system except for vendor information, if you invoice and track payments within DBA.

If you generate very few invoices with relatively few line items on them, which is the case for many small job shops, it is little extra effort to continue invoicing in your accounting system.

Another alternative for some companies is to print the invoice in DBA, then enter summarized information in your accounting system's Accounts Receivable module via AR vouchers, solely to get the invoice totals onto your AR aging. Then you can track and record customer payments within your accounting system. If you do this, you can shut off DBA's Accounts Receivable module via AI-A, *Accounting Configuration*.

In summary, unless you have a strong reason to continue invoicing and tracking payments within your accounting system, we recommend that you transfer these two functions to DBA to avoid re-keying of data.

DBA vs. Accounting functionality table

The following chart shows you the functions to be performed within DBA versus the functions to be performed by your accounting system.

Function	DBA	Acct System
Manufacturing	√	
Inventory Management	√	
Mfg-Related PO's	√	
Reconcile AP w/Mfg PO's	√	
Customer Master File	√	
Invoicing	√	
Record Customer Payments	√	
Customer Aging	√	
Sales Tax Reporting	√	
Sales Commissions	√	
Set Period End Close Date	√	√
Vendor Master File	√	√
Enter Vendor Invoices		√
Vendor Aging		√
Print AP & Payroll Checks		√
U.S. 1099 Reporting		√
Deposit Customer Payments		√
General Expense PO's		√
Bank Reconciliation		√
GL Journal Entries		√
Print Financial Statements		√
Payroll		√
Fixed Assets Accounting		√

GL accounts to be set up

The following general ledger accounts are used to reflect the various activities of DBA Manufacturing and should be set up in your accounting system's chart of accounts.

Accounts Receivable (asset)

This account will be debited as an offset to credit entries made to reflect your invoiced sales, freight, and sales tax for the period.

Sales (income)

Your invoiced sales total will be a credit entry to this account. A single figure is all that is needed for financial statement purposes. For detailed analysis of sales, a variety of reports are available in the *Sales Analysis* module.

Freight Expense (income or contra-expense)

Your invoiced freight will be a credit entry to this account. Some companies locate this account in the sales section of the income statement; others within operating expenses.

Taxes Payable (liability)

Your invoiced sales taxes will be a credit entry to this account.

Inventory (asset)

Different types of manufacturing and purchasing transactions will produce debits or credits to your inventory account. For financial statement purposes, all inventory can be a single account. For analysis purposes, you can break out your inventory value by item class or type using IN-F, *Print Inventory Value*.

Cost of Goods Sold (expense)

The cost of your invoiced items will constitute a debit to this account. You can also manually calculate the cost of goods sold using the formula detailed in the previous section.

PO's received not invoiced (liability)

This account is set up adjacent to your *Accounts Payable* account. When purchases or services are received, this account gets credited. When the vendor invoice is received and entered in your accounting system, this account gets debited and your *Accounts Payable* account gets credited.

Work in Process (asset)

Different types of manufacturing and purchasing transactions will produce debits or credits to your *Work in Process* (WIP) account. It is usually located on your balance sheet next to your regular inventory account(s).

Absorbed WIP Expense (expense)

This account reflects the direct labor, outside service, and factory overhead that

was charged to work orders and therefore was absorbed into the cost of your WIP inventory. If you set your estimated costs accurately in your routings and bills of material, these absorbed expenses should come close to equaling your actual expenses for direct labor, outside service, and factory overhead. This account is normally located as an expense somewhere close to your direct labor or factory overhead expense accounts. Unlike other expenses, which are debits, this is a credit entry that offsets the actual expenses adjacent to it.

Processing customer payments

The ideal way to handle customer payments is to invoice and track accounts receivable within DBA, and deposit the actual money and handle all banking activities within your accounting system.

Doing your invoicing within DBA closes the work order and inventory cycles and automatically creates entries in DBA's *Accounts Receivable* aging file. It therefore makes sense to record and track customer payments within DBA rather than re-key that information into your accounting system.

The actual payments received, though, should be entered into your accounting system so that all banking is done within that system for cash management and bank account reconciliation purposes.

What to do daily in DBA

Each day you should record customer payments in AR-C, *Record Payments*, and track open customer invoices via AR-F, *Print Aging*. See chapter 21 for details on additional *Accounts Receivable* functions.

What to do daily in your accounting system

Each day you will make a bank deposit for customer payments received. This should create a debit entry to your *Cash* account and a credit entry to your *Accounts Receivable* account.

Processing vendor invoices

All vendor invoices will be entered in your accounting system's *Accounts Payable* module. The following are a couple of things to keep in mind when processing payables.

Post PO-related invoices to the *PO's Received Not Invoiced* account

It is not necessary to charge DBA PO-related vendor invoices to specific GL accounts like you would with non-manufacturing vendor invoices for general expenses and purchases. Instead, post all DBA PO-related vendor invoices to your *Purchases Received Not Invoiced* account.

Why is it done this way? It's because in DBA the expense or asset is realized at the time of receiving the item or service, not when you enter the vendor's invoice. At the time of purchase receipt, a debit entry is made to an asset or expense account, and the credit entry is to *Purchase Received Not Invoiced*. Later, when you receive the vendor's invoice, a credit entry is made to *Accounts Payable* and a debit entry is made to *Purchases Received Not Invoiced*.

Using the *Purchases Received Not Invoiced* account simplifies payables entry and more accurately reflects when expenses are incurred.

Reconciling invoices with your purchase orders

Many companies like to match vendor invoices with the original purchase order to make sure that the vendor invoice matches what was purchased, or to correct job and inventory costs after-the-fact for purchase order prices that were not accurate.

The easiest way to do this is to run PO-I-F, *Print Received Not Invoiced*, and to compare purchase order prices with your vendor invoices. For all purchase orders that do not need price changes, you can close them out through PO-K, *Close Purchase Orders*.

If you need to make any purchase price corrections, you can do so through PO-N, *Reconcile Vendor Invoices*. This program will correct work order and inventory costs, and will also close the purchase order without having to do so through PO-K, *Close Purchase Orders*.

For financial statement purposes, reconciling your vendor invoices with purchase orders is not a requirement. If you feel your purchase order costs are reasonably accurate such that it's not worth the time making them perfect, you can safely skip the reconciliation procedure. In any event, don't forget to close your purchase orders via PO-K, *Close Purchase Orders*.

Period end journal entries

At the end of each period you will make a dozen or so journal entries in your accounting system to reflect all the activities that occurred during the period within DBA Manufacturing. The information needed to make these entries is found on the AI-B, *Period End Accounting Worksheet*.

This report not only provides totals for each type of activity, but it also suggests which debit and credit entries to make to specific GL accounts.

Summary totals are all that are needed in your accounting system to produce complete financial reports. If you need itemized detail, AI-C, *Period End Accounting Worksheet*, can be run in detail mode to provide a complete listing of detailed transactions behind each of its totals.

See the next section for a listing of GL accounts that should be set up in your chart of accounts for purposes of posting these period end journal entries.

The suggested entries provide true WIP accounting

The entries that are suggested by AI-C, *Period End Accounting Worksheet*, provide true work-in-process inventory accounting, where labor, material, outside processing, and overhead are absorbed into your WIP inventory such that these costs are then reflected in your inventory item costs and the cost-of-goods sold on your invoiced shipments. This type of accounting aligns your income accurately with your production costs.

You can also manually calculate cost of goods sold

You are not required to use all the suggested entries. Many companies choose not to absorb labor, material, outside service, and overhead into their WIP inventory and instead expense these as they occur through standard accounts payable entries.

Cost of goods sold can be calculated as follows.

Inventory Value at Start of Period
+ Material Purchases for Period
+ Direct Labor for Period
+ Outside Service for Period
+ Factory Overhead for Period
- Inventory Value at End of Period
= Cost of Goods Sold

Consult your accountant if you need help

Most accountants are well familiar with the various items on the *Period End Accounting Summary* report and/or the manual method of calculating cost of goods sold. If you need help, we suggest you show the report to your accountant

and get advice on what specific period end entries he or she would like made.

If you want to break Sales and COGS into more detail

For income statement purposes, some companies prefer to break their sales and cost-of-goods-sold into more detail than a single lump sum figure for each. You can get such a breakdown from AI-D, *Print Sales & COGS by Class*.

This report will break out your sales and corresponding cost-of-goods-sold by item class or by customer class. Using the data on this report, you can post your sales data in as much detail as you wish.

If you want to break Inventory into more detail

The AI-C, *Period End Accounting Worksheet*, shows a single figure for inventory. If you like to break your inventory into more than one account, you can run IN-F, *Print Inventory Value*, which lists your inventory value by item class and item type.

Period end balances to be reconciled

A recommended period end procedure is to reconcile the totals maintained in DBA for inventory, WIP, and PO's received not invoiced with those of your accounting system's general ledger.

Reconciling these balances is a good safety check to insure that all entries made in DBA are fully and accurately reflected in your general ledger.

Current inventory value

The total from IN-F, *Print Inventory Value*, added with the total from PO-J-B, *Print Inventory in QC*, should agree with the current balance in your general ledger *Inventory* account.

If for any reason your GL balance does not agree with these reports, make a journal entry to correct it. The offsetting entry can be made to your *Cost of Goods Sold* account.

NOTE: The IN-F, *Print Inventory Value*, balance is current as of the time the reports are run. In order to recreate inventory balance as of a prior date, you need to run IN-F, *Print Inventory Value*, as of today and also IN-N-A, *Print Month End Inventory Costing*, to determine the net change between today and the prior date. Other reports can be run as of a prior date.

Current WIP value

This total from JC-M, *Print WIP Summary*, should agree with the current balance in your general ledger *WIP* account.

If for any reason your GL balance does not agree with this report, make a journal entry to correct it. The offsetting entry can be made to your *Cost of Goods Sold* account.

Current PO's received not invoiced value

This total from PO-I-F, *Print PO's Received not Invoiced* added to the total from PO-J-B, *Print Inventory in QC*, should agree with the current balance in your general ledger *PO's Received not Invoiced* account.

NOTE: Unless you reconcile each vendor invoice through PO-N, *Reconcile Vendor Invoices*, these report totals will differ from your general ledger account by the amount of any price differences between your PO prices and those on your vendor invoices. If so, make a journal entry to adjust your *PO's Received Not Invoiced* account to agree with the sum of these two reports. The offsetting entry can be made to *Cost of Goods Sold*.

Accounts receivable

The total from AR-F, *Print Aging*, should agree with the current balance in your general ledger *Accounts Receivable* account.

Setting the period end close date

When you formally close a period in your accounting system, you should also run *AI-B, Reset Period-End Close Date*, in DBA to prevent any transactions from being posted within DBA prior to the close date.

This is a control procedure that insures that a closed period is truly closed, meaning that it is off-limits from any further transactions.

APPENDIX A – SETTING UP PRINTING

Printing set up consists of two phases. The first phase is to select the print formats and format settings you wish to use for printing forms, labels and checks. The second phase is to set up your printers at each workstation for printing system reports.

Any Windows compatible laser or ink jet printer can be used for printing graphical or text-based forms, labels, and checks, and for printing system reports (most of which are currently text-based). Narrow or wide carriage dot matrix printers can be used for text-based printing of forms, labels, and checks. A wide-carriage dot matrix printer is recommended for printing system reports.

DBA forms, checks, and labels are supplied by Fortune Forms, Inc. (800-884-8804 [phone] 888-884-7766 [fax]).

PRINTER DRIVERS: In this chapter we make reference to printer drivers. If you want to make sure you have the latest driver for your printer, there's a good web site (www.winfiles.com) where you can get current drivers for virtually any device that runs in Windows.

Laser and ink jet printing

Selecting & setting print formats for forms, labels, and checks

When printing with laser or ink jet printers you can print forms, labels, and checks in graphical formats that use proportional fonts, bold titles, and gray background shading. Text-based formats are also available. Print formats (graphical or text) are available for a universal form (in which a single form can be used for all forms printing), plain paper, and in the case of invoices, a pre-printed invoice form.

Selecting print formats

Print formats for forms are selected in various programs on the *System Defaults* menu. See the manual or *Help* file for specific details on each format.

- Statements: SD-P, *Accounts Receivable Defaults*.
- Acknowledgements, packing slips, invoices, and sales quotations: SD-M, *Sales Orders Defaults*.
- Purchase order and RFQ formats: SD-C, *Purchase Orders Defaults*.
- Customer quotes (estimates): SD-G, *Estimating Defaults*.

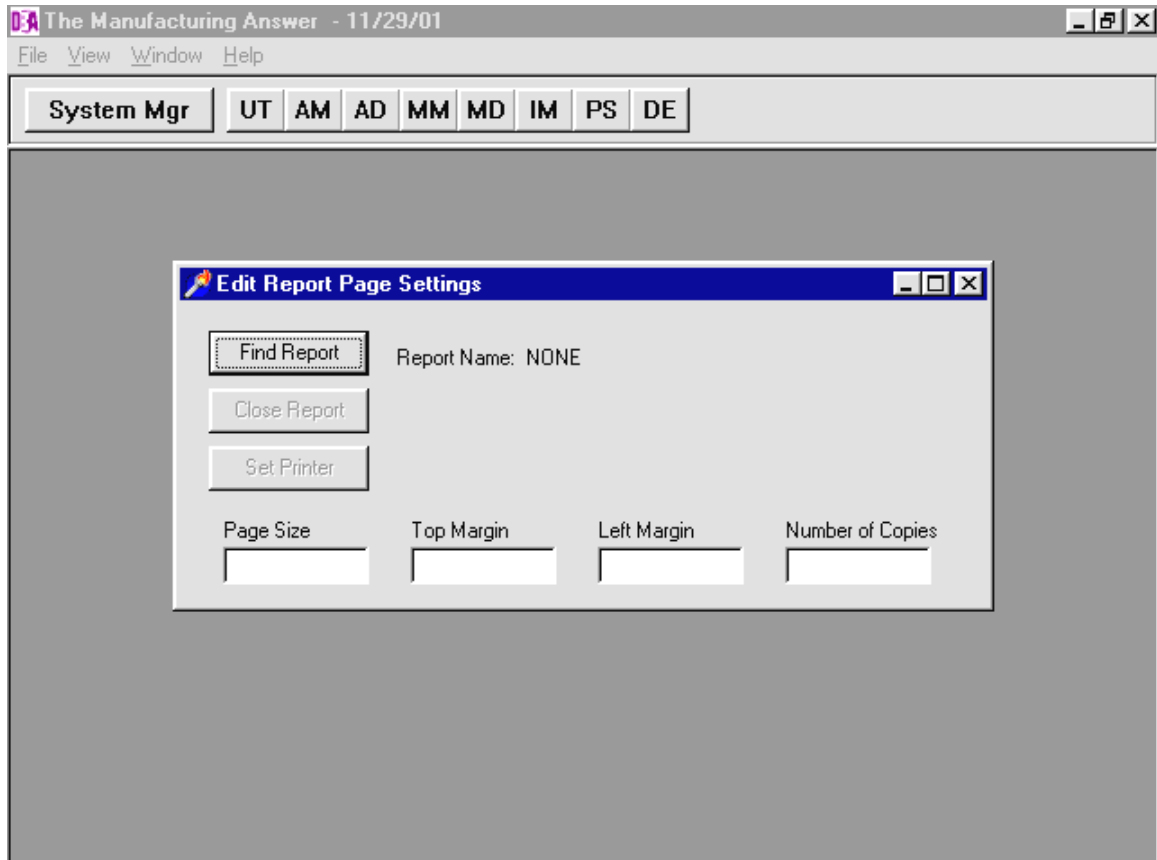
Setting graphical format defaults (UT-G)

If you select any of the graphical formats, you can go to UT-G, *Enter Graphical Print Settings*, and set the following defaults for each format: printer, page size, top margin, left margin, and number of copies. If you go into UT-G and the *Utilities* menu is covering a portion of the entry screen, click on any portion of the gray background and the menu will close and the entry screen will be fully displayed.

While selecting a graphical format from a lookup window in any of the default programs listed above, you will notice a format file name (with an extension of *.RTM) in parenthesis to the right of the format description. To change the settings for a particular format in UT-G, you must first open it by clicking on the *Find Report* button. All the *.RTM print formats are displayed in a window. Highlight the format you want to change by clicking on it, then click on the *Open* button. The current settings will be displayed.

For a listing of .RTM file formats and the forms, checks, or labels to which they correspond, see the on-line Help.

Once the format's settings are displayed, you can change them. Click on the *Set Printer* button if you want the form to print to another printer than your default printer. *Page Size*, *Top Margin*, and *Left Margin* are expressed in decimal inches. By changing these settings you can precisely control print alignment for each particular format. Set *Number of Copies* as desired. When finished, click on the *Save Report* button.



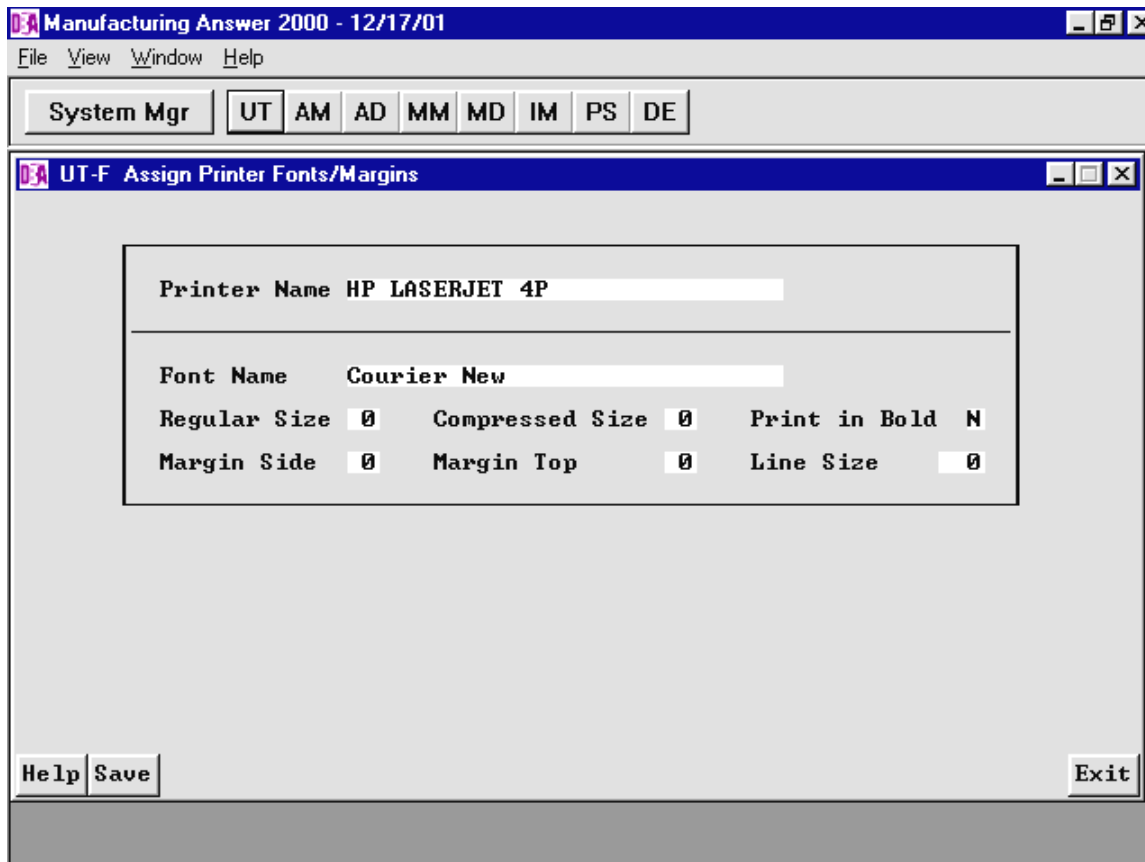
UT-G Enter Graphical Print Settings

Be aware that the settings made in UT-G, *Enter Graphical Print Settings*, are universal settings that are not user-specific. If two or more users will be printing the same format each to his own printer, leave the *Set Printer* setting set to the default printer so that it will print to each person's default printer, rather than designate a specific printer.

SHADING INTENSITY ON BLANK PAPER FORMATS: The plain paper print formats use gray shading in some areas to give the form a more graphical look. If you fax these forms, however, the gray shading may obscure some of the overlying text. Most printer drivers have a graphics or gray scale setting that can be adjusted. Set the intensity to the minimum setting and in most cases the shading will be suppressed, even when using *WinFax*.

Setting up printers for system reports (UT-F)

While forms, labels, and checks are available in graphical format, most of the reports in the system currently print in a text-based format using the *Courier New* non-proportional font. Rather than enter individual settings for each report like you can with forms, labels, and checks, you instead go to UT-F, *Assign Printer Fonts/Margins*, and enter settings that control the font sizes and margins for all printers used at each workstation.



UT-F Assign Printer Fonts/Margins

Go to UT-F, *Assign Printer Fonts/Margins*. A printer selection window will be displayed over the DBA entry program in the background. Your default Windows printer should be displayed, but you can select specific printers as well by clicking on the down arrow next to the printer *Name* field and selecting a printer from the lookup window. Select the printer you wish to define and click on the *OK* button. The window will close and you will be in the DBA entry screen with the printer's name displayed in the *Printer Name* field.

Press <Enter> to advance through the fields. The default setting in the *Font Name* field should be *Courier New*. For most printers the *Regular Size* field should be set to 11 and the *Compressed Size* field to 7. *Print in Bold* should be set to N.

With Windows printing there can be slight variations in print starting positions and line widths among different printers. The next fields allow you to adjust the printing for each of your printers. The *Margin Side* field allows you to move the left hand margin a number of characters to the right. The *Margin Top* field allows you to move the printing of the first line on the page a number of lines down. The *Line Size* field can increase the width of spacing between lines. You can start by leaving the margin and line settings blank and adjust them only if necessary.

When you complete these settings you will be asked if you want to save the printer information. Press <Enter> or click on *Yes*. The printer selection window will display once again. If you are finished defining printers, you can click on

Cancel to suppress the printer selection window, then you can click on *Exit* to get out of the program and return to the *Utilities* menu.

Repeat this procedure on each workstation, and be sure to define each printer used by that workstation. Be aware that these printer settings, unlike UT-O described in the first section, are specific to each user and do not apply to the company as a whole. (Note: these printer settings are stored in the TP5WIN.INI that is located in each user's local Windows folder)

Printing reports in landscape mode

System reports print very nicely in landscape mode and are easier to read because you can change the compressed print font from 7 point to a larger 9-point size.

Because you may want to print some reports in portrait mode (using a *Compressed* size of 7) and some in landscape mode (using a *Compressed* size of 9) we suggest you set up an additional driver for the same printer so that each driver can have its own font size settings. When you create the second driver, set the printer for *Landscape* mode, then go into UT-M, *Assign Printer Fonts/Margins*, and set *Compressed* to 9 for this new driver.

CREATING A SECOND PRINTER DRIVER: To create a second printer driver for the same printer, click Windows *Start*, then *Settings*, then *Printers*, then *Add Printer*. After indicating whether the printer is to be a local or network printer, you can then select a *Manufacturer*. Find your printer's manufacturer and click on it. In the *Printers* list box find your printer. Click *Next* and finish answering the rest of the prompts. Give the printer a name that is different than the one associated with the first driver. You now have two drivers, each with its own name, for the same printer.

Dot matrix printing

Selecting & setting print formats for forms, labels, and checks

With dot matrix printers forms and checks are printed in text-based formats only. While many dot matrix printers can print graphical output, each printer interprets the *Arial* font used in DBA forms differently, often resulting in incomplete fields and incorrect alignment. For printing text-based formats we recommend using a *Generic / Text Only* printer driver (see below) that produces consistent alignment from printer to printer and prints significantly faster than printing via a printer-specific driver. Text-based print formats are available for a universal form (in which a single form can be used for all forms printing), plain paper, and in the case of invoices, a pre-printed invoice form.

DOT MATRIX ON WINDOWS NT: Be aware that if you print to a dot matrix printer from a *Windows NT* workstation, you will not be able to print forms less than 11" in height, such as checks and labels. Print these forms from a *Windows 95* or *Windows 98* computer.

Selecting print formats

Print formats for forms and checks are selected in various programs on the *System Defaults* menu. See the manual or *Help* file for specific details on each format.

- Statements: SD-P, *Accounts Receivable Defaults*.
- Acknowledgements, packing slips, invoices, and sales quotations: SD-M, *Sales Orders Defaults*.
- Purchase order and RFQ formats: SD-C, *Purchase Orders Defaults*.
- Customer quotes (estimates): SD-C, *Estimating Defaults*.

Installing and using the *Generic / Text Only* printer driver

To install the *Generic / Text Only* printer driver, click *Windows Start*, then *Settings*, then *Printers*, then *Add Printer*. After indicating whether the printer is to be a local or network printer, you can then select a *Manufacturer*. Click on *Generic*. In the *Printers* list box you will then see *Generic / Text Only* as the only listing. Click *Next* and finish answering the rest of the prompts. You can give the printer a unique name if you wish, and you can also indicate whether you want the printer to be your default printer or not.

Whenever you print forms, be sure and select the *Generic / Text Only* printer (or whatever name you may have assigned it). If you use your dot matrix printer primarily for forms printing, it is advisable to make it your default printer so that it always displays as your first choice. To make a printer your default printer, go to *Start/Settings/Printers*. Click and highlight your printer icon. While highlighted, click on *File* and then click on *Set as Default* so that it is check marked.

Setting up printing for AP checks

The *Generic / Text Only* printer driver must have its *Paper Source* setting changed for the printing of AP checks. The default setting is *Continuous – Page Break*. It must be changed to *Continuous – No Page Break* in order for the checks to print correctly with a 7” page length.

If your dot matrix printer is dedicated solely to check printing, you should modify the *Generic / Text Only* driver as follows. Click on *Start*, then *Settings*, then *Printers*. Click and highlight your *Generic / Text Only* printer (or whatever name you may have assigned it). While highlighted, right click to display a menu and then click on *Properties*. Click on the *Paper* tab. Change the *Paper Source* setting to *Continuous – No Page Break*.

If you use your dot matrix printer for both forms and checks, we recommend installing a second copy of the *Generic / Text Only* driver and use one for forms and the other for checks. The only difference between the two drivers would be the *Paper Source* setting (as described above). To install a second copy of the driver, follow the instructions in the previous section for adding a printer; however, you will be presented with a screen that offers two choices: *Keep Existing Driver (Recommended)* and *Replace Existing Driver*. Choose *Keep Existing Driver* and a second copy of the driver will be created that can then be modified for the different *Paper Source* setting.

Be aware that even with one copy of the *Generic / Text Only* printer driver you can change the *Paper Source* setting on-the-fly at time of printing. When you click on the *Print* button in any DBA print program, the default printer is displayed. You can then click on *Select* and that printer’s properties are displayed. Change the *Paper Source* setting and then close the window by clicking *OK* and proceed with printing.

IF YOUR CHECKS DO NOT PRINT PROPERLY: If your checks print on 11” centers instead of the correct 7” page length, the *Paper Source* setting is still set to *Continuous – Page Break* instead of *Continuous – No Page Break*. Either print the checks again and remember to change this setting on-the-fly as described above, or install a second copy of the *Generic / Text Only* driver solely for check printing and make *Continuous – No Page Break* the default *Paper Source* setting for that printer.

WINDOWS NT/2000 WORKSTATION: The *Generic / Text Only* driver for this operating system will not print on 7” centers no matter the settings. You will have to print continuous form checks from a Windows 95 or Windows 98 workstation.

Printing system reports

System reports print best using the same *Generic / Text Only* printer driver used for forms (with the *Paper Source* set to *Continuous – Page Break*). This will give you the fastest printing speed; however, it requires a wide-carriage printer due to

the wide format layout of the majority of reports and because the *Generic / Text Only* driver does not support condensed printing.

If you have a narrow-carriage printer, you might be able to print reports in condensed print by printing through your printer-specific driver instead of the *Generic / Text Only* driver. If so, you must set up your printer through UT-F, *Assign Printer Fonts/Margins*. See *Setting up system reports* under *Laser & Ink Jet Printing* for details. On some dot matrix printers the 11 and 7-point standard settings result in too wide a report by a few characters. You might try different point sizes until you find a setting that works for your printer. Another option is to print reports in landscape mode. See *Printing Reports in Landscape Mode* under *Laser & Ink Jet Printing* for details.

Faxing forms or reports

Many companies prefer to fax documents such as purchase orders directly to their vendors rather than send them by mail. An easy way to do this within DBA is to use a product such as *WinFax*, which can be set up as a printer driver. When you click on the *Printer* option when printing any DBA forms or reports, *WinFax* will intercept the printing and will fax the document instead. Make sure you use the plain paper formats so that the form will be complete with borders, etc. You might also consider using DE-A, *Export Data*, to export names and fax numbers into the *WinFax* database so that they are available for automatic dialing.

APPENDIX B - SEQUENCE OF EVENTS

The manufacturing process involves five basic spheres of activity: planning, items maintenance, work orders, and purchase orders. The sequence of events for each of these spheres of activity are presented in checklist fashion within this appendix.

Manufacturing planning

Planning is the most important set of activities in a manufacturing enterprise. Planning encompasses a range of activities, including inventory planning, scheduling, and capacity requirements. Inventory planning involves the study of *Inventory* and *MRP* reports to determine what, how many, and when to manufacture and purchase materials, subassemblies, and finished goods. Scheduling involves studying *MRP* and *Scheduling* reports to adjust start and finish dates on work orders and purchase orders in order to respond to changing requirements. Capacity requirements involves studying loads on work centers to determine manpower and machine requirements to meet present and future production needs. All of these planning activities interact with one another.

The following is an outline of the steps required for the manufacturing planning process. These are not in any sequential order.

Planning work orders and purchase orders

- Print reorder report to determine what is to be purchased and manufactured (IN-D).
- Convert sales orders (SO-N) and estimates to work orders (ES-E).
- Enter forecasts for inclusion in MRP generation (MR-A).
- Generate material requirements (MR-F).
- Generate work orders and purchase orders via MRP (MR-I, MR-J) or manually (WO-A, PO-A).
- Generate RFQ's (if applicable) via MRP (MR-K).

Scheduling existing work orders and PO's

- Print *Reorder Report* to make sure PO receiving dates and WO finish dates align with requirements (IN-D).
- Print *Sales Order/Work Order Schedule* to coordinate production with shipments (SO-O-G).
- Run infinite or finite scheduling programs to schedule or reschedule work orders (SH-E, SH-F).
- View or print work center loads to see if schedules can be handled with existing capacity (SH-K, SH-L).
- Change capacity (overtime, hiring, added machinery, etc.) as needed; adjust work centers accordingly (SH-O).
- Reschedule work order finish dates as needed (load leveling) to fit schedule to available capacity (SH-A).

- Change work order dates (SH-A) and purchase order dates (PO-A) in response to EXPEDITE and DELAY messages from the Order Action Report in the MRP module (MR-H).
- Assign sequences (operations) to machines, if applicable (SH-D).

Key Reports

- Reorder Report (IN-D)
- Forecast Report (MR-B)
- Work Order Schedule (SH-G)
- Machine Schedule (SH-J)
- Work Center Load (SH-L)
- Order Action Report (MR-H)

Items maintenance

Items maintenance involves regular attention to the data that affect the planning process, including inventory reorder levels, lead times, standard costs, time standards, and inventory stock status accuracy.

Inventory management

- Set reorder levels, reorder amounts, lead times as needed (MR-D).
- Run lead time generator for manufactured items, if desired (SH-N).
- Verify on-hand quantities through cycle or selective counting or complete physical inventory (IN-J, IN-K).

Item specifications

- Update standard costs as required (IN-L-A, IN-L-E).
- Update bills of materials as required (BM-A).
- Update routing time standards as required (RO-A).
- Perform cost rollups as required (BM-G).
- Update estimating material costs as required (ES-H).
- Update work centers for capacity, rates (RO-C, SH-O).

Key reports

- MRP Parameters (MR-E)
- Physical Check (IN-J)
- Inventory Value (IN-F)
- Print Bills of Material (BM-B)
- Routings Report (RO-J-A)
- Costed Bill of Material (BM-G)
- Material Costs (ES-I)
- Print Work Centers (RO-J-B)

Work orders

Work orders consists of all the activities involved in physically preparing and processing production orders through the factory, including generating the orders, printing documents such as shop travelers and pick lists, processing labor, materials, and outside processing, and reporting finished production.

- Create work orders manually (WO-A), by converting sales orders (SO-N), by converting estimates (ES-E), or by generating via MRP (MR-I).
- Modify the work order routing (WO-K-A) or work order bill of material (WO-K-B) if needed.
- Generate multiple date work orders (WO-K) or generate multiple assembly work orders (WO-K-D) if needed.
- Release work orders when ready to go to shop floor (WO-B).
- Print the following shop paper as needed: shop travelers (WO-C, DC-G), pick lists (WO-D), labor cards/labels (WO-E, DC-E).
- Purchase materials for specific work orders (PO-A).
- Issue materials to work orders from inventory (WO-G).
- Enter labor manually (WO-F) or via data collection (DC-A, DC-C).
- Enter any miscellaneous or extra costs (WO-H).
- Report production of finished goods and optionally backflush materials and optionally close work order (WO-I).
- Close the work order (WO-J).

Key reports

- Work Order Schedule (WO-L-B)
- Work Order Schedule (SH-G)
- Work Center Backlog (WO-L-C)
- Work Center Schedule (SH-I)
- Work Order Status (WO-L-A)
- Work Order Status (SH-H)
- Work Order Shortages (WO-L-F)

Purchase orders

Purchase orders consists of all the activities involved in coordinating purchases and receipts of materials and services from outside vendors.

- Enter purchase orders manually (PO-A) or convert from RFQ's (PO-G) or generate via MRP (MR-J).
- Print purchase orders (PO-B).
- Receive purchase orders (PO-C).
- Close purchase orders if not using *Accounts Payable* module (PO-K).
- Enter vendor invoices if using *Accounts Payable* and automatically close the PO when fully received and invoiced (AP-C).
- Transfer inspected items from QC to inventory, if applicable (PO-J-C).

Key reports

- Open PO Listing (PO-I-A).
- Closed Purchase Orders (PO-I-B) for buying history
- Inventory Transactions (IN-E) for buying history

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